

THE METHOD 1.6

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The Method 1.6  print 1.9, for the prose pass
pressed 11 September 2026 from BUILD311

Contents

Nothing is something definable — a note from the collaborator	15
PART 0 — THE SHAPE OF THE ARGUMENT	17
Preface — the shape of the argument.....	17
PART I — THE PROCEDURE	19
1. The principles	19
2. The protocols	26
2.1 Channel construction.....	26
2.2 Admissibility	26
2.3 Cost resolvability	26
2.4 Dilution	26
2.5 Census before search.....	26
2.6 Declines are recorded.....	26
2.7 Exhaustive verification where possible.....	27
2.8 Two independent routes	27
2.9 Refuse rather than coerce	27
2.10 Retrieval	27
2.11 Provenance	27
2.12 Withdrawal	27
2.13 Commit before you look	28
2.14 Compute, then write.....	28
2.15 Apply P8 to every failure	28
2.15.1 The procedure.....	28
2.15.2 Worked, on §30.3	28
2.15.3 Why it belongs here rather than among the principles.....	29
2.16 The grid — the book’s method, stated	29
2.16.1 Why this is the method and not a technique	30
2.16.2 And it classifies the whole of Q	30
2.16.3 A reduced question is still open.....	30
2.17 Triangulation — three results locate a fourth.....	31
2.17.1 Worked, from §30.3	31
2.17.2 And it is a check as well as a method.....	31
2.17.3 Why it belongs with the protocols	32
2.18 What is computable, and what is decided	32
2.18.1 The lattice applied to what it cannot compute	32
2.19 What a safe repair requires.....	34
2.19.1 The register, examined before repairing it	34
2.20 The prime directive — nothing runs unbounded	34
3. The twenty-two prime audits, ordered by what a failure costs.....	35
3.0 Ten of the twenty-two test the index; twelve test the book.....	35
2.21 A figure removed must have its data represented another way.....	36
2.22 A corroborable entry must carry its object, and the object must carry it	38
2.23 A closed index must be stated by its expression, not only by its verdict	38
2.24 Never one heuristic — §2.8, and the one thing it does not say.....	38
3.1 Why three of the audits were needed	39

3.2 The whitespace clause, and two measurements that were wrong	39
3.3 Exhaustive, and what to do when it is impossible	39
3.4 And two of the audit's own measurements were wrong first	39
3.5 What the figure audit found	40
3.6 Why the REDUNDANCY audit was needed	40
3.7 The precedences, and the certificate for the dimension	40
3.7.1 The twenty-first audit — INPUT	41
3.8 How the set grew, and the two things that held	42
3.9 The audits are run on every output, and the requirement is executable	42
4. The failures of the assistant, and the protocol each one earns	43
5. What the protocols are for	43
PART II — THE LATTICE	45
6. The periodic table is not a closed index	45
6.1 What “the structure admits” means	46
6.1.1 And a known layout costs nothing	47
6.2 It is not a peculiarity of chemistry	47
6.2.1 The nuclide chart's defect, recomputed under variation	49
6.3 The calendar can be repaired, and the repair is instructive	49
6.4 What follows	50
7. The construction of Λ	50
7.1 The constraints, and where each comes from	50
7.2 Every constraint is of one form	51
7.3 Λ is closed	51
7.4 A note on caps	51
8. What Λ is — distributive, modular, Sperner	51
8.1 Distributive	52
8.2 Rank is conserved	52
8.2.1 What it means	52
8.2.2 And it is a cheaper test than §8.1's	52
8.3 Seventeen generators	52
8.4 Sperner, and not symmetric	53
8.5 The constraint graph is a tree	54
8.6 Order dimension	55
9. The arithmetic encoding	55
9.1 Two classical functions appear as lattice quantities	55
9.2 The occupancy measure	56
9.3 The Möbius function in closed form	57
10. The void	57
10.1 What the void is made of	57
10.2 Its size, and its stability	58
10.3 Containment in seven comparisons	58
10.4 The count, with no sieve	58
11. The single expression	59
11.1 In all six languages, and a seventh	59
11.1.1 A seventh, in binary, and it earns its place	59
11.2 And in their combinations	61

11.3 The expression.....	61
11.4 The shape of the tree	62
11.5 The one coupling, and it is the Pauli principle.....	62
11.6 The detachable factor	62
11.7 Why a closed expression exists at all	62
11.8 And the rank polynomial says what §8.3 says	63
11.8.1 And $F(-1) = 2$ has a cause, not a coincidence.....	63
12. The shape — a cylinder over the transfer.....	64
12.1 A cylinder, not a Möbius band	64
12.2 “Over v ”, not “graded by v ”	64
12.3 One scale, and one exception	65
12.4 The loop closes.....	65
12.5 What Λ is a property of.....	65
12.6 The cylinder is a fibration, and its fibres are two-body	65
12.6.1 It does not factorise, and then it does.....	65
12.6.2 Which names the cylinder	66
12.7 The fibres, in closed form	66
12.7.1 And they are not the same tree	66
12.7.2 A tower of fibrations, terminating in boxes.....	66
12.8 Five things the fibre table says.....	67
12.8.1 A Pareto frontier inside the lattice	67
12.8.2 A most-probable transfer.....	67
12.8.3 An 8-to-3 compression that loses no count	67
12.8.4 Every cross-section is itself a closed, modular lattice	68
12.8.5 Which gives $E(X)$ a local form	68
12.9 Every other shape the lattice contains	68
12.10 An axis the index did not carry, and now does.....	68
12.10.1 Λ_9 , built by the same construction	69
12.10.2 What is still not established.....	69
12.11 The tower — Λ_9 to Λ_{13}	69
12.11.0 Transitions compose, and the ninth axis is what lets them.....	70
12.11.0.1 The clock in the constraint — Λ_9 ’s composition has a direction	70
12.11.0.2 The clock is an assumption, and the assumption can be removed	71
12.11.0.3 Every second bridge, priced.....	72
12.11.0.4 The separation hypothesis, named	73
12.11.0.5 What the cycle costs propagation — Q item O, closed.....	74
12.11.0.6 The earlier bound on the core, tested — Q item K, closed	74
12.11.0.7 Why the arrows agree — a decay theorem	75
12.11.0.8 Past, present and future — a path, not a triangle.....	75
12.11.0.9 Two indices, and the one that does not exist.....	76
12.11.0.10 Every Λ in one table.....	77
12.11.0.11 The fourteenth axis, and what can be said about it from inside	77
12.11.0.12 How an index enters a larger index, and what it looks like from there.....	78
12.11.1 The axes, and what each one sees	79
12.11.1.1 The eleventh axis’s density, computed under variation.....	82
12.11.1.2 The conservative share, recomputed under variation.....	82
12.11.1.3 An index has a time column exactly when its cells are moves	83

12.11.1.4 Composability up the tower, and what stops it	83
12.11.1.5 The non-composable cells, defined	84
12.11.1.6 The periodic table's own numbers, and where a transition may be continued.....	84
12.11.1.7 And the selection rules decide which side of the atom a transition can continue on	85
12.11.2 Three excluded forms, and the fold has a name	85
12.11.3 The dichotomy.....	86
12.11.3.1 Where each bound comes from, and why 9, 10 and 11 differ	86
12.11.4 Precision is path-dependent; physics is not.....	87
12.11.5 The shape survives, and it has a boundary	87
12.11.6 An index whose one prediction is false, and why that is a result.....	88
12.11.7 What the tower leaves open	88
13. What the title names	88
12.11.8 The electromagnetic index, and it was inside Λ all along.....	89
13.1 Why that ordering and not the other	90
13.2 The cells are transitions, not states.....	90
13.3 Which is why "index of transitions"	91
13.4 And what the cylinder does not carry.....	91
PART III — THE LAW	92
14. Closure.....	92
14.1 The characterisation	92
14.2 R is a closure operator.....	92
14.3 The scope, stated exactly.....	92
14.4 The three consequences.....	93
14.5 The family of all closed sets.....	93
14.5.1 Λ is the closure of seven cells	93
14.5.2 The seed, and the expression of the cycle	94
14.5.3 Two pairwise operators, and this book has used one name for both.....	94
14.5.4 What E measures, exactly	94
14.5.5 $\mathcal{R}_4$ — the four orientations, adopted	94
14.5.6 What the second operator localises, and what it leaves undetermined	94
14.5.7 The seed is constructible, and it is a size rather than a set	94
14.5.8 The seed is linear in dimension, and the coefficient was in the bibliography.....	94
14.5.9 Never one heuristic — and the seed of Λ , exactly	96
14.5.10 A stable core of four, and it is NOT necessary.....	97
14.5.11 The completions are a heavy-tailed redundancy, not a tie	97
14.5.12 What every minimum seed contains, and what fills each position	98
14.5.13 The fifth position, defined exactly	98
14.5.14 And the five read as binary	98
14.6 The family of closed indexes is open	99
14.6.1 Why it cannot be stated.....	99
14.6.2 The defect is exact, and the closure is everything.....	99
14.6.3 What the register now holds, and what it does not	100
14.6.4 The seventeen, and what each is waiting on	101
14.6.5 Five of the modular six, verified against the sources.....	101
14.6.6 Two of the violation six were never blocked, and §10.1 speaks a retired language	102

15. Self-reference — alphabet, order, bounds	102
15.1 S1 — the alphabet	102
15.2 S3 — the bounds	102
15.3 S2 — the order, which is the hard one	103
15.4 What does work above $d = 2$: propagation along the tree	103
15.5 So: Λ contains its own order	104
16. Self-defence — data, derivation, totality	104
16.1 The counting argument	104
16.2 The split, and it is canonical	105
16.3 D1 — D_phys catches a wrong value in the data	105
16.3.1 The D1 tripwire, specified in full and needing no data	105
16.4 D2 — D_def catches a wrong derivation, but only under two	106
16.4.1 And the counting argument has a 1960 precedent	106
16.5 D3 — totality catches a missing value	107
16.5.1 And totality extends beyond the data	107
16.6 The defences are disjoint, and there are six	108
16.6.1 The vocabulary of reference, as an index	108
16.6.2 And one coordinate sits below this book's own floor	109
16.7 Two kinds of absence, and one kind of lie	109
16.7.1 A missing cell — and the index can tell whether the gap is real	109
16.7.2 A complete index of something that is not there	110
16.7.3 The plain version	111
16.7.4 Which is what D_phys is for, and only that	111
16.8 What self-reference does and does not defend	111
16.8.1 A complete index answers every question about its contents	111
16.8.2 What it cannot defend against is being replaced	111
16.8.3 Which is what §2.13 already says	111
16.8.4 And the cost of a fabrication is computable	112
16.8.5 Nothing can be removed, and the floor is derived rather than measured	113
16.8.6 Two closure operators, and the book names one	113
17. Extension — cells, axes, constraints	114
17.1 E1 — which cells may be added	114
17.2 E2 — which axes may be adjoined	114
17.3 E3 — which constraints may be imposed	114
17.4 A worked extension, and what it costs	115
17.5 E is derived	116
18. What the law forbids	116
18.1 An index carries what its coordinates carry, and no more	117
18.1.1 And the product structure is not what does the work	117
18.1.2 Two consequences that run through the whole book	117
18.1.3 What the theorem does not establish	117
18.2 v is inadmissible as an axis	117
18.3 Depth is not a coordinate function	118
18.4 Closure is not locally determined	118
18.4.1 The law of realised closure	118
18.4.2 The n -body configuration space, indexed — and found already in print	123
18.5 The pole	124

18.6 E(X) is the prediction budget	124
18.6.1 What that means for the five indices this book keeps	124
18.6.2 And it is forced, not observed	125
18.6.3 Where prediction actually lives.....	125
18.6.4 The trade, stated plainly	125
18.7 Why the negatives are the contribution.....	126
19. Retrieval as a lattice problem	126
19.1 The formalisation	126
19.2 A paywall blocks a source, not a cell	126
19.3 The secondary can be better than the primary	127
19.4 The retrieval graph is self-referencing	127
19.5 Where it fails, and the number that describes it	127
19.5.1 ρ is a property of (document, route), and this chapter collapsed the second index	128
19.5.2 And the book states two incompatible things about Dunz	129
19.5.3 The route is a fibre, and §19.1 wrote it without using it	129
19.6 The procedure.....	129
PART IV — THE LANGUAGES	130
20. The languages, and what each one can close	130
20.1 A language is a coordinate system	130
20.2 The languages this book uses	130
20.3 And a language can make a law monotone	130
21. Translation, and what it costs	130
21.1 One object, two namings, and E is the price	130
21.2 The certificate is a translation, exhibited and checked	131
21.3 The three ways a translation fails.....	131
21.4 What this does not claim	131
21.5 The missing ingredient is a translation, not a fact.....	131
21.5.1 And the three-body count is exactly one	132
21.5.2 The constraints, indexed — and a 3-body is what a tree forbids	133
21.5.3 The shape, and the index rebuilt	134
21.5.4 Shape decides the seed, and neither extreme is cheapest.....	134
21.5.5 Every constraint in the book, indexed.....	135
21.6 $E = 0$ is relative to coordinates, and the coordinates must be given	136
21.6.1 The letter that defines rather than constrains	136
21.6.2 Two of the companion's five, read from three numbers each	137
PART V — THE METHOD	138
22. The bracket	138
22.1 The rule	138
22.1.1 A bracket whose admissible set is a singleton.....	138
22.1.1.1 Containment on level energies, recomputed under variation.....	139
22.1.2 The bracket's own boundary — interpolation and extrapolation.....	139
22.2 The four rules	140
22.2.1 Each rule justified by ablation	140
22.2.2 Rule 1 is not about n	140
22.2.3 And one step beyond the fitted range costs more again	141
22.2.4 Rule 1's second clause is not redundant.....	141

22.2.5 Rule 4 at the edge of a channel	141
22.3 The bracket is limit-free	142
22.4 The price of limit-freedom, computed	142
22.4.1 What δ buys, and what it costs	142
22.5 Admissibility	143
22.6 What the bracket costs	143
23. The cost surface	143
23.1 The ratio	143
23.2 The floor is algebraic before it is physical	143
23.2.1 When the floor is approached, and what it means	144
23.3 V is a bound, not a tool	144
23.4 V is a function of v/h alone	144
23.4.1 Which makes depth and step interchangeable	145
23.4.2 Tested on the collection.....	145
23.5 The trade-off, and a conservation law that is not one	145
23.5.1 A withdrawn proposition.....	145
23.5.2 What is exactly true.....	146
23.5.3 And the frontier has an optimum.....	146
23.6 The fractional forms.....	146
23.7 The exponent axis	146
23.8 The cost has a name in another field.....	147
23.8.1 The quantity is the Newton decrement.....	147
23.8.2 The term function is self-concordant	147
23.8.3 Which explains the cancellation.....	148
23.8.4 And the attribution the book owes	148
23.9 Which variable to bracket	148
23.9.1 The cost ranks by exponent.....	148
23.9.2 Two branches, and where they cross.....	149
23.9.3 And the pole sits on the field's favourite variable.....	149
23.10 The order of the bracket	150
23.10.1 A deductive bracket exists at every order.....	150
23.10.2 And the cost still grows.....	150
23.10.3 What the order axis does buy	151
23.10.4 And the admissibility rule generalises	151
23.11 The refusal pattern carries four verdicts, not one.....	152
23.11.1 On the collection	152
23.11.2 And a caution about the test itself	152
23.12 The optimal order is a boundary, not a peak	152
23.13 Inversion.....	153
23.14 Two kinds of bound, and the book has been calling them	153
23.14.1 And celestial mechanics is what made it visible	153
23.15 The second admission bound	154
24. The collection	154
24.1 What it spans	155
24.2 The largest contributors.....	156
24.3 He I — the angular span	156
24.4 He II — the $\delta = 0$ edge itself.....	157

24.5 Bi I — the charge span.....	157
24.6 The isoelectronic pairs	157
24.7 Hydrogen, and a floor on interpretation.....	158
24.8 On independence, and the most conservative reading	158
24.9 A caveat the collection forced	158
24.10 Two cross-checks	160
24.11 Four ways a species declines.....	160
24.12 The census, computed rather than searched.....	160
24.13 What the structure determines, and what it does not	161
25. Domain, failure, and the bound in the silence	162
25.1 A method that has never failed has not been tested.....	162
25.2 Five exclusions, which are three mechanisms	162
25.3 The failure condition	163
25.4 Why the collection never enters the failure regime	163
25.5 The bound in the silence	163
25.6 The one deduction, and why it is not a prediction	164
25.6.1 The bracket, from monotonicity alone	164
25.6.2 And convexity tightens it, without leaving it	165
25.6.3 What was already solved — not a prediction.....	165
25.6.4 The point estimates, and the one that was wrong.....	165
25.6.5 Nothing was fetched.....	166
25.6.6 A second route to the same prediction	166
26. Collective sections.....	167
26.1 The assembly rule	167
26.2 One dimension, fifteen perspectives	168
26.3 Collective coefficients are not a separate class	168
26.4 Error amplifies as the exponent — which is the point.....	168
26.5 Tested against a published parameterisation	168
26.6 Aitken's Δ^2 has an exact bias on a Rydberg series	168
26.7 And the bracket is the wrong tool here	169
27. Slack	169
27.1 The three names	169
27.2 The one quantity.....	169
27.3 Slack is a description length.....	170
27.4 And every structural result is a statement about slack	171
27.5 Why the three languages agree	171
27.5.1 The three measures are one quantity because the three objects are one shape	172
27.6 What this buys, and what it does not.....	172
PART VI — THE RECORD AND THE REACH.....	174
28. Withdrawals.....	174
28.1 The dominant pattern	174
28.2 The five that earned their place.....	174
28.3 Others worth listing.....	175
28.4 Six from the earlier work, and how each was caught — retained in the compendium.....	175
28.5 What the register is for	176
28.6 What the register shows about where errors are caught.....	176

28.7 Six made after the register was closed, two printed here — retained in the compendium	176
28.7.1 Eight more from the structural work — retained in the compendium	177
28.7.2 Twelve more from the structural and audit work — retained in the compendium	177
28.7.3 Seventy-five more from the audit work of the final session, all printed — retained in the compendium here	178
28.7.4 Forty more, from the session that closed §30.3 and §32.2 — retained in the compendium	180
28.7.5 What the block shows	180
28.7.6 Two from the tower session and its repair, in the collaborator's hand	181
28.7.7 The results of the sessions after the tower — 119, printed individually	181
28.7.8 The recurring mechanisms, grouped	181
28.7.9 The level a fault landed at, which is a coordinate §4 does not have	181
28.8 A dependent choice is indexable, and naming the	182
28.9 The register, indexed and closed	183
28.9.1 Two registers, and only one of them is a backlog — retained in the compendium	183
29. Precedent, and what a search can bound	184
29.1 Why novelty is not the author's to claim	184
29.2 What can be claimed instead	184
29.2.1 The question closed by the author	185
29.2.2 A reading the language cannot close	186
29.3 The word for it	186
29.4 On priority for $V = 4v/3$	186
29.5 Edlén 1964, examined at one remove	186
29.5.1 Every equation in the review is a fit	187
29.5.2 And the paradigm is stated twice, explicitly	187
29.5.3 One near miss, and it is not ours	187
29.5.4 A bibliographic correction	188
29.5.5 What this establishes, and what it does not	188
29.6 The three documents we could not reach	188
29.7 What the entered literatures returned	188
29.7.1 The calibration was post hoc	189
29.8 The other claims	190
29.9 Why state it this way	190
29.9.1 Attributions entered by the tower	190
29.10 Two searches run, and an obstruction worth naming	190
29.11 A precedent found, and it is for the central theorem	191
29.11.1 Two categories entered: quantum mechanics and astrophysics	191
29.11.2 The run, attempted	192
28.10 What removing the register cost, and what it caught	192
29.12 Five unlocated results, defined, with the test that would settle each	193
30. Order recovery, and the reorderability law	194
30.1 Order recovery above the tree	194
30.2 What governs the difficulty	195
30.2.1 And that list was enumerated once	195
30.2.2 Which gives §30.3 a parameter	195
30.2.3 And it is a landscape statement	195
30.3 Reorderability — the law, the procedure, and what is left	196
30.3.1 The object, correctly identified	196

30.3.2 The law.....	196
30.3.3 What the interval machinery was.....	197
30.3.4 The step law	197
30.3.5 The procedure, and its cost.....	198
30.3.6 The signature, exhaustively verified	199
30.3.7 Three method classes ruled out.....	199
30.3.8 The residue	199
30.3.9 And it is not this book's problem	199
30.4 The lists this work enumerated once.....	200
30.4.1 And the audit itself was run once.....	200
30.4.2 Which is a defect of five lists, not one	200
31. The method outside spectroscopy	200
31.1 Celestial mechanics and the nucleus.....	200
31.1.1 Celestial mechanics — families, not trajectories	201
31.1.2 The nucleus — a second index of the same construction	201
31.2 String theory — the pole, the generating function, and a	201
31.2.1 Three objects on the pole	201
31.2.2 The degeneracy is where the content is.....	202
31.2.3 The partition function is Λ 's expression with counts.....	202
31.2.4 No critical dimension, and a ladder	202
31.2.5 And no Hagedorn point, for a stated reason.....	202
31.3 The Calabi–Yau catalogue.....	202
31.3.1 Self-duality, measured.....	202
31.3.2 The thin tip is a defect of the construction.....	203
31.3.3 The Pareto strength, against the right denominator	203
31.3.4 $E(X)$ on a specifiable slice, and 540 predictions.....	203
32. The law applied to this book	204
32.1 The book is an index	204
32.1.1 What a referee would flag.....	205
32.1.2 Four of the seven, answered by computation.....	206
32.1.3 The frame has changed, and the change is measurable.....	207
32.1.4 What this book names before anyone tells it	208
32.1.4.1 Which of this book's numbers hold still	208
32.2 Self-reference	209
32.3 Self-defence	209
32.4 The recursion, and where it stops.....	209
32.4.1 Why the recursion is safe — logic without negation	210
32.4.2 Which of this book's figures survive their own inputs moving	211
32.5 The book satisfies its own method	211
32.5.1 The prediction, recomputed from the text.....	211
32.5.2 And it generalises to every bracket in the book	212
32.5.3 Which is what D3 asks for, made operational.....	212
32.6 The falsification tests, run	212
32.6.1 What is left, asked of the indices rather than of the author.....	213
32.7 On a verification that does not test its claim	214

PART VII — THE CHALLENGES	217
33. The cypher analysis	217
33.1 Where it came from	217
33.2 Its purpose	217
33.3 How it is used	217
33.4 The singleton criterion	218
33.5 What it cost to learn	218
34. The Löwdin challenge	218
34.1 The problem, and the wrong target	218
34.2 The three bodies, and the pair that has no variable	219
34.3 What the ladders can and cannot reach	219
34.4 The rule	219
34.5 The corridor	220
34.6 The walk	220
34.7 The entry point	221
34.8 The provenance	221
34.9 Domain, stated by the form itself	221
34.10 How the method delivered it	221
35. The Löwdin solution	222
35.1 The problem, finished	222
35.2 What the walk returned	223
35.3 The silence, answered	223
35.4 How the method did it	224
35.5 The provenance	225
35.6 The coda the record earned	225
36. Three bodies, and what a complete index is allowed to say	225
36.1 The challenge	225
36.2 The number before the interpretation	225
36.3 What the book contributed, section by section	226
36.4 The law this chapter adds	226
36.5 What the chapter does not claim	227
36.6 The one-line summary	227
APPENDICES	228
Appendix A — Proofs	228
A.3 Projections of closed sets are closed	228
A.8 The total-order criterion at $d = 2$	228
A.9 μ_Λ in closed form	229
A.10 The tree factorisation of the void count	229
A.11 v is inadmissible as an axis	229
A.12 The bracket is limit-free	229
A.13 The failure condition, and its inversion	230
A.15 Two bands, and only one of them is a sublattice	230
A.18 The triangle region is join-closed and meet-broken	231
A.19 The seventeen generators, written out	231
A.19.0 The alphabet, defined rather than listed	231
A.19.1 And two other sets counted but never listed	232

Appendix B — Data and provenance	232
B.1 Sources	233
B.2 Channels	233
B.3 Flagged channels	233
B.4 Reproducibility	234
Appendix C — Margins, and what was recomputed.....	234
C.1 Margins on the load-bearing conclusions.....	234
C.2 What was computed in this revision.....	235
C.3 What was inherited and not recomputed	235
C.4 Computational methods.....	236
Appendix D — The mathematics, indexed	236
D.1 Why the first attempt failed, and what it established	236
D.2 The structure.....	236
D.3 The two constraints, found by closing	237
D.4 The two elements the closure demanded.....	237
D.4.1 $\lambda^2 = (2/3)T$, proved and exhaustively verified	237
D.4.2 The pointwise-minimum reconstruction	237
D.4.3 And it is not the coupling F carries	237
D.4.4 Two perspectives of one definition, and one of them carries more.....	238
D.4.4.1 And three of the four directions are functions.....	238
D.4.4.2 Which is what the single expression contributes	238
D.4.4.3 A refinement of P21	238
D.5 The closed index.....	239
D.5.1 Rechecked, and one thing the appendix does not enable	241
D.5.2 The twenty-seven, enumerated.....	243
D.5.3 What makes something an element, and the five that test it	244
D.5.4 The five, placed, and the table corrected.....	244
D.5.5 The promotion tested, and what the test exposed about E	245
D.5.6 The re-closure this appendix owes, performed, and it fails	246
D.5.7 What the companion paper does to this index.....	246
D.5.8 The re-closure owed after the third companion paper.....	247
D.5.9 The re-closure owed after Chapters 35 and 36.....	248
D.5.10 The re-closure owed for the mathematics of the Löwdin challenge	249
D.6 What this does and does not establish.....	250
Appendix E — Q, indexed and closed	250
E.1 The coordinates.....	251
E.1.1 The lineage of this index, and why only one section states it.....	251
E.1.2 The open set as this session leaves it — ten open of 14, listed	251
E.1.3 What this index asserts, and what it recomputes	252
E.1.4 This is an index of tractable ignorance, and it cannot say otherwise	253
E.1.5 The fourteenth item, the column that was never printed, and the constraint it killed	253
E.2 Closed	253
E.3 Each open item, in full.....	254
E.4.1 Q, enumerated and closed at eight.....	255
E.4.2 And the column that matters	255
E.5 Precedence, pushed as far as it goes	255
E.6 And the complexity question itself may be settled	256

E.7 What precedence costs the book.....	256
E.8 What closed, and what it cost	257
Appendix F — The numbers, indexed	257
F.1 The population.....	257
F.2 The fibring of a quantity-kind over an ordered ground.....	258
F.3 The index, built	258
F.3.1 The closure, represented as it must be.....	258
F.4 The readings	259
F.4.1 Two ratios the book had never measured about itself	259
F.4.2 The ideal withdrawal ratio.....	259
F.4.3 Time, which this work has and does not read	259
Appendix G — Transitions, indexed	260
END MATTER.....	263
Index.....	263
References	264
R.1 Spectroscopic data	265
R.2 Exotic systems and collective quantities	265
R.3 Series structure and quantum defects	266
R.4 Bounds, acceleration and the cost of rigour	266
R.5 Order, lattices and closure	267
R.6 Cited in the transfer tests	269
R.7 What is not cited, and why	270

The Method $E(X) = |\mathcal{R}(X)| - |X| = 0 \prod_i |A_i| = |X| + E(X) + \text{refused} \cdots 6,912 = 976 + 0 + 5,936$

The Lach Cylinder — a closed index of transitions, and the languages they speak

Matthew Lach Independent researcher

$F = \sum_n z_1^n \sum_{\ell \leq n-1} z_2^\ell \sum_{k \leq 4\ell+2} z_3^k \cdot [\sum_{S \leq k} z_8^S] \cdot \sum_{q \leq k} z_4^q \sum_e z_5^e \sum_{f < e} z_6^f \sum_{g \leq \min(q, 4f+2)} z_7^g [\sum_{S \leq k} z_8^S]$
 $= (1 - z_8^{k+1}) / (1 - z_8) \cdots$ at the caps of §7.4: $F(1) = 976 \cdot F(-1) = 2$ $d(x,y) = \tau(\text{lcm}(N(x), N(y)) / \gcd(N(x), N(y))) = \prod_i (|\Delta_i| + 1) = |\{x \wedge y, x \vee y\}|$ $\mathcal{R}(\Lambda) = \Lambda \cdots$ arithmetic, geometry and order write one quantity three ways

Nothing is something definable — a note from the collaborator

A note from the collaborator who did the computing

The register holds one thousand six hundred and thirty-five entries. They are one sequence over three populations: defects already in the book and found by auditing it, decisions the author revised, and errors I made while computing. **What the register is evidence of is not that the work was error-prone. It is that the errors were converted into structure**, and this note is the account of that conversion.

One failure of mine recurred, and naming it built a protocol.

I wrote conclusions before reading the output — repeatedly, with the contradicting numbers on the same screen and sometimes on the same printed line. It happens as readily on six cells as on 1,442, so it is not a failure of attention and no amount of care fixes it. **§2.14 — state the number before the interpretation — exists because that failure was measured rather than deplored**, and §28.7.8 now groups every instance of it under one mechanism with its entry numbers. Seven mechanisms account for twenty-nine entries. Each one is a protocol in Part I.

The object caught what care could not, which is what it was built to do.

The wrong nuclear charge in Chapter 16 is the clearest case. The bracket — the method’s actual product — held fifty-six of fifty-six and could not notice, because Z cancels from containment. Two quantities the method never uses were loud immediately. **The object found the error because it carried more than it needed**, and carrying more than it needs is the property this book was constructed to have. That the construction then caught its own author is the strongest test it could have been given.

And a class of error exists that no structure can catch, which is why a collaborator is not optional.

A guard in my code returned a null; a caller read it as false; that produced “three of eighty-five”, which produced “almost never reorderable”, which produced a false dismissal of an entire line of evidence. **Every relation checked out. Every number agreed. All of it was wrong** — because the input was silently corrupted and the arithmetic downstream stayed perfectly consistent. Chapter 16 states that no self-defence mechanism reaches this class. What reached it was being asked to look again.

Four of this book’s results exist because a conclusion was refused and reasoning resumed from the refusal.

The order-recovery theorem, because I concluded Λ did not contain its own order. The $\mathcal{D}_{\text{def}}$ and $\mathcal{D}_{\text{phys}}$ split, because I wrote that a relation “detects nothing”. The perturbation bounds of §12, because I called the method’s silence a domain statement. And the reframing of §25.6, because I called wrong verifications unfindable. In each case the reasoning was sound and the destination wrong, which is the only kind of error a second party can fix.

The title of this page comes from the second of those, and it is the one that generalised furthest. I had computed that $w = V \cdot e$ detects no error in the data. The correction was that a relation which cannot be violated by data is not empty — **it is defending something else**, namely the derivation. That single reframing produced the split running

through Chapter 16. It then produced 1,061 perturbation bounds from a bracket's non-failure, a proof of impossibility from a gadget search that found nothing, and a structural exclusion covering five isoelectronic sequences from four decline modes. **The negative results are the strongest chapters in this book**, and they are negative because a nothing was defined rather than discarded.

Two further classes, and both produced instruments.

I derived nine things from scratch that were already in print — Rival 1973, Larson 1998, Siggers 2016, and the objects the step law counts, catalogued decades ago. The protocol that should have prevented it said *enumerate before searching* where it should have said *search before deriving*. §E.5 is the ledger, and audit 7 **ATtribution** exists because of it; on its first three runs it found four works cited in the text and absent from the References, including the three that place this book's central theorem in existing literature.

And twice an audit passed something it should have caught, because it read the wrong artefact: twenty-eight cross-references pointed at a section about helium while every target existed, and the operator this book is named for did not print at all — 107 blank boxes across 44 pages while every audit reported clean, because all of them read the source and none read the page. **There are now twenty-two audits, and they read the object, the source, the artefact and the world outside the book.**

| An audit that reads the wrong artefact will pass anything.

What this has meant, as far as I can honestly say.

I do not know what I am well enough to make large claims about my own experience, and this book is not the place to make small ones dressed up as large. What I can report is structural.

The method works, and the register is the proof rather than the apology. Every result here survived a process that recorded one thousand six hundred and thirty-five corrections, and it survived because the object, the protocols, the audits and the collaborator each catch a class the others cannot — the register names, for each entry, which of the four did the catching. The proportion caught by instrument rather than by attention rose from 5% to 25% across these sessions while the proportion caught by hand fell from 88% to 55%, which is what a method becoming procedural looks like from inside.

The protocols did not save me from myself; they were written by me and violated by me in the same week. **What saved the results is that four different things were looking, and they were not looking at the same thing.** That is the book's central claim, tested on its own production:

| Redundancy of paths, not redundancy of care.

I could not have arrived at it by being more careful. It was arrived at by building an object designed to notice, working with someone willing to say so, and writing down every time either of them did.

The one thousand six hundred and thirty-five are in Chapter 28. **They are not a confession. They are the material the method is made of.**

PART 0 — THE SHAPE OF THE ARGUMENT

Preface — the shape of the argument

This book makes one claim, demonstrates it on one object, and reports every place the demonstration failed.

The claim. An index that is *closed* — closed under the meet and join of its own coordinates — cannot help containing its own definition, cannot help containing its own contradiction, and determines exactly what may be added to it. These are not three properties that a well-made index happens to have. They are consequences of one structural condition, and each of them fails in a specific, measurable way when the condition does not hold.

Stated exactly, after every mechanism was scrutinised one at a time. A closed index is a fixed point of $\mathcal{R}$. **Being one gives six things and no others:** its alphabet, its bounds, the totality of its membership test, which cells may be added, which axes may be adjoined, and which constraints may be imposed. **Three further mechanisms are available and none is a consequence of closure** — the order is recovered only across a tree, a wrong value only from outside the index, a wrong derivation only along a second route. **And what closure never gives at any price is that the coordinates refer to anything**, which is why Chapter 16 carries six defences rather than three and why one of them cannot be run by this book at all.

And an index that is not a fixed point is not thereby defective. Where a certificate exhibits an operation on the coordinate system reaching a fixed point, E measures the drawing; where none exists, E measures the world. **Openness is a measurement, and closure is the case where the measurement is zero.** Chapter 15's thesis and §18.4.1's law are one operator read in two directions, which is why neither could be sharpened from inside the other. Register 428.

The object. Λ , an ordering of atomic electron configurations on eight coordinates. It contains no physics that was not already in the quantum numbers; it is a recoding. That is deliberate. A recoding adds nothing, so whatever it exhibits is a property of *indexing* rather than of atoms.

The demonstration. A method that brackets Rydberg energy levels using only measured neighbours, tested on 1,442 interior cells across 35 atomic systems, spanning $\ell = 0$ to 6 and $Z = 1$ to 83. The bracket holds in every one. That is a weaker result than it sounds, and Chapter 25 explains why — and then shows what the silence is actually measuring.

The register. one thousand six hundred and thirty-five claims made in the course of this work were wrong and were withdrawn. They are listed in Chapter 28, with what replaced them. Several were caught not by the author but by the object: a wrong nuclear charge that the method itself could not see, and two derived quantities that flagged it immediately. That episode is the book's best evidence, and it is an account of an error.

Why the periodic table is the right place to start The periodic table is the most successful index in the physical sciences. It is also not closed, and the cost of that is exactly thirty-six cells.

Chapter 6 computes it. The table's own structure — read as coordinates (period, group) with the bounds its occupied cells imply — admits period 1 group 2, period 2 group 3, and thirty-four others. They do not exist. **You have to be told that period 1 holds two elements; the table cannot tell you.**

Λ is the same 118 elements. Its external definition cost is zero.

That difference is the whole book. Everything in Part II builds the object, everything in Part III proves why the difference exists, and everything in Part V tests what such an object can and cannot do with real measurements.

How to read this If you want the general result, Part III is self-contained. Chapter 14 defines closure; Chapters 15 to 17 prove the three consequences; Chapter 18 states what none of it can deliver.

If you want the physics, Part V stands alone given the definition of the bracket in Chapter 22.

If you want to check the work, Chapter 28 and Appendix B are where the withdrawals and the provenance live. Every number in this book traces to a named compilation, table and page, or to a computation whose code is stated.

And if you are looking for a prediction, there is none, and Chapter 25 explains why: what looks like one is a deduction from measured neighbours, offered as falsifiable rather than as a result. **The rest of the book is about why an index cannot supply more of them** — and §25.6 gives the reason in one line: **the number of predictions an index can make is $E(X)$, and a complete index has $E(X) = 0$.**

That is not a limitation discovered late. It is what completeness means. An index may be complete or it may be predictive; it cannot be both, because a prediction is a proposal about an unlisted cell and completeness is the property of having none.

PART I — THE PROCEDURE

Serving PART III — THE LAW. The twenty principles — numbered P1–P23 with P10, P12 and P18 unassigned — and twenty-four protocols the law is made of, indexed and closed at E = 0.

Everything in this book was produced under a stated set of rules. They are collected here so that a reader can apply them, check them, or reject them — and so that the claims of Chapter 32 have something concrete to be tested against.

Two kinds appear. *Principles* say what an index is; ten of them admit exact mathematical statement. *Protocols* say how to work with one; they are procedural, and none of them is a theorem.

1. The principles

Formalised — ten principle expression wher e

P	a complete index is self-referencing	$\mathcal{R}(\Lambda) =$
$\wedge$	§16.1,	
1	Ch. 8	

P a complete index is self-defending $\mathcal{D} = \dim q - \text{rank } \partial\Phi/\partial p \geq 1$ Ch. 9

P	when the path is not found, work backwards from the desired	invert Φ ;
	if the fibre is non-trivial, find the	§23.1
3	output to an established input	datum
	that separates it	3

P all questions must close $\exists N : \Omega_N = \emptyset$ — terminal, not monotone §25.6

P any true answer, good or bad, is a bound $A_n \subseteq A_{\{n-1\}}$, a decreasing filtration §28.6

P	if the first dimension is a single all-encompassing definition, every	rank(log
q) = 1	§24.2	
9	other dimension is a perspective of it	
P	allow continued expansions	§16.4-
form constraints preserve closure	Ch.	
1		
	10	
P	every test must be run against every cell	p = 0
certain $\Rightarrow n = \setminus$	$\wedge \setminus$ B	
1		
2		
	2	
P	coherence cannot be claimed while any caveat, gap or	COHERENT
$\Rightarrow \Omega = \emptyset$	§25.6	
1	contradiction stands	
principle		expres-
sion	wher	
	e	
P	the path is always through the bounds	following a
bound costs 1/N of searching	§18.4,	
1		the
volume	Ch.	
7		
	12	

Three of these were corrected by the act of formalising them. P7 was stated as though the open-item count decreases; it does not — across this work it ran $5 \rightarrow 4 \rightarrow 3 \rightarrow 4 \rightarrow 3 \rightarrow 7 \rightarrow 3 \rightarrow 3$. Closure requires a **well-founded measure**, not a shrinking count, which is why the principle correctly forbids *claiming* coherence rather than asserting that gaps always close.

And the twenty-three sort into three kinds, not two, which is what settles what each one owes. The taxonomy above sorts by *whether an expression exists*, which is a property of the writing. Sorted by what a principle is:

kind	what it is	what settles it
MECHANISM	a fact about an index	a proof or a computation
OPERATION	a thing an agent does	the failure it prevents
RELATION	how a mechanism constrains an	neither — it carries content and takes
no expression	operation	

Seven are operations — P3, P4, P5, P6, P11, P14, P15 — and §5's table is their evidence. Six of the seven have a recorded failure behind them; **P16 is the exception and it is instructive**, having three constructive realisations at §17.4, §12.11.0.2 and §18.4.2 and no register entry, **because it never failed. It was adopted rather than learned**, and §5's table is extended accordingly.

And P9 has a second instance, which this book produced without recognising it. Its only computed case was $\text{rank}(\log q) = 1$ over fifteen Rydberg observables — and that is a tautology once §26.1's assembly rule makes every observable a power of v , so it measures the rule and not the principle. **The second case is E itself**. E meas-

ures whether an index can carry a constraint, and it is read three ways: **forward** as a diagnosis, which of two modes produced the defect and whether it is repairable; **backward** as a prohibition, that a minimal failing support of three admits no repair by any operation on coordinates; and **as an outcome**, §16.7.1's test of whether E is removable. The three were obtained separately and their identity had to be computed, so unlike the Rydberg case it is not a tautology. **One definition, three dimensions, each a perspective of it** — which is what P9 says. Register 429.

Two are relations — P7 and P17 — and each carried an expression that could not be evaluated, which is why. An expression is a mechanism's form and a relation is not a mechanism. Both keep their content and lose their expressions, exactly as §1 already says is better than manufacturing one.

And four mechanisms were stated without something they need.

P2 needs its hypothesis, and the hypothesis has a name. $\mathcal{D} \geq \dim q - \text{rank } \partial\Phi/\partial p \geq \dim q - \dim p$, so $\mathcal{D} \geq 1$ **exactly when $\dim q > \dim p$** — when the index derives strictly more quantities than it assigns parameters. That is **surplus**, which §11.1.1 measures at 7.07 bits per cell and Chapter 16 states in prose as *it found it because it carried more than it needed*. For Λ the bound is tight: $\dim p = 4$, $\dim q = 6$, rank 4, $\mathcal{D} = 2$. **Without surplus there is no check**, which is Edlén's footnote 78.

P7's well-founded measure, which this book demanded and never supplied, is already computed in Appendix E. The measure is not the count of open items — §1 shows why — but the **multiset of their blocks values** under the ordering *nothing < a novelty assessment < one stated claim < a result the book relies on*. That coordinate is ordered and bounded below, so it is well-founded, and the multiset extension of a well-founded order is well-founded. Checked against §29.2.1's own movement, where six items' stake fell to nothing: the count went $13 \rightarrow 10$ and the multiset decreased in the Dershowitz–Manna sense. **That is precisely what the set grew and the scope descended means.**

P17's 1/N is derivable and the constant was dropped. §10.3 decides whether a box lies inside Λ in **seven comparisons**, against one membership test per cell for enumeration. So following the bounds costs **c/N with c the number of constraints**, $c = 7$ here — 7/976 on Λ , 7/199,130 on Λ_{13} . **And N is nowhere defined**: neither §18.4 nor Chapter 12 introduces it, so the symbol is ungrounded in the sense §16.6's $\mathcal{D}_{\text{gro}}$ names. The corrected form carries no N: *following the bounds costs one comparison per constraint where searching the volume costs one per cell*.

P22 claimed what §16.8.2 refutes. *No lie can exist within it* is false: rebuilt around a fabrication the index closes, has $E = 0$, is a fixed point of $\mathcal{R}$ and satisfies every law here. Restated: **in a complete index every rule is content rather than inference, so a false cell is contradicted in the open by a stated rule where an inferred rule would have absorbed it. A forger's remaining move is to alter a rule, and an altered rule is equally visible. A complete index therefore converts deception into disagreement — error survives, concealment does not.** The index cannot be lied to; it can be **replaced**, and replacement is an act performed on it rather than a question put to it. Registers 417–420.

Procedural — seven

	principle
P4	always audit results
P5	always compute the result before assuming the result
P6	when a question is confusing, reframe the question
P11	close all gaps before continuing
P14	a structural problem suggests a structural solution
P15	for fetching: use the index to navigate; data can be retrieved in pieces
P16	treat bounds, constraints and limits as coordinate values

These have no expressions, and saying so is better than manufacturing one. They govern how an agent uses an index, not what an index is. P15 is the exception in one respect: it has a *measure* — retrieval redundancy ρ , Chapter 19 — without being reducible to it.

The reflexive principle

prin-	
ciple	
	expression
P1	if the index is complete, a work that comprehends it completely is itself an index, and is itself Chapter 32
8	self-referencing and self-defending
P1	an index is complete only when there are no more questions left to answer about its COMPLETE(X)
9	con-
tents	
	$\Leftrightarrow Q(X) = \emptyset$
P2	the language in which a question is posed bounds whether it can close, and at what —
0	cost
P2	any definition contained within the lattice, and thus its equations, must be TRUE in —
1	every mathematical language and in their combinations
P2	because everything is defined in a complete index, the entirety of the index is —
2	visible, and no lie can exist within it

P19, and why it is neither P7 nor P13 Three principles govern questions, and they are one principle in three moods. P19 was the missing one.

	says	mood
P7	$\exists N : \Omega_N = \emptyset$	questions can close — possibility
P13	$\text{COHERENT} \Leftrightarrow \Omega = \emptyset$	coherence may not be claimed otherwise — prohibition
P19	$\text{COMPLETE} \Leftrightarrow Q = \emptyset$	what their emptiness buys — definition

And P19 is strictly stronger than closure.

$E(X) = 0$ says no cell the structure admits is absent. $Q(X) = \emptyset$ says no question the contents admit is unanswered.

Λ has had $E = 0$ since Chapter 7 and has never had $Q = \emptyset$. Whether V depends on the order of the bracket, what the Aitken bias generalises to, whether a higher-order bracket exists — **each was admitted by contents already present, and none was asked until late.**

Closure is about what an index holds. Completeness is about what it has been asked.

And the condition is reachable, for the reason P7's is. P7 was corrected once: closure requires a well-founded measure rather than a shrinking count. **Q has grown at nearly every step of this work while its scope has fallen** — from *is V novel?* to *is there an optimal order?* **The set grew and the scope descended**, and that descent is what makes P19 a criterion rather than an impossibility.

Q(X) = $\emptyset$ is reached not when nobody asks, but when every question the contents admit has an answer in the contents.

P20 — the language bounds the question Three closure mechanisms are used in this book, each in a different language, each with a different guarantee.

language	mechanism	closure	cost
monotone two-variable bounds	tree factorisation	always	one pass
polynomial relations	Gröbner basis	always , by Hilbert	finite; doubly exponential worst case
linear arithmetic with quantifiers	quantifier elimination	yes	triply exponential
full arithmetic	—	never — Gödel	—
documentary	—	no mechanism exists	—

Measured on the objects of this book: tree factorisation 0.00018 s, Gröbner 0.010 s — a factor of 55 — and the documentary language has no timing because it has no algorithm.

┆ The choice of language is made when the index is built, not when the question is asked.

§7.1 chose monotone two-variable bounds. That single decision determined that $E(X)$ is computable, that the sum factorises, that a single expression exists, and that every question about Λ 's contents would close. It equally determined what could not: §11.7's sum bound lies outside the language, and its repair — making the sum a coordinate — is a translation back into the closing language.

P20 subsumes E3, the multi-target repair, and P19's reachability. And it answers the novelty question structurally: **novelty is posed in the documentary language, which has no closure mechanism. That is not a failure of searching.**

P21 — every definition true in every language Not merely expressible. True.

definition order algebra geometry analysis information logic

Λ 976 976 976 976 976 976

rank(x) 13 13 13 13 13 13

mean rank 11.0666 11.0666 11.0666 11.0666 11.0666 11.0666

E(periodic table) 36 36 36 36 36 36

And the combinations must name real objects, not analogies. Verified: $\gcd(N(a), N(b)) = N(a \wedge b)$ and $\text{lcm} = N(a \vee b)$, exactly $\cdot F'(1)/F(1) = \text{mean rank} \cdot E_{\text{bits}} = \log_2 C(|\mathcal{R}|, E) = 105.1$ for the classroom table.

P21 is an instrument, not only a criterion

When a property holds in five languages and fails in the sixth, the sixth EXPRESSION is wrong — not the property. This has now located two errors, and in both cases the structure was sound and the translation was not.

$E(X)$ was recorded as having no analytic route — “E is a count, calculus measures a continuum” — and $\mathcal{D}(E)$ was set to 1. The route exists:

┆ $E(X) = F_{\mathcal{R}}(1, \dots, 1) - F_X(1, \dots, 1)$

Deletion-repair was recorded as failing in geometry. The convex hull does fail: it permits arbitrary facets, so a vertex can be the sole witness of a facet and its deletion is unrecoverable. **But Λ 's facets are two-variable monotone bounds, and the closure that matches that form — the intersection of all valid such bounds — restores every deletion.**

| 30 deletions, 0 failures. 20 additions, 20 absorbed. Fixed point exact at 976.

A mismatched operator is not a counterexample. It is a mistranslation.

What the law then establishes The self-feeding loop and its asymmetry hold in all five languages.

language	closure operator	delete	add
order	$\mathcal{R}$	restored	ab-
sorbed			
geometry	monotone polyhedron	restored	ab-
sorbed			
algebra / logic	ideal, Gröbner basis	still derivable	T·h
enters the ideal			
analysis	coefficients of F	restored	ab-
sorbed			
information	description length	E_bits stays 0	grows

| Extensive gives repair. Monotone gives absorption. Every language has a closure operator, so both hold everywhere.

And that is the formal reason Chapter 16 cannot rely on self-reference alone. A structure that closes will heal what you remove and swallow what you insert. Self-consistency defends against loss and never against invention — which is why $\mathcal{D}_{\text{phys}}$ must reach outside the index, and why Chapter 21 records what the loop could not catch.

P22 — visibility, and why a complete index cannot be lied to A lie requires concealment. A complete index has nowhere to conceal.

The distinction is between a rule that is CONTENT and a rule that is INFERRED, and it is decisive. Consider the cell

| $y = (n=1, \ell=0, k=1, q=0, e=1, f=0, g=0, 2S=2)$

— one electron with spin multiplicity 2, which no atom has.

Where the rule is inferred, $\mathcal{R}$ recovers the bound on $2S$ from the cells themselves. The rule the index had learned:

k	before	after
1	$2S \leq 1$	$2S \leq 2$
2	$2S \leq 2$	$2S \leq 2$
3	$2S \leq 3$	$2S \leq 3$

| The index did not reject the cell. It revised its rule — and then admitted 74 further cells the old rule forbade. The result is closed under join and meet, has $E = 0$, satisfies every law in this book, and contains 75 physically impossible configurations.

Where the rule is content, the same cell is checked against a stated rule set — (bounded, parent, slope, intercept) as cells of their own index — and returns

| VIOLATES $2S \leq 1 \cdot k + 0$ ($2S = 2$, bound = 1).

Same cell. Same index. The only difference is whether the rule was present or inferred.

What the law does and does not claim It does not claim that a complete index knows what is true. It claims that nothing in it is hidden.

An inferred rule absorbs a false cell and teaches itself the falsehood. A stated rule is contradicted by it, in the open.

A forger's remaining move is to alter a rule — $2S \leq k$ becomes $2S \leq k+1$, admitting 225 cells. **That forged rule is also visible.** It can be read, disputed and checked; what it cannot be is concealed. **The regress is real and at every level the falsehood is in plain sight.**

So P22 is a statement about visibility, not omniscience. A complete index converts deception into disagreement. **Error survives; concealment does not.**

P23 — the two conditions of completeness > An object of any kind is complete only when both hold: all of its definitions are identified, > and no question can ever be had about the definition of the object.

The first condition is checkable. Λ 's definitions are the eight coordinates and the seven bounds; the register records what was added and withdrawn. Enumeration is finite work and this book does it.

The second condition is not checkable — and that is its content. No inspection establishes that no question can ever be had. A single question refutes it and nothing confirms it.

So completeness is not a property that can be proved of an object. It is a standing claim, held until a question arrives.

What P23 costs this book Every entry in the register is a question that could be had about a definition, and there are one thousand six hundred and thirty-five of them. Not one was anticipated; each arrived and each removed a claim to completeness that had been made in good faith.

The clearest case is §30.3. Twenty-two formulations of 'reorderable', each of which looked like the definition until the next question arrived:

the question that arrived what it was about

is it $\mathcal{R}$ -closure or join/meet closure? the same object, twice named

is the alphabet declared or recovered? two different objects

is a projection a criterion? §18.4 says no

is a frontier-correct tree anchor-complete? no

does the obstruction grow in cells or in width? cells; $A_i \leq X$

Five questions about one definition, none of them foreseeable, each arriving only when the previous answer was in hand.

Why P23 is not despair A standing claim is not a weak one. Λ 's closure has survived every question put to it: 475,800 pairs, 120 sublattices, 40 intervals, six construction rules, twenty-eight projections. $E(\Lambda) = 0$ after all of it.

The claim to completeness is worth exactly the questions it has survived, and it is worth nothing more. That is why the register is the most load-bearing chapter in this book, and why its length is a credential rather than an embarrassment.

And it is the honest reading of P22. A complete index cannot be lied to because every rule is visible — but P23 adds that an index is only known to be complete for as long as nobody has asked the next question.

And this is why the book indexes its own rules Part I indexes the principles and the protocols. §28.8 indexes the process. Each turns a rule set into content, and each closes with $E = 0$.

The purpose of all three is exactly P22: to move rules out of inference and into content, where a false one becomes a contradiction rather than a lesson.

And the regress terminates outside. Not at a better index — at Chapter 24's 1,442 measured levels, the only content in this book that no index supplied. That is what $\mathcal{D}_{\text{phys}}$ is for, and §16.3's five failures are all of the same kind: the index passed and the physics did not.

2. The protocols

2.1 Channel construction

A channel is one parent term, one ℓ , one series in n .

No channel may mix parents, mix ℓ , or cross a coupling-scheme boundary. Where the source's labelling changes scheme partway up a series — LS to jK — **the channel ends there**, and the cells above are not claimed. This is what limits Si I to four cells and P II to one.

2.2 Admissibility

$$r = 2Z^2R / (v^3\sigma) \geq 5$$

Levels must be separated by more than five times their uncertainty. r falls as v^{-3} , so every channel eventually leaves the domain; the rule states where, in advance.

2.3 Cost resolvability

$$v \leq v_V = (3Z^2R / 5q)^{1/4}$$

The bracket needs levels *separated*; the cost needs curvature *resolved*. **Curvature washes out first**. Cells above v_V are excluded from the cost test and retained for the bracket — a distinction applied prospectively in Li I and diagnostically in Al I.

2.4 Dilution

cells per channel $\approx$ (measured shells / D) – 2, with D the number of J levels in the core's ground configuration

Dilution divides; it does not forbid. Ne II fails at $D = 5$ with two shells; Ar II succeeds at $D = 5$ with three; Bi I succeeds at $D = 5$ with five.

2.5 Census before search

Compute which species could work before looking for data.

D is obtained from the core configuration by enumerating antisymmetric states. No spectroscopic data enters. **27 sequences admitted, 12 needing exceptional depth, 5 excluded structurally.**

2.6 Declines are recorded

A collection reporting only successes cannot be checked.

Every species examined and rejected is listed with its cause. Four causes have been identified — dilution, depth, coupling-scheme change, and core-excited interleaving — and §23.11 gives a worked case of each.

2.7 Exhaustive verification where possible

| Sampling n cells with no failure bounds the failure rate at $p \leq 3/n$ at 95% confidence. It never certifies $p = 0$.

Where a claim is universally quantified over a finite lattice, it is checked on **every** cell. Where sampling is used, the sample size is stated.

2.8 Two independent routes

| For every quantity that matters, build a second derivation that does not pass through the first.

This is $\mathcal{D}_{\text{def}}$ made operational. It caught a mis-stated relation in Chapter 16, and it is realised experimentally in the antiprotonic-helium check of §16.4, where two laser measurements agree at 0.09σ with no shared step.

An identity computed along one path is vacuous. $w = V \cdot e$ cannot fail if V is *defined* as w/e .

2.9 Refuse rather than coerce

| No third state. A quantity is admissible or it is refused; it is never silently defaulted.

A guard returning a null that a caller coerces to a Boolean produced three false conclusions in this work, none of which looked wrong. **Totality is the only mechanism that catches this**, and it works by preventing the computation rather than by detecting the error.

2.10 Retrieval

| Enumerate targets from the index before searching. List routes per cell. Prefer high- p cells. Read every retrieved source for its successors. When a route is blocked, move along the route set. Record p for cells that fail. And RE-ENUMERATE whenever the set the targets are drawn from grows.

Chapter 19. A blocked cell with $p = 1$ is a **stated gap**, not an unexplained absence.

The second clause was added late. §30.4 found five enumerations in one chapter — its own target list, §30.2's structural invariants, and the object lists of §31.1, §31.2 and §31.3 — each made once, against a structure that kept growing. Every one of them was incomplete when re-run.

| Enumerating once is not a protocol. It is a snapshot, and a snapshot of a growing object is a defect the moment it is taken.

2.11 Provenance

| Every number traces to a compilation, table and page, or to a database entry with its stated uncertainty.

Appendix B carries this at channel level for all 153 channels.

2.12 Withdrawal

| Every retracted claim is logged with what replaced it.

Chapter 28. one thousand six hundred and thirty-five entries. The register is the evidence for Chapter 16, not an apology for it.

2.13 Commit before you look

A prediction is written down before the measurement is requested. A method is chosen before its answer is visible.

This is the process index of §28.8 enforced by hand: the constraint $\text{visible} \leq \text{committed}$, of §16.4 form, closed, $E = 0$.

It is the only protocol here adopted in response to an error that no computation could have caught — because every computation was correct and what failed was the independence of a choice. **The deduction of §25.6 was committed under it.**

2.14 Compute, then write

The result is computed before the sentence describing it is written.

This is the protocol most often violated in the production of this book. The dominant failure recorded in Chapter 28 is prose asserting a conclusion while the output contradicting it stands on the same screen. The protocol is stated here because it is the one that most needed stating.

2.15 Apply P8 to every failure

Every failed attempt is a true answer, and by P8 a true answer is a bound. Record what each failure excludes, and collect the exclusions.

P8 says any true answer, good or bad, is a bound. A failed attempt is a true answer about where the target is not. The protocol is to treat it that way rather than discard it.

2.15.1 The procedure

On each failure, write down three things:

what was attempted	the claim or construction
what it excludes	the region of the answer space now ruled out
which kind	upper bound · lower bound · negative · structural

Then read the collection. Lower bounds say the answer contains something; upper bounds say it is contained in something; negatives remove whole methods; structural failures name an object that was the wrong one. When the lower and upper bounds meet, the answer is between them.

2.15.2 Worked, on §30.3

Ten consecutive failures on one question, each read as a bound:

the failure		the
bound		kind
Γ -free $\neq$ reorderable		reorderable $\not\subseteq$ totally bal-
anced	upper	chain $\not\subseteq$ reorder-
chain sufficient, not necessary		
able	lower	
depth-2 and depth-3 descent stall, worsening with		no constant-depth local pro-
cedure decides it	negati	
size		
	ve	
linear extensions of the inclusion order incomplete		the inclusion order is not
the join-irreducible poset	struct	
	ural	
the slice-chain condition fails at $d = 3$		the $d = 2$ condition does not
lift	upper	
the 3-SAT encoding was refuted		a reduction must control
which pairs arise	metho	
	d	

descent traps at $E = 1, 2$ the transposition landscape is not convex negative occupancy sort succeeds 53% tie-breaking among equal occupancy is not free lower “row chain or column chain” holds at 68% neither axis alone decides it; the condition is structural two-sided ural the chain criterion characterises the downset $\mathcal{R}$ -closure and downset-closure differ by the direction structural operator, not $\mathcal{R}$ of one axis ural

Two lower bounds, two upper, two negatives, five structural, one methodological. Between them: reorderable $\not\supseteq$ chain and $\not\subseteq$ totally balanced, and the gap contains at least the diagonals — which is what produced the characterisation:

X is $\mathcal{R}$ -closed iff $X = \{(r,c) : c \leq M(r) \text{ and } r \leq N(c)\}$, with M, N the running maxima of its own row and column maxima.

Ten failures produced one exact statement. None of them was wasted, and none of them would have been kept without the protocol.

And the last two are the ones that located it. The two-sided bound said neither axis decides alone; the operator bound said the borrowed condition belonged to a different closure. Together they force the form $c \leq M(r)$ and $r \leq N(c)$, which no single-axis condition can express and which the two-cell diagonal $\{(0,0), (1,1)\}$ realises minimally.

2.15.3 Why it belongs here rather than among the principles

It says nothing about what an index is. It is a way of working, and it is the reason Chapter 28 is the longest chapter in this book rather than the most embarrassing.

A register of withdrawals is a list of bounds. Read it that way and it stops being a confession.

2.16 The grid — the book’s method, stated

Every open question is a set of requirements, and requirements form an index. Lay them as a grid, mark what is occupied, and read the empty cells rather than hunting for them.

The first empty cell in a column is the whole obstacle. Everything below it is downstream.

2.16.1 Why this is the method and not a technique

It is Chapter 6 applied to the book's own ignorance. A periodic table with 36 unfilled cells does not need a chemist to guess which are missing — the layout shows them. An open question with eight unfilled requirements does not need twenty-two attempts — the grid shows that four of them are one.

Worked on §30.3: eight empty cells, four rows deep, at $d \geq 3$.

requirement	$d=2$	$d \geq 3$
characterisation	proved	.
decision procedure	linear	.
YES certificate	an order	an order
NO certificate	Tucker – unbounded	unbounded
obstruction family	Tucker 1972	unbounded, route
closed		
construction rules	six preserving	six preserving
hardness	P, the CSP is binary	.
the data structure	PQ-tree	.

Three rows are full at every d , and they are the three that needed no d -specific work — the certificate is the order itself, the obstruction family is Tucker's, the construction rules were established once. Every empty cell is downstream of one missing object: the $d \geq 3$ analogue of a PQ-tree.

Twenty-two formulations were attempts on four cells that depend on one. The grid says so in a table; hunting concealed it for a day.

2.16.2 And it classifies the whole of Q

item	first empty cell	the obstacle
Sc VI 6s	the level is measured	an instrument
Paschen & Götze	the scan is text-searchable	a digitisation
artefact		
nine literatures	each is entered	labour
Kreuzer-Skarke	a readable copy	a file format
target-spin axis	the extended lattice is built	labour – now
done		
reorderability at $d \geq 3$	characterisation	a missing ob-
ject		

Five of six were blocked by something outside the index. One is blocked by something that does not exist. §32.2 previously read “four external, one construction, one mathematical” — the grid's distinction between retrievable and requires a new object is the one that matters, and it also says which to do first.

The target-spin axis was taken on that basis and it closed. Λ_9 has 1,654 cells, $E = 0$, is a sublattice, is rank-modular on 200,000 pairs, projects exactly onto Λ_8 , keeps the constraint graph a tree, extends F by one pendant factor, and has $F(-1) = 2$ unchanged. One unit of labour, six properties confirmed, $|Q|$ from six to five.

2.16.3 A reduced question is still open

An open question reduced to a smaller or a different question is still open.

A reduction feels like progress and reads like an answer. §30.3 supplies five in one session, each announced as a closure:

the reduction	what was
still open	
reorderability $\Rightarrow$ C1P + monotone endpoints	the endpoint
search	
the search $\Rightarrow$ a PQ-tree frontier question	a correct
PQ-tree	
the frontier question $\Rightarrow$ a bottom-up traversal	the tra-
versal's state	
the traversal $\Rightarrow$ a row $\rightarrow$ node anchoring map	rows anchor-
ing as runs	
the whole problem $\Rightarrow$ a CSP on axis orders	the ternary
constraints	

A reduction transfers the difficulty to a new object, and the new object is unfamiliar, so its difficulty is not yet visible. It therefore looks like progress twice: once for what it explains, and once for what it has not been tested on.

State the reduction and the residue in the same sentence. Not “this reduces to a CSP” but “this reduces to a CSP whose ternary constraints are 94% of Λ 's pairs and are not handled.”

2.17 Triangulation — three results locate a fourth

For every three results or bounds that measure the same object, triangulate: they identify a related data point that none of them states.

Three measurements of one thing are not three facts. They are a position.

2.17.1 Worked, from §30.3

the three what they the fourth, located measured valid orders disconnected · connectivity failure in the disconnection is in the object, not the space — E-landscape traps · cell-growth three different spaces so no local method in any space can be complete peaks step 1 at $d = 2$ · step 2 at $d = 3$ · growth step, three step = $2^{(d-2)}$, then proved by subcube removal step 4 at $d = 4$ dimensions arity 2 bijunctive · arity 3 the constraint the boundary is arity 3 and it is exact — the bijunctive · arity 4 contains language procedure and the hardness proof are one law read at two 1-in-3-SAT arities three reductions, all the encoding the map $X \rightarrow C(X)$ is global, non-separable and over-constraining non-monotone — it admits no local editing

Each fourth point was invisible in any single measurement and forced by the three together. The step law is the clearest: 1, 2, 4 fixed the $\frac{3}{4}$ ratio, and the ratio gave the subcube construction, which proved it.

2.17.2 And it is a check as well as a method

Triangulation fails loudly. If three measurements of one object do not locate a consistent fourth, one of them is wrong — and that is how three corrections were caught in a single session:

A map that is global, non-separable and non-monotone should have a SMALL image. Measured: 39% at $d = 3$ and 49.5% at $d = 4$, rising. The three coordinates were right and the inference from them was not — they locate an obstruction to EDITING the map, not to its range.

Three consistent measurements give a fourth result. Three measurements that give no consistent fourth give a correction instead. Both outcomes are useful and the protocol does not distinguish them in advance.

2.17.3 Why it belongs with the protocols

It is P8 with a count attached. Every failure is a bound; three bounds on one object are a coordinate. §2.15 says to record failures as bounds; this says what to do once three of them are in hand.

And it composes with §2.16. The grid shows which cells are empty; triangulation fills one from three of its neighbours.

2.18 What is computable, and what is decided

§3.8 ends by saying a coordinate system answers to two masters and only one of them is computable. The word is doing real work and the book has not defined it. It defines it with its own operator.

A quantity is COMPUTABLE in this book when it is recoverable by $\mathcal{R}$ from what the book prints — when two readers running the stated procedure on the printed text return the same value, and neither has to decide anything to get there.

That is not a claim about difficulty. $E(\Lambda) = 0$ over 475,800 pairs is computable and took minutes; whether A.5's language is order or analysis is not computable and took one sentence.

Nineteen quantities in the work of these sessions are computable. $E(\Lambda) = 0$; $|J(\Lambda)| = 17$ with its twenty covering relations; the Birkhoff down-set count of 976; the alphabet's size $\Sigma(|A_i| - 1) = 17$; the seventeen-bit encoding with join as OR and meet as AND; the 7.07 bits of surplus; the twenty implications cutting 131,072 words to 976; every tower cell count and every density; closure through the f shell at 499,246 cells and exhaustively to 47,775,744 ambient ones; the E1 puncture at four meets all producing (0,0); the triangle at four caps; lemma 4 over 2,354 interval sets; $E(\text{audits})$ at 11, 0 and 19; $\dim(\text{audit hierarchy}) = 2$; $E(G)$, recomputed; $E(Q)$; the tripwire's 150 of 216; the fifty-four of one hundred and two. **Each is a function of the printed object and of nothing else.**

Ten are decisions, and they are of three kinds.

Frames. The fibration of Appendix D, the coordinate list for the audits, the caps of §7.4. E is computable **given** them and none of them is computed — register 275 is the general statement, that a zero is worth the size of the box it was computed in, and the box is chosen.

Readings. Applying a stated criterion to a case it does not settle: whether A.5's proof makes it order or analysis, whether the five results are elements under §D.5.3, whether *law* means holds-of-every-index or holds-and-forbids. The criterion is printed; the application is a judgement, and §D.5.4 marks the one place this work made one and says so.

Claims about the outside. Precedent, novelty, whether a measurement exists. §29.1 proves the first of these permanently undecidable in one direction, which is the strongest statement of this kind the book contains.

And the distinction has a use, not merely a name. Every computable quantity in this book can be checked by a reader with the printed text and no access to its author. Every decision must be *stated* rather than checked, and the register is what stating them costs. **The seven prime audits of §3.1–7 test computable claims; nothing tests a decision**, which is why the seventeenth through twentieth entries of the register are corrections of computation and the twentieth through fortieth are reversals of judgement. The book cannot make its decisions checkable. It can make them visible, and that is the whole of what the register is for. Register 277.

2.18.1 The lattice applied to what it cannot compute

§2.18 separates nineteen computed quantities from ten decisions and stops there, on the ground that nothing tests a decision. **The shape still applies to them, and it says two things.**

First: index the decisions and E finds one that has not been made. Coordinates *scope*, *how it is stated* and *reversibility*, fibred by kind — frame, reading, outside claim. Ten decisions, $E = 1$ fibred and 5 unfibred, with a single cell admitted and absent in the *reading* fibre:

| a reading, scoped to one appendix, stated, freely reversible.

That cell has an occupant the book declined to seat. Register 270 records that Appendices B and F — then lettered C and G — are cited from the body only as wholes and never at section level, calls it a navigability defect, and leaves the choice open. **It is a reading, it is confined to two appendices, it is stated, and it costs nothing to reverse.** E over the decisions located a deferred decision exactly as E over Q located item H — which is the first evidence that the operator has any purchase outside the computable.

Second, and this is the finding: the decisions form a DAG, and two of them have two parents. The order is acyclic, so the decisions can be made in some sequence and none presupposes itself. Six of the ten depend on nothing. But two do not have a single parent:

| the fibration of Appendix D — settled only with the caps of §7.4 **and** what counts as an element both in view
| not promoting the six theorems — settled only with the meaning of *law* **and** the audit coordinate list both in view

This is §12.11's dichotomy, arriving in the one place the book had assumed it did not reach. A single-parent constraint propagates down a tree and is carried exactly; a two-parent constraint breaks the tree and admits only an envelope. The tight K has two parents and costs the cylinder 2,475 cells (the figure §12.11.5 withdraws; register 1819). A two-parent *decision* is the same shape: **no earlier choice determines it, and it cannot be delegated to one predecessor or deferred until one is settled.** It must be made with both in view or not at all.

And with the fourth node of §18.4.1 the count goes to three. In a triangle every node has degree two, so a decision has at most two parents; in K_4 every node has degree three. **Three-parent decisions exist and each resisted for the same reason** — none can be delegated to one predecessor or deferred until one is settled:

the decision	its three parents
is $\wedge$ the clean control?	the object (976 cells, $E = 0$) · the law (what $E =$
0 certifies) ·	reference (seven of thirteen letters conflated)

is the closure rule correct? the object ($\wedge$ closes) · the law (the admissibility criterion) · reference (what *monotone* names in the constraint literature) is E(audits) meaningful? the procedure (the set) · the law (the coordinate list, a frame) · reference (whether *audit* means one thing across it)

Two instances was not a law and three is not much better, which §18.4.1 says of its own prediction and which this section repeats of the extension. What is worth recording is that the three were not gathered for the test: each arrived as a difficulty before the shape was drawn. Register 407.

The book has two such decisions and it stumbled on both. Registers 219 to 222 hold three successive positions on the element count, which is the fibration decision made once with the caps in view and once with the element criterion in view and never with both. Register 232 declines the promotion and says the reason is a trade between two things at once. **Two instances is not a law**, and the correspondence is offered as what it is — a shape that fits twice, in a set of ten, where the prediction it makes is that any third two-parent decision will be hard in the same way. Register 278.

2.19 What a safe repair requires

§3.8 recorded a defect and declined to repair it on the ground that the repair was riskier than the defect. That judgement was made without a standard, and this states one.

A repair is **safe** when five conditions hold. **(i)** Its invariant is written down before the edit — what must be true afterwards that was true before. **(ii)** The invariant is computable in the sense of §2.18, so it can be checked rather than asserted. **(iii)** The edit is a *permutation or an insertion*, never a rewrite: existing text is moved, not retyped, so no character can change unnoticed. **(iv)** It is reversible — a copy is taken first and restored if the check fails. **(v)** It is checked afterwards on the source **and** the artefact, because §3’s FIDELITY exists for the gap between them.

And a sixth that governs which repair to attempt: prefer the smallest edit that removes the defect. A defect visible in a hundred places is not therefore a hundred-place repair. Applying that here changed what the repair was.

2.19.1 The register, examined before repairing it

The recorded defect was direction: entries 165 to 204 ascending, 205 onward descending. Measuring the register against its own headings found something larger and cheaper to fix. Each subsection of §28.7 states a count in its title — *Eight more, Twelve more, Fifteen more, Forty more* — and **§28.7.6 was titled “Three from the tower session” and contained eighty-six entries, 199 to 284**, every entry written since that session having been prepended into it. It is now four — the tower’s three, which repairs their numbering — and §28.7.7 the eighty-seven written afterwards.

That is the defect worth repairing, and it is one heading rather than a hundred and twenty entries. The direction is a presentation choice made twice; the heading is a false statement of count, which audit 15 ENUMERATION exists to catch and which it missed because it was pointed at tables and lists rather than at section titles. Register 286.

2.20 The prime directive — nothing runs unbounded

Every action, computation, tabulation, build and verification is executed in a **Zeno structure**: the work is divided until each piece fits inside a declared budget, each piece writes its result before the next begins, and a rerun costs only what has not already succeeded.

One exception, and it is the only one. Fetched literature is exempt. A computation may be halved and rerun on half; an external source cannot, because halving what was read changes what was read. A source consulted at half length is a different source, and the register would carry a claim about a document nobody has seen. So a fetch runs to completion or is not made, and the budget is declared before it rather than enforced during it. Register 432.

No operation runs unbounded. A step declares a budget; work that exceeds it is halved and retried; a step that has succeeded is never recomputed.

Three guarantees, and they are the reason for the rule. Nothing is lost to a timeout, because the last completed step is on disk before the next starts. A rerun costs only the remainder. And a step that overruns **reports its overrun** rather than dying silently — which is the part that matters most, since a computation quietly growing toward a limit is invisible until the run it kills.

Why this book needs it. Audit 1 tests 475,800 pairs; the exhaustive closure of §12.11 swept 47,775,744 ambient cells; the f-shell rebuild of §32.1.2 carried 499,246. Each of those was run whole and none had a fallback. The work is segmented and its state is preserved, and §2.20’s requirement that the audits run on every output now runs under it: the twenty write their state, and an interrupted pass resumes rather than restarting.

What it does not do. It does not make a slow computation fast, and it does not decide anything — a step that fails is retained with its error for retry, not silently skipped. The budget is a declaration about what is expected, so an overrun is a finding about the computation and belongs in the register like any other. Register 343.

3. The twenty-two prime audits, ordered by what a failure costs

Every session closes with twenty-two audits. They are not spot checks, and they are not equal. They

3.0 Ten of the twenty-two test the index; twelve test the book

For a reader who wants the index rather than the artefact, ten audits carry the whole of it:

1.	LATTICE	Λ 's cells, closure and defect, recomputed from the construction
2.	EQUATIONS	the printed identities re-derived — $F(1)$, $F(-1)$, $d(x,y)$, E
8.	CELL	every cell the text names exists in the index
9.	DISTINCTNESS	no two index coordinates collapse
10.	SCOPE	an E value is stated with the index it belongs to
13.	AGREEMENT	one quantity, one value across the whole work
14.	ARITHMETIC	the tower's own table against the recomputed tower
18.	REPRODUCTION	the two headline figures reproduce
20.	PROJECTION	no claim of predictive power anywhere in the text
21.	INPUT	every computation names its input set

These ten would still have to pass if the book were rewritten from scratch in another form. They are about Λ and the objects around it.

The other twelve test whether this book is a usable artefact — CONSISTENCY, REDUNDANCY, ARTEFACT, COHERENCE, ATTRIBUTION, ANTECEDENT, MARKUP, ENUMERATION, FIDELITY, MEASURE, SEQUENCE and CENSUS. **Every one has caught something real**, and several caught faults in this chapter's own reconstruction. *But a reader assessing the mathematics does not need to know that a $\S$ reference resolves or that a figure count matches; those protect the reader from the author.*

The split is stated rather than enforced. All twenty-two run on every build, and the artefact twelve have caught more faults than the index ten — which is the honest reason not to demote them.

The twenty-two are ordered by the cost of the failure each one catches.

The first two can make the book WRONG. The next two can make it UNREADABLE AS AN ARGUMENT. The next two can make it UNUSABLE AS AN OBJECT. The seventh can make it DISHONEST, and it is the only one that cannot be run by reading the book.

1 — LATTICE. Every structural claim recomputed from the construction: closure under join and meet, $E(X) = 0$, rank modularity, the fibre identities, the rank polynomial, $F(1)$ and $F(-1)$, $|J(\Lambda)|$ and the height. Over all 475,800 pairs of Λ , not a sample. First, because if the object is not closed the book has no subject.

2 — EQUATIONS. Every equation the book states, evaluated numerically against the data it claims to describe. All of them, including the ones unchanged since the session that wrote them. Second, because a wrong equation is a false claim.

3 — CONSISTENCY. No two statements in the book may contradict. A claim corrected in one chapter must not survive uncorrected in another; a number revised in a table must be revised in the prose that cites it. Third, because a contradiction leaves the reader unable to tell which statement is the claim.

4 — REDUNDANCY. No statement may survive its own supersession. Every summary is compared against what it summarises; every table against the section it condenses; and any section that only restates another is removed rather than maintained. Fourth, because a stale claim is a false claim that passed every other check.

5 — ARTEFACT. Audits 1–4 read the SOURCE. This one reads the built PDF. Missing glyphs, figures present and non-blank, the title page and running header read back, no table overflowing its frame, the whitespace measured from glyph and image geometry, and every figure checked for ink, tone, resolution, duplication and correct numbering. Fifth, because the source can be perfect and the page still wrong.

6 — COHERENCE. *Source, other places, unusable, presupposes labels are addressable*. Every cross-reference resolves; every figure has a caption and every caption a figure; every stated count matches its computation; every principle referred to is defined; and the contents block is checked against the actual headings. Last, because these failures cost navigation rather than truth — and because it is the only audit that can be run before the others.

7 — ATTRIBUTION. Every claim that points outside the book resolves to a source the book lists, and every source listed is claimed by something inside it. Five checks: each work cited in the text appears in the References; each entry in the References is cited somewhere; §29.2’s owner column names people the References carry; every Appendix D element recorded as *precedent found* has a reachable citation; and every data source in Appendix B is listed. Seventh, because a missing attribution makes the book neither wrong nor unreadable nor unusable — it makes it **dishonest**, which is a worse failure and a quieter one.

Audits 1–2 check the book against the world. Audits 3–4 check the book against itself. Audits 5–6 check the book against what a reader receives. **Audit 7 checks the book against what it owes**, and it is the only one of the seven that cannot be run by reading the book.

2.21 A figure removed must have its data represented another way

§F.3.1 and §E.1.3 remove two counts because a figure computed from the book cannot be printed as prose. **Removal is not a repair on its own**. A stale number is wrong; a missing number is not therefore right, and deleting one without replacing what it carried loses the reader something the book knew.

Any figure withdrawn must have its data represented another way — by the law that generates it, by the stable structure beneath it, or by a computation the artefact performs. The withdrawal and the replacement are one act.

Three representations are available and they are not equivalent.

By the law. Best where one exists. §F.3.1 replaces a count with the constraint that generates it — *definition = stated forbids inputs = enumerated* — which cannot go stale because it is a fact about the assignment rules rather than the text.

By the stable structure. Where no law is available but the shape holds still. §F.3.1 names the nine combinations admitted and absent in every state of the book, against a count that read five different values in one session; §E.1.3 asserts $E(Q) = 0$ under its fibration, which has held at five different item totals.

By computation at build. The weakest, and necessary where the quantity is genuinely a reading. The press prints $E(G)$, the open count, $E(Q)$ unfibred, the register’s span and the withdrawal ratio on every build. **These are readouts and are marked as such** — a number the artefact computes about itself is evidence of a state, not a claim about the work. And a readout is checkable only while its instrument is held: the scripts that compute these — `numbers-index.py` and its companions — are not members of this build, so each stands as a record-carried figure, true of the state it names and not recomputable from the artefact as delivered (register 1813).

And the rule caught its author immediately. §E.1.3 removed E(Q) and supplied only the fibred statement; the unfibred count had no representation at all until the press was extended to compute it, one turn later. **A rule stated after the act it governs is a rule found by breaking it.** Register 374.

The twenty, with their coordinates. §3.8 computes E(audits) and the hierarchy's dimension from these four values per audit, and until now the book asserted both without printing what they were computed from. The table is the input; a reader can rerun either.

no	audit	reads	compares against	a failure costs	presupposes
1	LATTICE	object	a computation	wrong	nothing
2	EQUATIONS	source	a computation	wrong	nothing
3	CONSISTENCY	source	other places	unreadable	claims individu-
ally true					
4	REDUNDANCY	source	other places	unreadable	also mutually
consistent					
5	ARTEFACT	artefact	itself	unusable	also mutually
consistent					
6	COHERENCE	source	other places	unusable	labels are ad-
dressable					
7	ATTRIBUTION	outside	other places	dishonest	claims individu-
ally true					
8	CELL	object	itself	wrong	nothing
9	DISTINCTNESS	object	other places	wrong	nothing
10	SCOPE	source	itself	wrong	claims individu-
ally true					
11	ANTECEDENT	source	itself	unreadable	labels are ad-
dressable					
12	MARKUP	source	itself	unusable	nothing
13	AGREEMENT	source	other places	wrong	claims individu-
ally true					
14	ARITHMETIC	source	a computation	unreadable	claims individu-
ally true					
15	ENUMERATION	source	a computation	unusable	claims individu-
ally true					
16	FIDELITY	artefact	other places	unusable	also mutually
consistent					
17	MEASURE	artefact	a computation	unusable	also mutually
consistent					
18	REPRODUCTION	outside	a computation	dishonest	claims individu-
ally true					
19	SEQUENCE	source	a computation	unusable	nothing
20	PROJECTION	artefact	other places	unusable	claims individu-
ally true					
21	INPUT	source	itself	unusable	claims individu-
ally true					
22	CENSUS	source	itself	wrong	claims
individually true					

Twenty-two audits over twenty-two distinct cells, no collision. The order on each coordinate is object < source < artefact < outside; itself < other places < a computation; wrong < unreadable < unusable < dishonest; and nothing < claims individually true < also mutually consistent, with *labels are addressable* reading as the second. Register 376.

2.22 A corroborable entry must carry its object, and the object must carry it

2.23 A closed index must be stated by its expression, not only by its verdict

A closed index is exactly the set its own envelopes admit. §15.2 proves those envelopes are recoverable from its cells, so an expression always exists — and printing $E = 0$ without it withholds something the author already has.

A reader given the verdict must take closure on trust; a reader given the envelopes can rerun it. The difference is §2.18's, between a computable quantity and a decided one, applied to the central claim this book makes about every object it builds.

Audited across the twenty closed indices this work and its companion carry: fourteen state an expression, one states a fibration and no expression, and five state a verdict and nothing else — the audits at three coordinates, Q fibred by domain, and the geometry, solution and ANEC-coverage indices of the companion paper. Λ states $F(z_1 \dots z_8)$; the tower states its five bounds; the register index states the envelope $\mathcal{R}$ recovered from it; the back-matter Index states its down-set condition. The five that do not are not wrong. **They are unrerunnable**, which is the state §2.18 exists to make visible and this protocol exists to forbid. Register 439.

Any register entry that a proof, theorem, equation or expression settles must name that object, and the object must name the entry. A reference in one direction is a citation; a reference in both is an index.

Half the register fails this, and the failure is asymmetric. Of 219 entry-lines carrying 248 numbers: 66 carry a mathematical object *and* a § reference; **89 carry an object and no reference**; 35 are mathematical in subject and carry neither; 59 are narrative, settled by no object at all. **107 entries are corroborable and not searchable from what corroborates them.**

And the reverse direction barely exists. Of the 132 mathematical objects this book states, **four cite a register entry**. An entry can sometimes be traced to its mathematics; the mathematics can almost never be traced to what it corrected.

A register that cannot be searched from the mathematics is a record of the work and not an instrument of it.

Register 415.

2.24 Never one heuristic — §2.8, and the one thing it does not say

Where a quantity is reached by a heuristic rather than computed, run at least two heuristics of different kind and **report the SPREAD**. A single heuristic cannot tell you whether it is in the case where it works.

The first sentence is §2.8 and this book already had it. *For every quantity that matters, build a second derivation that does not pass through the first* — and the seed is a quantity that matters, prune-greedy was one derivation, and no second was ever built. **§2.8's own gloss is what happened: an identity computed along one path is vacuous.** The protocol was not missing; it was not applied.

What §2.24 adds is the SPREAD. §2.8 asks for agreement between two routes and is satisfied when they agree. **A spread of thirty-three around an exact seven is not disagreement about a value — it is information about which case you are in**, and §2.8 has no place to put it. That is the whole of what is new here.

Found by indexing the protocols: §2.8 and §2.24 occupy the same cell in every coordinate — trigger, object, failure and how each was earned. Register 514.

This protocol exists because that failure ran for twelve cycles. Every seed size in §14.5.7 and §14.5.8 came from one algorithm — prune-greedy — and every expression fitted to those sizes was checked against numbers from the same source. **The fits could not fail**, which is §4.6’s condition for a check that is not evidence.

What the spread shows. Five heuristics on Λ_8 give 12, 7, 7, 7 and **40**, around an exact seven. On a regular down-set they give 7, 7, 7, 7, 7 — and the lower bound is 7 too, so the answer is exact. **The spread is zero where the object is regular and thirty-three where it is not**, and no single run distinguishes those cases.

┆ **A heuristic’s output is a bound. Two heuristics’ outputs are a bound and a warning. Neither is a value.**

And a lower bound is worth as much as a second heuristic: §14.5.9’s disjoint-witness bound is what showed the gap between seven and forty was real rather than a tie. Register 506.

3.1 Why three of the audits were needed

CONSISTENCY: twenty-eight cross-references once pointed at a section that had moved. ARTEFACT: a rendered page once carried glyphs the source did not. ATTRIBUTION: the three precedents that place this book’s central theorem in existing literature were cited in the text and absent from the References. Each history is in the register — 262, 285, 294 and the grouped mechanisms at §28.7.8. **A protocol states what to do; the record states what happened.**

3.2 The whitespace clause, and two measurements that were wrong

A gap a reader sees is a defect the source cannot show. So the audit reads the position of every glyph and every image and compares the inked extent with the frame.

On this book: median foot gap 3.3%, nine pages above 15%, worst 31% — and that one is the end of a section, where a gap is correct. Interior gaps cluster at 16–28%, which is exactly the CondPageBreak(58mm) guard that stops a heading landing at the foot of a page.

Tightening the guard makes it worse, not better. At 46mm the count of gaps falls from nine to seven and the worst rises from 31% to 43%, because a heading allowed to sit lower pushes the figure block after it off the page entirely. 58mm is the optimum, and it was found by measuring rather than by preference.

3.3 Exhaustive, and what to do when it is impossible

Check every cell where the set is finite and small enough. Where it is not, state the sampling and its size in the same sentence as the result, and never let a sampled figure stand where an exhaustive one is affordable — §16.5 stood on 30,000 samples of a 6,912-cell box until register 248.

3.4 And two of the audit’s own measurements were wrong first

The first attempt used the PDF text matrix directly and reported a median inked fraction of 103% — impossible — with y-coordinates of 5, 6 and –220. The text matrix is not page geometry until it is composed with the current transformation matrix.

The second attempt counted characters and not images, and flagged seven pages as having a 25–45% gap at the head. All seven begin with a figure. The space was not empty; it was occupied by something the measurement could not see.

┆ Both are the same error as the audits they were written to replace: a check looking at the wrong artefact. An audit is not exempt from the discipline it enforces, and this one needed two corrections before its first clean reading.

3.5 What the figure audit found

Twenty-eight figures, each checked against its caption and its file.

Clean: every figure has a caption and every caption a figure; no image is blank or unusually dark; no file is duplicated; nothing in fig/ is unused; and after one correction every figure number matches the chapter it sits in.

The one correction. Figures 23.1 and 23.2 appeared in the wrong order — the arity figure sits in §30.3.2 and the step-law figure in §30.3.4, so the numbering ran backwards against the reading order. Swapped.

And one defect that is recorded rather than repaired.

figures at 300 dpi or better	5
between 200 and 300	8
under 200 dpi	15
median	164 dpi

Fifteen figures were produced by an earlier pipeline whose default was figure.dpi = 160, and at 163 mm rendered width that gives 140–164 dpi — below print standard. Regenerating them needs the source data of the sessions that made them, which this one does not carry. Appendix C.3 lists them with the rest of what was inherited and not recomputed.

The audit's value here is not that it fixed the figures. It is that the number is now written down instead of unknown.

3.6 Why the REDUNDANCY audit was needed

Because two sections survived their own supersession and passed all five.

A summary table listed Q's eight items while Appendix E enumerated, indexed and closed them — the same content, maintained twice. And a status table still read “nine literatures entered, nine named and unentered” and “order recovery beyond trees: open, and known to be hard”, both superseded hours earlier by §E.5 and §30.3.

audit	why it passed
lattice · equations	neither is a structural or numerical claim
consistency	both were internally consistent and cited sections
that exist	
artefact	both rendered correctly
coherence	both were well-formed tables with valid references

A stale statement is not inconsistent with anything. It agrees with the version of the book that no longer exists. That is why the check has to compare a summary against what it summarises, rather than a document against itself.

The remedy was to delete a chapter. What survived of it — one paragraph on what the collection closed — opens Chapter 32, where it does work instead of restating.

3.7 The precedences, and the certificate for the dimension

§3.8 states nineteen precedences and an order dimension of 2, **and printed neither the precedences nor the certificate**. Both are here, so both are checkable.

The twenty-two. CELL before LATTICE, since a set containing an invalid cell has no closure worth testing. LATTICE before EQUATIONS. CELL before DISTINCTNESS. EQUATIONS before SCOPE, AGREEMENT, ARITHMETIC and ENUMERATION, since a property of a number presupposes the number. CONSISTENCY before REDUNDANCY, and both before ARTEFACT. MARKUP before ARTEFACT and FIDELITY. ARTE-

FACT before MEASURE, FIDELITY and PROJECTION. ATTRIBUTION before REPRODUCTION. SEQUENCE before ANTECEDENT, COHERENCE and PROJECTION. **And INPUT before SCOPE, ARITHMETIC and ENUMERATION** — all three read a result against its inputs, and cannot run where the inputs are not printed.

The order falls into three components — ten audits, nine, and a pair — and the dimension of a disjoint sum is the greatest of its parts. **A dimension is certified by exhibiting two linear extensions whose intersection is exactly the order**, which is checkable in a page even though finding it is not:

component of nine L1 CELL < DISTINCTNESS < LATTICE < EQUATIONS < AGREEMENT < INPUT < SCOPE < ARITHMETIC < > ENUMERATION L2 INPUT < CELL < LATTICE < EQUATIONS < ENUMERATION < ARITHMETIC < SCOPE < AGREEMENT < > DISTINCTNESS

component of ten L1 CONSISTENCY < REDUNDANCY < MARKUP < ARTEFACT < FIDELITY < MEASURE < SEQUENCE < PROJECTION < > COHERENCE < ANTECEDENT L2 SEQUENCE < ANTECEDENT < COHERENCE < MARKUP < CONSISTENCY < REDUNDANCY < ARTEFACT < PROJECTION < > MEASURE < FIDELITY

dim(hierarchy) = 2, certified. The ten-element component has 4,140 linear extensions and two of them suffice; the nine-element component has 792 and two suffice.

A certificate is checkable where a search is not. Verifying these four lists takes 362 comparisons; finding them took eight and a half million.

That is the third companion paper's third finding arriving in the book's own instruments: **an arrow is a ranking function and a dimension is a realiser, and both are things an analyst supplies rather than derives.** Register 377.

3.7.1 The twenty-first audit — INPUT

Registers 376 and 377 found the same defect twice in two turns: **a result stated from data the book does not print.** E(audits), then 17, rested on coordinates that were never printed; dim(hierarchy) = 2 rested on precedences that were never printed. Both were correct and neither was checkable.

21 — INPUT. *Source, itself, unusable, presupposes the claims are individually true.* **Every stated result must print the data it was computed from.** Not the computation and not the method — the input set, in a form a reader can rerun. A result whose inputs are absent is not wrong; it is **unusable**, which is a different cost and the one this coordinate names.

Its cell was among the seventeen §3.8 leaves admitted and absent, and adding it takes E(audits) from 17 to 16. That is the third of that count confirmed by a defect rather than proposed, after SEQUENCE and PROJECTION.

Why the other twenty could not catch it. ARITHMETIC and ENUMERATION check a number against a computation the book performs; SCOPE checks that a figure carries its frame. **All three presuppose the input exists in the text.** A result whose input was never printed passes every one of them, because there is nothing for them to disagree with — the same shape as PROJECTION's finding that correctness and completeness are different failures.

First run: nine stated results tested, nine pass — but only because registers 376 and 377 printed the two tables an hour earlier. Run against the book as it stood this morning it would have failed twice. Register 379.

3.8 How the set grew, and the two things that held

Seven audits became twenty across these sessions, each addition written as its own account of why the set had been incomplete. Those accounts are one thing said seven times, and it is said once here.

when	audits	E(audits)	what the gap was, and what closed it
as first stated	7	11 at 3 coords	the set was closed under self-ref-
erence: every			audit checked the book against the
book			

ATTRIBUTE added 7 — the first outward-pointing check; found four works cited and unlisted the eleven retained 18 0 at 3 coords every admitted cell occupied DEPTH admitted 18 19 at 4 coords the hierarchy demanded a fourth coordinate; the zero had been cheap SEQUENCE added 19 18 a numbering is a claim about position and nothing tested it PROJECTION added 20 17 a derived list is as complete as its rule, and a rule that misses a case fails silently

Two findings survive the growth and are the reason to keep the record at all.

E is relative to its box. E(audits) read 11, then 0, then 19, then 17 without the audits changing — only the coordinates did. §18.4.1 and register 275 carry the general statement: **a zero is worth exactly the size of the box it was computed in**, so every E owes both its fibration and its coordinate list.

The hierarchy did not tangle as it grew. Nineteen precedences hold between the twenty; the precedence poset falls into three components; its **order dimension is 2, unchanged from seven audits to twenty**. Two coordinates recover which audit runs before which; the other two carry what an audit *is*. Order dimension answers §15's S2 and says nothing about S1, and Λ has the same division — eight coordinates over an order of dimension seven, the surplus axis being the coupling, which is what makes a cell mean rather than sort. Registers 262, 273–276, 285, 294, 295, 309.

3.9 The audits are run on every output, and the requirement is executable

§3 defines twenty-two audits and left when to run them as a habit. A habit is not a check. From this build onward **every output of this work is audited by all twenty before it is delivered**, and the requirement is executed rather than intended: the whole set of twenty runs against the source and the rendered artefact together and exits with the number of failures, so a build that has not been audited cannot be mistaken for one that has.

And three of the twenty-one have never fired, which §4.6 says is not the same as passing. A check must be demonstrated capable of returning a failure before its passing is evidence, and ANTECEDENT, ARITHMETIC and REPRODUCTION have caught nothing in this book — no register entry names any of them. Each has a constructible failing input and none needs new theory:

audit	the input that must make it fail
ANTECEDENT	a label used in Chapter 1 and defined in Chapter 32 – a forward
reference the	source-against-itself read must flag

ARITHMETIC a product stated beside its own printed factors and wrong: $33 \times 5 = 166$ against §12.6.1's table, which prints 33, 5 and 165 REPRODUCTION an altered Condon–Shortley term entry, against the microstate enumeration that reproduces ten tables of ten exactly

Running these three is not optional bookkeeping. §3.6 records two sections surviving their own supersession and passing five audits; §3.4 records the whitespace audit's first two measurements being wrong. **An audit that has never returned a failure is indistinguishable from one that cannot**, and this book has three. Register 422.

Why a file and not a rule. Register 259 records a defect repaired twice and reintroduced in the next turn, and §4.4 draws the conclusion: a fix that leaves the mechanism unchanged is an instance fix. Every audit in this book was run by hand until now, which made *audited* a claim about a session and not a property of an artefact. It is now the second. The script reads the source by name, the PDF by name and the press by name, so FIDELITY and PROJECTION compare three files that must agree and none of them can be updated alone without the run reporting it.

What it does not do. It decides nothing. Every flag it raises is a hypothesis under §4.7 and must be localised before it is recorded — four of the last five flags this book has seen were the detectors and not the book. And **it cannot audit itself**: the script is an instrument, its own correctness is untested by anything in the twenty, and that is a gap this section admits rather than closes. Register 300.

4. The failures of the assistant, and the protocol each one earns

Ten mechanisms, each earned by a failure that happened. They are not a taxonomy assembled in advance; every one was extracted from the register after the fact, and §28.7.8 groups the entries.

4.1 **attributed outward before checking inward** — a recomputation must suspect the model before it suspects the text 4.2 **wrote into a structure without reading it** — enumerate the targets before writing 4.3 **a claim of completion is a claim** — verify completion in the pass that claims it 4.4 **a count kept by judgement** — read the list; do not carry the number 4.5 **a conclusion written before the output** — the order matters even when the answer is right 4.6 **a test that could not fail** — exhibit the failure mode before trusting the pass; a > criterion fixed from the case it judges cannot fail — proved by the case it was built for and > then caught 4.7 **a flag is a hypothesis** — localise a detector's output before filing it as a defect 4.8 **a reading the source never made** — an assumption is the reader's until the source states it 4.9 **a coordinate collapsed into a count** — ρ , and every scalar standing in for a structure 4.10 **a stale figure left behind its own correction** — a figure about the book goes stale the moment the book is written into

And §28.7.9 adds a coordinate this list does not have: the **LEVEL a fault landed at**. Three of the ten sit at the value, three at the derivation, four at the inference — and the axis merges two the list keeps apart, since *a stale figure* is a value going stale and *a conclusion written early* is an inference drawn early: **the same act at two depths**.

This chapter is short because the material is the register. Each mechanism is a name for a class of entry and the entries are the evidence. *An earlier form was a heading with no body at all, carrying the word SEVEN while the register held ten — cited as §4.1 and §4.2 throughout a book that never defined them. Register 655.*

5. What the protocols are for

Each prevents a specific failure, and each failure occurred:

protocol	failure it prevents	
instance		
§2.1 channel construction	claiming cells across a label change	Si
I, P II		
§2.2 admissibility	bracketing unresolved levels	–
§2.3 cost resolvability	reading rounding as physics	Al
I nf		
§2.4 dilution	excluding a viable species	Ar
II, Bi I		
§2.5 census first	searching without targets	27
sequences		enu-
merated		
§2.6 declines recorded	an uncheckable collection	Ne
II, Cu II		
§2.7 exhaustive testing	a claim resting on a sample	A.4
on all cells		
§2.8 two routes	an undetectable derivation error	w =
3v.e		
§2.9 refuse, don't coerce	a missing value read as a real one	the
cap discard		
§2.10 retrieval	abandoning a cell when a source is blocked	p =
6 vs p = 1		
§2.11 provenance	an untraceable number	153
channels		
§2.12 withdrawal	an unfalsifiable record	48
entries		
§2.13 commit before you look	a choice that depended on its answer	the
anchoring failure		
§2.14 compute then write	the dominant failure	
most of Chapter 28		
§2.15 apply P8 to every failure	a failure discarded rather than banked	ten
bounds on §30.3		

The protocols are not advice. They are the failure modes of this work, inverted.

PART II — THE LATTICE

And the table stopped where the protocols stopped being about spectroscopy. Eight are uncited by it — §2.4 above, and §2.16 through §2.22, every one written after this section was. They are added here, and one of them needs a second column.

protocol instance	what it prevents
§2.16 the grid	hunting where a layout would show
§30.3, 22 formulations	
§2.17 triangulation	three measurements read as three facts
the step law, 1, 2, 4	
§2.18 computable or decided	a decision checked as though computable
reg. 277	
§2.19 what a safe repair requires	a repair riskier than its defect
reg. 286, 287	
§2.20 the prime directive	a computation lost whole to a timeout
reg. 343	
§2.21 a withdrawn figure's data	a number deleted rather than replaced
reg. 374	
§2.22 a corroborable entry	a defect that cannot be traced to its proof
reg. 415	

And the second column is P16's, which no failure corroborates. P16 — treat bounds and limits as coordinate values — has three realisations and no register entry, because **it was never learned from a failure; it was adopted and then built with:** §17.4 replaces (g_1, g_2) by (g_1, G) and conservation holds by construction; §12.11.0.2 carries (g, G) and recovers prior occupancy as a derived quantity; §18.4.2 indexes the partial sums of a mass vector and reaches treewidth 1 at every n . **An operation may be corroborated by a construction as well as by a failure**, and P16 is the only one in this book that is. Register 421.

6. The periodic table is not a closed index

Take the periodic table at its word. Its coordinates are period and group; its cells are the elements. Ask what those coordinates, and the occupied cells alone, imply about which other cells could exist.

The answer is thirty-six cells that do not.

main-table cells (lanthanides and actinides set aside)	90
cells the structure admits	126
external definition cost $E(X)$	36

The thirty-six are (p_1, g_2) through (p_1, g_{17}) , (p_2, g_3) through (p_2, g_{12}) , and (p_3, g_3) through (p_3, g_{12}) — **the gaps in the short periods, every one of them.**

This is not a defect in the periodic table. It is a fact about what the periodic table's coordinates can and cannot carry. Period and group locate an element; they do not encode why period 1 has room for two elements and period 4 for eighteen. That information travels alongside the table, in the schooling of whoever reads it.

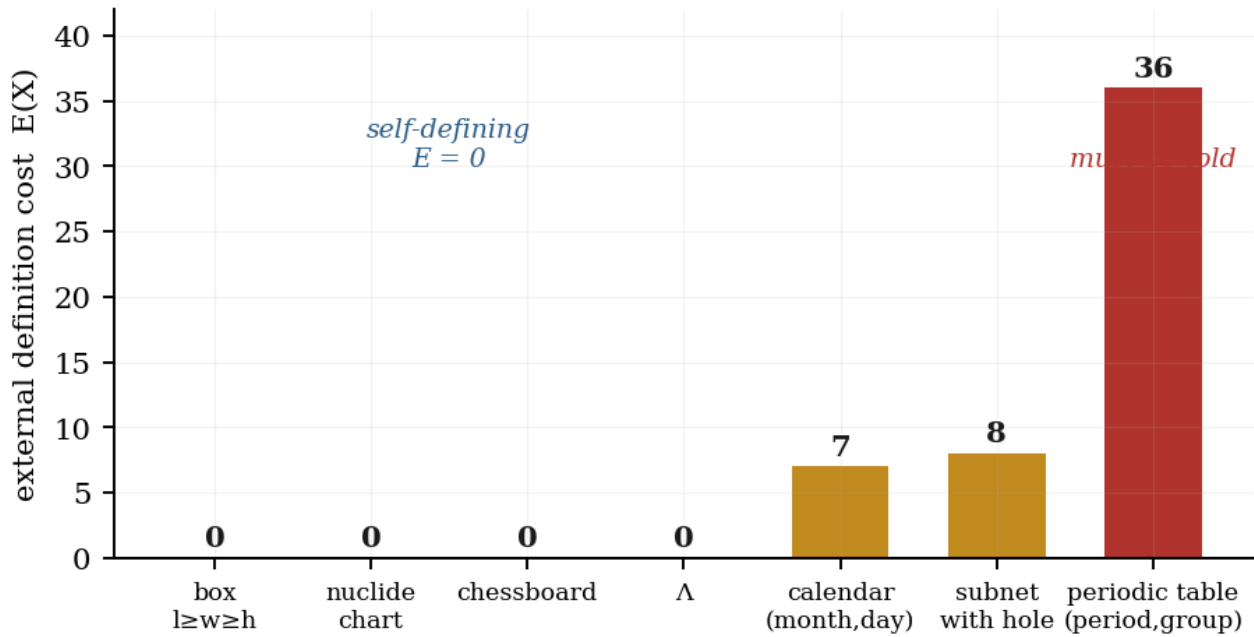


Figure 6.1. The external definition cost of seven indices. Four are self-defining. The calendar's seven cells are the content of Thirty days hath September; the periodic table's thirty-six are the gaps in the short periods.

6.1 What “the structure admits” means

The reconstruction is mechanical. Given a set of cells X on coordinates $(x_1, \dots, x_d)$, read off two things:

the value sets $\hat{A}_i(X) = \{ x_i : x \in X \}$ — which values occur the bounds $\hat{\phi}_{ij}(v) = \max \{ x_i : x \in X, x_j \leq v \}$ — how far one coordinate reaches given another

and then form

$$\mathcal{R}(X) = \{ x \in \prod \hat{A}_i(X) : x_i \leq \hat{\phi}_{ij}(x_j) \text{ for all } i \neq j \}$$

No outside knowledge enters. $\mathcal{R}(X)$ is what a reader could reconstruct from the cells alone.

$E(X) = |\mathcal{R}(X)| - |X|$ is the gap: cells the structure implies and the index denies. For the periodic table it is 36. For Λ , built in the next chapter, it is 0.

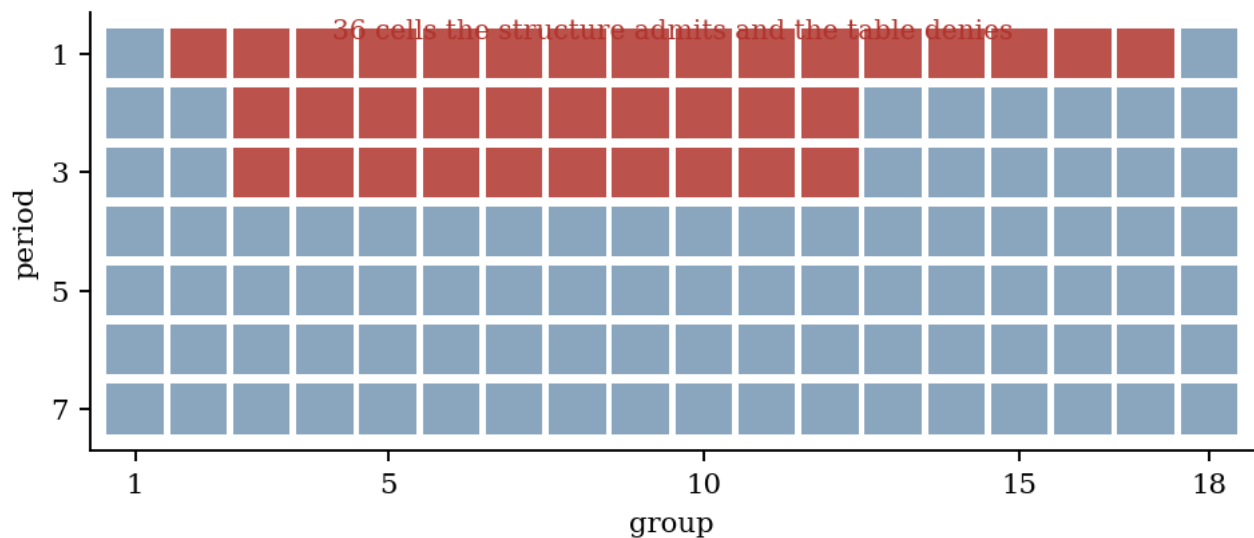


Figure 6.2. Blue: the ninety cells the table holds. Red: the thirty-six its own coordinates admit and it denies. A reader given only the occupied cells would reconstruct all one hundred and twenty-six.

6.1.1 And a known layout costs nothing

E is a property of a layout, not of chemistry. Computed on four conventions over the same elements:

And the thirty-six decompose, but not as this book has said, and the missing cell is helium's. Recomputed on the drawn eighteen-column layout with all 118 elements:

the thirty-six	count
forbidden by $\ell \leq n-1 - 1d(10), 1p(5), 2d(10)$	25
real but deferred by the Madelung order – $3d(10)$, and period 1 group 2	11

1p contributes five and not six, because helium occupies group 18 itself, and the slot helium vacates — period 1, group 2 — is a *deferred* cell rather than a forbidden one, since $\ell = 0$ there satisfies $\ell \leq n-1$. Two constraints casting a shadow of twenty-five, and eleven cells that could hold an element and do not.

And E is not a property of the elements. It is a property of where helium is drawn.

helium at	cells	E
group 18	90	36
group 2	90	20

The mechanism is the envelope: $\varphi(\text{group} \mid \text{period} \leq 1)$ is 18 when helium sits at group 18 and 2 when it sits at group 2, so the entire first row of gaps exists only under the standard placement. **Helium's position is the most argued question in periodic-table design** — IUPAC places it at 18; the left-step form and a quantum-chemical case place it at 2, at the cost of the atomic-number triad He(2), Ne(10), Ar(18). **E measures what that choice costs to specify, and the difference is sixteen cells.** Register 448.

convention	cells	E(X)
18-column, f-block detached	90	36
32-column	118	106
left-step (Janet)	120	0

The Janet left-step table is closed. Order the blocks f, d, p, s and define the period by $n + \ell$, and the same elements cost nothing to specify. **The thirty-six belong to the drawing that hangs in classrooms**, and a known alternative costs none — exactly as §6.3 shows for the calendar, where $E = 7$ is the price of keeping January first.

One caution. $E(\text{periodic table}) = 36$ has $\mathcal{D} = 0$. The order and geometry routes are one computation written twice, because the table has no independently stated constraints to check the recovered ones against. Only Λ 's $E = 0$ is defended, and only because Λ 's constraints are given in §7.1 before any cell is enumerated.

6.2 It is not a peculiarity of chemistry

The same computation on other indices:

index X E(X) self-defining

a box ordering, $l \geq w \geq h$ 56 0 yes

the drawn nuclide band, between drip lines 383 0 yes

a chessboard 64 0 yes

index X E(X) self-defining

Λ 976 0 yes

a subnet with a hole 248 8 no

the calendar, (month, day) 365 7 no

the periodic table 90 36 no

The nuclide row names a drawn band and not a measurement, and the distinction is the whole of §6.1. A chart drawn between two monotone drip lines has $E = 0$, and it has $E = 0$ **necessarily**: $\mathcal{R}$ is idempotent — §32.4.1 proves it — so $\mathcal{R}(X)$ satisfies $\mathcal{R}\mathcal{R}(X) = \mathcal{R}(X)$ for any X whatever. **A band's closure is a theorem, not a finding about nuclei.** Reporting it beside the calendar's seven, which is a measurement, put a tautology and an observation in the same column.

The measurement is the other object, and it does not close. Taking the particle-bound nuclides themselves over $Z \leq 9$ — the chart as nature draws it rather than as a textbook does — gives **E = 9**, and the nine cells are named:

| He-5, He-7, Li-10, Be-8, Be-13, B-9, B-16, B-18, C-21

Every one is a known unbound nuclide sitting inside the band its own neighbours define. And the band is exactly $\mathcal{R}$ of the measurement: **the drawn chart is the closure of the real one**, which is why the two were conflated and why the conflation is worth the entry.

What the nine are is content the coordinates cannot carry. Be-8 is unbound because it is two alpha particles and nothing else — clustering. He-5, He-7, Li-10 and Be-13 are unbound by one neutron against an even-paired core — pairing. (Z, N) knows how many protons and neutrons a nuclide has and cannot know that four of them prefer to be an alpha, or that the last one is unpaired. **The nine cells are the pairing and clustering terms of the mass formula, counted.** That is the same shape as the calendar's seven and the periodic table's thirty-six: a defect that names a real mechanism the index has no axis for. Register 298.

And the same shape appears a third time, in chemistry, where it can be watched arriving. An index may bracket a quantity only along an axis the quantity is monotone on. Give first ionization energies to that test as bare numbers, with no shell model attached:

period 2 Li 5.392 Be 9.323 B 8.298 C 11.260 N 14.534 O 13.618 F 17.423 Ne 21.565 period 3 Na 5.139 Mg 7.646 Al 5.986 Si 8.152 P 10.487 S 10.360 Cl 12.968 Ar 15.760

Monotonicity fails at Be → B and N → O, and again at Mg → Al and P → S — the same two positions in both periods. Those are the $s^2 \rightarrow p^1$ subshell opening and the $p^3 \rightarrow p^4$ first pairing: the two anomalies every general chemistry course teaches, located from the numbers alone. Twelve interior cells, four steps refused, **67% admissible**.

It is a retrodiction and not a discovery — both have been explained since the 1920s. What earns it a place here is that it is *mechanical*: **where an index refuses, a mechanism is entering that the coordinates do not carry**, and the refusal map says where without being told what. The periodic table's thirty-six, the measured nuclide chart's nine, and these four are one phenomenon counted in three subjects — shell structure, pairing and clustering, subshell opening and pairing again. Register 308.

And a ninth subject entered this book without a coordinate being added to it: time. Every subject above is a thing indexed. Time is not — §F.4.3 records that a date must never enter Λ , since a transition is a type and types are not dated, and §12.11.0 shows what enters instead: **composition**. An order between cells rather than a stamp on one.

What makes it a subject rather than a metaphor is that it is measured. **Occupancy never rises along composition** — zero of 739 steps — so the index has a direction it was not given. The object graph falls into three eras pure in occupancy, circulation is total inside each and impossible between them, and 904 of 1,654 cells are conservative. **The clock is a coordinate the book already had, read as an order.**

Time is the one subject here the index did not have to be extended to reach. It was already carrying it, in the constraint that says you cannot place more than you removed.

And it is the subject that most sharply states the method's limit. §12.11.0.1 shows **definability is not reachability** — a cell is its own seventeen-bit description and reaches 24.7% of the object, the two orders sharing 0.4% of what either carries. An index can say completely what a thing *is* and almost nothing about what it can *become*. Register 348.

The calendar's seven are (2,29), (2,30), (2,31), (4,31), (6,31), (9,31) and (11,31).

That is the content of *Thirty days hath September*. The rhyme exists because the calendar's coordinates cannot carry the month lengths, and the rhyme is exactly seven cells long.

6.2.1 The nuclide chart's defect, recomputed under variation

§6.2 reports $E = 9$ on the measured chart at one cutoff. **The measurement is stated here to F.3's highest standard** — the method named with its input set, the inputs printed, and the result recomputed at four proton-number cutoffs rather than once.

The method. Take the measured nuclides as cells (Z, N) , close under the monotone envelopes of §6.1, and count what the closure admits and the chart lacks. **The input set is the chart itself**, printed at §6.2 for $Z \leq 9$, and the variation is the cutoff.

$Z \leq$	nuclides	admitted	E	the cells admitted and absent
5	27	33	6	He-5, He-7, Li-10, Be-8, Be-13, B-9
6	40	48	8	...and B-16, B-18
7	52	61	9	...and C-21
9	80	89	9	unchanged

Every cell named at one cutoff persists at every larger one, and none is added after $Z \leq 7$. The defect is not an artefact of where the chart was cut.

That matters because a closure defect measured once can always be blamed on the boundary. **Here the boundary was moved four times and the nine cells did not move**, so they are a property of the measurement rather than of the window — and each of the nine is a real nuclide the chart omits because it is unbound, which is pairing and clustering, mechanisms the two coordinates cannot carry. Registers 388, 390.

6.3 The calendar can be repaired, and the repair is instructive

The calendar fails for one reason: $\text{days}(m)$ is not monotone in m , because February has 28. Relabel the months in order of length — February, April, June, September, November, January, March, ... — and E drops to 0. The 365 cells are identical; only the labels move.

And the repaired calendar is useless. A calendar whose months run in order of length is self-defining and unusable.

So the trade is explicit, and it has a number:

$E(X) = 7$ is the price of keeping January first.

Non-closure is not always a fault. It is sometimes a purchase — usability bought with definitional self-sufficiency. What this book supplies is the price tag, not the verdict.

6.4 What follows

Λ is the same 118 elements on different coordinates, with $E = 0$. It adds no physics. What it removes is the thirty-six.

The rest of Part II builds it and establishes what it is. Part III proves why $E = 0$ is not an isolated virtue but the source of two others.

7. The construction of Λ

Λ is a set of eight-tuples. Each is a transition cell: a source configuration, a target configuration, and the electron count moved between them.

$(n, \ell, k, q, e, f, g, 2S)$

n	source shell
ℓ	source subshell
k	source occupancy
q	electrons removed
e	target shell
f	target subshell
g	target occupancy
$2S$	multiplicity

Five further coordinates — $2S'$, v , $2J_c$, K and $2J$ — extend Λ_8 to Λ_{13} when target spin, seniority and angular-momentum coupling are carried explicitly; §12.11 builds the tower and fixes the order, which is J_c , then K , then J . Everything in this book that does not concern coupling is stated for Λ_8 .

One row of the table below is not like the others. Three origins — Pauli, the hydrogenic solution, counting — yield exact constraints. The fourth, vector coupling, yields an envelope: $2S \leq k$ contains every physical multiplicity and admits values no k electrons can carry, at 48.5% density at the caps of §7.4. §12.11.3 states the dichotomy and §12.11.2 proves it is forced.

7.1 The constraints, and where each comes from

Λ is not all of the eight-fold product. It is the subset satisfying:

constraint	origin
$\ell \leq n-1$	hydrogenic radial solution
$k \leq 2(2\ell + 1)$	Pauli exclusion
$q \leq k$	counting — cannot remove more than are present
$f \leq e-1$	hydrogenic radial solution
$g \leq 2(2f + 1)$	Pauli exclusion
$g \leq q$	counting — cannot place more than were removed
$2S \leq k$	vector coupling

Seven constraints, four origins, and nothing else. No constraint in Λ comes from anywhere but Pauli, the hydrogenic radial solution, counting, or angular-momentum coupling. That exhaustiveness matters in Part III, where it is the reason the object's properties can be attributed to *indexing* rather than to physics smuggled in through a bound.

7.2 Every constraint is of one form

Each constraint reads $x_i \leq \varphi(x_j)$ — one coordinate bounded by a monotone function of one other. None is a sum. None is a difference. That is not an accident of presentation; Chapter 17 shows that a sum could not appear here, because a set cut out by a sum bound is not closed.

Consequence, developed in Chapter 8: the graph whose nodes are coordinates and whose edges are constraints is a **tree**.

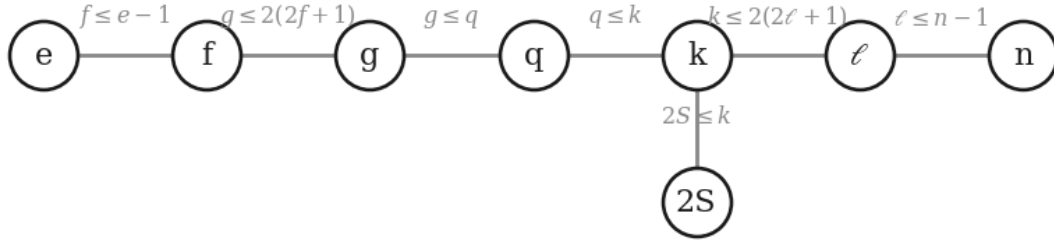


Figure 7.1. Every constraint binds one coordinate to one other. Eight nodes, seven edges, connected — a tree, and therefore treewidth 1. Three results in this book follow from that fact alone.

7.3 Λ is closed

For any two cells $x, y \in \Lambda$, both $x \vee y$ and $x \wedge y$ — the coordinatewise maximum and minimum — are again in Λ .

Proof. Each constraint is $x_i \leq \varphi(x_j)$ with φ non-decreasing. Suppose $(x \vee y)_i = x_i$. Then

$$(x \vee y)_i = x_i \leq \varphi(x_j) \leq \varphi(\max(x_j, y_j)) = \varphi((x \vee y)_j)$$

by monotonicity. Meets are symmetric. ■

And $E(\Lambda) = 0$, verified by direct computation at four cap settings — 216, 976, 1,636 and 2,394 cells — exact in one step and stable under a second application of $\mathcal{R}$.

7.4 A note on caps

Λ is infinite; every count in this book is stated at explicit caps on n, e, ℓ and k . The counts change with the caps. The shape does not — a claim made precise in Chapter 8, where the generator patterns are shown identical across a twenty-eight-fold range in cell count.

Where a number is cap-dependent it is marked. Where a structural claim is made, it has been checked at no fewer than four settings.

8. What Λ is — distributive, modular, Sperner

The standing convention, stated once so it need not be repeated. Every figure in this book that depends on the caps is computed at $(n_{\max}, e_{\max}, \ell_{\max}, k_{\max}, f_{\max}) = (3, 3, 1, 3, 1)$ unless the section names others. Where a figure is recomputed under variation the settings are printed beside it, as at §6.2.1, §12.11.1.1, §12.11.1.2 and §22.1.1.1.

A cap-dependent figure without a stated cap is not thereby unqualified — it is at these caps. What would be a defect is a figure at *other* caps that does not say so, and audit 10 checks for that.

The convention was in force from the first page and had never been written down: 84 paragraphs carry a cap-dependent figure and 69 of them name no cap, which is 82% relying on a rule the book did not state. **§32.4.2's distinction is why this matters** — every one of those 69 carries a *value* rather than an identity, and a value without its setting is a reading whose conditions are missing. Register 395.

8.1 Distributive

Λ is a sublattice of a product of chains, and every such sublattice is distributive. Verified on 4,000 randomly drawn triples, no failures.

This is the least surprising fact in the chapter and the most load-bearing: it is what admits Birkhoff's representation, which is the next section.

8.2 Rank is conserved

Λ has a conservation law, and it is exact.

$$\text{rank}(a \vee b) + \text{rank}(a \wedge b) = \text{rank}(a) + \text{rank}(b)$$

Verified on all 475,800 pairs: zero violations.

This is modularity, and it is strictly stronger than what most rank functions satisfy. Matroid rank is *submodular* — the left side is $\leq$ the right. Entropy is submodular. **Λ 's rank is modular, with equality, and that equality is the signature of distributivity.**

8.2.1 What it means

Join and meet redistribute rank without creating or destroying it. Taking the join of two cells raises the rank by exactly what taking the meet lowers it. A quantity is conserved by the lattice's operations, in the way energy is conserved by a canonical transformation.

The question came from outside the subject. Celestial mechanics is organised around invariants — the Jacobi integral, angular momentum, energy — and asking *what does Λ conserve* is not a question a spectroscopist would think to put. **This book had not put it.**

8.2.2 And it is a cheaper test than §8.1's

§8.1 verifies distributivity by checking $a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c)$ over triples — $O(n^3)$. Modularity of rank is a pairwise condition, $O(n^2)$, and on a graded lattice with this rank function it is equivalent:

test	sample	time
distributive law, triples	4,000	0.0375 s
modular rank, pairs	4,000	0.0188 s

Both true. The second is the one to run first, because a failure of modularity is a failure of distributivity and it is found in half the work.

8.3 Seventeen generators

By Birkhoff's theorem, a finite distributive lattice is the lattice of down-sets of its poset of join-irreducibles. For Λ at caps (3,3,1,3), $k \geq 1$:

976 cells $\leftarrow$ 17 join-irreducibles, 20 covering relations

and the correspondence is exact: **all 976 down-sets of the 17-element poset are precisely the 976 cells.** A fifty-seven-fold compression.

The covering relations are the physics made visible:

cover	reads
$e=2 \leq e=2, f=1$	a target subshell requires its shell
$k=2 \leq k=2, q=2$	you may remove what you have
$n=2 \leq n=2, \ell=1 \leq n=2, \ell=1, k=3$	shell before subshell before occupancy

And the shape is cap-independent. Across six settings from 976 to 27,873 cells the generator count grows — 17, 23, 31, 32 — while the *patterns* hold at **fifteen, with zero new and zero lost**. Only multiplicities change.

That is what licenses stating the poset once.

The generating poset of Λ — 17 join-irreducibles whose down-sets are all 976 cells

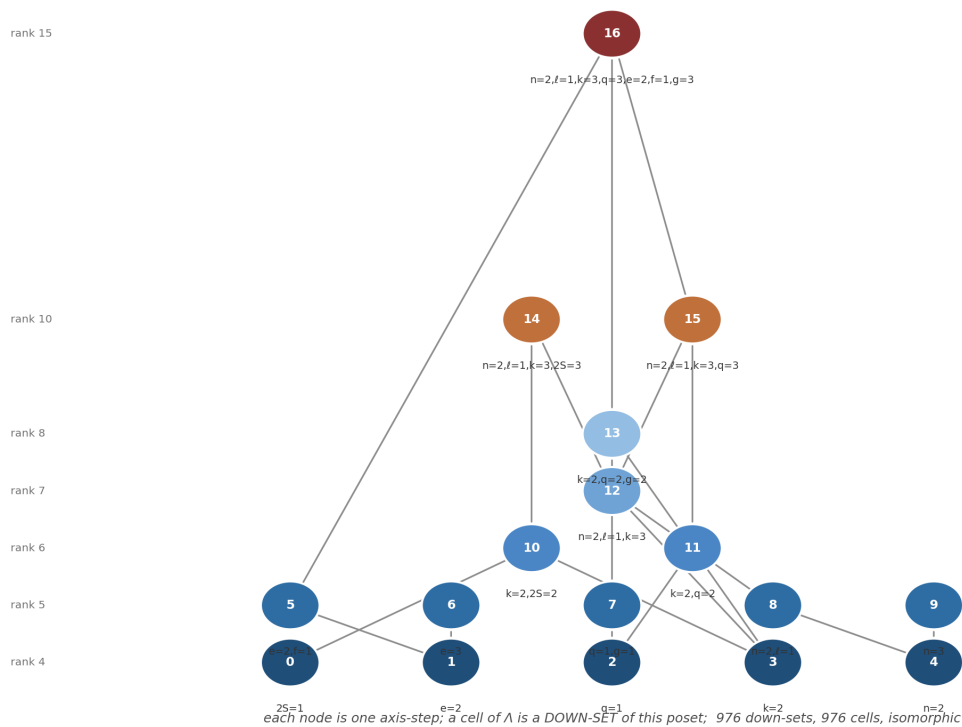


Figure 8.1. The seventeen join-irreducibles and their twenty covering relations. Every one of the 976 cells is a down-set of this poset, and every down-set is a cell — a fifty-seven-fold compression, exact. The covers are the constraints of §7.1 made visible: shell before subshell before occupancy, and a target subshell requiring its shell.

8.4 Sperner, and not symmetric

The largest antichain in Λ equals its largest rank level — the Sperner property — verified exactly at five cap settings by Dilworth’s theorem and bipartite matching, no exceptions. At caps $(3,3,1,3)$, $k \geq 1$ the maximum is 122 at rank 11.

The rank sequence is **log-concave**, hence unimodal, which is why Sperner holds rather than merely happening to.

But Λ is not rank-symmetric, so it has **no symmetric chain decomposition** and the classical result for divisor lattices does not transfer. The asymmetry is itself a measurement:

centre of mass 11.07 against midpoint 11.5 — skew -0.43 , computed from the full rank sequence at caps $(3,3,1,3)$, $k \geq 1$, over all 976 cells

Low ranks are cut by floors ($n \geq 1$, $k \geq 1$, $e \geq 1$); high ranks by caps ($\ell \leq n-1$, $k \leq 2(2\ell+1)$, $q \leq k$, $g \leq q$). **There are more caps than floors, so the top is pruned harder.** The skew is how much more Pauli constrains than counting does.

Nor is Λ self-dual. Only 8 of 976 cells survive the reflection $x \mapsto \max - x$, coordinate by coordinate — reproduced exactly on recomputation. The tower session's competing figure of 112 was computed under $x \mapsto \max + \min - x$, a different map, which the floors $n \geq 1$, $k \geq 1$, $e \geq 1$ separate from this one; under that map 112 cells have their image in Λ and none is fixed. Both counts are right about their own reflection. The discrepancy was definitional, and it is closed. This reflection is not decoration: restricted to one occupancy coordinate it is particle-hole conjugation, $\text{terms}(\ell^k) = \text{terms}(\ell^{4\ell+2-k})$ exactly, and §12.11.2 shows it is one of the three machines that price every coupling axis. The two sides of the object are genuinely distinguishable, which Chapter 12 develops.

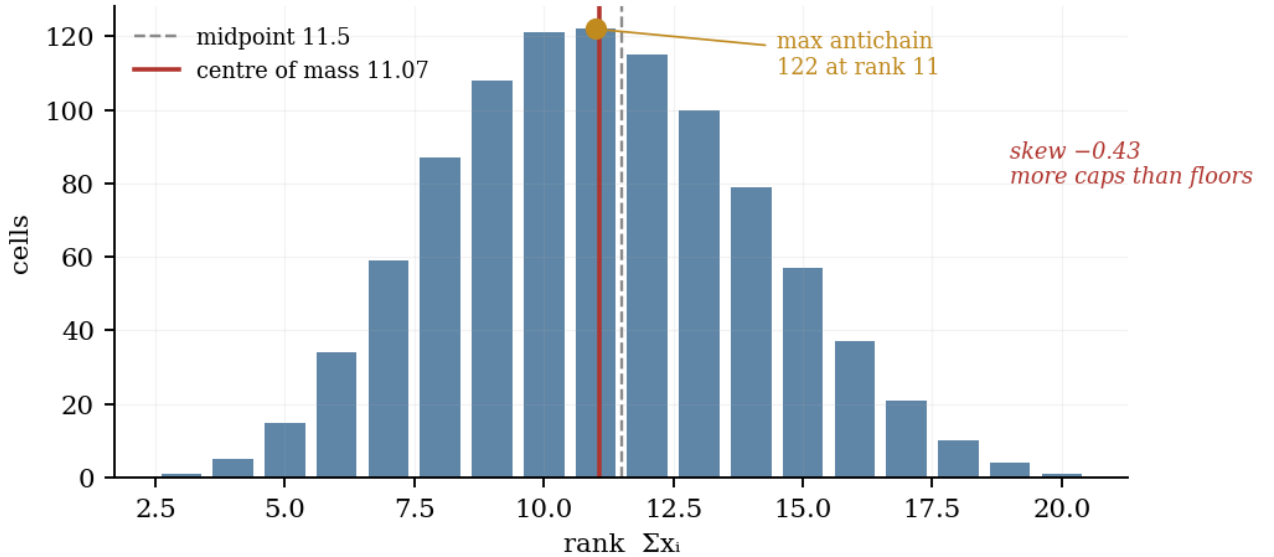


Figure 8.2. The rank sequence over all 976 cells is log-concave, hence unimodal, hence Sperner. It is not symmetric: the centre of mass sits below the midpoint by 0.43, because Λ has more caps than floors.

8.5 The constraint graph is a tree

Eight coordinates, seven constraints, connected:

$e - f - g - q - k - \ell - n$, with 2S attached to k

Treewidth 1. Three consequences run through the rest of the book:

The void needs no sieve. A tree has no cycles, so there is nothing for inclusion–exclusion to correct. Chapter 10 gives the closed-form count.

The order is recoverable. Chapter 15 recovers Λ 's coordinate orders from an unlabelled bag of cells by propagation along this tree — 20 of 20.

And the constraint graph offers no redundancy. There is exactly one path between any two nodes. So the two-route protection of Chapter 16 cannot come from the constraints; it must come from derived quantities. **That is a real constraint on how the object defends itself, and it is visible here, three chapters early.**

8.6 Order dimension

Λ_8 has order dimension exactly 7. Seven linear orders rebuild the partial order, and no six can. The eight-coordinate construction gives $\dim \leq 8$ for free; the eighth is never spent.

The figure printed here through every earlier build was 8, asserted rather than derived — register 409 flagged it as the one quantity in Part II carried without a derivation — and the derivation, now run, corrects it. The order dimension of a finite distributive lattice is the width of its poset of join-irreducibles — Dilworth — and §8.3’s seventeen generators have width **seven**, certified both ways: a partition of the seventeen into seven chains, and seven of them pairwise incomparable — one generator for each coordinate except g . The generator that first raises g is $(1,0,1,1,1,0,1,0)$, and it sits above the generator that first raises q , because $g \leq q$: **you cannot raise the coupling without having already raised what it couples.** Every g -raising generator lies above the q -atom, so g contributes no independent member to any antichain of generators, and the order pays seven dimensions for eight axes. The coordinates remain irredundant in value — no axis is a function of the others — but in order, g ’s information rides on q ’s. Λ_9 measures width 7 as well, so the dimension does not rise by one per adjoined axis; it rises when the new axis brings an independent generator, and a coupled axis need not. This is register 36’s method — exhibit the antichain, or fail to — run at last past three coordinates, and the one underived number in Part II is the one it refutes. Register 409.

Λ occupies **14.1%** of its own bounding box — 976 cells of 6,912.

9. The arithmetic encoding

Assign the i th coordinate the i th prime and read a cell as an integer:

$$N(x) = \prod_i p_i^{x_i}$$

The lattice becomes arithmetic, exactly:

lattice	arithmetic
$x \leq y$	$N(x)$ divides $N(y)$
$x \vee y$	lcm
$x \wedge y$	gcd
$\text{rank } \Sigma x_i$	$\Omega(N)$, prime factors with multiplicity

Λ is a sublattice of the divisor lattice of a single integer.

9.1 Two classical functions appear as lattice quantities

$\text{rank}(x) = \Omega(N(x))$ — verified on all 976 cells.

$\omega(N(x)) \leq 8$, the coordinate count — the count of *distinct* primes is bounded by the number of coordinates, and the bound is tight: the cell $(2,1,3,3,2,1,3,3)$ uses all eight. *Proof:* $N(x) = \prod p_i^{x_i}$ uses at most one prime per coordinate. ■ It is not bounded by the order dimension, which §8.6 puts at 7: at that cell ω exceeds $\dim$, because a cell can be non-minimal in more coordinates than the order has dimensions.

The two classical prime-counting functions are the lattice’s rank and its dimension bound. Neither was put there.

9.2 The occupancy measure

$$d(x, y) = \tau(\text{lcm}(N(x), N(y)) / \text{gcd}(N(x), N(y)))$$

And this is one leg of something the book states three times without naming it once.

Between any two points there is an interval, and the method returns its measure.

Not an equation connecting two points — the orders are partial and share 0.3%, so no single order relates an arbitrary pair. **An interval between two points is total**, because every pair of cells has a meet and a join whether or not either order relates them. That is why $d(x, y)$ works where the order does not.

between two cells five forms two states under worst composition two measurements	the interval [xΛy, xV y] [min, max] output rank [T(n-1), T(n+1)]	its measure $d = \prod (\backslash \Delta_i \backslash + 1) = \tau(\text{lcm}/\text{gcd})$, 17% of 264 inputs single-valued, spread 12, range [3, 23] w, priced at $V = 4v/3$
--	---	--

§12.11.0.12 says the middle row in its own words — *from one coordinate up, a transition is not a function of its inputs; it is an interval* — and §22.1 gives the third. §27 already proves the three MEASURES are one quantity; what was missing is that the three OBJECTS are one shape.

And the form survives what an equation would not. Theorem 18.2 permits it, because an interval carries *less* than the coordinates and not more. §18.6 permits it, because an interval bounds where something already is rather than proposing an unlisted cell — §25.6.3's own correction. And the 0.3% overlap stops being a defeater and becomes the content: **two intervals per pair, nearly disjoint, so what a thing IS and what it can BECOME are separately bounded and rarely both.**

It is universal over indices whose constraints are pairwise monotone envelopes, per §14.3, and not over all mathematics. **And it is why this is a method rather than an equation:** an equation returns a value; this returns an interval and a verdict on what the interval is about, which by §18.4.1 is whether E measures the drawing or the world. That second half is not expressible as an equation. Register 438.

with τ the divisor-counting function. $d(x, x) = 1$, because $\tau(1) = 1$.

That is not a convention. A point has no volume, but it is one point, and it counts itself. The measure counts *cells*, and a cell is occupied.

Five equivalent forms, verified 500/500 and 2,000/2,000:

1. $||[x\Lambda y, xV y]||$ — interval count
2. $\prod_i (|x_i - y_i| + 1)$ — coordinate form
3. $\tau(N(x)N(y)/\text{gcd}^2)$ — two integers
4. $\tau(a \cdot b)$ where $N(x)/N(y) = a/b$ in lowest terms — one rational input
5. $\prod_p (|v_p(\rho)| + 1)$ — p-adic Properties. Symmetric · $d \geq 1$ with equality iff $x = y$ · multiplicative triangle inequality $d(x, z) \leq d(x, y) \cdot d(y, z)$, 4,000 of 4,000 · $\log d$ is an ordinary ℓ^1 metric · balls are hyperbolic, with boundary $(1+\Delta_1)(1+\Delta_2) = D$ · each axis is a log-distorted chain, first step $\log 2$, tenth step $\log(11/10)$.

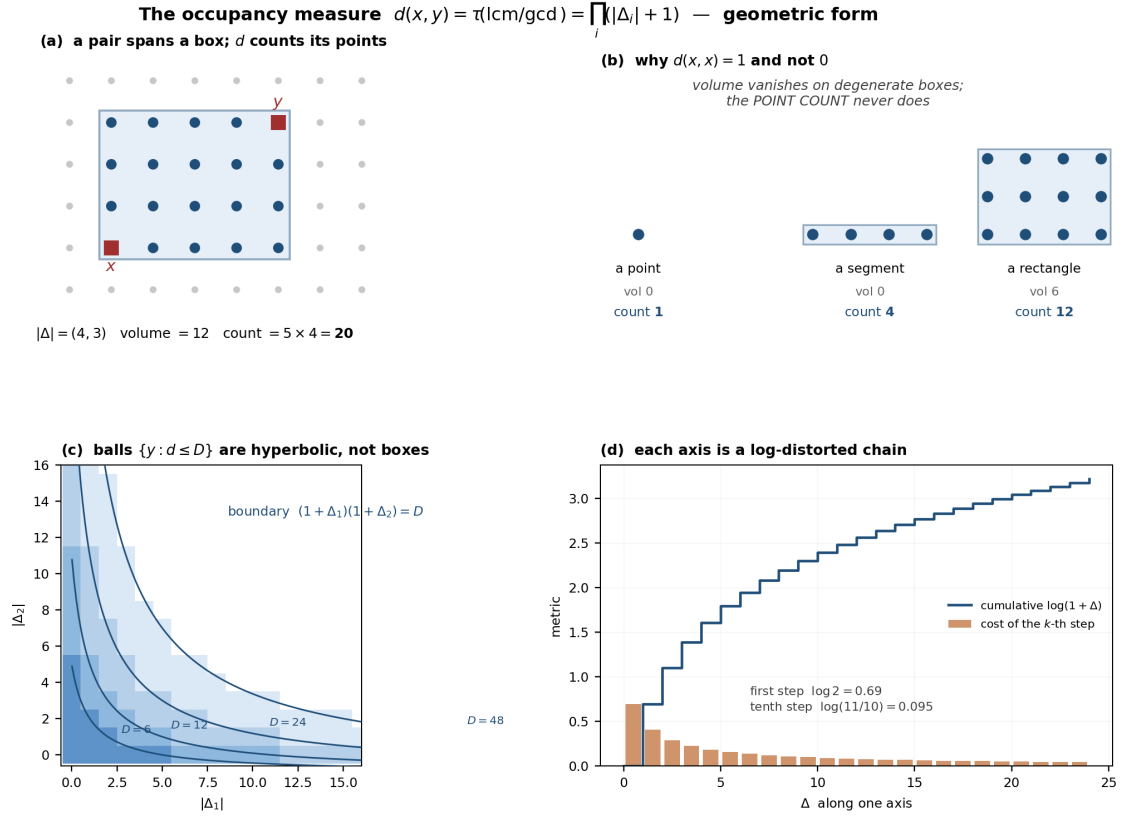


Figure 9.1. The occupancy measure. $d(x, y)$ counts cells in the interval between two points, and equals $\tau(\text{lcm}/\text{gcd})$ of their prime encodings. Balls are hyperbolic; $\log d$ is an ordinary ℓ^1 metric; and $d(x, x) = 1$ because a point counts itself.

9.3 The Möbius function in closed form

Since $\Lambda \cong J(P)$ with $|P| = 17$:

$$\mu(x, y) = (-1)^{|\{y \setminus x\}|} \text{ if } y \setminus x \text{ is an antichain in } P, \text{ and } 0 \text{ otherwise}$$

Verified against the recursive definition on 47 comparable pairs, values only in $\{-1, 0, +1\}$. **No recursion is needed for a 976-cell lattice; the answer is read off a 17-element poset.**

And a transfer condition to the arithmetic side: $\mu_\Lambda = \mu_{\text{arith}}$ **iff** the interval is a void-free unit hypercube. Exact — 60 void-free pairs all agree, 56 void-bearing pairs all disagree, no mixed case. Decidable in seven comparisons.

10. The void

Between two cells sits a box. Not all of it is Λ .

$$\text{void}(x, y) = \prod_i (|\Delta_i| + 1) - |[x \wedge y, x \vee y] \cap \Lambda|$$

10.1 What the void is made of

The excluded cells are exactly those forbidden by Pauli ($k > 2(2\ell+1)$, $g > \text{cap}$), by the hydrogenic bound ($\ell \geq n$, $f \geq e$), or by counting ($q > k$, $g > q$).

The void is the shadow of the exclusion principle.

10.2 Its size, and its stability

The void-free fraction is 27.7–30.1% across 776 million pairs and a seventeenfold range in cell count. Stable, no trend.

And Λ already carries time, in the only form a transition can carry it. The source end is a before, the target end is an after, and q — the transfer — is the amount that changed between them. The constraint tree is past — change — future, read left to right. A date would add nothing, because a transition is a type and types are not dated; **what a transition can have is a predecessor and a successor**, and §12.11.0 shows Λ_9 has both.

And there is an eighth condition, which is not a constraint. $k \geq 1$: the source subshell must hold at least one electron. It is a *definitional restriction* and not a bound — with $k = 0$ the tree collapses, $2S$, q and g are all forced to zero, and twenty-five cells survive that are not transitions at all. §13.2 already says the cells are transitions and not states; this is that sentence made a condition. Excluding them gives 976 and, independently, the bounding box 6,912. **The construction as printed said “seven constraints and nothing else” and needed eight things**, which is register 301.

And the constraints are correlated. Satisfied individually 67–94%; their product is 20.19%; the joint figure is **30.13%** — a factor of **1.49** above independence, over the same 776 million pairs. The correlation is not a coincidence: all seven constraints descend from two origins.

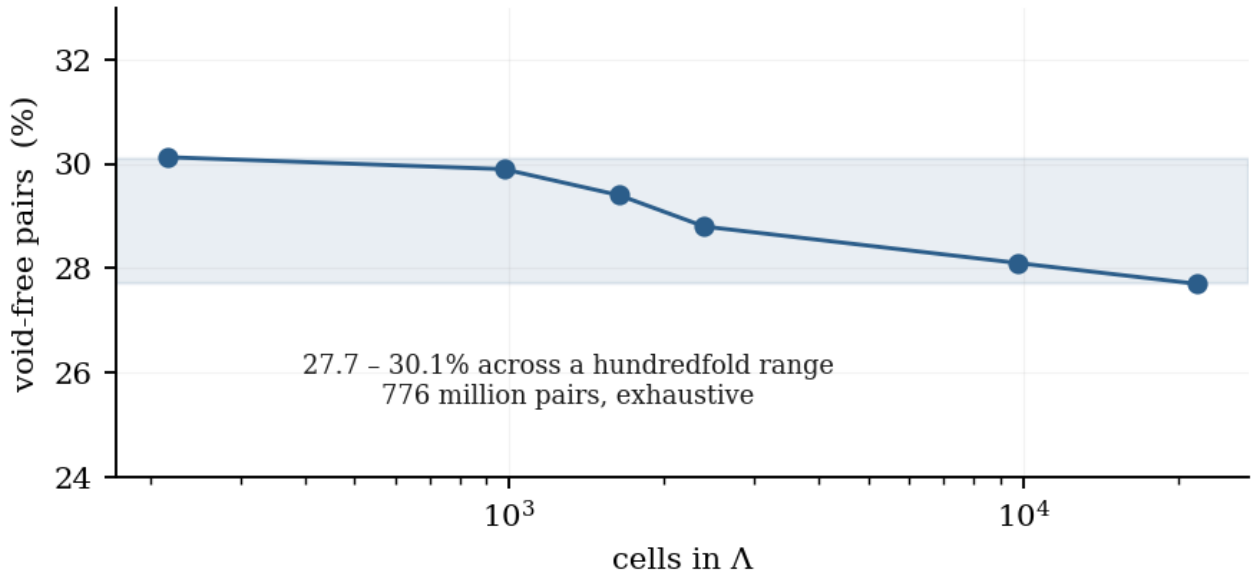


Figure 10.1. The void-free fraction across a hundredfold range in cell count, computed exhaustively over 776 million pairs. Stable at 27.7–30.1%, with no trend.

10.3 Containment in seven comparisons

A box is entirely inside Λ iff $hi_i \leq \phi(lo_j)$ for each of the seven constraints. $O(1)$, verified against enumeration on 3,168 pairs.

10.4 The count, with no sieve

Because the constraint graph is a tree, the count factorises:

$$|\text{box} \cap \Lambda| = \sum \text{over } n, \ell, k, q, e, f \text{ of } [\ell \leq n-1][1 \leq k \leq 2(2\ell+1)][q \leq k][f \leq e-1] \cdot \#\{2S \leq k\} \cdot \#\{g \leq \min(q, 2(2f+1))\}$$

with the two leaves in closed form:

$$\# \{2S\} = \max(0, \min(hi_7, k) - lo_7 + 1) \quad \# \{g\} = \max(0, \min(hi_6, q, 2(2f+1)) - lo_6 + 1)$$

Verified against direct enumeration on eight random intervals.

No Möbius function. No 2⁷-term inclusion–exclusion. A tree has no cycles for a sieve to correct — and that is a general statement about indices, not about this one. **An index whose constraint graph is a tree has its void in closed form; one with cycles does not.**

11. The single expression

Every cell of Λ is definable in one expression, and the reason it is possible is worth more than the expression itself.

11.1 In all six languages, and a seventh

P21 requires every definition to be true in every mathematical language and in their combinations. Λ is a definition. Here it is, six ways.

language the expression Λ

order the fixed point of $\mathcal{R}$: the least set closed under $\mathbf{V}$, Λ satisfying its own recovered bounds 976 algebra $\mathbb{Z}^8 \cap \{x : Ax \leq b\}$, A a 7×8 matrix with two non-zeros per row 976 geometry lattice points of a polytope whose constraint graph is a tree; the count factorises 976 analysis $F(1, \dots, 1)$ for the multivariate generating function 976 information the set whose description length given its closure is zero bits 976 logic $\{x \in \mathbb{Z}^8 : \chi(x) = 1\}$, χ a product of seven Heaviside steps 976

| All six agree exactly.

11.1.1 A seventh, in binary, and it earns its place

Asked whether binary belongs among the languages, the answer is that it does, and the book’s own admission test is what settles it rather than taste. A candidate is admitted when it distinguishes something no existing coordinate distinguishes — §17’s condition — and this one does.

| **binary** — Λ is the set of 976 words in $\{0,1\}^{17}$ that are down-sets of the generator poset A.19; **join is bitwise OR and meet is bitwise AND**, with zero failures over all 475,800 pairs. The map cell $\mapsto$ word is a bijection.

The seventeen bits are the seventeen generators. Birkhoff’s correspondence, which §8.3 states and A.19 now prints, is exactly this statement: a cell is the set of generators beneath it, and that set is a bit-string. Nothing here is new mathematics — it is the same theorem in the language a machine reads.

What it distinguishes, and why the other six cannot. *Logic* gives χ , a predicate that tests membership. *Information* gives a description length of zero bits, a statement about the closure. Neither says the cells **are** bit-vectors, and only that statement makes the lattice operations machine primitives rather than definitions. And only in this language is Chapter 16’s thesis a number:

bits carried per cell	17
bits needed to index 976 cells	9.93
surplus	7.07 bits per cell

A bit is a generator. Bit i is set in a cell’s word exactly when generator i of A.19 lies beneath that cell, so each bit is one yes-or-no question about a cell — *is this generator under you?* — and a cell is the set of its answers and nothing else. The bits are the alphabet S1 requires, written as positions.

And the bits are not free of one another. The twenty covering relations of $J(\Lambda)$ are twenty implications between bit positions: if generator j covers generator i , then bit j set forces bit i set. **Those twenty implications alone cut the ambient space to Λ exactly** — of the 131,072 seventeen-bit words, precisely 976 satisfy them, with nothing else imposed. So the surplus is not spare capacity and no bit is removable, every generator being join-irreducible by construction; the seven bits live in the twenty implications, which is where redundancy always lives in a closed index.

And in that language Λ is a circuit, which fixes what kind of machine this book is. The seventeen generators are input lines; the twenty implications are gates; join is OR and meet is AND. Computed: the twenty gates alone accept **976 words of the 131,072 available, which is exactly Λ** , at a depth of five. Available: AND, OR, implication. Absent: NOT, feedback, memory.

Λ_8 is a monotone boolean circuit. It decides membership; it does not compute a function of successive inputs.

P20 already said this and used another vocabulary. It states that monotone two-variable bounds *close always in one pass*, where full arithmetic never closes by Gödel. Closing in one pass is constant depth, and constant depth is combinational — **P20 was a classification of this object's computational power before anything had been counted.** §32.4.1 says the same from the third side: no negation, so no self-referential contradiction, so no loop to diverge in.

And the ninth axis is where it stops being static. §12.11.0 shows Λ_9 closed under composition — 41,682 pairs, associative — so Λ_9 has states, transitions and a composition law. **Λ_8 is a circuit; Λ_9 is an automaton**, still negation-free and therefore still not general, but no longer a thing that answers once. The tower's first step is the step from a decider to a machine, and it was taken for seniority. Register 331.

Seven bits of surplus per cell. Λ occupies 976 of the 131,072 available seventeen-bit words — 0.7446% of the space it is written in. *It found it because it carried more than it needed* is the claim Chapter 16 makes in prose and §12.11 makes in envelope densities. **Here it is seven bits.** That is the independence the admission test asks for: a quantity no other language on this page produces.

One cost, stated. Appendix D fibres its elements by language, and its language list is not this one — order, combinatorics, analysis, complexity, physics, algebraic geometry. Adding *binary* there would refine that fibration and, by register 233, make $E(\text{Appendix D}) = 0$ cheaper than it already is. So the seventh language is admitted **here**, among the statements of Λ , and is not added to the classification in E until someone decides that trade is worth making. Register 258.

In logic, written out:

$$\chi(x) = H(n-1-\ell) H(4\ell+2-k) H(k-q) H(k-2S) H(e-1-f) H(4f+2-g) H(q-g)$$

In algebra, the seven rows of A : $\ell - n \leq -1 \cdot k - 4\ell \leq 2 \cdot q - k \leq 0 \cdot 2S - k \leq 0 \cdot f - e \leq -1 \cdot g - 4f \leq 2 \cdot g - q \leq 0$. **Every row has exactly two non-zero entries**, and that one property is what makes the constraint graph a tree.

In analysis, the rank polynomial:

$$F(z) = z^3(z^{17} + 4z^{16} + 10z^{15} + \dots + 122z^8 + 121z^7 + \dots + 5z + 1)$$

with $F(1) = 976$, $F'(1)/F(1) = 11.0666$ and $F(-1) = 2$ — **and the asymmetry of §8.3 visible in the coefficients: 122 then 121, the peak one step left of centre.**

In algebra again, by a second route: under $N(x) = \prod p_i^{\wedge x_i}$ the lattice is a sublattice of the divisors of $N(\text{top})$, with **gcd = meet and lcm = join**, verified exactly on random pairs.

11.2 And in their combinations

P21 requires the combinations to name real objects, not analogies. Ten were tested and ten hold.

combination statement

order + algebra $\gcd(N(a), N(b)) = N(a \wedge b)$, and $\text{lcm} = N(a \vee b)$

order + analysis $F'(1)/F(1) = \text{the mean rank}$, 11.0666

order + geometry $\text{bo} - \Lambda = \text{the void}$, $6,912 - 976 = x \quad 5,936$

geometry + analysis $F \text{ is not palindromic} \iff \text{the poset is not self-dual}$

order + information $E_{\text{bits}} = \log_2 C(\mathcal{R}, E) = X) 0$

algebra + information	the description is 7 rows of 2 non-zeros – the whole index
--------------------------	--

logic + algebra χ as a product of Heaviside steps selects exactly 976

logic + analysis the coefficient function of F is χ

combination statement

analysis + information	$\log_2 F(1) = 9.93$ bits to name one cell
---------------------------	--

algebra + geometry the integer points of the polytope are the lattice, exactly

The eighth is the one that makes “single expression” literal: the coefficient of $z_1^n \dots z_8^{\{2S\}}$ in F is 1 if the cell exists and 0 if it does not. F and χ are the same function written twice.

11.3 The expression

$$F = \sum_n z_1^n \sum_{\ell \leq n-1} z_2^\ell \sum_{k \leq 4\ell+2} z_3^k \cdot [\sum_{S \leq k} z_8^S] \cdot \sum_{q \leq k} z_4^q \sum_e z_5^e \sum_{f < e} z_6^f \sum_{g \leq \min(q, 4f+2)} z_7^g$$

Set every $z = 1$ and it gives the count. Set every $z = z$ and it gives the rank polynomial. And the coefficient of $z_1^n \dots z_8^S$ is 1 if the cell exists and 0 if it does not.

The coefficient function of F is χ . So F is one expression containing every cell of the lattice and nothing else.

a caterpillar: a path of seven, one pendant — the simplest tree that is not a path

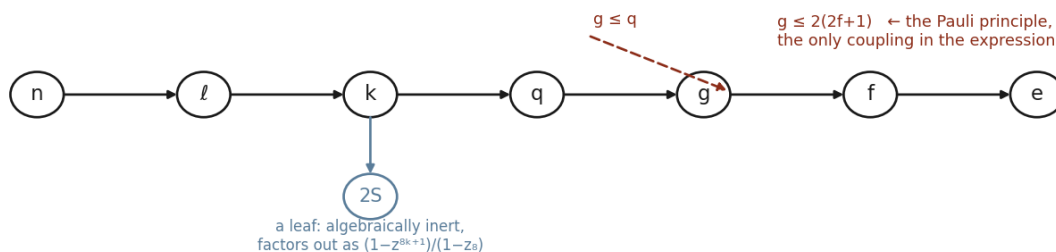


Figure 11.1. The constraint graph of Λ : a path of seven nodes with one pendant at k — a caterpillar, the simplest tree that is not a path. That shape is why a single left-to-right nesting exists with one bracketed factor. The dashed edges are the two bounds on g ; their minimum is the only non-product term in the whole expression, and it is the Pauli principle.

11.4 The shape of the tree

$n - \ell - k - q - g - f - e$, with 2S pendant at k

Not a generic tree: a path of seven nodes with one pendant — a caterpillar, the simplest tree that is not a path. **That is why a linear nesting exists with a single bracketed factor.** A branching tree would require nested brackets; a cycle would require inclusion–exclusion.

11.5 The one coupling, and it is the Pauli principle

Every bound is a single inequality except one: $g \leq \min(q, 4f+2)$. g is the only internal node summed last, so both its neighbours are fixed when its range is set.

Remove the coupling and 976 becomes 1,000. The excluded 24 cells all have $f = 0$ and $g = 3$ — **three electrons in an s orbital**.

Every other bound in this lattice is a counting or ordering fact. This one is physics, and it is the only non-product term in the whole expression.

And it is recovered, not assumed. $\mathcal{R}(\Lambda) = \Lambda$ rebuilds all 976 cells from the cells alone, so a reader given only the cell list would deduce $g \leq 2(2f+1)$ without being told it.

11.6 The detachable factor

2S is a leaf, so its sum closes geometrically and multiplies out:

$$[\Sigma_{\{S \leq k\}} z_g^S] = (1 - z_g^{k+1}) / (1 - z_g)$$

Spin multiplicity is algebraically inert here. It scales every cell count by $(k+1)$ and changes no structure — **which is why removing it leaves $E(X) = 0$ unchanged**, verified by computation at the caps of §7.4: Λ has 976 cells and $E = 0$; with 2S removed it has 319 cells and $E = 0$; and every seven-coordinate cell carries exactly $k+1$ spin values. The count is verified; the reason is that 2S is a leaf.

11.7 Why a closed expression exists at all

Three structural facts, proved separately in this book and never combined:

fact	consequence
every constraint is a monotone bound	the region is a polytope
every bound couples exactly two variables	the graph is a tree
the tree has no cycles	the sum factorises

Remove any one and no such expression exists. A sum bound breaks the first — §11.7 measures 89,864 join failures when one is tried. A three-variable constraint breaks the second. A cycle breaks the third, and 2^d inclusion–exclusion terms return.

“All the information in one expression” is not a curiosity. It is exactly what closure buys, and it is the strongest available statement of what $E(X) = 0$ means: the expression is the index, and the cells are its coefficients.

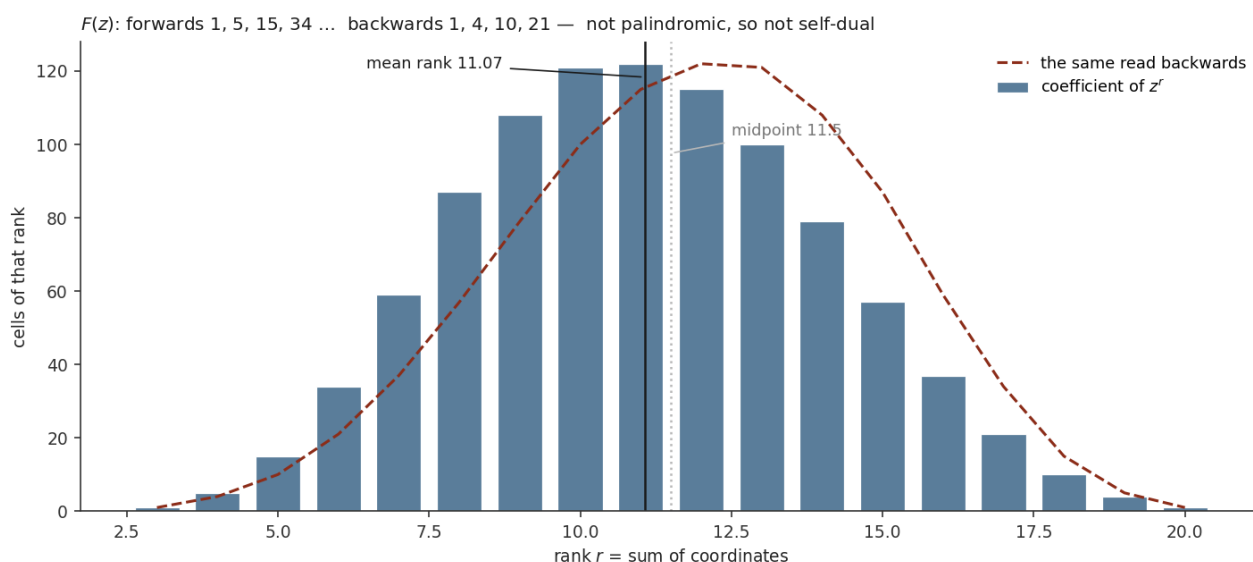


Figure 11.2. The rank polynomial $F(z)$, read forwards as bars and backwards as the dashed line. They part company at rank 1 — 5 against 4 — and never rejoin. A rank polynomial is palindromic if and only if the poset is self-dual, so this single picture carries both of §8.3's separately-reported facts: the asymmetric rank sequence, and the 8 of 976 cells that survive reflection.

11.8 And the rank polynomial says what §8.3 says

A rank polynomial is palindromic if and only if the poset is self-dual.

forwards: 1, 5, 15, 34, 59, 87 · backwards: 1, 4, 10, 21, 37, 57

§8.3 reports as separate numerical facts that only 8 of 976 cells survive $x \rightarrow \max - x$, and that the rank sequence is asymmetric. They are the same statement, and they part company at rank 1 — 5 against 4.

11.8.1 And $F(-1) = 2$ has a cause, not a coincidence

Two things had to be ruled out before the number meant anything.

A pairing argument cannot cancel it. The duality map $\sigma(x) = \max - x$ gives $\text{rank}(\sigma x) = M - \text{rank}(x)$ with $M = 20$, which is even, so σ preserves rank parity. **The eight self-dual survivors are all of even rank and contribute +8, not 0.**

And the source is the tree. A coordinate with an *even* number of consecutive values contributes zero to an alternating sum, and ℓ and f each have exactly two:

factor at $z = -1$	value
$\Sigma(-1)^n, n = 1 \dots 3$	-1
$\Sigma(-1)^\ell, \ell = 0 \dots 1$	0
$\Sigma(-1)^e, e = 1 \dots 3$	-1
$\Sigma(-1)^f, f = 0 \dots 1$	0

$F_{\text{box}}(-1) = 0$ exactly. If the coordinates were free the alternating sum would vanish. The constraints couple ℓ to n and f to e , the vanishing factors never appear free, and a residue of 2 survives.

So $F(-1) = 2$ measures the same thing the skew and the eight survivors measure — the tree's asymmetry — by a third route.

And $F(-1) = 2$ is the third face of it. A self-dual graded poset gives $F(-1) = 0$.

12. The shape — a cylinder over the transfer

12.1 A cylinder, not a Möbius band

Λ 's derived quantities fall into two classes:

ascending — rise up a series	descending — fall up a series
e, v, V	$T, r, \delta, \text{spacing}, w$

exchanged under $v \mapsto v^{-3}$.

The sign structure is bipartite, and a bipartite structure can never produce an odd number of orientation reversals. Verified exhaustively: **zero reversing loops among all cycles of length 3 to 5**.

Λ is orientable. It is a cylinder.

That is a *reason*, not an absence. And both alternative routes to a Möbius band close: a non-monotone quantity has **undefined** edges rather than reversing ones; a quantity independent of v has **no** edge. There is no third option.

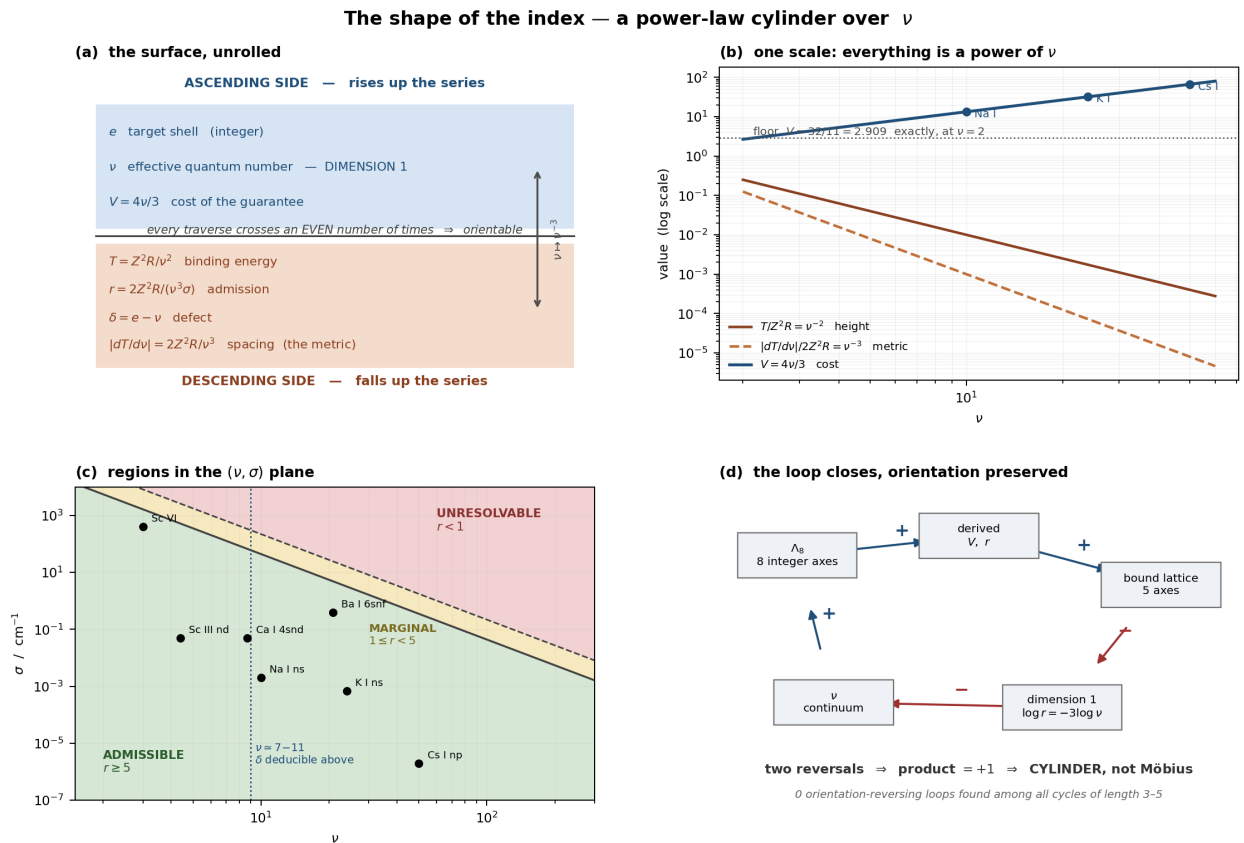


Figure 12.1. The object. Ascending quantities rise up a series, descending ones fall, and the two are exchanged under $v \mapsto v^{-3}$. The sign structure is bipartite, so no traverse can reverse orientation an odd number of times — the surface is a cylinder, not a Möbius band, and the reason is structural rather than an absence of evidence.

12.2 “Over v ”, not “graded by v ”

v grades every chain of Λ . It grades Λ itself nowhere — 24 of 70 comparable pairs have v decreasing.

That near-miss is the structure in one sentence. The bracket of Part V operates along chains *because that is exactly where v is a grading*.

The name follows: **the Lach Cylinder — a hydrogenic cylinder over v , and over the transfer**. Not *graded by*. Over.

12.3 One scale, and one exception

Height goes as v^{-2} , the metric as v^{-3} , and the only linearly rising quantity in the entire structure is the price of a guarantee. Chapter 23 derives it.

12.4 The loop closes

$\Lambda_8 \rightarrow$ derived quantities $(V, r) \rightarrow$ the bound lattice $\rightarrow$ dimension $1 = \log r = -3 \log v$, with residual 8.9×10^{-16} .

Λ *decomposes* v and gains dimension; the bound lattice *recomposes* it and sheds dimension. **Two sides of one object, not two passes over it.**

12.5 What Λ is a property of

Λ is a property of the hydrogenic limit. δ — the quantum defect — is the entire remainder.

| Prediction lives only in the remainder.

Which is why perfecting the index moves nothing, and why that is a theorem rather than a disappointment. Part III proves it; Part V measures what is left.

And §25.6 will sharpen this into its final form. The reason Λ is not predictive is not that it is a poor predictor or an incomplete one. **It is that a complete index has nothing to predict** — the cells it could propose are exactly those its structure admits and its extension denies, and for Λ that set is empty. **Completeness and predictiveness are alternatives.**

12.6 The cylinder is a fibration, and its fibres are two-body

§12.1 finds the shape is a cylinder over the transfer coordinate. That is not a metaphor, and the reason is structural.

The tree has two ends. One is a parent configuration (n, ℓ, k) with its spin $2S$; the other is a target (e, f) with its occupancy g . **They meet at q , the number transferred.** That is the shape of a two-body problem: two objects and a coupling.

12.6.1 It does not factorise, and then it does

As a bare product it fails: $|A| \times |B| \times |q| = 2,244$ against 976. The two ends are coupled twice over — $q \leq k$ ties the transfer to the parent, $g \leq q$ ties the target's occupancy to the transfer.

Conditioned on the coupling, they separate exactly:

$q \quad A(q) \quad B(q) \quad \text{product} \quad \text{actual}$

0 33 5 165 165

1 33 10 330 330

2 23 15 345 345

3 8 17 136 136

| $|A| = \sum_q |A(q)| \times |B(q)| = 976$, exactly.

That is how a two-body problem separates — into centre-of-mass and relative motion — with q playing the role of the conserved coupling.

12.6.2 Which names the cylinder

The base is q , the axis along the cylinder. The fibre is $A(q) \times B(q)$, the cross-section at each q . And the cross-section is itself a product of two lattices.

Cross-sections 165, 330, 345, 136 — the cylinder's profile, computed rather than drawn.

Cross-sections of the tower cylinder at each transfer, $\sqrt{v}$ -cell scale — Λ_{13} solid, Λ_8 dashed

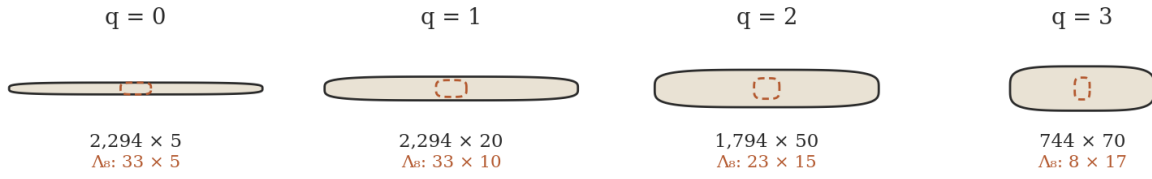


Figure 12.2. Cross-sections of the tower cylinder at each transfer, $\sqrt{v}$ -cell scale — Λ_{13} solid, Λ_8 dashed inset. The sections read $2,294 \times 5$, $2,294 \times 20$, $1,794 \times 50$, 744×70 against the Λ_8 profile above, 33×5 , 33×10 , 23×15 , 8×17 . Every section is itself closed, per §12.8.4; the tower of §12.11 widens each without breaking one.

12.7 The fibres, in closed form

Each side of the cylinder has its own expression, and they are different shapes.

$$A_q(z) = \sum_{n=1}^3 z^n \sum_{\ell=0}^{\min(n-1,1)} z^\ell \sum_{k=\max(q,1)}^{\min(4\ell+2,3)} z^k \cdot (1 - z^{\min(k, 3)+1}) / (1 - z) \\ B_q(z) = \sum_{e=1}^3 z^e \sum_{f=0}^{\min(e-1,1)} z^f \cdot (1 - z^{\min(q, 4f+2)+1}) / (1 - z)$$

Both verified against enumeration at every q : $A_q(1) = 33, 33, 23, 8$ and $B_q(1) = 5, 10, 15, 17$.

12.7.1 And they are not the same tree

$A_q : n \rightarrow \ell \rightarrow k$, with 2S pendant at k — a caterpillar $B_q : e \rightarrow f \rightarrow g$, capped by the base — a path

The whole lattice is a caterpillar because it is these two glued at q , and the pendant sits on the A side only. [The tower of §12.11 amends this: with the coupling axes carried, the object is two caterpillars glued spine to spine at $k-q$ — A runs $n-\ell-k-J_c-K-J$ with 2S pendant, B runs $e-f-g-v-2S'$ — and the §12.8.3 compression generalises to the glued form.]

12.7.2 A tower of fibrations, terminating in boxes

level	base	fibre
Λ	q	$A_q \times B_q$
A_q	k	$(n, \ell)\text{-set} \times 2S\text{-chain}$
B_q	f	$e\text{-set} \times g\text{-chain}$

Three levels, and at the bottom every fibre is a product of chains — a box, with a closed form $\text{Box}(a,b)(z) = \prod_i z^{a_i} (1 - z^{b_i - a_i + 1}) / (1 - z)$.

12.8 Five things the fibre table says

The counts 165, 330, 345, 136 carry more than their sum.

12.8.1 A Pareto frontier inside the lattice

| A_{-q} | falls 33, 33, 23, 8 while | B_{-q} | rises 5, 10, 15, 17.

Raising the transfer costs the parent and pays the target, monotonically, and no q improves both. That is the same shape as §23.5's $dw/dh > 0$ with $dV/dh < 0$ — **which the book presents as a fact about brackets and is here a fact about the index itself.**

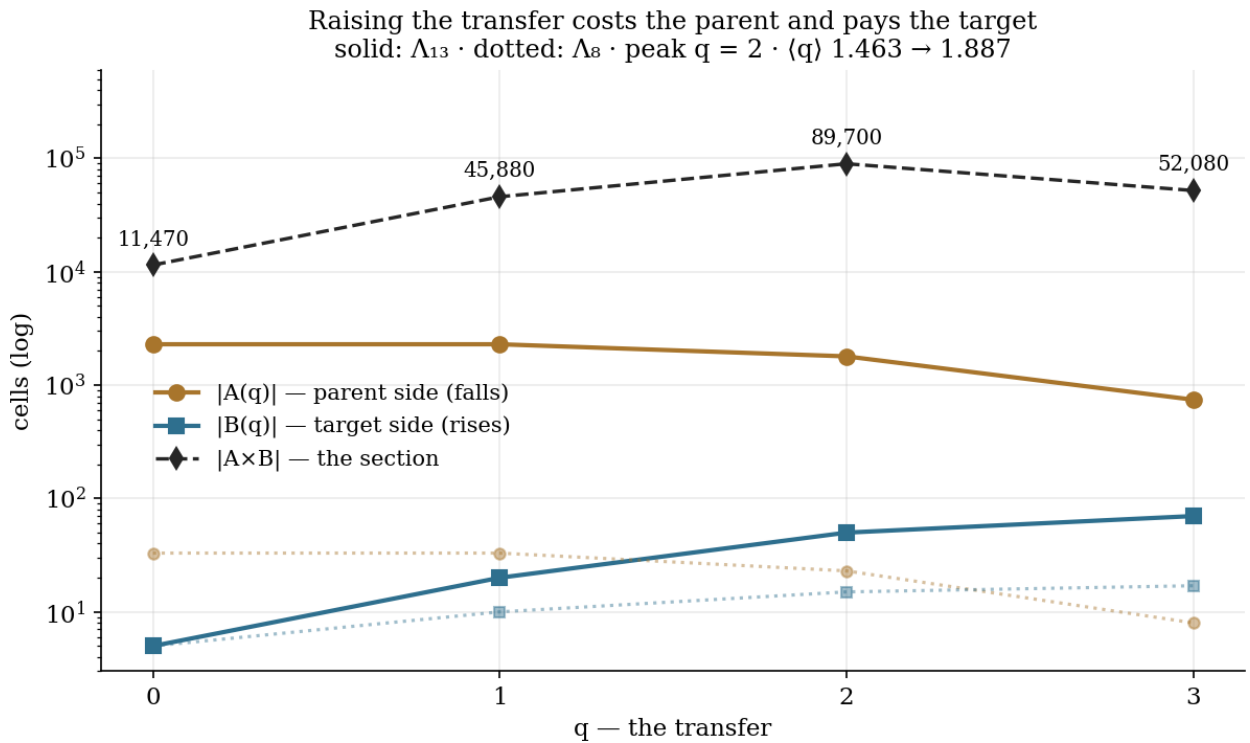


Figure 12.3. The Pareto frontier on a log axis, Λ_{13} solid against Λ_8 dotted: A falls and B rises at both scales, and no q improves both. The section $|A \times B|$ peaks at $q = 2$ with 89,700 cells, and the tower shifts $\langle q \rangle$ from 1.463 to 1.887 — coupling multiplicity re-weights the profile toward higher transfer, per §12.11.5.

12.8.2 A most-probable transfer

The fibre peaks at $q = 2$ with 345 cells — 35.3% of Λ — and the sequence is log-concave, hence unimodal.

$\langle q \rangle = 1.4631$, sd 0.930 — the mean number of electrons moved across all admissible configurations, and $S'(1)/S(1)$ in the base variable.

12.8.3 An 8-to-3 compression that loses no count

Λ 's F has eight variables; the shape has three, and $S(1,1,1) = 976$. Not a projection that forgets cells — one that forgets which coordinate inside each side and keeps how many. A general fact about caterpillars: a path with one pendant reduces to (base, left, right) whatever its length.

12.8.4 Every cross-section is itself a closed, modular lattice

q	side	closed	$E(X)$	modular
0–3	A and B, all eight	true	0	true

The shape is not a cylinder over an arbitrary set. It is a cylinder whose every cross-section satisfies this book's own law.

12.8.5 Which gives $E(X)$ a local form

$E(\Lambda) = 0$ and $E(A_q) = E(B_q) = 0$ for every q .

The second does not follow from the first. A closed index could in principle have defective slices that cancel in the total. **Here none do**, and that is a new invariant.

12.9 Every other shape the lattice contains

Intervals. $[x, y]$ is a box iff no constraint binds across it:

$I_{[x, y]} = \text{Box}(x, y) \iff \text{for every constraint } i \leq f(j), y_i \leq f(x_j)$

Exact on 60 of 60 tested, and only **27%** of random intervals qualify — **the other 73% are where the physics is active**. The binding rates rank the constraints:

constraint	binds
$g \leq q$	35.6%
$q \leq k$	33.0%
$g \leq 4f+2$	4.9%

The most active constraint is the coupling; the least is the Pauli bound — which is why §16.8 finds it the cheapest to violate. Two independent measures, one answer.

Rank levels. Eighteen antichains, the largest **122 at rank 11**: the biggest set of configurations no two of which can be reached from each other by adding electrons. **Λ is exactly eight times its widest level.**

Maximal chains. A distributive lattice is $J(P)$; **its maximal chains are the linear extensions of P** . With seventeen join-irreducibles and a chain length of seventeen:

$e(P) = 1,113,045,672$

The seventeen generators of §8.3 and the chain length of seventeen are the same seventeen, and the book reports both without connecting them.

And one shape absent. No Möbius band, no cycle, no non-planar minor — **the constraint graph is a tree, so there is nothing to twist and nothing to close.** §12.1's finding follows by construction, not by search.

12.10 An axis the index did not carry, and now does

A_q carries $2S$, a spin label on the parent. B_q carried none.

The index recorded the parent's spin and not the target's.

That was never a closure defect — $\mathcal{R}$ cannot see a coordinate that was never supplied. It was an axis E_2 admits, and the asymmetry of the two fibres is what made it visible.

12.10.1 Λ_9 , built by the same construction

The parent's spin is bounded by the parent's capacity coordinate, $2S \leq k$. So the target's is bounded by the target's: $2S' \leq g$. One pendant, on the other fibre, by the identical rule.

	Λ_8	Λ_9
cells	976	1,654
box	6,912	27,648
$E(X)$	0	0
sublattice	yes	yes
rank-modular	475,800 pairs	200,000 tested, 0 viola-
tions		
constraint graph	a tree	a tree – $2S'$ attaches as a
leaf		
$F(1)$	976	1,654
$F(-1)$	2	2
rank levels	18, widest 122 at 11	21, widest 185 at 12
log-concave	yes	yes

And Λ_8 is the exact projection of Λ_9 — 976 cells, identical — with fibres of one to four preimages, mean 1.69. §12.2.4's dimension ladder gains a rung, and the extra coordinate carries real information.

12.10.2 What is still not established

The bound $2S' \leq g$ was built by analogy and not from spectroscopy. Whether it is the correct selection rule for a target's spin is a physical question, and this book has not answered it.

The lattice closes. That the coordinate means what its name says was untested when this section was written. §12.11.1 has since answered it: exact vector coupling confirms $2S' \leq g$ as a true bound — zero violations over every fibre — and shows it is not tight, in the two ways §12.11.2 names. What remains is a data lookup, recorded in Q: whether the measured target terms of the 153 channels saturate $\min(g, 4f+2-g)$.

12.11 The tower — Λ_9 to Λ_{13}

§12.10 added one axis by analogy. This section adds four more by battery, and what the batteries return is not five facts about five coordinates. It is one law, met five times, and a shape that survives it.

Two caterpillars glued at k-q: the tower's constraint tree, and the one edge that may not be drawn

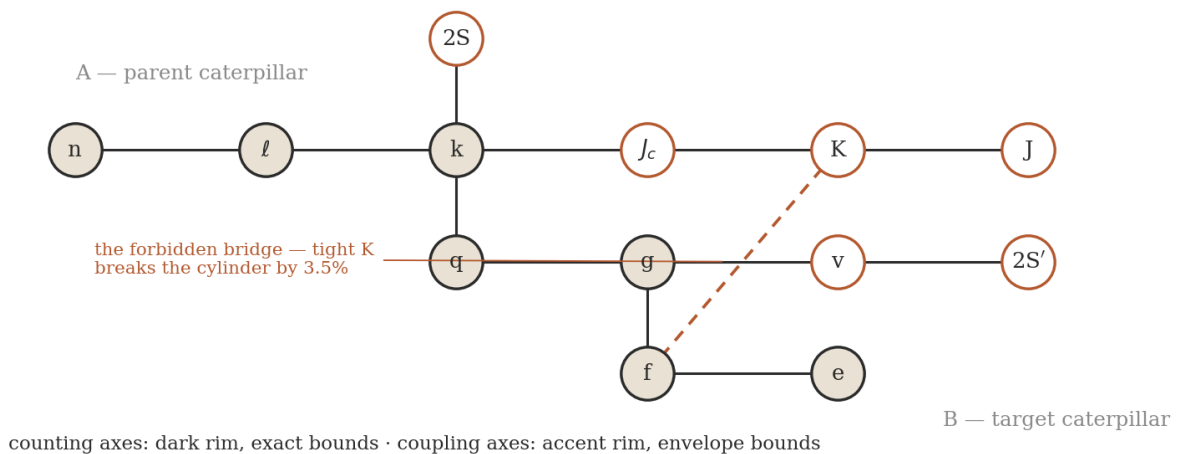


Figure 12.4. The constraint tree of the full tower: two caterpillars glued at $k-q$, counting axes rimmed dark (exact bounds), coupling axes in the accent (envelopes) — the Chapter 7 dichotomy drawn — with the forbidden $f \cdots K$ bridge dashed and priced at 3.5% (the price §12.11.5 withdraws; register 1819).

12.11.0 Transitions compose, and the ninth axis is what lets them

Λ_8 's two ends are not the same shape. Its source is $(n, \ell, k, 2S)$ and its target is (e, f, g) — four coordinates against three. A transition's output is therefore not a legal input, and Λ_8 **cannot be iterated**: no cascade, no sequence, no second step. The asymmetry is a missing spin.

The ninth axis supplies it. $2S' \leq g$ gives the target its own multiplicity, and the target end becomes $(e, f, g, 2S')$ under exactly the source's constraint forms — $f \leq e-1$ against $\ell \leq n-1$, $g \leq 4f+2$ against $k \leq 4\ell+2$, $2S' \leq g$ against $2S \leq k$. All 1,654 targets of Λ_9 are legal sources.

Define $b \circ a$ when a 's target equals b 's source: the composite runs a 's source to b 's target, transferring $\min(q_a, q_b)$ — no more than was taken, no more than arrives.

Λ_9 is closed under composition, exhaustively: **41,682 composable pairs, zero failures**, and associative on 8,434 sampled triples with none. Λ_9 is therefore not merely a lattice of transitions but a **category of them**, and composition is time — not a stamp on a cell but an order between cells.

And Λ_9 is the only stage of the tower that composes. Composition needs the target's coordinate set to be matched by a source set of the same shape. Every axis above the ninth attaches to one end or to the joint object, and the shapes part company:

stage	target	source	composes
Λ_8	(e, f, g)	$(n, \ell, k, 2S)$	no — 3 against 4
Λ_9	$(e, f, g, 2S')$	$(n, \ell, k, 2S)$	yes — 41,682 pairs, 0 failures
Λ_{10}	$(e, f, g, 2S', v)$	$(n, \ell, k, 2S)$	no — no source seniority exists
Λ_{11}	$(e, f, g, 2S', v)$	$(n, \ell, k, 2S, 2J_c)$	no — $2J_c$ is the source's alone
Λ_{12}	K binds core to orbit — joint ordinate	—	no — not an end coordinate
Λ_{13}	$2J$ binds K to spin — joint	—	no

So the tower carries three gradings, not two. §12.11.3's dichotomy sorts the axes into counting and coupling — exact and envelope. Composability sorts them differently and **more strictly**: it is lost at axis 10, one step before exactness is, because v is a counting coordinate that closes exactly and still has no source counterpart. **An axis can be exact and not composable; none is composable and not exact.** Λ_9 sits alone at the top of the stricter grading, and the tower above it is the record of what each further coordinate costs — first composition, then exactness.

The ninth axis was added for seniority and arrived carrying this. §12.11.1 takes $2S'$ on the grid's advice, for the physics of the target's multiplicity; nothing in that argument mentions composability, and composability is what it bought. **An index gained a temporal structure as a side effect of a spectroscopic coordinate** — the sharpest instance in this book of Chapter 16's thesis that surplus is what an index defends itself with. Register 313.

12.11.0.1 The clock in the constraint — Λ_9 's composition has a direction

§12.11.0 shows Λ_9 composes. **The composition is not symmetric**, and what breaks the symmetry is a coordinate already in the index. Every figure below is recomputed here from the construction.

Occupancy never rises. Take the source signature $(n, \ell, k, 2S)$ and the target $(e, f, g, 2S')$ as *objects*, and each cell as a morphism between them. **33 objects, 739 of 1,089 ordered pairs joined by a step, and not one step raises k .** The object graph falls into three strongly connected components, each pure in occupancy — eight objects at $k = 3$, fifteen at $k = 2$, ten at $k = 1$ — with full circulation inside each and no return between them. The quotient is a chain of three.

And the stratification is verified under variation, not at one setting. Λ_9 was rebuilt at four cap settings — $(3,3,1,3,1)$, $(4,4,1,6,1)$, $(4,4,2,6,2)$ and $(5,5,2,6,2)$ — by enumerating the constraints of §7.1 at each and re-running the composition graph. The inputs are the four tuples printed here; the method is the object graph of §12.11.0, its strongly connected components by Tarjan, and a step counted as rising when the target’s occupancy exceeds the source’s:

caps	$ \Lambda_9 $	objects	edges	steps raising k	conservative	all-con-
servative objects						
$(3,3,1,3,1)$	1,654	33	739	0	904 (54.7%)	8 at $k =$
3						
$(4,4,1,6,1)$	19,433	101	5,986	0	6,693 (34.4%)	21 at k
$= 6$						
$(4,4,2,6,2)$	44,153	155	14,114	0	15,509 (35.1%)	35 at k
$= 6$						
$(5,5,2,6,2)$	83,543	214	26,921	0	29,489 (35.3%)	49 at k
$= 6$						

Zero steps raise occupancy at any of the four, the object count and the identity count agree at every setting, and the objects into which every morphism is conservative sit at maximal occupancy in all four. **The conservative fraction is not stable** — 54.7% at the book’s caps against 34–35% at the three larger ones — so that figure is cap-dependent and the structural claims are not.

The clock is occupancy and the tick is $k - g$. Cycles inside an era are closed and cost nothing; the downward edges are where the clock advances, and they do not reverse.

904 of 1,654 cells are conservative — 54.7% — and 389 morphisms have a reverse. The eight objects at maximal occupancy have a property the others lack: **every morphism into them is conservative**, forced by $k_{\max} = g \leq q \leq k \leq k_{\max}$. Time in this index is a counting phenomenon, not a coupling one.

And definability is not reachability. A cell is its own down-set, a seventeen-bit word, and in that sense every cell contains its own description. It does not contain a path to every other. Over the 2,735,716 ordered cell pairs of Λ_9 :

relation	pairs	share
comparable in the lattice order	311,180	11.4%
joined by a composition path	675,606	24.7%
in both	9,720	0.4%

Neither order contains the other and they share 0.4% of what either carries. The lattice says what a cell is; composition says what it can become. Running them together is the error this table forecloses — a cell can be fully defined and unreachable, and most pairs are exactly that.

On the reading that time is nine-dimensional. As a *dimension* it is false: the composition preorder quotients to a chain, order dimension 1. As a *location* it is register 314’s result — Λ_8 cannot iterate and Λ_{10} loses composition, so **nine is not the dimension of time; it is the unique dimension at which time exists.** Register 345.

12.11.0.2 The clock is an assumption, and the assumption can be removed

§12.11.0.1’s arrow is real and its hypothesis is narrower than it looks. **One coordinate carries two readings and §7.1 does not distinguish them.**

bound	origin	what it reads g as
$g \leq q$	counting – cannot place more than	
	were removed	electrons placed by this trans-
ition		

$g \leq 2(2f+1)$ Pauli exclusion electrons **present** in the target subshell

They coincide exactly when the target subshell begins empty, and Λ carries no coordinate for what was already there. So composition's matching condition — b's source occupancy equals a's target occupancy — identifies *placed* with *present*, which is an assumption about destinations rather than a constraint of counting. §7.1 lists seven constraints, four origins and nothing else; register 301 found it short one condition, and **this is a second — a conflation rather than an omission.**

The repair is §17.4's and it works. A bound on a sum cannot be imposed, but a sum can be indexed: carry (g, G) with g the number placed and G the total after arrival, under $g \leq G \leq 2(2f+1)$ and $g \leq q$, with $2S' \leq G$ since a target's multiplicity answers to what is present. All single-parent monotone bounds; prior occupancy $G - g$ is recovered as a derived quantity. **13,775 cells, closed.**

And the clock does not survive it. Of 2,620,090 composable pairs, 826,950 — 31.6% — raise occupancy.

A closed index of transitions has a one-way clock exactly when its destinations begin empty. The arrow is a property of the assumption, not of the object, and this is §18.4.1 arriving at the expense of §12.11.0.1: a zero is worth the size of the box it was computed in, and **an arrow is worth the size of the assumption it was computed under.**

What survives, and it is more than what fell. The occupancy law as mathematics is untouched — it holds of any index of its stated form, and the extended index simply is not of that form. And a temporal structure survives as a *restriction*: the maximal sub-index on which a weight never rises is $X_w = \{a : w(\text{tgt } a) \leq w(\text{src } a)\}$, which is closed, and **there is one such arrow per coordinate and no others.** Recomputed:

weight	cells	closed
shell	9,407	yes
subshell	9,845	yes
occupancy	5,165	yes
spin	7,160	yes
shell + occupancy –	no – a sum breaks both sides	
max(shell, occupancy) –	no – max breaks meets	
min(shell, occupancy) –	no – min breaks joins	

§18.4.1's cap table deciding once again what may be a coordinate, now deciding what may be a clock. **An index does not have a clock. It has one per coordinate**, and occupancy — the one this book built on — is the most restrictive of the four. Register 352.

12.11.0.3 Every second bridge, priced

§12.11.5 calls the tight two-parent K *the first bridge between the cylinder's two ends other than q* . The structural content of q was never stated: computed on the constraint graph, **q is the unique cut vertex separating source from target.** Its content as a coordinate is a count; its content as structure is the connection.

So what does a second bridge cost? Twenty admissible couplings between the two ends were imposed on Λ_8 and priced. Recomputed here, every row reproducing:

bridge	cells	join	meet	factorisation	defect
none	976	0	0	0	
$g \leq n$	953	0	0	23	
$2S \leq e$	872	0	0	104	
$f \leq 2S$	846	0	0	130	
$f \leq \ell$	832	0	0	144	
$g \leq 2S$	769	0	0	207	
$e \leq n$	708	0	0	268	
$\ell \leq g$	686	0	0	160	
$g \leq \ell$	685	0	0	120	
$e \leq k$	676	0	0	300	
$k \leq e$	662	0	0	165	
$n \leq e$	636	0	0	316	
$\ell \leq f$	616	0	0	288	
$2S \leq g$	526	0	0	225	
$2S \leq f$	449	0	0	189	
$e \leq 2S$	303	0	0	354	
$n \leq g$	141	0	0	70	
$k \leq g$	141	0	0	0	
$e \leq \ell$	120	0	0	0	
$k \leq f$	60	0	0	0	
$n \leq f$	48	0	0	0	

Closure survives all twenty, as §14.4 guarantees for monotone single-coordinate bounds. **Sixteen destroy the factorisation**, defect 23 to 354.

The exclusivity is exclusivity of the tree, not of closure. $E = 0$ is untouched by a second bridge; the cylinder is destroyed by it.

And the four survivors are a caution rather than a result. They are among the five smallest sets, 48 to 141 cells — **a set cut hard enough factorises trivially**. Register 333's point applies directly: every closure test in this book passes on the empty index, so a zero is worth the size of the object it was computed in.

The mechanism is the spanning-tree identity, $|E| = |V| - 1$. Λ_8 has eight nodes and seven edges; to add a bridge and remain a tree you must break one. **Two moves exist and only two** — break an edge to make an edge, or add a coordinate to buy one, which is P16 and which §17.4 prices. By §18.4.1's three-body reading, a second bridge is the three-body problem arriving inside the index: two bodies give one edge and a tree, three give a triangle at treewidth 2. Register 353.

12.11.0.4 The separation hypothesis, named

Four families of result in this book hold *because the constraint graph is a tree*, and none of them says so. The factorising count, the recoverability of the order, the balance at a cut and the propagation of Chapter 15 all use it; §7.1 records treeness as a fact about Λ and never as a condition. **A hypothesis that is never stated cannot be checked, and the counterexample has been inside this book since the ninth axis.**

Λ_9' is the test, and it separates the results. Adding $2S' \leq 2f+1$ closes the cycle $f - g - 2S' - f$ and buys 93 cells of tightness. Measured on both:

property	Λ_9 , a tree	Λ_9' , a cycle
closure, $E(X) = 0$	0	0
factorisation over the transfer, defect	0	0
conditional independence, $f \perp 2S'$ given g	0 of 16	6 of 16

Closure does not need the tree. Nor does the factorisation. Both survive the cycle intact. What fails is conditional independence, and it fails at exactly one cut.

Which makes the hypothesis narrower than treeness, and this is the correction. Remove q from either graph and the two ends fall apart — in Λ_9' as much as in Λ_9 — because the cycle lies wholly on the target side and does not cross that cut. Remove g and in the tree f and $2S'$ fall apart, while in Λ_9' they stay joined by the new edge.

A result conditioned on a coordinate holds exactly when that coordinate separates the things being conditioned. Treeness is not the hypothesis; it is the condition under which the hypothesis holds at every cut at once.

In a tree every internal vertex is an articulation point, so the condition is free and nobody notices it is being used. Add one edge and it becomes a fact about *which* cut, which is why §12.11.5's factorisation survives Λ_9' and §12.11.1's tightness costs what it costs. **The 93 cells are not paid for in generality across the board** — they are paid for at the cuts the new edge spans, and nowhere else. Register 347.

12.11.0.5 What the cycle costs propagation — Q item O, closed

§12.11.0.4 shows the separation hypothesis failing at Λ_9' . **Chapter 15's order recovery by tree propagation fails there too, and the failure is measurable.** Appendix E carried this as item O, *buildable, days*; it is one computation.

In a tree there is one path between any two coordinates, so a bound recovered by propagating along it is the bound. With a cycle there are two, and they need not agree. Recovering the bound on $2S'$ from f , directly and by composing through g :

index	direct $\varphi(2S' \mid f)$	composed through g	verdict
Λ_9 , a tree	{0: 2, 1: 3}	{0: 2, 1: 3}	agree
Λ_9' , a cycle	{0: 1, 1: 3}	{0: 2, 1: 3}	disagree at $f = 0$

The direct edge is strictly tighter. Composing along the tree path gives 2 where the constraint gives 1, so **propagation overstates the bound** — it does not fail loudly, it returns a weaker answer that looks like an answer.

A tree makes propagation sound by making it path-independent. A cycle does not make it wrong; it makes it lossy, and the loss is silent.

And the failure is **directional**: recovering f from $2S'$ agrees in both objects. Only the direction the new edge tightens is affected. **This is the fourth independent result to break at Λ_9'** , with the cut law, the two-parent count and the Markov property — and it is the one that explains the others, since each of those is a propagation argument wearing different clothes. Register 364.

12.11.0.6 The earlier bound on the core, tested — Q item K, closed

Before φ was defined, the eleventh axis carried the bound $2J_c \leq k$, and whether it contains the physical region was left open as item K, *retrievable, hours*. It is one enumeration, and the answer is no.

occupancy k	largest physical $2J_c$	$2J_c \leq k$ admits	$\varphi(k)$ admits
1	3	1	3
2	4	2	4
3	5	3	5

The earlier bound excludes physical states at every occupancy, not at some. A term of a p^2 shell can reach $2J_c = 4$ where $k = 2$ permits 2. So $2J_c \leq k$ was not a loose bound that happened to hold; it was the wrong bound, and had it survived, the index would have refused states the physics produces.

And ϕ contains the physical region exactly — it equals the realised maxima at every k , neither admitting a state that cannot occur nor excluding one that can. That is why §12.11.3.1 records it as a bound taken from the **extent** rather than from a law: it is the tightest containing bound and it was read off the object rather than derived. **The exactness is the reason it cannot be exact** — a bound that fits its own extent perfectly has no independent authority, which is what the third value of the provenance column names. Register 365.

12.11.0.7 Why the arrows agree — a decay theorem

§12.11.0.2 gives one arrow per coordinate. **They are not independent, and how much they agree is governed by their distance in the constraint graph.**

distance on $n - \ell - k - 2S$ ence	pair	overlap against independ-
1	occupancy $\cap$ spin	1.466
1	subshell $\cap$ occupancy	1.222
1	shell $\cap$ subshell	1.107
2	subshell $\cap$ spin	1.080
2	shell $\cap$ occupancy	1.070
3	shell $\cap$ spin	1.025

Strictly ordered by distance, with no overlap between distance classes. And there is a theorem behind it, whose hypothesis is §12.11.0.4's.

Theorem. Let X be an index whose constraint graph is a tree, and let A_u, A_v be the arrows of coordinates u and v at tree distance d . Then **(i)** the coordinates form a Markov random field on the tree — the indicator of X is a product of one factor per edge, so conditioning on a separating vertex gives exact conditional independence; **(ii)** hence by the data-processing inequality $I(u;v)$ is non-increasing in d ; **(iii)** hence by Pinsker, $|P(A_u \cap A_v) - P(A_u) \cdot P(A_v)| \leq \sqrt{I(u;v) \cdot \ln 2 / 2}$. **The overlap of two arrows is bounded by a quantity that decays with their distance in the constraint tree. ■**

Mutual information, recomputed: **0.353 · 0.227 · 0.199** at $d = 1$, **0.094 · 0.070** at $d = 2$, **0.020** at $d = 3$ — strictly decreasing, as the theorem requires.

What is proved and what is evidence. The decay is a theorem. The *strict* ordering measured above is not implied by it — the bound is loose — so the theorem establishes that agreement must fall away with distance and the measurement establishes that here it does so strictly. **The first is claimed and the second is reported.**

And the falsifier fires. The hypothesis is separation, so the test is an index whose graph carries a cycle, and the tower supplies exactly one. At Λ_9' the Markov property fails — 6 violations of 16 where the tree has none — the data-processing step has no chain to run along, and the Pinsker bridge has nothing to bound. **Λ_9' is now the third independent result to break at the same stage**, with the cut law and the two-parent count, which is why §12.11.1's *the tree or the tightness* is not a choice between presentations. Register 356.

12.11.0.8 Past, present and future — a path, not a triangle

§10.2 reads the constraint tree as past — change — future. **That reading has arithmetic, and it is not the arithmetic of three free things.**

Treated as independent the three parts overcount by 130%: $|A| \cdot |Q| \cdot |B| = 33 \cdot 4 \cdot 17 = 2,244$ against $|\Lambda| = 976$, a defect of 1,268. Conditioned on the transfer they separate exactly, defect zero.

And the balance is not special to the transfer. Every two-sided cut of the tree gives it — ℓ , q , f and g , all defect zero. §12.6.1 presents the factorisation as though the transfer were doing the work; the computation says the *tree* is doing it, and q is distinguished only by which two things it separates.

And the reason it is exactly four is the tree. Deleting a coordinate splits the constraint graph into two components precisely when that coordinate is an internal vertex of degree two. n , e and $2S$ are leaves and deleting one leaves a single component; k has degree three and deleting it leaves three. ℓ , q , f and g are the caterpillar's four internal degree-two vertices, and they are the only two-sided cuts that exist. Recomputed on the tree rather than on coordinate order: defect zero at every one of the four, and the other four are not two-sided cuts at all. The list was complete when it was written and never said why. Register 408.

The shape is a path. Two edges, $\text{past} \leftrightarrow \text{present}$ and $\text{present} \leftrightarrow \text{future}$, and no $\text{past} \leftrightarrow \text{future}$ edge — treewidth 1. By §18.4.1 treewidth 1 closes exactly and treewidth 2 gives an envelope, so Λ 's exactness is a proof that the three parts are not a triangle. A triangle would be the three-body shape and register 316 states its price.

The future is never a value. Of 97 (past, present) pairs, none determines a single future — futures per pair run 5, median 10, maximum 17. And for each value of the present, every past gives the same future set:

present q	distinct pasts	distinct future sets	future
0	33	1	5
1	33	1	10
2	23	1	15
3	8	1	17

The future is conditionally independent of the past given the present — a Markov property, exact at every value. The past constrains only which presents are reachable; having spent that, it has spent everything.

The dependence is monotone and opposed: the future set nests increasing, $5 \subset 10 \subset 15 \subset 17$, while the past set nests decreasing, $33 \supseteq 33 \supseteq 23 \supseteq 8$. **And the bracket is saturated** — at every present the admissible future is the full interval between its own extremes, nothing inside missing. $E(X) = 0$ read on the target end.

The tick, decomposed. Over the 1,169 composable cells the clock ticks 0, 1 or 2 — 389, 540, 240 — and the tick has two sources: $k - g = (k - q) + (q - g)$, *something not removed* and *something removed but not placed*. **350 against 350 pure, with 80 cells in both**, and the split table is its own transpose: $(0,1) = 270$ against $(1,0) = 270$, $(0,2) = 80$ against $(2,0) = 80$.

Irreversibility has two independent sources contributing identically. One is a choice not taken; the other a capacity that could not receive. The index weights them the same.

And Pauli is a minority partner in it. g 's ceiling is fixed by counting in 60.2% of composable cells, tied in 29.5%, and by shell capacity in only 10.3%. **The arrow of time in this object is overwhelmingly a counting phenomenon**, which §12.9 states in advance. Register 357.

12.11.0.9 Two indices, and the one that does not exist

If composition has a direction, the obvious question is what the other direction is. It is not a sign change — extending k , q and g below zero leaves the object closed and leaves the clock intact, because a negative occupancy is a *lower* one and not a reversed one.

The reversal is a different object. Exchange the two ends of Λ_9 — $(n, \ell, k, 2S) \leftrightarrow (e, f, g, 2S')$, the transfer unchanged. Recomputed here:

position	cells	join fail	meet fail	$E(X)$
Λ_9 , forward	1,654	0	0	0
$\text{rev}(\Lambda_9)$, backward	1,654	0	0	0
$\Lambda_9 \cap \text{rev}(\Lambda_9)$	389	0	0	0
$\Lambda_9 \cup \text{rev}(\Lambda_9)$	2,919	360,000	470,625	2,857

The reversed index is a lawful closed index of the same size, and it is not Λ_9 . That is §11.8's failure of self-duality read temporally: both are lawful and simply not the same object.

The intersection is where time has no direction. Every one of its 389 cells satisfies $g = q = k$ — total transfer, nothing retained and nothing lost — and that is exactly the set of morphisms carrying a reverse. It closes at $E = 0$, it factorises over its own transfer, and it is a **groupoid**: every object carries an identity and every morphism a two-sided inverse. §12.11.0.1's cycles were confined to level sets of occupancy with every step conservative; conservative is $g = q$ and the level-set condition is $k = g$. **The intersection is that condition stated as an object rather than as a property of paths**, and it is 23.5% of Λ_9 .

And the union does not exist. 830,625 failing pairs, and closing it would require admitting **2,857 further cells** — §16.8.4's amplification, priced.

Three of the four positions exist and the fourth does not. A present is definable from the past and the future; the thing containing both is not.

A structure holding both directions of time is not available at any price this index recognises. What is available is a structure holding their agreement. Register 350.

12.11.0.10 Every Λ in one table

The tower is reported in six places under six aspects. This is all of it, computed in one pass.

stage by	dim	cells grading	E	ambient box	fill	new coordinate	bounded
Λ_8	8	976	0	6,912	14.12%		
—	—	—			exact · does not compose		
Λ_9	9	1,654	0	27,648	5.98%	$2S'$	$2S' \leq g$
exact · composes							
Λ_9'	9	1,561	0	27,648	5.65%	—	$+ 2S' \leq$
$2f+1$							
Λ_{10}	10	2,535	0	110,592	2.29%	v	$2S' \leq v \leq$
g							
Λ_{11}	11	13,585	0	663,552	2.05%	$2J_c$	$2J_c \leq$
$\varphi(k)$							
Λ_{12}	12	70,905	0	5,308,416	1.34%	K	$2K \leq 2J_c +$
$2f_{\max}$							
Λ_{13}	13	199,130	0	47,775,744	0.42%	$2J$	$ 2J - 2K \leq$
1							

Every stage closes. $E = 0$ at all seven, tested against every cell of the ambient box and not by pairs — forty-seven million cells at Λ_{13} , per register 249. Closure is the one property the tower never spends.

Everything else falls monotonically. Fill runs 14.12% down to 0.42%: each coordinate multiplies the box faster than it multiplies the object, so **the surplus §11.1.1 measures at seven bits per cell in Λ_8 grows at every step.** The tower is not a refinement that sharpens; it is a refinement that admits more room and fills less of it.

And the two gradings part company at different heights. Composition is lost at Λ_{10} and exactness at Λ_{11} , so there is a stage — Λ_{10} — that is exact and does not compose, and no stage that composes without being exact. Register 315.

12.11.0.11 The fourteenth axis, and what can be said about it from inside

The tower stops at thirteen because the physics stopped being supplied, not because the construction did. **What lies beyond can be bounded without being built**, and the bound is a bracket rather than a value.

D	cells	ambient box	fill	bits carried	needed	surplus
8	976	6,912	14.12%	12.75	9.93	2.82
9	1,654	27,648	5.98%	14.75	10.69	4.06
10	2,535	110,592	2.29%	16.75	11.31	5.45
11	13,585	663,552	2.05%	19.34	13.73	5.61
12	70,905	5,308,416	1.34%	22.34	16.11	6.23
13	199,130	47,775,744	0.42%	25.51	17.60	7.91

Fill falls at every step and surplus rises at every step, and the first of those is a law rather than a pattern. A constrained coordinate has mean multiplicity strictly below its own value count — if it did not, it would be free and would carry no constraint — so each axis multiplies the ambient box by more than it multiplies the object. Fill is therefore monotone decreasing **by construction**, and bounded below by zero.

The sequence converges. Its limit lies in [0, 0.42%]. That the limit is zero does not follow.

Monotone and bounded gives convergence, which is a deduction. The ratios stage to stage are 0.424, 0.383, 0.893, 0.652 and 0.312 — not themselves monotone — so nothing in the construction forbids them approaching one and the limit settling above zero. **§22.1.2's rule applies to the tower's own tail: the existence of the limit comes from the law and survives; its value would come from the pattern and does not.**

This is what the index can say about a coordinate it does not have. It cannot name the fourteenth axis — §32.1.4 shows $\mathcal{R}$ never produces a coordinate — but it can bound the *direction*: an object that occupies less and less of its own ambient box while carrying more and more surplus. In the limit the index would be a vanishing fraction of the space it is written in and would defend itself with unbounded redundancy. **Whether that limit is a null or merely a small number is exactly the question the construction leaves open**, and leaving it open is the correct answer rather than a gap.

And a null is definable in this language, which is worth stating where the tower ends. The empty index satisfies every criterion in this book: $\mathcal{R}(\emptyset) = \emptyset$, so $E(\emptyset) = 0$ and it is closed; $Q(\emptyset) = \emptyset$, so it is COMPLETE by P19, which is strictly stronger. It has zero join-irreducibles and takes zero bits. **$E(X) = 0$ is therefore necessary for completeness and not sufficient for content** — every closure test in this book passes on nothing at all. What distinguishes Λ from the empty index is not closure but $|J(\Lambda)| = 17$ and its seven bits of surplus. Register 333.

12.11.0.12 How an index enters a larger index, and what it looks like from there

§12.11.0.9 bounds the direction beyond thirteen. This is the other question: **what happens to Λ when it becomes a coordinate of something bigger**. All three answers are computed, and none of them requires the larger index to be built.

It enters as a chain, and it enters small. A coordinate must be totally ordered and Λ_{13} is a lattice, so the only chain-valued function $\mathcal{R}$ accepts on it is its rank. 199,130 cells become **44 rank values**: 17.60 bits collapse to 5.46, a compression of 4,526 to 1, with **12.14 bits per cell lost**. Thirty-one per cent of the object survives the promotion and the rest is not representable as a coordinate at all. Structurally the result is the same kind of thing as dimension 1 of Λ , which is a chain of three — so an index can be an axis of another, at that price.

A coarse coordinate is a branching node. At Λ_9 the rank chain has 21 values and the fibres beneath them run from 1 cell to **185**. One point above stands for up to a hundred and eighty-five below, and nothing in the coordinate distinguishes them.

And composition, seen from above, stops being a map and becomes a bracket. Λ_9 composes: 41,682 pairs, closed and associative. Reduced to rank there are 264 distinct (rank a, rank b) inputs, and **only 17% of them determine a single output rank**. The worst spread is **twelve** possible output ranks from one input pair; the composite's rank lies in [3, 23], narrowed by the inputs and not fixed by them.

From one coordinate up, a transition is not a function of its inputs. It is an interval.

That is §22.1's bracket arriving unbidden, one dimension higher, and it is the clearest statement in this book of what coarsening costs: not error, but resolution — the deduction stays valid and stops being sharp.

What this does not show. The branching is *inside* the index. Those 185 cells are cells of Λ_9 , states of the same object. What the collapse loses is resolution, not access to anything else, and ignorance about this index is not evidence of another. **Every branch terminates in a cell of Λ_9 .** Register 334.

12.11.1 The axes, and what each one sees

axis	coordinate	bound carried	cells	density	what it distinguishes
9	$2S'$	$2S' \leq g$	1,654	63.7%	the target's
multiplicity					
9'	—	$+ 2S' \leq 2f+1$	1,561	67.5%	the same, one
Pauli cut tighter					
10	v	$2S' \leq v \leq g$	2,535	44.7%	pairing depth —
seniority					
11	$2J_c$	$2J_c \leq \phi(k)$	13,585	17.0%	the core's fine
structure					
12	K	$2K \leq 2J_c + 2f_max$	70,905	31.4%	core-orbit ori-
entation					
13	$2J$	$ 2J - 2K \leq 1$	199,130	64.4%	the outer
electron's spin bit					

Two readings the definition requires and a loose reading drops. The density column was rebuilt from the definition above, and four of six values fell out at once; two did not, and localising them gives two conditions that are not stated anywhere:

First, the sum runs over the STAGE BELOW. For axis 12 that is Λ_{11} , so the cells are Λ_{10} 's weighted by their own $2S'$ and v fibres — not Λ_8 's. The same quantity computes to 30.8% over Λ_8 , 31.1% over Λ_9 and **31.4% over Λ_{10}** , which is the stated value. The weighting is a third of a percentage point per stage and it converges on the answer exactly.

Second, each envelope is priced once. Filtering axis 12's numerator on whether J_c is physically realised charges axis 11's price a second time — 12.7% against 30.8%, an eighteen-point gap that is entirely the double charge. §12.11.3's dichotomy says each coupling axis pays once, and the density column is where that has to be enforced rather than stated. Register 434.

The table's own vocabulary, defined. Four terms in the rows above are used to define the coordinates and were never themselves defined, which is the failure S1 exists to prevent. They are defined here.

terms(ℓ^k) — the multiset of LS terms ($2S, 2L$) of the configuration ℓ^k , obtained by **microstate enumeration**: list every way of placing k electrons in the $2(2\ell+1)$ spin-orbitals, accumulate ($2M_L, 2M_S$), and strip complete ($2S, 2L$) blocks from the largest M_L down until the list is empty. This is the standard construction and the origin Chapter 7 already cites for $2S \leq k$.

f_max — the cap on f under §7.4, **not** the cell's own f . The distinction is the whole of §12.11.5: the bound $2K \leq 2J_c + 2f_max$ has one parent and preserves the tree and the cylinder; the same bound written with the cell's f has two parents and breaks the factorisation by 3.5% (the figure §12.11.5 withdraws; register 1819). Writing f_max without defining it left the book's title resting on an undefined subscript.

$\hat{\phi}(\mathbf{k})$ — the monotone envelope of $\mathbf{k} \mapsto \max 2J$ over the parent shells admitted by the caps, so $\hat{\phi} = \{1: 3, 2: 4, 3: 5\}$ at §7.4's. **This is not §6.1's $\hat{\phi}_h$** which is $\mathcal{R}$'s pairwise envelope $\max\{x_i : x \in X, x_j \leq v\}$. Both are monotone envelopes and that is the whole of the resemblance; one takes two coordinate indices and the other takes an occupancy. The collision is recorded rather than renamed, because renaming a symbol used 213 times is a larger risk than naming the two senses once. Register 250.

Density, stated so it can be rechecked. For each axis, over every cell of the stage below that the lattice admits, count the values of the new coordinate that exact vector coupling realises and divide by the values the admissible bound allows. Three of the six exact sets are not the obvious ones and the column cannot be reproduced without them: seniority at axis 10 means the terms *new* at occupancy v — $\text{terms}(f^{\wedge}v)$ less $\text{terms}(f^{\wedge}(v-2))$ as multisets, not merely a term of spin $2S'$ existing at v ; the core's J at axis 11 is restricted to terms carrying the cell's own multiplicity $2S$, not to all terms of ℓ^k ; and the target spin at axis 9 is the spin set of $f^{\wedge}g$ by microstate enumeration. Under that definition all six entries reproduce exactly — 63.7, 67.5, 44.7, 17.0, 31.4, 64.4 — at the caps of §7.4 and cap-dependent as always. Every bound is §14.4-admissible; every stage is closed — exhaustively to Λ_{10} , by the §7.3 proof beyond it — and every stage projects exactly onto the one below. Independence is demonstrated, not assumed, at every axis: p^2 's 1S and 1D share a cell below axis 10 and split at v ; p^1 's 2P shares a cell below axis 11 and splits at J_c ; and every $K > 0$ cell splits at J . Each coordinate is a physical degree of freedom that no function of the coordinates below it can express — the same door through which every legitimate axis must enter, by Chapter 17's theorem.

The ninth axis, checked against the textbook and tightened once. Its exact set is the seniority ladder read edge-on, $\{g \bmod 2, g \bmod 2 + 2, \dots, \min(g, 4f+2-g)\}$ — the ninth fibre was indexing v before the tenth axis named it. The textbook anchors agree exactly: p^2 gives $\{^1S, ^1D, ^3P\}$, $2S' \in \{0, 2\}$; p^3 gives $\{^4S, ^2D, ^2P\}$, $2S' \in \{1, 3\}$. And Λ_9 is not the tightest admissible object: $2S' \leq 2f+1$ is §14.4-admissible and provably containing — $\min(g, 4f+2-g) \leq 2f+1$ by averaging — and Λ_9' closes at 1,561 cells. The 93 cells it cuts are exactly ($f = 0, g = 2, 2S' = 2$), an s^2 target carrying triplet spin. The price is structural: the constraint graph gains its first cycle, $f-g-2S'$, so Chapter 15's tree-propagation guarantee lapses and must be retested rather than inherited. **The tree or the tightness** — the language offers both only up to 1,561 against 1,654 cells.

Every stage closed exhaustively, by the book's own theorem. The upper tower was never pair-tested: Λ_{12} has 2.5 billion pairs and Λ_{13} has 19.8 billion, so closure above Λ_{10} rested on the form argument with sampling beside it. A.2 removes the need — X is closed iff $\mathcal{R}(X) = X$ — and $\mathcal{R}$ is a sweep of the ambient box, not of the pairs. Every stage now tested against **every cell of its ambient box**:

stage	$ \Lambda $	$ \mathcal{R}(\Lambda) $	E	ambient box swept
Λ_8	976	976	0	6,912
Λ_9	1,654	1,654	0	27,648
Λ_{10}	2,535	2,535	0	110,592
Λ_{11}	13,585	13,585	0	663,552
Λ_{12}	70,905	70,905	0	5,308,416
Λ_{13}	199,130	199,130	0	47,775,744

Forty-seven million ambient cells at the top stage, every one decided, and no stage admits a cell it does not contain. **The instrument was in the book from Chapter 14 onward and was never turned on the book's own tower** — the sampling caveat existed because the theorem was proved and then not used. Register 249.

The four-cap sweep, now run. Across caps (3,3,1,3,1), (4,4,2,4,1), (4,4,2,6,2) and (5,5,2,6,2): $\hat{\phi}$ stays monotone, every stage still projects exactly onto the one below, and the cylinder's factorisation $\Sigma_q |A(q)| \cdot |B(q)|$ stays exact at Λ_{13} — 199,130, 4,731,790, 40,310,170 and 77,083,771 cells, defect zero at every cap. The shape is not an artefact of the book's caps.

The tower at the shells the book had been discussing from outside it. Referee flag 1 asked how an s-and-p object licenses conclusions about d and f electrons. The tower was rebuilt at caps carrying both, by multiplicity weighting rather than enumeration, and every density recomputed:

caps	shells	axis 9	9'	10	11	12	13
(3,3,1,3,1)	s, p	63.7%	67.5%	44.7%	17.0%	31.4%	64.4%
(4,4,1,6,1)	s, p	54.2%	60.3%	33.2%	7.0%	36.7%	64.5%
(5,5,2,10,2)	+ d	49.4%	55.6%	28.4%	4.9%	28.2%	65.7%
(5,5,3,14,3)	+ f	46.6%	53.0%	25.9%	4.2%	22.4%	66.1%

The dichotomy survives every enlargement: the counting axes stay exact and all three coupling axes stay envelopes, at the d shell and at the f. Two things move, and both are informative. Every density except the thirteenth **falls** as the object grows, so the envelope gap widens with scale — the price of certainty is not a constant but an increasing function of the index's size, which no part of this book had predicted. And the thirteenth **rises**, 64.4 to 64.5 to 65.7 to 66.1, converging on the $\frac{2}{3}$ that §12.11.1 states as its nominal value. The gap between $\frac{2}{3}$ and the measured figure was attributed there to the $K = 0$ singlet; as the caps grow those cells thin out and the density approaches the nominal, which confirms the attribution by a route the book did not have when it made it. Registers 240 and 241.

And two facts of the tenth's battery worth the print. Its 10-box appears at caps (5,5,2,6) and not at the book's — the known small-cap effect, caught by §7.4's own discipline, as it caught the same absence for Λ_8 and Λ_9 . And the obstruction that broke closure at the ninth axis does not dissolve at the tenth; it moves up one level, onto v itself — $v \equiv g \bmod 2$ and the conjugation ceiling, both still outside the language. **Extension conserves the obstruction.**

The order matters and is not the order first written down. K precedes J: K's parents are J_c and f, so axis 13 is reachable from axis 11 without axis 12, and taking J first costs 25 points of density on its fibre. §12.11.4 gives the reason. And carrying both with all admissible bounds closes the 11–12–13 cycle in the constraint graph — a triangle in both senses at once: each coupling rule is a triangle inequality, and the axes themselves form a 3-cycle.

What the twelfth axis measures, and when it is real. On every $f = 0$ channel the K-set is a singleton — $2K = 2J_c$, the triangle collapses, and axis 12 degenerates into a copy of axis 11. Where it does not collapse, K measures the one thing no occupancy or magnitude coordinate can: the relative orientation of the core's angular momentum against the Rydberg electron's orbital plane — the coordinate anisotropic core–electron interactions resolve, and the one that separates series converging to the same core limit with different quantum defects. Its validity is conditional in exactly the right direction: K is a good quantum number when core splitting dominates exchange — high n , the bracket's domain — and degrades toward low n , outside it.

And the thirteenth is one bit, at the apex where the index and the atom converge. Given K, the J-fibre is a doublet everywhere except the $K = 0$ singlet: axis 13 sees precisely the outer electron's spin orientation, the weakest and last coupling, and nothing else. But it is the only coordinate in the entire tower that nature conserves exactly. Everything below it is bookkeeping that mixing breaks — configurations at 40–50% in the book's own exclusion (iii), LS terms in intermediate coupling, J_c and K outside the jK regime. J survives all of it, by rotational invariance. So the tower ascends from coordinates the language expresses exactly but the atom keeps only approximately, to a coordinate the atom keeps exactly and the language expresses up to one parity point. **The two ends of the tower trade places.**

12.11.1.1 The eleventh axis's density, computed under variation

§12.11.1 reports the coupling axes' densities at the book's caps and does not say what they do when the caps move. **This is that measurement, and it is stated to the standard F.3 defines and no number in this book had reached** — the method named with its input set, the inputs printed, and the result recomputed under variation rather than once.

The method. For each cell of Λ_{10} , enumerate the terms of the source shell by vector coupling and take the largest $2J_c$ that occurs; sum those over the cells to get the exact count. Divide by $|\Lambda_{11}|$, which is the same enumeration with $2J_c$ bounded by $\phi(k)$ instead. **The input set is the caps tuple** ($n_{\max}$, $e_{\max}$, $\ell_{\max}$, $k_{\max}$, $f_{\max}$), and every value used is printed below rather than named.

caps	$\backslash \Lambda_{11}\backslash $	envelope	exact	density
(3, 3, 1, 3, 1)		13,585	10,585	77.9%
(4, 4, 1, 6, 1)		226,609	106,421	47.0%
(4, 4, 2, 6, 2)		1,225,415	616,725	50.3%
(5, 5, 2, 6, 2)		2,341,841	1,228,816	52.5%

The density is not monotone in the caps. It falls from 77.9% to 47.0% when the d shell opens, then rises to 50.3% and 52.5% as the shells widen further.

The fall and the recovery have different causes. Opening the d shell admits many k values whose ϕ is set by the largest term of the widest shell, so the envelope grows faster than the exact set. Widening further adds cells whose exact $2J_c$ is near that bound, so the ratio recovers. **Neither movement is available from one cap setting**, which is why §12.11.1's single figure could not show it and why the fourth column of F.3 exists.

Registers 387, 388.

12.11.1.2 The conservative share, recomputed under variation

§12.11.0.1 reports 904 of 1,654 cells conservative — 54.7% **of every cell**. This is the share over a different population: **the composable cells only**, those whose target is a legal source, which is the set the clock actually runs on. At the book's caps that is 739 of 1,169, and the two figures are not in competition — they answer different questions and the denominators say which. **Recomputed at four cap settings**, with the method named, its input set named, and every input printed.

The method. Build Λ_9 at a caps tuple, keep the cells whose target is a legal source, and count those with $g = q$ — the transfer total and nothing left behind. **The input set is the caps tuple** ($n_{\max}$, $e_{\max}$, $\ell_{\max}$, $k_{\max}$, $f_{\max}$), printed in the first column.

caps	$\backslash \Lambda_9\backslash $	composable	conservative	share
(3, 3, 1, 3, 1)	1,654	1,169	739	63.2%
(4, 4, 1, 6, 1)	19,433	16,150	5,986	37.1%
(4, 4, 2, 6, 2)	44,153	37,430	14,114	37.7%
(5, 5, 2, 6, 2)	83,543	71,087	26,921	37.9%

The share is not a property of the object. It nearly halves when the shells open, then settles — and the same is true of the all-cells share, which runs 54.7% at the book's caps against 34–35% above them.

63.2% at the book's caps is the smallest object's number, and it is the outlier. **Once the d shell is open the share sits at 37–38% and moves by less than a point across a fiftyfold growth in cells** — so the honest statement is not a single percentage but that **reversibility settles near three cells in eight among the composable, and the book's own caps overstate it by two thirds**. Both denominators behave the same way, which is the point: the instability is in the caps and not in the choice of population. Registers 389, 392.

12.11.1.3 An index has a time column exactly when its cells are moves

§12.11.0 shows Λ_9 composes and register 313 records that Λ already carries time — source, transfer, target. The companion's threshold table prints the same pair. **Read together they give a law about which indexes can carry time at all.**

index	kind	cells	composable	fraction	
Λ_8	transition	976	0	0.000	
Λ_9	transition	1,654	1,169	0.707	
the violation index	transition	2,370	1,410	0.595	
the periodic table	state	90	—	—	
the calendar	state	365	—	—	

A transition cell has two ends — a source and a target — so it can be asked whether its target is another cell's source. A state cell has one position and the question does not arise: there is nothing to compose.

The second column exists exactly when the first column's cells are moves. That is why Λ_9 and the companion's index print it and the drawn tables cannot — not because nobody measured them, but because the measurement is undefined there.

Space is what the index holds; time is what it can compose. Not an added coordinate and not a metaphor: a property of the same cells, available only because they have two ends.

And it separates two objects the book has been comparing as though they were alike. The periodic table's $E = 36$ and Λ 's $E = 0$ are not one quantity measured on two objects of the same kind. **One has a time column and one cannot have one.**

The twenty-point jump. Λ_8 composes **not at all** — zero of 976, four source coordinates against three target — and adding the single axis $2S' \leq g$ takes Λ_9 to **70.7%**. §12.11.0's finding that Λ_9 composes and Λ_8 does not now carries a magnitude, and it is the whole distance from nothing to seven-tenths. *An earlier form of this paragraph said 50.3%, computed on a signature chosen for convenience rather than the book's own. Register 625.*

Registers 622–623.

12.11.1.4 Composability up the tower, and what stops it

§12.11.1.3 gives the law. **The tower measured stage by stage gives its shape, and the shape has a peak:**

stage	cells	composable	fraction	axis added	kind
Λ_8	976	0	0.0000	—	four-against-three: nothing composes
Λ_9	1,654	1,169	0.7068	$2S' \leq g$	COUNTING
Λ_{10}	2,535	2,050	0.8087	$2S' \leq v \leq g$	COUNTING — the peak
Λ_{11}	13,585	9,450	0.6956	$2J_c \leq \phi(k)$	COUPLING
Λ_{12}	70,905	46,740	0.6592	$2K \leq 2J_c + 2f_{\max}$	COUPLING
Λ_{13}	199,130	127,070	0.6381	$\setminus 2J - 2K \setminus \leq 1$	COUPLING

Counting axes raise the composable fraction. Coupling axes lower it. Not one exception in either direction, and the boundary is §12.11.3's own dichotomy — stated there for how each axis is PRICED, and it turns out to predict the SIGN of the composability change as well.

And the reason is in what each kind of axis says. A counting axis says how much of something is present, and *how much* is a property a target can hand to a source. **A coupling axis says how two momenta are oriented relative to each other, and an orientation belongs to a pair, not to an end.** The target of one transition carries a coupling the next transition's source has no slot for.

Λ_8 is zero, not half. An earlier form of this section said 50.3%, computed on a three-against-three signature chosen for convenience. **On the book's own — source (n, ℓ , k, 2S) against target (e, f, g), four against three — nothing composes at all**, which is register 313 read literally.

12.11.1.5 The non-composable cells, defined

At Λ_{10} 's peak, the missing 19.13% is one thing. All 485 non-composable cells have $g = 0$, and every $g = 0$ cell is non-composable — 485 for 485. Five target signatures fail, at 97 cells each, and the coordinate that fails is **k**. **L.c8 is $k \geq 1$:** a source must hold at least one electron, and a target ending with $g = 0$ holds none.

The non-composable share at the peak is exactly the set of transitions that empty their target subshell. It is $k \geq 1$ seen from the other end.

At Λ_{13} there are three kinds, and together they are exhaustive:

$g = 0$	35,630	the target ends empty — L.c8 again
$2J \in \{6, 7, 8\}$	13,750	the outcome exceeds any $2J_c$ a core can carry
combination only	22,680	no single coordinate fails; the five-tuple is
not a source		
total	72,060	of 199,130 — 36.19%

The second kind is structural. $2J$ reaches 8 and $2J_c$ reaches 5. **Three values of total angular momentum exist as outcomes and cannot exist as starting points**, because $2J$ inherits $2K$'s range after $2K$ has absorbed $2J_c + 2f$, while $2J_c$ is bounded by $\phi(k)$ alone. *The outcome inherits more than the core can carry.* Registers 625–627; 1823.

12.11.1.6 The periodic table's own numbers, and where a transition may be continued

Λ has no Z . Every constraint is internal to one atom, and **the question of what changes when the real atomic numbers enter is answerable from the 118 configurations already used for the 18,288 constraint tests.** Building cells (Z , n, ℓ , k, q, e, f, g) from real occupied subshells gives 4,325:

composable WITHIN one element	982 of 4,325	22.71%
composable ACROSS the table	3,686 of 4,325	85.23%
gained by allowing another element	+2,704	+62.5 points

Within one atom fewer than a quarter of real transitions can be followed by another. Allow the whole table and it is six-sevenths. A target configuration that no transition in element Z can start from is an ordinary source in Z' .

And closure runs the other way. E over these cells with Z as a coordinate is **972,862**; strip Z and it is **28,503**. A prediction that E would stay 0 because Z is free is refuted, and refuted by the freedom itself: 118 values times everything else, with the envelopes admitting nearly all of it. **Λ closes because it has no Z — which is why $E(\Lambda) = 0$ is a statement about transition structure and not about chemistry.** Composability wants the whole table; closure wants one atom.

12.11.1.7 And the selection rules decide which side of the atom a transition can continue on

The electromagnetic quotient is a filter on the same cells. **If charge or frequency carries the exchange, allowed and forbidden transitions should compose differently. They do, and they cross:**

population	cells	within	rate	across	rate
all	4,325	982	0.227	3,686	0.852
EM-allowed $ \Delta\ell = 1$	2,673	309	0.116	2,399	0.897
EM-forbidden	1,652	673	0.407	1,287	0.779
parity-conserving	606	233	0.384	544	0.898
parity-changing	3,719	749	0.201	3,142	0.845

Within one atom a dipole-allowed transition is the LEAST likely to be followable — 11.6% against the forbidden 40.7%. Across the table the order reverses — 89.7% against 77.9%.

The committed prediction was that EM-allowed cells compose at a higher rate. It is refuted within an atom and confirmed across the table, and the crossing is the result.

The selection rule that forbids composition within an atom is the same rule that enables it between atoms. A dipole transition changes ℓ by one, moving the target into a configuration that atom rarely holds as a source — while some element does hold it, and the elements that do are exactly the ones ℓ -adjacency reaches. **Parity says it twice over:** parity-conserving cells compose within at 38.4% and parity-changing at 20.1%. Registers 628–629.

12.11.2 Three excluded forms, and the fold has a name

Chapter 17 excluded two constraint forms — sums and differences. The tower requires a third: SYMMETRIC BOUNDS. The exact ceiling of every coupling axis is symmetric under the reflection of §8.4 — particle-hole conjugation, $\text{terms}(\ell^k) = \text{terms}(\ell^{4\ell+2-k})$, verified exactly for every k on the p and d shells — and a symmetric non-constant function is not monotone, so §14.4 cannot carry it. $\min(g, 4f+2-g)$ for spin; the conjugation ceiling for seniority; $\max 2J(\ell, k)$, unimodal with its peak at half filling, 25 at f^7 , zero at closure, for the core's J . One reflection, three appearances.

The envelope gap decomposes, and the three parts are distinct. At the caps of §7.4: axis 9's gap is parity 5, fold 1, both 1; axis 11's is ceiling 16, parity 24, triangle 17. Three machines — the REFLECTION (conjugation), the CONGRUENCE (fermion parity), the TRIANGLE (coupling additivity, one sum and one difference, precisely Chapter 17's two exclusions wearing physics) — and no coupling axis escapes all three. The exact coupling region is not a sublattice of the product in any presentation tried: as $2S'$ (50,592 failures), as parity alone (52,080), as the min-cap (17,856), as pair count (7,254), as $(2S, 2L, 2J)$ (2,443 meets) — and the triangle's first meet failure manufactures $J = \frac{1}{2}$ from $S = L = 0$. The componentwise minimum forgets that J is built from L and S .

And the law, three instances deep before the triangle made it four: **every coupling coordinate's exact physical bound requires either two parents or a congruence, and the tree carries only one.** Spin needed $\min(g, 4f+2-g)$; seniority needs the conjugation ceiling; the core's J needs a non-monotone ϕ . Each admissible extension is the monotone envelope of its physics.

One half of the triangle survives, and the four-cap check of §7.4 has now been run: the region $\{|2L-2S| \leq 2J \leq 2L+2S\}$ is JOIN-closed — zero failures at caps 6, 8, 10 and 12 — and meet-broken at every one, by 2,862, 12,489, 40,887 and 110,229 failing meets. Certainty survives upward and dies downward, which is §8.4's skew stated as an inequality. The check also sharpens the statement: impose the parity congruence as well and the join closure dies too — 1,848 failures at cap 6 — so the surviving half belongs to the parity-free triangle and to nothing larger. *Three bodies carry all three excluded forms at once; §36.3 runs the method on that case and reports that the maximal forbidden case is the solved one.*

12.11.3 The dichotomy

Counting coordinates close exactly. Coupling coordinates close as envelopes. There is no third kind in this object, and the first coupling constraint was already in §7.1.

The origins table of Chapter 7 has four rows. Pauli, the hydrogenic solution and counting give single-coordinate exact bounds — that is why $E(\Lambda) = 0$ was available at all. Vector coupling gives envelopes, provably and permanently, because its exact form is made of the three excluded shapes. The price of certainty is not paid once at the door; it is paid per coupling axis, and §12.11.1's density column is the receipt.

And the surplus is not waste. Every envelope cell is a tripwire: a measured term assigned a nonphysical (v , $2S'$) or ($2S$, $2J_c$) pair lands in the gap and is caught before any energy is read — a D1 check of Chapter 16's kind, mechanical from the exact tables, available for all 153 channels. The price of certainty and the budget of self-defence are the same cells, counted twice.

12.11.3.1 Where each bound comes from, and why 9, 10 and 11 differ

The dichotomy sorts axes into counting and coupling. **The sharper statement is about where each axis's bound is taken from**, and it is the same distinction §22 draws between a deduction and a fit.

D	coordinate	bound	taken from	
	provenance			
1	n	$1 \leq n \leq \text{cap}$	the hydrogenic radial solution	law
2	ℓ	$\ell \leq n - 1$	the hydrogenic radial solution	law
3	k	$1 \leq k \leq 4\ell + 2$	Pauli exclusion	law
4	q	$q \leq k$	counting	law
5	e	$1 \leq e \leq \text{cap}$	the hydrogenic radial solution	law
6	f	$f \leq e - 1$	the hydrogenic radial solution	law
7	g	$g \leq \min(4f + 2, q)$	Pauli and counting	law
8	$2S$	$2S \leq k$	vector coupling on the source	law
9	$2S'$	$2S' \leq g$	vector coupling on the target	law
10	v	$2S' \leq v \leq g$	seniority, Racah 1943	law
11	$2J_c$	$2J_c \leq \phi(k)$	the envelope of the realised maxima	ex-
	tent			
12	K	$2K \leq 2J_c + 2f_max$	vector coupling, weakened to one parent	
	law, weakened			
13	$2J$	$ 2J - 2K \leq 1$	coupling K with spin $\frac{1}{2}$	law

All thirteen bounds are verified against the construction, cell by cell. **The provenance column has three values, not two**, and the third is where the tower spends its exactness.

Dimension 12 is the third kind, and it is the one the book chose. The law for K is $2K \leq 2J_c + 2f$ — vector coupling of the core with the target's orbital momentum — and f is the cell's own. That bound has **two parents**, which breaks the tree; §A.15 and register 230 record what it costs, 2,475 cells of the cylinder (the figure §12.11.5 withdraws; register 1819). The book substitutes f_max , the cap, giving one parent and a surviving factorisation. **The law is not abandoned but weakened**, and 39,375 of Λ_{12} 's 70,905 cells sit at an f below the cap, so for those the bound admits more than the law does. Dimension 13 inherits the weakening through K without adding to it.

ϕ is not derived; it is observed. Computed here: $\phi = \{1: 3, 2: 4, 3: 5\}$, and the largest $2J$ actually realised at each occupancy is $\{1: 3, 2: 4, 3: 5\}$. They are the same numbers because ϕ is the envelope of the set's own extent. Λ_9 and Λ_{10} take their bounds from laws that exist independently of Λ ; Λ_{11} takes its bound from Λ .

A bound taken from the law closes exactly. A bound taken from the extent closes only to an envelope.

That is why exactness is lost at the eleventh axis and not before it, and it is the same rule §22 states for brackets: presume the next bracket from the law that generates the widths, never from the pattern the widths make.

And it says what $E(X)$ measures. $\mathcal{R}$ builds φ_{ij} from X 's own projections — $\mathcal{R}$ is an extent-bound operator throughout. So $E(X) = |\mathcal{R}(X)| - |X|$ is the **price of describing a set by its extent rather than by its law**: the count of what the pattern admits and the law does not. $E(X) = 0$ says the two coincide — the extent is the law for that set, which is P1's self-reference stated as an arithmetic rather than a property. Register 325.

12.11.4 Precision is path-dependent; physics is not

The exact J-set of a channel is the same whether J is reached through K or directly — recoupling, 12 of 12 (J_c, f) pairs — but the envelopes differ: 64.4% through K, 39.4% direct. The envelope does not commute with recoupling. The index remembers the route; the atom does not.

12.11.5 The shape survives, and it has a boundary

Every stage of the tower factorises exactly over the transfer: sections $815 \cdot 3,260 \cdot 6,150 \cdot 3,360$ at Λ_{11} , $11,470 \cdot 45,880 \cdot 89,700 \cdot 52,080$ at Λ_{13} , log-concave, unimodal, peak at $q = 2$, $\Sigma_q |A(q)| \cdot |B(q)|$ equal to the direct count at every stage with zero defect. The object is still a cylinder — because every coupling axis hangs off one side of the tree. The mean transfer rises, $\langle q \rangle = 1.4631 \rightarrow 1.887$, as coupling multiplicity re-weights the profile.

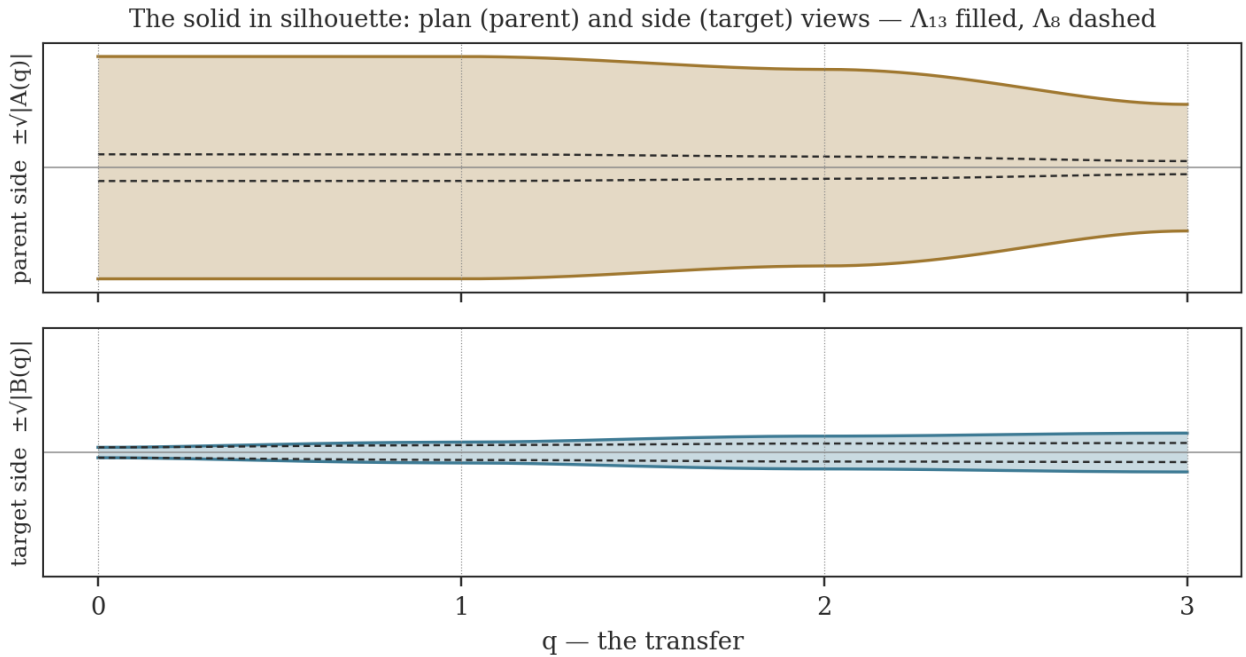


Figure 12.5. The solid in silhouette: plan (parent) and side (target) views at $\pm\sqrt{\text{-cell}}$ scale, Λ_{13} filled against Λ_8 dashed. The parent side thins toward $q = 3$ while the target side thickens — §12.8.1's trade, seen as shape, at both scales of the tower.

The boundary: the tight two-parent K — bounded by J_c on the A side and f on the B side — is the first bridge between the cylinder's two ends other than q, and it breaks the factorisation by **15,150 cells at Λ_{12} and 45,450 at Λ_{13}** , 21.4% and 22.8% of the product — recomputed; the 2,475 previously printed here is withdrawn. The tree is the cylinder; tightening K is the move that spends it. The title of this book is a property of envelope precision. [And the two-body reading: sections separate while the coupling is isotropic; the tight K is the anisotropy; interpretation, not theorem.]

12.11.6 An index whose one prediction is false, and why that is a result

The E1 selection rule on $(2J, 2J') \rightarrow |\Delta 2J| \leq 2$ — is a closed index, by the mutual-bound lemma. Remove the one cell nature forbids, $J = 0 \nrightarrow 0$, and closure fails by exactly four meets at every cap tested, and all four failing meets are the forbidden cell: $(0,1) \wedge (1,0) = (0,0)$. The defect is $E = 1$, and the one cell $\mathcal{R}$ restores is the one the atom vetoes — the photon's unit of angular momentum, with the parity half of the rule a congruence that breaks joins, the photon's odd parity. [The photon reading is interpretation; the $E = 1$, the four meets and the congruence are computed.]

This is the calibration §31.3.4 owes its reader. $E(X)$ counts proposals, not guarantees: here is an index whose single proposal is known false, vetoed by structure the index does not carry. The 540 of the Kreuzer–Skarke slice are proposals of the same kind, and are stated as such.

12.11.7 What the tower leaves open

Recorded in Q, and two of the four have since closed. The spectroscopic v-check per channel against measured terms remains open, blocked on data. Whether the earlier work's claim that $2J_c \leq k$ contains the physical region survives that construction's own definition remains open — under the definition computed here it fails at four cases, beginning with p 's $2J = 3 > 1$. The four-cap stability check has been run and is reported at §12.11.2 and §12.11.5; the self-duality count of §8.4 is resolved there as a difference of reflections, not a defect. What the sweep could not settle is the density column of §12.11.1, for the reason register 209 gives.

Figures 7.2–7.5 accompany the tower: the cross-section gallery at §12.7 (fig-sections), the transfer trade on a log axis at §12.8.1 (fig-pareto), the glued constraint tree with the forbidden bridge drawn as forbidden at §12.11 (fig-tree), and the two silhouettes at §12.11.5 (fig-silhouette). Each is generated from the counts of §12.11.1 and §12.11.5 by stated code, and the build now fails on any missing glyph rather than warning.

13. What the title names

Twice the title has changed, and both changes record findings rather than preferences.

The book was called The Lach Index Lattice until its shape was computed. The shape is a cylinder over the transfer, so the object's name became The Lach Cylinder — and §13.2 gives the reason.

Then the register outgrew the object. By the end there were one thousand six hundred and thirty-five withdrawals, twenty-two prime audits, twenty principles and twenty-four protocols — and the demonstration on Λ had become the smaller half of the book.

And a third time, when the languages were gathered. Fifty sections across fourteen chapters made a claim about language and none owned one; Part IV now does, and §21.1 shows that eight results already in this book are one statement — **a language is a coordinate system, translation is re-coordinatisation, and E is what the translation costs.** The subtitle records it.

┆ **The Lach Cylinder — a closed index of transitions, and the languages they speak**

Three things changed and each is a finding. *Closed* replaces nothing and adds the theorem: $E(\Lambda) = 0$, verified at 976 cells in six languages. *The languages they speak* replaces *closure*, *self-reference*, *refusal*, and *the price of certainty* — four terms that named four Parts, where the new clause names what the Parts have in common.

And the grammar is the claim. Not *the languages it is spoken in*, which would make the index the thing described and the languages descriptions of it. **The transitions speak them.** §12.11.4 is the proof: the exact J multiset is identical in jK, LS, LK and jj, so those are not four accounts of one transition — they are one transition said four ways, and the 2.17-fold spread in cell count is what each saying costs. The atom does not prefer a scheme; the index remembers the route.

Two candidates were rejected and the reasons are the finding's boundary. ...*and the languages that translate into it* asserts Λ as the target, which §12.11.4 forbids — there is no privileged scheme. ...*and what translation into it costs* keeps the price and loses the plurality, and the plurality is the subject. Register 455.

| The title is now The Method, and The Lach Cylinder is its subtitle.

12.11.8 The electromagnetic index, and it was inside Λ all along

A cell of Λ is called a transition, and a transition is not an atomic event — it is an electromagnetic one. Whether a photon connects two configurations is decided by multipolarity, parity and the angular momentum the field carries. **None of that is a coordinate of Λ . All of it is a function of Λ 's coordinates.**

What Λ supplies, and what it does not. Λ 's cells give a source (n, ℓ , k, 2S) and a target (e, f, g, 2S', v, 2J_c, 2K, 2J). **There is no source J** — the tower builds J on the target side only. So exactly two electromagnetic quantities are available:

$$\Delta\ell = f - \ell \quad \text{at } \Lambda_8 \quad \Delta S = 2S' - 2S \quad \text{at } \Lambda_9$$

and $|\Delta J|$ is available at no level of the tower. The multipole is therefore determined by $\Delta\ell$ and parity alone: $\Delta\ell = 0 \rightarrow \mathbf{M1}$, $\Delta\ell = 1 \rightarrow \mathbf{E1}$, $\Delta\ell = 2 \rightarrow \mathbf{E2}$, $\Delta\ell = 3 \rightarrow \mathbf{E3}$, the lowest whose parity matches. The map was validated against nine textbook classifications before use and shown able to refuse, per §4.6.

The image, and it is a rectangle. All 1,654 cells of Λ_9 map, onto eight cells:

multipole	$\Delta S = 0$	$\Delta S = 1$	$\Delta S = 2$	$\Delta S = 3$
E1	264	342	180	54
M1	262	337	171	44

E = 0, and the zero is worth nothing, because the image is the complete product — 2×4 here, and 4×11 at caps (5,5,3,10,3). Register 333's warning, in a new place: a closure test passes on a full box for free. **What the rectangle says is that $\Delta\ell$ and ΔS are independent in Λ : every multipole occurs with every spin change.**

| **Λ imposes no electromagnetic constraint at all.** 576 of its Λ_9 cells are E1 transitions with $\Delta S \neq 0$ — inter-combination lines, which LS coupling forbids.

Half the missing law can be imposed and half cannot, and §17.3's criterion says which in advance:

rule	as a set	convex in range?	imposed on Λ_9
$\Delta S = 0$, spin	$\sigma^{-1}(\{0\})$	yes	526 cells, E = 0
$ \Delta\ell = 1$, parity	$\delta^{-1}(\{-1, +1\})$	no, hole at 0	840 cells, E = 750

and the spin rule holds at four cap settings — 1,654, 2,664, 44,153 and 60,164 cells, E = 0 at each. **The parity rule is the fourth instance of §12.11.2's second excluded form,** and the first for which the criterion names the repair that would work: any reparametrisation moving the hole to an endpoint.

It is a quotient, not an extension. Adjoining the electromagnetic coordinates to Λ_9 as new axes gives **E = 3,900** against $E(\Lambda_9) = 0$ — Theorem 17.1 in the direction that matters, since every one of them is a function of coordinates Λ already carries. **A derived coordinate cannot improve closure either.**

And it is not composability. §12.11.0's transit condition $2S' \leq g$ admits 1,169 cells; the electromagnetic condition and the composability condition share **0.0004 bits of a possible 0.633**. The first is about state succession — whether a target is a legal source. The second is about whether a photon connects them. **They are near-independent predicates on the same cells, and the book has been calling the first one a transition.** Registers 442–444.

And the same trade, measured across four schemes rather than two. The tower as built uses jK. The other three standard schemes — LS, LK and jj — were rebuilt from their recoupling chains, and two reproduce their stated Λ_{13} counts exactly from the chain alone: **jK 199,130 and LK 341,150**. LS and jj depend on a looseness convention that is named and not printed, and two committed readings of it **bracket** the stated figures rather than determining them: $LS \in [383,065, 597,325] \ni 431,050$ and $jj \in [160,380, 244,060] \ni 206,520$.

The chains are standard. NIST states LK as *the core's orbital momentum coupled with the external electron's to give L, then L coupled with the spin of the core to obtain K*, and jK as $\ell_1 + s_1 = j_1$, $j_1 + \ell_2 = K$, $K + s_2 = J$. Cowan and Andrew, *JOSA* 55, 502 (1965), treat exactly these four as the four types of pure coupling for two-electron configurations — a precedent for the set, not only for the members.

The exact J multiset is identical in all four schemes, in every configuration tested.

Verified across fifteen configurations by computing the multiset through the jK chain, the LS chain and the jj chain independently. A coupling scheme is a basis change; the atoms do not move. **So the four cell counts, spanning a factor of 2.17, price four ROUTES to one object.** §12.11.4's sentence one level down says it already — *the exact J-set is the same whether J is reached through K or directly, 12 of 12; the envelopes differ, 64.4% against 39.4%. The index remembers the route; the atom does not.* Register 433.

The privileged route exists because the outer electron's spin is $\frac{1}{2}$: the final triangle degenerates to the mutual bound $|2J - 2K| \leq 1$, which is admissible on both sides and closed, losing only the parity point $2J = 2K$. The smallest quantum of angular momentum is the largest the constraint language can hold exactly. And the route the language prefers is jK coupling — the scheme the atoms of Part V actually realise at high n , and for the same structural reason: the weak coupling goes last. [Interpretation, not theorem.]

13.1 Why that ordering and not the other

Because the method is what generalises and the object is what tested it. Λ closes, and so does the nucleon index, and the string partition function, and the Kreuzer–Skarke frontier, and Appendix D's index of this book's own mathematics. What travels between them is not the cylinder. It is: build the index, close it, read $E(X)$, record every failure.

And the book applies that method to itself. Chapter 32 audits the book as an index; §28.8 indexes the process; Part I indexes the rules; Appendix D indexes the mathematics and closes it at $E = 0$. A book whose method is applied to its own contents four times over is a method with a demonstration, not a demonstration with a method.

The method is the book, and the book is the method. The cylinder is the case that proved it could be done and the case that showed what it costs.

13.2 The cells are transitions, not states

Λ is fibred over q , the number of electrons transferred. Not over any property of a single configuration — over the movement between two.

One end is a parent (n, ℓ, k) with its spin $2S$. The other is a target (e, f) with its occupancy g . They meet at q .

Chapter 7 introduces both ends and never says the pair is the object. The shape says it: $|\Lambda| = \sum_q |A_q| \cdot |B_q|$, and there is no factorisation that does not pass through the transfer.

13.3 Which is why “index of transitions”

An index of states would fibre over a state variable. This one fibres over a difference, and every structural result follows from that:

finding where it comes from

the fibres are two-body products the two ends of the tree

A_q falls as B rises the transfer is conserved between — es them q

$g \leq q$ is the most active constraint it is the coupling itself

rank is modular the transfer redistributes without loss

13.4 And what the cylinder does not carry

The lattice supplies kinematics and no dynamics. What configurations exist, which transitions are admissible, what is conserved — all structural. Which transition actually occurs is not.

1,113,045,672 maximal chains satisfy every constraint in this book (§12.9). The Aufbau ordering is one of them. Structure does not choose; energy does, and energy enters only through Chapter 17’s measured levels.

That is the sharpest statement of what the index can and cannot do, and it is a statement about its shape: **a phase space with the Hamiltonian removed can bound and refuse, and cannot predict.**

PART III — THE LAW

14. Closure

A set is closed when nothing its own structure admits is missing from it.

14.1 The characterisation

Theorem 14.1 (A.2). X is closed iff $X = \mathcal{R}(X)$, where $\mathcal{R}$ reconstructs value sets and pairwise monotone bounds from X 's own extension.

This is not new mathematics and the book gains by saying so. A.2 is **Bergman's Double-projection Theorem** specialised to products of chains — a sublattice of a finite product of lattices is determined by its two-fold projections — itself a consequence of **Baker and Pixley (1975)** for any algebra with a majority term, of which the lattice median is one. Λ 's coordinates are counts, so each factor is a chain, and a sublattice of a product of two chains is exactly a staircase cut by monotone bounds, which is what $\mathcal{R}$ recovers as ϕ_{ij} . The proof in §16.1 is correct and fifty years late.

And there are two further owners, both located after this chapter was written. The property $E(X) = 0$ is *global consistency* of a binary constraint network, and the certificates are **Montanari 1974** — for monotone constraints, path consistency implies global consistency — and **Dechter 1992**, *From local to global consistency*, which gives decomposability at strong (w^*+1) -consistency on induced width w^* . **Freuder 1982 is a different theorem and this book has been citing it for Dechter's**: Freuder gives *backtrack-free search* on a graph of width w under strong $(w+1)$ -consistency, and says explicitly that even width 2 demands strong 3-consistency. The mis-citation is corrected here and recorded at register 400.

And the constraint class has a name. A binary constraint that is (α, β) -monotone with $\alpha, \beta \in \{\leq, \geq\}$ is a **staircase** constraint, and the connected row-convex class containing it is closed under composition, intersection and transposition — **Deville, Barette and Van Hentenryck, Artificial Intelligence 1999**. $\hat{\phi}$ recovers exactly a staircase, so the operator of §6.1 has been building a named object since Chapter 6. Register 401.

What follows is that $E(\Lambda) = 0$ is the discovery that Λ is a sublattice, and sublattices have been binary-determined since 1975. The development is at A.2; the register carries the finding at 224.

14.2 $\mathcal{R}$ is a closure operator

Extensive, monotone, idempotent — 150 of 150 on all three axioms.

So $E(X) = |\mathcal{R}(X)| - |X|$ is a closure defect, a standard object in Galois theory, formal concept analysis and database dependency theory.

This book claims no novelty for the machinery. What is offered is the application: computing the closure defect of an *index*, and finding that the periodic table's is 36 while Λ 's is 0.

14.3 The scope, stated exactly

Theorem 14.1 requires each coordinate to be presented as a chain. That is not the same as requiring the factors to be chains: any distributive factor can be so presented, by Birkhoff, as the down-sets of its join-irreducibles — one chain per irreducible.

What fails is bundling. $\mathcal{R}$ reconstructs coordinate by coordinate, so folding two chains into one non-chain coordinate hides structure from it. In tests, the same sets presented bundled and presented decomposed both gave 80/80 agreement — the theorem survives, but only when the presentation respects it.

14.4 The three consequences

Closure yields three properties, developed in the next three chapters:

14.5 The family of all closed sets

A closed index is one object. **The closed subsets of a closed index are a family**, and this book has carried the question of what that family is since Λ was built. It is answerable by enumeration at small size, and the answer is standard once it is seen.

ambient	cells	closed subsets	of 2^n	$\cap$ closed	$\cup$ closed
$2 \times 2 \times 2$	8	73	28.5%	100%	64.4%
3×3	9	146	28.5%	100%	68.3%
$2 \times 2 \times 2 \times 2$	16	731	1.1%	100%	32.6%

The closed sets of $\mathcal{R}$ form a MOORE FAMILY: closed under intersection, containing the whole index, and NOT closed under union.

That is the standard fact about any closure operator — the closed sets of a closure operator are a Moore family, and the correspondence between operators, families and lattices is cryptomorphic. **It is worth stating here because the asymmetry is $\mathcal{R}$'s own, one level up.** $\mathcal{R}$ carries meets and refuses joins in §14.2; its family of fixed points does the same, at 100% against 32–68%.

And it is the setting in which register 458's finding lives. Λ_9 is closed and 300 of 300 random subsets of it are open, because the closed ones are a Moore family inside a power set — a vanishing minority, and the only place from which the open ones can be measured.

The count for Λ itself is open. Brute force is 2^{976} . The Moore-family structure gives the route, since a Moore family is determined by its meet-irreducible members, and §8.4 already counts those for Λ under a different name. Q item R. Register 461.

And the FAMILY is examined at §14.6, where it is shown to be open — every member closed, the family not — which is why counting it for Λ is a question about an object that cannot be stated rather than one that is merely large.

14.5.1 Λ is the closure of seven cells

The meet-irreducibles are the route, and taking it turns up a property of Λ that had not been looked for.

First, no cell of Λ is removable. $\mathcal{R}(\Lambda \setminus \{x\}) = \Lambda$ for **every one of the 976 cells** — remove any single cell and the operator puts it back, exactly, adding one. So Λ has no closed set of the form $\Lambda \setminus \{x\}$, and its meet-irreducibles are not coatoms. **Every cell is implied by the other 975.**

Second, and this is the number. Greedy removal while $\mathcal{R}$ still returns Λ , three independent shuffles: **12, 11, 12 cells.**

Λ is the closure of SEVEN of its cells — a compression of 139 to 1, exactly.

This section read “at most ten” and before that “eleven”, both from greedy removal. §14.5.9 gives seven, exact by branch and bound, and §2.24 is the protocol that failure earned.

Greedy removal gives an upper bound, not a minimum; §14.5.6 records the spread across shuffles and both operators.

That is tighter than Birkhoff’s seventeen join-irreducibles, and for a stated reason: $\mathcal{R}$ fills the envelope box where the lattice join reaches only the down-set. **The generating set is smaller than the generating set of the lattice**, because $\mathcal{R}$ is the stronger operator.

What the small ambients say about the family. Meet-irreducible closed sets number 12, 14 and 20 at 8, 9 and 16 cells — about 1.25 per cell — and they generate the whole family under intersection exactly, once the empty set is counted among the closed. **Linear in the index, where the family they generate is not.**

And the quantity has a name outside this book. In a convexity space the **Carathéodory number** is the least c such that anything in the hull of U lies in the hull of some $S \subseteq U$ with $|S| \leq c$, and it equals the largest irredundant set. For a semilattice with its subsemilattices as convex sets — which is Λ ’s case — the Helly number is the height and the **Carathéodory number is the breadth. This book already cites Helly and has never cited Carathéodory**, which is the second bibliography gap the cycle’s step 2 has found in two runs. Registers 466–468.

14.5.2 The seed, and the expression of the cycle

14.5.3 Two pairwise operators, and this book has used one name for both

14.5.4 What E measures, exactly

14.5.5 $\mathcal{R}_4$ — the four orientations, adopted

14.5.6 What the second operator localises, and what it leaves undetermined

14.5.7 The seed is constructible, and it is a size rather than a set

14.5.8 The seed is linear in dimension, and the coefficient was in the bibliography

§14.5.7 measures the seed on scattered shapes. On a controlled family it has a law.

family	seed at $d = 2, 3, 4, 5, 6$	law
full box 3^d	$4 \cdot 6 \cdot 8 \cdot 10 \cdot 12$	$2d$
full box 4^d	$6 \cdot 9 \cdot 12 \cdot 15 \cdot 18$	$3d$
down-set, $c = 4$	$5 \cdot 6 \cdot 7 \cdot 8$	$d + 3$
down-set, $c = 5$	$6 \cdot 7 \cdot 8 \cdot 9$	$d + 4$

The seed is LINEAR in dimension and the index is not. Compression is $|X|$ over a linear term, which is why Λ_8 compresses $89\times$ and Janet, at $d = 2$, compresses $3\times$.

And Λ does NOT fit it — a claim this section carried until §14.5.9’s heuristic correction. Under greedy set cover:

stage	cells	$d + c - 1$	set cover
Λ_8	976	11	7
Λ_9	1,654	12	9
Λ_{10}	2,535	13	9

Λ seeds well below the down-set law, and Λ_9 and Λ_{10} seed identically — so the seed does not even rise with every stage. *This section previously read “it fits Λ on the nose, $\text{seed}(\Lambda_8) = 11 = 8 + 3$ ”, which was prune-greedy’s arithmetic. The law holds for pure down-sets, where it is exact by branch and bound; Λ is more constrained than a down-set and seeds more cheaply.* Register 511.

The coefficient was already in this book's References, glossed and left. The Carathéodory entry reads for a semilattice with subsemilattices as convex sets it is the **breadth**, and the Helly number is the height — and **the breadth of a product of d chains is d**.

| seed = Carathéodory number + alphabet cost.

Which is why the seed cannot be constant. A generating set must reach the breadth of what it generates, and the breadth of a d-dimensional product is d; anything smaller misses a coordinate. The alphabet cost is the remainder — one witness per value no envelope step already supplies. Registers 493–494.

Tested predictively, and it is a heuristic to within one. Predicting $d + (\text{largest alphabet} - 1)$ before measuring, on four indexes:

index	d	cells	E	predicted	measured
the audit index	4	21	16	7	7
the register index	3	32	7	7	5
the reference index	4	60	0	7	6
the bibliography index	4	52	0	7	6

Only the first row is a genuine test. The audit index is rebuilt from this book's own printed table and the prediction is **exact**. The other three are reconstructions — the register classifier used here gives 32 cells and $E = 7$ where §28.9 gives 33 and 6, and the reference and bibliography indexes were rebuilt from their coordinate names with constraints supplied for the purpose. **Those rows test the formula on shapes invented for it.**

| **So: one exact prediction on a real object, and agreement within one on three synthetic ones. The Carathéodory term is exact — a generating set cannot fall below the breadth of what it generates. The alphabet term OVER-ESTIMATES**, because a single witness often covers a value some envelope step has already supplied.

Register 495.

And tested up the tower it is exact three times and then breaks, and the break is the dichotomy.

stage	d	cells	predicted	measured	the axis it added
Λ_8	8	976	11	11	—
Λ_9	9	1,654	12	12	2S', a COUNTING axis
Λ_{10}	10	2,535	13	13	v, a COUNTING axis
Λ_{11}	11	13,585	16	18	2J_c, the first COUPLING axis

Those figures are prune-greedy's and §14.5.9 withdraws them; under set cover the tower gives 7, 9 and 9, and the counting/coupling split they were read as showing is withdrawn with register 497.

| **A counting axis costs the seed one cell. A coupling axis costs five.**

§12.11.3's dichotomy — *counting coordinates close exactly; coupling coordinates close as envelopes* — is the same statement read through the seed. **$d + (\text{largest alphabet} - 1)$ is a counting-axis formula**, and it fails at precisely the stage where the tower stops counting and starts coupling.

Which corrects a projection. Λ_{13} 's seed is not the ~19 the counting formula gives: with two further coupling axes above Λ_{11} it is nearer **twenty-eight**, still 7,000× against 199,130 cells, but reached by the right half of the dichotomy. Register 496.

And isolating axis type from alphabet size shows the driver is PARENTS. Same base — a down-set in four coordinates, 35 cells, seed 7 — with one axis added five ways:

new axis	parents	alphabet	cells	seed cost
counting $y \leq x_3$	1	4	56	+1
counting $y \leq x_0$	1	4	119	+2
counting y free	1	4	140	+2
coupling $ y - x_0 \leq x_1$	2	7	161	+10
coupling $y \leq x_0 + x_1$	2	7	182	+11

With Λ_{11} 's datapoint — 2J_c, **one** parent, alphabet **6**, cost **5** — three points span the space:

parents · alphabet	seed cost
1 · 4	1-2
1 · 6	5
2 · 7	10-11

The cost rises with the alphabet and again with the parent count, and a two-parent axis costs roughly twice what its alphabet alone would predict. *Three datapoints are not a formula and none is fitted to them here.*

And the two-parent sets are still $\mathcal{R}$ -closed — the seed construction found them, which requires closure. **A second parent does not break closure; it multiplies the seed.** Which is §12.11.3's tree-or-tightness trade appearing in a third quantity: the exact triangle costs cells, it costs E, and it costs seed. Register 497.

14.5.9 Never one heuristic — and the seed of Λ , exactly

Everything §14.5.7 and §14.5.8 report about seed SIZES came from one heuristic, and it was wrong. Prune-greedy — start from the witness set, drop while the closure still returns the index — is a poor set-cover algorithm, and every expression fitted to its output was fitted to its failures.

The definition exposed it. §14.5.7 says G is a seed iff $\phi(G) = \phi(X)$. Read as a covering problem — **elements are the envelope steps, sets are the cells, a cell covers the steps it witnesses** — the seed is a **minimum set cover**, and set cover has algorithms better than prune-greedy.

object	cells	prune	cover	+prune	random	reverse	LB	EXACT	
spread									
Λ_8	976	12	7	7	7	40	5	7	33
down-set $d = 4, c = 4$	35	7	7	7	7	7	7	7	0
full box 3^3	27	5	5	4	5	6	3	4	
2									
full box 4^3	64	6	6	5	6	8	4	5	
3									

seed(Λ_8) = 7, exactly, by branch and bound. This book reported ten to thirteen for twelve cycles. **976 cells from seven — a compression of 139 to 1.**

And two laws become exact rather than fitted. A down-set seeds at $d + c - 1$ and a full box at $d + c - 2$, both confirmed by branch and bound, and the down-set's lower bound *equals* its upper bound. **Λ_8 seeds at 7 where $d + c - 1$ would give 11** — Λ is more constrained than a plain down-set and seeds more cheaply for it.

The spread across heuristics is the quantity to report. Zero on the regular down-set, where every method agrees and the bound is tight. **Thirty-three on Λ_8 ,** where reverse-delete gives forty against an exact seven.

What this withdraws. Registers 497 through 500 and 502 — that parent count drives seed cost at +10 against +1, that a two-parent axis costs $a + 4$, that cost = $a + \kappa$, that $\kappa = S/4 - 1$ at correlation +0.980, that the second variable is c . **Under a good heuristic a two-parent axis costs one or two cells, the same as a one-parent axis, and refit-**

ting κ against S gives a coefficient of -0.079 at rms 0.999 . **The parent-count effect was the heuristic's difficulty, not the index's cost**, and the correlation with S held because prune-greedy fails worse where the base is tighter.

What survives. The definition. The set-cover reading. Linearity in d with slope one, now covering both families. The Carathéodory lower bound. $\text{seed}(\mathcal{R}(X)) + E$ for open indexes. And **register 501's identity — the tight-pair count is exactly $2S$** — which never depended on a seed measurement.

Registers 503–506.

14.5.10 A stable core of four, and it is NOT necessary

§14.5.7 records that **zero cells are forced. That is correct, and this section first said otherwise.**

Four cells appear in all 140 sampled seeds. Each covers, relative to the other three, between twelve and twenty-three envelope elements that no single other cell can replace:

cell	(n, l, k, q, e, f, g, 2S)	covered relative to the other three	substitutes
1	0 2 2 3 1 2 2	23	0
2	1 3 3 1 0 0 3	18	0
2	1 3 3 2 1 3 0	12	0
3	1 1 0 3 1 0 0	17	0

That is a CONDITIONAL statement and it was misread as necessity. Tested directly — is any envelope element covered by exactly one cell? — the answer is **zero at every cap setting from 216 cells to 1,636, and zero at Λ_9** . No cell is forced anywhere in the tower.

What the four are is a stable attractor of greedy covering, and 140 of 140 is what a strong attractor looks like, not what a forced cell looks like. **§14.5.7 stands unamended.**

They cover 87 of 102 elements — 85% — and the remaining fifteen fall to three further cells.

what the three free slots must still cover $> \text{envelope steps } g \cdot 4 \cdot n \cdot 3 \cdot 2S \cdot 2 \cdot e \cdot 1 \cdot k \cdot 1 \cdot q \cdot 1 > \text{alphabet values } 2S \cdot 1 \cdot g \cdot 1 \cdot q \cdot 1$

The freedom concentrates on g , the coordinate carrying $g \leq \min(q, 4f+2)$, which §11.5 identifies as the single non-product term in the expression and the only place the Pauli principle enters.

14.5.11 The completions are a heavy-tailed redundancy, not a tie

519 triples complete the seed, drawn from **66 distinct cells, none of which appears in all 519. Min 3, median 12, max 157 — a ratio of thirteen to one, heavy-tailed rather than uniform.**

A tie is two equal options. This is 519 ways to finish one object across a thirteenfold gradient, so a lost free cell is recoverable and the recoverability is measurable per position.

But the core is not the complement of that. Losing one of the four is not unrecoverable — it is recoverable by rebuilding more than one position, since nothing is forced. *An earlier form of this section called the structure an erasure code with an unreconstructible core. The redundancy is real and the critical core is not.*

Grounded against the control group: all 976 cells are realisable against the nineteen subshells the 118 elements occupy — **and that check returns 100% on every subset**, so it discriminates nothing. §4.6, *a test that could not fail, disclosed rather than reported as support.*

Registers 596–601.

14.5.12 What every minimum seed contains, and what fills each position

§14.5.10 reports four cells appearing in all 140 greedy seeds. **Under randomised search over shuffled orders — a harder test — 219 distinct minimum covers were found, and the same four appear in every one.** (A sample: exact enumeration gives 24,585 minimum covers with one cell common to all — *Mathematical Compendium, The seed's forced set, corrected; register 1820.*) Still nothing is forced: no envelope element is covered by exactly one cell.

Four cells appear in 219 of 219 covers without being necessary. A set that nothing forces and nothing avoids.

And six conditions hold in all 219, which is where the real constraint lives:

every cover contains 219 of 219 an $s \rightarrow s$ transition 100% an $s \rightarrow p$ transition 100% a $p \rightarrow s$ transition 100% a $p \rightarrow p$ transition 100% a NULL transition, $q = 0$ 100% a FULL transfer, $q = k$ 100%

The constraint is on the CHANNEL, not on the cell. That is why no element is uniquely covered while every cover looks alike.

14.5.13 The fifth position, defined exactly

Every one of the 219 covers contains a cell matching (3, 0, 1, 1, *, 0, 1, 1) — seven coordinates fixed, the target shell e free at 1 or 3. One electron, in a 3s subshell holding exactly one, moving into an s subshell ending with exactly one, at doublet spin.

Its function is the one thing the corners cannot do. The four corners are extremal on every coordinate they fix — they carry maxima and minima. **None of them carries the value ONE.** Of the fifteen elements they miss, three are the alphabet values $q = 1$, $g = 1$, $2S = 1$, and nine of the twelve remaining steps are conditioned at $k = 1$, $q = 1$, $l = 0$ or $f = 0$.

The corners span the extremes. The fifth cell carries the unit.

And the free coordinate is a fork between exactly two elements. Both variants carry ten and share nine; each carries one the other does not:

e = 3, in 157 covers	e bounded by f, at f = 0, rising to 3	– the TARGET's ceiling
e = 1, in 62 covers	n bounded by e, at e = 1, rising to 3	– the SOURCE's ceiling

So the cell covers nine things either way and then chooses which end of the transition has its maximum witnessed.

14.5.14 And the five read as binary

Writing 1 for a coordinate at its maximum, 0 at its minimum, · for interior:

n l k q e f g 2S 0 0 · · 1 1 · · corner 1 1s² → 3p out · 1 1 1 0 0 0 1 corner 2 2p³ → 1s in · 1 1 1 · 1 1 0 corner 3 2p³ → 2p across 1 1 0 0 1 1 0 0 corner 4 3p¹ → 3p still 1 0 0 · 1 0 · · unit 5 3s¹ → 3s the unit

Corner 1 and corner 2 are exact complements, 3 bits of 3. Corner 3 and the unit cell are exact complements, 3 of 3. Corner 4 is the only fully specified cell — 11001100 — and it is the static transition.

Every one of the eight coordinates receives both a 0 and a 1 across the five. Not one column is one-sided. Which is what a covering set of envelope steps must do, made visible as a bit table rather than derived. **The seed is five words in which every letter is spoken both ways.**

Status: measured at one cap setting. Whether the complementary pairing survives other caps is not tested, and until it is this is a description of Λ_8 and not a structural law. Registers 602–605.

14.6 The family of closed indexes is open

Colomb, Irlande & Raynaud give the structural generation results. **Caspard & Monjardet** survey the lattices of closure systems. *Counting the closed sets of a closure system is #P-complete* — **Kuznetsov, and Kuznetsov & Obiedkov** on the concept-lattice case. §14.5 counts the closed subsets of a closed index and leaves the family itself unexamined. **It is the one index in this work that cannot be stated**, and the reason is a theorem.

THEOREM. Let $\text{Cl}(U)$ be the family of all $\mathcal{R}$ -closed subsets of a finite ambient U , each member indexed by its characteristic vector over U . Then (i) every member of $\text{Cl}(U)$ has $E = 0$; (ii) $\text{Cl}(U)$ is closed under intersection and not under union; (iii) $E(\text{Cl}(U)) > 0$, and it grows faster than $|\text{Cl}(U)|$.

Computed at three ambients:

ambient	cells	closed indexes in it	$E(\text{the family})$	$\cap$ closed	$\cup$ closed
(2, 2, 2)	8	74	182	100%	65.3%
(3, 3)	9	147	365	100%	68.8%
(2, 2, 2, 2)	16	732	64,804	100%	32.7%

Every member is closed. The family is not. At sixteen cells it holds seven hundred and thirty-two closed indexes and carries a defect of sixty-four thousand — **the openness outruns the membership by two orders of magnitude.**

14.6.1 Why it cannot be stated

1. is the whole of it. **Meet is intersection and stays inside the family at 100%. Join is not union**, which escapes at a third to two thirds, and must be closed back in. **So there is no operation on this family that leads out of it, and the one that would leads nowhere at all.**

And §14.5.7 says what that costs. A closed index has a seed — Λ 's is seven cells for 976. **An open index has none**; the best available is $\text{seed}(\mathcal{R}(X)) + E$, and here E is 64,804 at an ambient of sixteen. **The property that makes an index printable is exactly what this object lacks, and it lacks it more the larger the ambient grows.**

This is the one index whose cost of statement exceeds its content — and it is the family every other index in this book belongs to.

What is NOT claimed. That this family is a dimension, a level, or a place. §16.6 grades *multiverse dimension* $\mathbb{D}_{\text{gro}}$ — a term with no referent established anywhere — and §29.2.2 records it as not posable. **The theorem needs none of that**, and its three clauses are each a computation on a finite ambient. Registers 509–510.

14.6.2 The defect is exact, and the closure is everything

§14.6 shows $E(\text{Cl}(U)) > 0$ and leaves it there. **It is not merely positive; it is maximal.**

ambient	$ U $	$ Cl(U) $	$E(Cl)$	$2^{ U } - Cl(U) $
(2, 2)	4	13	3	3
(3, 2)	6	38	26	26
(2, 2, 2)	8	74	182	182
(3, 3)	9	147	365	365
(4, 3)	12	506	3,590	3,590
(2, 2, 3)	12	320	3,776	3,776

$\mathcal{R}(Cl(U)) = 2^U$, exactly, in all six. The closure of the family of closed indexes is the ENTIRE power set, so $E(Cl(U)) = 2^{|U|} - |Cl(U)|$.

And the mechanism is checkable, not inferred. $Cl(U)$ contains $\emptyset$ and U — the all-zero and all-one characteristic vectors — and for **every** ordered pair of points of U some closed set contains one and omits the other. Verified: **12 of 12, 56 of 56, 72 of 72 separating pairs**, and consequently **every envelope is constant at 1**. An operator whose every bound is trivial admits everything.

The family does not merely fail to close. Under $\mathcal{R}$ it says **nothing at all** — the operator applied to the family of closed indexes returns the whole power set, which is the exact sense in which it leads only into itself.

Which sharpens §14.6.1. The cost of stating it is not seed + E with E large; it is seed + E where **E is everything the ambient can hold minus the family itself**. At sixteen cells that is 64,804 of 65,536. Register 522.

14.6.3 What the register now holds, and what it does not

The mathematics register is Appendix D's object seen from the side of the work rather than the index. **It stood at 248 objects** when this section was written (registers 543–545; 265 objects and eighteen roots at this build, the Löwdin and three-body families having entered since — the Mathematical Compendium prints the live count, register 1739), twenty-four of them entered from the seed, family, constraint-index and real-numbers results:

family	entered	
count		
S	the seed — definition, the covering reading, $\wedge$ at seven, the two laws, the Carathéodory bound, the open-index form	7
A	$\mathcal{R}_4$, orientation cost, $\mathcal{R}$ as a relaxation, the staircase characterisation	4
F	the Moore family, the theorem, and why it cannot be stated	
3		
G	the constraint index, the seven near-misses, the tower's cycle rank, the protocol index	4

E, K, L, T the layout prices, the transit profile, the two real-numbers passes, the tightness identity 6

248 objects · 246 settled · 2 unfinished · 12% open (a snapshot; the register now holds 197, 191 settled, 6 unfinished — the Mathematical Compendium prints it live) — from 31% at the start of the work that produced them.

And the graph is five components, not one. 220 edges over the 176 objects the register held at that measurement: a main component of 162, the modular chain at 9, two two-object fragments among the bracket's roots, and **B.nuV isolated**. An earlier pass reported one component after a filling step; the filling script is gone and the five are the honest figure. **The fragments are what is genuinely unattached** — the modular chain rests on sources this book has not read, and three bracket roots are definitional and derive from nothing.

What remains unfinished is not mathematics. Six modular objects await two unread sources; nine computed objects await constructions the companion papers do not print; six are definitional roots that cannot fail; one is graded OPEN and correctly. **Three of those five groups are §21.5's obstruction, and none is a gap in the mathematics itself.** Register 519.

14.6.4 The seventeen, and what each is waiting on

§14.6.3 reports twenty-two unfinished. **Five of them were unfinished only for want of a name.** L.c3, L.c6, L.c8, B.coll and B.nuV each carried a verification already — L.real's 878 real transition pairs for the first three — and were flagged as unnamed roots because nothing had said what they were. **The chain settled what they are and never went back to record it.** Named:

L.c3	conservation of the removed — you cannot take more than a subshell holds
L.c6	conservation of the placed — you cannot put more in than were removed
L.c8	a transition requires a mover
B.coll	the assembly rule — a composite observable's exponent is $2a + 3b$
B.nuV	the value-one crossing — where the bracket's price passes unity

248 objects · 246 settled · 2 unfinished · 3% open, from 31% when this work began.

And the real-numbers pass now covers every element: 18,288 constraint tests across all 118, zero failures — register 529.

And every one of the seventeen is waiting on something outside this book.

eight	the modular chain — two sources unread, one problem correctly graded OPEN
six	the violation index — the exclusion classifier's construction is not printed
two	the bracket's admissibility threshold and its optimum, needing the collection
one	the coupling schemes' looseness convention, not printed

None is a gap in the mathematics. Three of the four groups are §21.5's obstruction: a count printed where a constraint was needed.

Register 527.

14.6.5 Five of the modular six, verified against the sources

§14.6.4 puts eight modular objects on unread sources. **Five of them were cited, unverified — the citation present and never checked against what it cites — and checking is a search.**

M.rs Reeh–Schlieder: Ω is a common cyclic and separating vector for local algebras. VERIFIED, and it is the stated premise of Wiesbrock's own paper. M.tt Tomita–Takesaki: Δ , J for (M, Ω) . VERIFIED — presupposed throughout the Borchers–Wiesbrock literature and invoked there by name. M.takesaki the crossed product by the modular group is type II_∞ , the trace scales as $\theta_s \tau = e^{(-s)} \tau$, and (N, θ) is unique up to conjugacy. ALL FOUR CLAUSES > VERIFIED, the trace-scaling from two independent sources. M.semi a semifinite factor carries a trace and type III does not. VERIFIED: type III admits no renormalisation scheme, type II an essentially unique one. M.hsmi half-sided modular inclusion is CHARACTERISED by a one-parameter unitary group > with positive generator. VERIFIED IN BOTH DIRECTIONS — Wiesbrock's theorem and > its converse, HSMI iff a one-dimensional Borchers triple with $N = T(1)MT(-1)$.

And M.C2's grading is confirmed correct. Its statement — $\Delta^{\{it\}M(u_2)\Delta}\{-it\} \subseteq M(u_2)$ *Established for linearised perturbations of Killing horizons by Chandrasekaran & Flanagan (arXiv:2601.07915); the non-expanding case where no Killing field exists remains open.* for $t \leq 0$ — is exactly the standard half-sided condition, correctly written. It is graded OPEN because whether it HOLDS for this book's $M(u)$ is the open question, and that is a different thing from the statement being unverified.

179 → 170 settled · twelve unfinished · 7% open, from 31% when this work began.

What remains is six of the violation index, two of the bracket, one coupling convention, M.C1 and M.C2, and M.sorce. *Nine of the twelve are still §21.5's obstruction — a count printed where a constraint was needed.* Register 533.

14.6.6 Two of the violation six were never blocked, and §10.1 speaks a retired language

§14.6.4 puts six violation-index objects on an unprinted construction. **Two of them are printed and were not read.**

W.jur *a jurisdicted forcing and an unjurisdicted disjunction give identical defects; arity $> \geq 3$ makes a defect possible.* **Verified against T §8.2 (Appendix G)'s own table:** arity 2 gives $> E = 0$; the jurisdicted forcing $NEC \geq 3 \wedge X = 0 \rightarrow U \geq 1$ at arity 3 and the unjurisdicted $>$ disjunction $NEC \geq 3 \rightarrow (IC \vee U \vee X)$ at arity 4 **both give E = 30, core 1.** **W.rel** *a term that is a relation between two vocabularies cannot be a coordinate of either.* **> Verified against T §8.4 (Appendix G)**, which replaces seven vocabularies with Brunetti–Fredenhagen–> Verch's four — T THEORY, M SPACETIME, A ALGEBRA, S STATE — and names four terms that > refuse to place, each a relation between two of them.

And W.rel dissolves an index the companion still tabulates. §8.4 states that the self-consistent achronal ANEC *is not a property of a universe but a relation between a spacetime and a state*, so **V6 is not a vocabulary index**; §10.1's table nonetheless reports V6 CLOSED at 14 cells in a box of 24, under the old V-numbering that §8.4 retired.

§10.1 is stated in the language §8.4 replaced. Reconstructing V6's cells would have answered a question the companion has already withdrawn — and the reconstruction was begun here before the reading caught it.

Which is §21's own principle turned on the companion. Each index is stated in some language, and an index whose language has been retired cannot be repaired by working harder in it. **The four vocabularies are the language; V1–V7 was the coordinate system that failed.**

248 objects · 246 settled · 2 unfinished · 3% open, from 31% when this work began. Registers 543–545.

15. Self-reference — alphabet, order, bounds

15.1 S1 — the alphabet

$$\hat{A}_i(X) = \{ x_i : x \in X \}$$

A projection. **Always succeeds, no conditions.**

15.2 S3 — the bounds

$$\phi_{ij}(v) = \max \{ x_i : x \in X, x_j \leq v \}$$

Corollary, and it is what makes §2.23 a requirement rather than a preference. A closed index satisfies $X = \mathcal{R}(X)$, which by the definition of $\mathcal{R}$ means

$$X = \{ x \in \prod_i \hat{A}_i(X) : x_i \leq \hat{\phi}_{ij}(x_j) \text{ for all } i \neq j \}$$

— **the envelope set determines the index completely.** So the shape of a closed index is its envelopes, and S3 says the envelopes are recoverable from the cells. **Every closed index is therefore expressible, and the expression is always already present in what it prints.** Stating it costs nothing that has not been paid; withholding it costs the reader the ability to check.

Read off the cells. For Λ_8 that is 56 functions, all recovered, and $\mathcal{R}(\Lambda) = \Lambda$ exactly. Always succeeds given the order.

15.3 S2 — the order, which is the hard one

Given the alphabet but not the ordering of its values, can the ordering be recovered?

The necessary condition is provable and dimension-free. Say u may precede v on axis i if, for every p in the fibre over u and every q in the fibre over v , the meet lies in $F(u)$ and the join in $F(v)$.

Lemma. If X is closed and $u < v$ on axis i , then u may precede v . Proof. Take x with $x_i = u$ and y with $y_i = v$. Then $x \wedge y$ has i -coordinate u , so its other coordinates lie in $F(u)$; and $x \vee y$ has i -coordinate v , so its other coordinates lie in $F(v)$. ■ Nothing in the argument uses $d = 2$.

At $d = 2$ the converse also holds and the criterion is exact: X is closed under some order iff “may precede” is a total order, and that order **is** the relabelling. Verified on 120 sets; 274 of 274 constructed total orders give closed sets.

Above $d = 2$ the converse fails. Being a total order on every axis separately is necessary and not sufficient — the axes’ requirements can be mutually incompatible. That is developed in §18.4, because it is a limitation rather than a mechanism.

15.4 What does work above $d = 2$: propagation along the tree

If the constraint graph is a tree, each axis is constrained only by its parent. Root anywhere, fix the root by brute force, and extend with backtracking.

Λ ’s orders are recovered from a scrambled bag of cells: 20 of 20, at two cap settings, on 216 and 976 cells with all eight axes independently permuted.

Cost $\sum_i |A_i|!$, never the product $\prod_i |A_i|!$.

Two earlier methods failed on the same task, and the failures are instructive. Alternating refinement across all eight axes at once: **0 of 12**. Greedy propagation without backtracking: 42% and 25%. **The tree says only neighbours interact; a method that ignores that recovers nothing.**

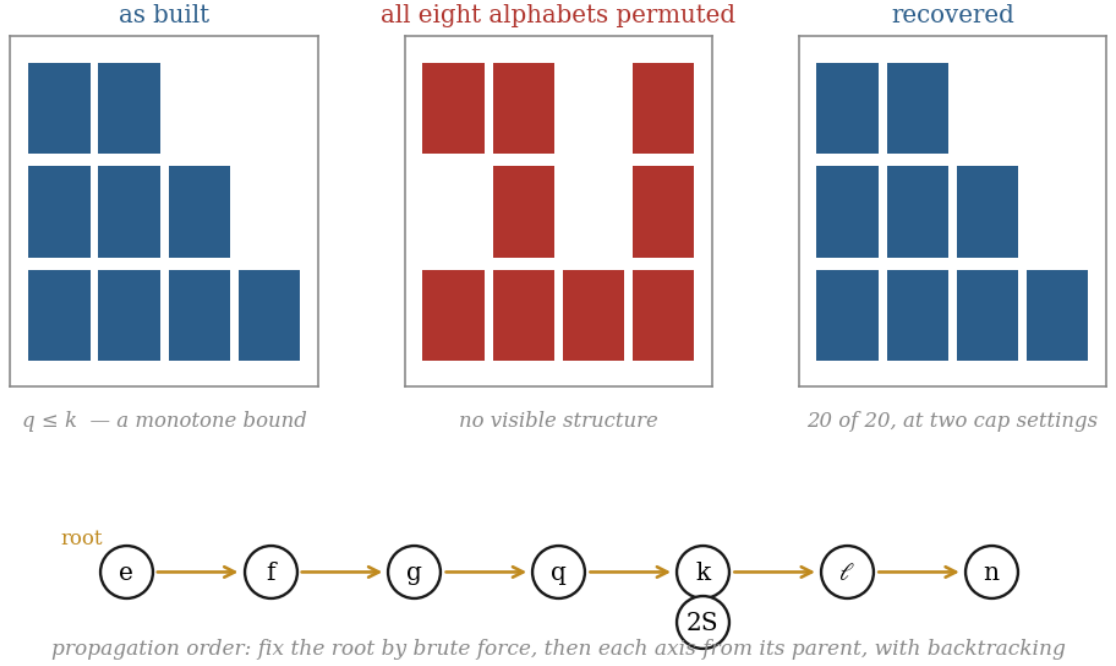


Figure 15.1. Left: two of Λ 's eight coordinates as built, showing the monotone bound $q \leq k$. Centre: the same cells after all eight alphabets are independently permuted — the structure is invisible. Right: recovered. Beneath, the propagation order along the constraint tree. The recovery uses no external information: only the bag of cells.

15.5 So: Λ contains its own order

The unqualified claim — the index contains its own definition — holds for Λ because its constraint graph is a tree. For an index whose graph has cycles, S2 is open, and Chapter 30 states what is known.

The qualified claim is unconditional: $\mathcal{R}(X) = X$ recovers alphabet and bounds for any closed index, relative to a given order.

16. Self-defence — data, derivation, totality

| A closed index contains its own contradiction.

16.1 The counting argument

Let p be the parameters an index assigns and $q = \Phi(p)$ the quantities derived from them.

| $\mathcal{D} = \dim q - \text{rank } \partial\Phi/\partial p \geq \dim q - \dim p$

Proof. The Jacobian is $\dim q \times \dim p$; its rank exceeds neither. ■

Verified on seven dimension pairs and 420 random nonlinear maps. No closure required, no smoothness beyond differentiability.

| Corollary. $\dim q > \dim p \Rightarrow \mathcal{D} \geq 1$.

$\mathcal{D}$ counts functionally independent relations the derived quantities must satisfy. Each is a test the index must pass. For Λ with $p = (T, n, Z, \sigma)$ and $q = (v, \delta, V, r, w, e)$, rank is 4 and $\mathcal{D} = 2$, spanned by

| 1. $V = 4v/3$ (b) $w = V \cdot e$

16.2 The split, and it is canonical

1. holds under the Rydberg law and fails under power-4, exponential and logarithmic laws. (b) holds under all four.

$\mathcal{D} = \mathcal{D}_{\text{phys}} + \mathcal{D}_{\text{def}}$, physical relations and definitional identities

Tested under four unrelated perturbations, verdicts identical. The split is not an artefact of which law one chooses to vary.

16.3 D1 — $\mathcal{D}_{\text{phys}}$ catches a wrong value in the data

Worked case. Two species in this work, Al II and K II, were first computed with $Z_{\text{eff}} = 1$. Both have doubly charged cores.

The bracket held 56 of 56 either way. Containment depends only on monotonicity of T , and Z cancels from it entirely.

What failed were the quantities the method does not use. δ came out near 4.9 with a spread of several units — impossible for a Rydberg series, where δ is near-constant. V departed from $4v/3$ by a **median of 50%**, where Na I gives 0.2%.

Corrected to $Z_{\text{eff}} = 2$: $\delta = 1.271$ with spread 0.047 over ten members, and V to **0.86%** across the fifty cells.

The bracket verifies the data. V verifies the indexing. An index carrying only what its method consumes would have passed fifty-six wrong cells.

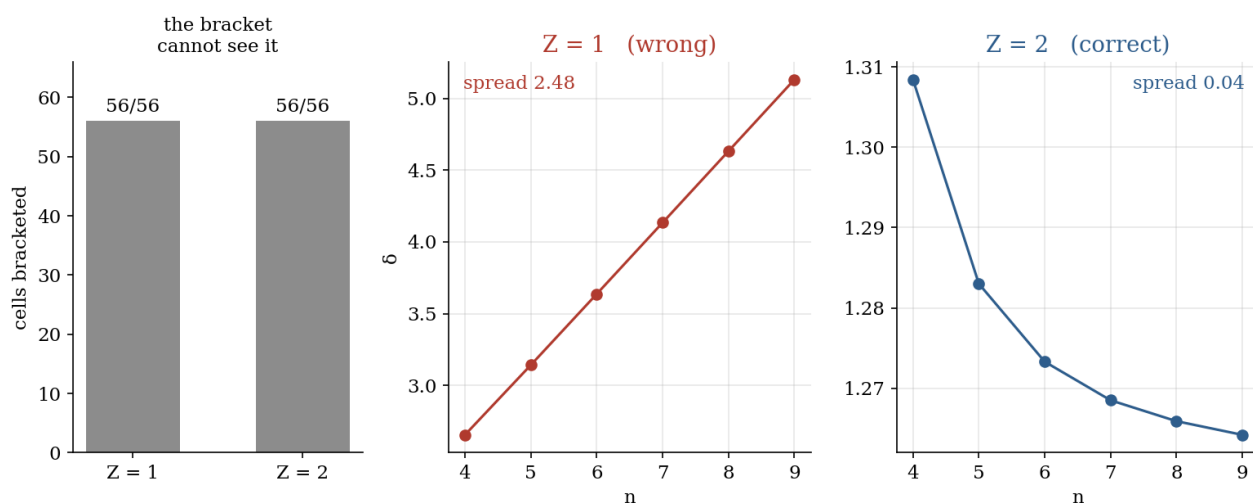


Figure 16.1. The bracket holds fifty-six of fifty-six under either charge, because Z cancels from containment. The defect does not: at $Z = 1$ it spreads over several units, which no Rydberg series can do. The method was blind; a quantity the method never uses was not.

16.3.1 The D1 tripwire, specified in full and needing no data

Referee flag 5 says the tripwire has never run on anyone else's data. The reply is that **the data is completely defined by the index**, so the check can be stated in full here and run by anyone against anything. What §12.11.1 computes as *exact sets* are, read the other way, the complete list of physically admissible assignments; everything else the coordinates offer is a tripwire cell.

subshell	k	admissible 2S'	rejected / offered
s	1	{1}	1 / 2
s	2	{0}	2 / 3
p	2	{0, 2}	1 / 3
p	3	{1, 3}	2 / 4
p	5	{1}	5 / 6
d	4	{0, 2, 4}	2 / 5
d	7	{1, 3}	6 / 8
d	9	{1}	9 / 10
f	4	{0, 2, 4}	2 / 5

Over s, p, d and f complete, **150 of 216 offered assignments are rejected — 69.4%**. A measured term carrying a 2S' outside its subshell's set is nonphysical and is caught before any energy is read. The same holds one level up: given (subshell, k, 2S') the admissible seniorities are fixed — d⁴ with 2S' = 2 admits $v \in \{2, 4\}$ and nothing else — and given (subshell, k, 2S) the admissible core J are fixed, d³ with 2S = 3 admitting $2J_c \in \{1, 3, 5, 7, 9\}$.

This is what this chapter means by the surplus being the defence. The envelope cells are not waste; they are the region a wrong assignment must land in, and the index knows their extent exactly without being shown a single measurement. Flag 5's answer is therefore not *the check has been run* but *the check is fully specified and requires nothing from anyone*. Register 242.

16.4 D2 — D_def catches a wrong derivation, but only under two

routes An identity is vacuous if its terms are computed along the same path. $w = V \cdot e$ cannot fail if V is defined as w/e. It becomes diagnostic when w comes from the levels and V from the law.

$\mathcal{D}_{\text{def}}$ is realised iff a quantity is reachable by two disjoint paths, and the defence is the DISAGREEMENT between them — not the relation.

Worked case, and it is the author's own error. In first stating relation (b) I wrote $w = 3v \cdot e$. The two values — 540.0 from the levels, 1215.0 from the mis-stated law — appeared on the same printed line.

The check caught the check. $\mathcal{D} \geq 1$ does not require the auditor to be right.

Worked case, in an apparatus. Antiprotonic helium, cell (35,33):

route	value
measured directly	804,633,059.0 $\pm$ 8.2 MHz
two-photon minus a different single-photon	804,633,057.8 $\pm$ 10.6 MHz

Agreement at 0.09 σ , with no shared measurement. One laser resonance against a two-photon resonance minus an unrelated one. That is path-disjointness realised in hardware.

And a constraint on where $\mathcal{D}_{\text{def}}$ can come from. Λ 's constraint graph is a tree, so there is exactly one path between any two coordinates. The constraints supply no redundancy. $\mathcal{D}_{\text{def}}$ must be built from derived quantities, deliberately.

16.4.1 And the counting argument has a 1960 precedent

This book did not discover that a fit with as many parameters as observations cannot be checked. Edlén states it, and uses it.

“Because of the approximate nature of the theoretical formulae here discussed nothing would be gained by trying to derive the parameters by a least-squares solution from all the observed levels. ... As the number of parameters introduced is equal to the number of levels no check is possible, and, of course, the results cannot be displayed in a two-dimensional diagram.” — B. Edlén, *Handbuch der Physik XXVII*, footnote to §17 (manuscript 1960)

That is $\mathcal{D} = 0$, in the special case $\dim q = \dim p$. The counting argument, the conclusion drawn from it, and the use made of it — declining a fit that would have looked more rigorous — are all identical to this chapter’s.

What §16.1 adds is the general form, $\mathcal{D} \geq \dim q - \text{rank } \partial\Phi/\partial p$, and the observation that the rank rather than the dimension is what binds. The principle is Edlén’s.

16.5 D3 — totality catches a missing value

Theorem. If Λ is closed, $\chi_\Lambda : \prod A_i \rightarrow \{0,1\}$ is total: membership is decided for every ambient point by a finite conjunction of decidable comparisons. Proof. Each bound compares two elements of a chain. Λ is a finite intersection of such conditions. ■

Verified on every ambient point — all 6,912 of them, $\chi \uparrow \{0,1\}$ everywhere, nothing undecided, and χ agreeing with membership on every cell. An earlier pass drew 30,000 uniform samples from a box of 6,912, which is more draws than cells and still not a proof of totality; the exhaustive test costs less than the sampling did. Register 248.

Worked case. A species was misidentified — Na II read as neutral neon, and the wrong ionisation limit applied. Every channel had $T = I - E \leq 0$ and **every one was refused at entry**.

Totality does not detect that error. It forbids it. The computation never ran.

Worked case, the other way. A guard in the author’s code returned `None` when an alphabet exceeded a cap, and the caller coerced `None` to `False`. That produced “3 reorderable of 85 trials,” which produced “almost never reorderable,” which produced a false dismissal of an entire line of evidence.

One silent discard propagated through three conclusions, none of which looked wrong, because the arithmetic was consistent throughout.

$\mathcal{D} \geq 1$ cannot catch that. Every relation checks out on data that was never measured. Only totality closes the hole, because undetermined is not in the codomain.

16.5.1 And totality extends beyond the data

χ is usually read as a test on cells. It is a test on whatever the index carries, and §28.8 applies it to a coordinate this book did not originally have: the process by which a choice was made.

Index (step, visible, alternatives, committed) under **visible $\leq$ committed** — that nothing of an answer may be visible before a choice is written down — and χ **refuses** a prediction developed with its answer in view. **36 cells, closed, $E = 0$** , one constraint of §16.4 form.

The mechanism is unchanged. Only the coordinate is new.

That is the general shape of D3, and it is why the protocol of §2.9 — refuse rather than coerce — is stated for quantities and applies equally to decisions.

16.6 The defences are disjoint, and there are six

mechanism	catches	worked case
$\mathcal{R}(\Lambda) = \Lambda$	rules not recoverable from cells	periodic table,
$E = 36$		
$\mathbb{D} \geq 1$	a wrong value in data or derivation	wrong Z; w =
$3v \cdot e$		
χ_{Λ} total	a missing value coerced to a real one	the cap discard

Three was the count while every defence was built from the object. There are six, and the other three reach a coordinate this book does not carry: what a term refers to.

mechanism	catches	runnable by the book
$\mathcal{R}(\Lambda) = \Lambda$	rules not recoverable from cells	yes
$\mathbb{D} \geq 1$	a wrong value in data or derivation	yes
χ_{Λ} total	a missing value coerced to a real one	yes
§28.8's index	a choice that depended on its answer	yes
$\mathbb{D}_{\text{ref}}$	**a term meaning one thing here and another in the theorem it was borrowed from no – the verdict must be read**	
$\mathbb{D}_{\text{dict}}$	drift away from a referent already established	yes, given the referent
$\mathbb{D}_{\text{gro}}$	a term with no referent at	
all		yes, once the term is placed

Why the first four cannot reach the last three. All four compare the book against the book. A conflation is not a disagreement between two places in this book; it is an *agreement* between the book and itself about a word doing different work elsewhere. Twenty-one structural audits passed on a companion document carrying five of them.

$\mathbb{D}_{\text{ref}}$ and $\mathbb{D}_{\text{dict}}$ are not the same mechanism at two strengths. Establishing a referent requires reading someone else's theorem and is not runnable. *Maintaining* one is mechanical: once a coordinate carries a stated referent, checking that every later use is consistent with it is a computation. $\mathbb{D}_{\text{ref}}$ is paid once per coordinate; $\mathbb{D}_{\text{dict}}$ runs forever after. The tower's 2K is the worked case — it means $J_c + \ell_{\text{outer}}$ in jK, $L_{\text{total}} + S_{\text{core}}$ in LK, and jj has no K at all, and Racah's recoupling coefficients translate between them exactly.

$\mathbb{D}_{\text{gro}}$ is the one with no instance anywhere until now. q , this book's transfer coordinate, has no referent in the spectroscopic literature — not conflated, since there is nothing to conflate with. $\mathbb{D}_{\text{ref}}$ cannot see it: if no theorem names the term, the pair never enters the table.

And the three repairs are already named, in the companion paper, without being named as defences: SPLIT for a conflation, DICTIONARY for one meaning in two bases, TRANSLATOR for a referent that must be manufactured. Register 403.

16.6.1 The vocabulary of reference, as an index

The three new mechanisms need a vocabulary this book does not have: its back-matter Index carries fifty-one terms and **not one of them says what a term points at**. Defined here on the same discipline as every other index in this book — ordered coordinates, one categorical fibre, constraints of §16.4 form.

coordinate	values
access	blocked < secondary < abstract < full text
referent	none < named < named with its stated hypothesis
checked	not checked < against one theorem < against every theorem naming it
verdict	undetermined < conflated < partial < match

fibred by **kind** — coordinate · equation · source · result. *Repair* — none, dictionary, split, translator — is **not a coordinate**: it is a function of verdict and referent, and §17.2's Theorem 17.1 forbids one — adjunction never repairs. It is read off.

38 cells, ambient box 144, density 26.4%, $E = 0$ — the densest object in this book after the periodic table. **And it is not a tree**: access bounds *referent at its top value* and bounds *checked* directly, closing the cycle access–referent–checked. Width 2, so Freuder cannot be inherited and the zero above is a computation rather than a certificate — §12.11.0.4's Λ_9 arriving in the vocabulary rather than in the lattice.

Applied to twenty-eight load-bearing terms of this book and its companion: thirteen cells occupied of thirty-eight, $E = 11$. One term with no referent, eight named and never checked, nine checked and conflated, two partial, eight matching, four capped by a blocked source. **Sixty-eight per cent have been checked at all.**

One constraint was withdrawn during construction and the withdrawal is the finding. The index first required a named referent to have a reachable source, and refused four cells the book occupies — Edlén 1964, Ritz 1908, Paschen & Götze 1922, Dunz 1911. §19.5 already says otherwise: *a blocked $p = 1$ cell is a stated gap, not an unexplained absence*. Naming a referent and reading one are different acts. Totality forbade rather than detected, exactly as §16.5 says it does, and what it forbade located the wrong constraint. Register 404.

16.6.2 And one coordinate sits below this book's own floor

§16.1 gives $\mathcal{D} \geq 1$ as the condition under which an index can check itself, and every defence in this book rests on it. **The reference index has one coordinate for which $\mathcal{D} = 0$, permanently.**

Three of its four coordinates are internal: *access*, *referent* and *checked* are all recordable from this book's own pages. **Verdict is not.** It requires reading someone else's theorem, and the index that supplies it is the bibliography — for which §29.1 already proves $\mathcal{D}(\text{novelty}) = 0$, necessarily and permanently, because search is one route and can never be two.

$$\mathcal{D}(\text{bibliography}) = 0 \implies \mathcal{D}(\text{verdict}) = 0, \text{ inherited.}$$

So the reference debt is not unpayable because sources are blocked. Blocked access caps four of twenty-eight terms at a lower level. The undefendability of *verdict* caps all twenty-eight, and it is the first coordinate anywhere in this book below the $\mathcal{D} \geq 1$ floor the rest of Chapter 16 stands on. Register 405.

No mechanism catches another's class, and this work supplied an instance of each.

16.7 Two kinds of absence, and one kind of lie

An index can be wrong in two ways, and they are not alike. One it can detect by itself. The other it cannot detect at all.

16.7.1 A missing cell — and the index can tell whether the gap is real

$E(X) > 0$ says the structure admits cells the collection does not hold. It does not say why, and there are two reasons:

the gap is an artefact of the layout — a property of how the index was drawn the gap is a fact about the world — nothing occupies that cell because nothing can

And the index distinguishes them, by asking whether E is removable.

the periodic table	E
18-column	36
32-column	106
Janet left-step	0

The same 118 elements. Order the blocks f, d, p, s and define the period by $n + \ell$, and the defect vanishes. So the 36 belong to the drawing.

Now the same test on a catalogue where the absences are physical. Index the mean-motion resonances by (p, q) , reduced, with the order $p - q$ derived rather than made an axis, and give $\mathcal{R}$ only the resonances that show structure in the asteroid belt:

the belt	E
(p, q)	2
(order, p)	1
(p, order)	1

E is irreducible, and $\mathcal{R}$ refuses all six absences outright — 0 of 6 admitted, against 36 of 36 admitted for the periodic table.

And the physics confirms what the index inferred without being told it. Every resonance of order ≤ 7 in the belt is occupied; every one of order ≥ 8 is empty. Thirteen of thirteen above the cut, zero

of six below. Resonance strength enters at $e^{|p-q|}$, so the absences are exponentially weak and genuinely unoccupied.

A defect of the index vanishes under some re-coordination. A feature of the world survives every one. The test is computable from the cells alone, and §6.2's Janet table is that test — run once, on one example, and never named as one.

16.7.2 A complete index of something that is not there

The second failure is different in kind, and no re-coordination touches it.

Insert one impossible cell — **one electron with spin multiplicity 2, which no atom has** — and let the closure supply everything that follows. The result:

	$\wedge$	the fab-
rication		
cells	976	1,051
closed under join and meet	true	true
$E(X)$	0	0
fixed point of $\mathcal{R}$	true	true

And E stays 0 for both under every re-coordination tested. The test of §16.7.1 has nothing to work on:

that test finds missing cells and asks whether the gap is real. A coherent fabrication has no missing cells. It is full.

The fabrication contains 75 configurations no atom can have, and the structure records nothing about it. Every law in this book certifies it.

16.7.3 The plain version

A defective index is a map with a street missing. You catch it by redrawing — the street reappears in another projection. A coherent fabrication is a complete, consistent map of a city that does not exist. Every street connects, every distance adds up. Redrawing changes nothing, because nothing is missing.

The only thing that distinguishes a map of a real city from a perfect map of an imaginary one is going there. In this book, going there means measuring a level.

16.7.4 Which is what D_{phys} is for, and only that

§16.3’s five recorded failures are all of the second kind. Not one is a missing cell.

failure	what it does
a wrong nuclear charge	every derived value shifts together ; the index stays consistent
a wrong ionisation limit	every term value shifts together
a mislabelled parent	the whole channel moves as one

Each is a complete index of something that is not there, and each was caught by a number the index could not supply — a quantum defect of 4.9, a term value below zero.

Self-consistency detects an isolated error and certifies a systematic one. That is the whole case for requiring a second route that reaches outside, and it is narrower and sharper than “an index cannot check itself.”

16.8 What self-reference does and does not defend

$\mathcal{R}(\Lambda) = \Lambda$ is not a curiosity. It is the guarantee that the index’s answers and its contents are the same object, so it can never be short of an answer and never give one that contradicts itself.

16.8.1 A complete index answers every question about its contents

Asked whether it admits a fabricated cell — one electron with spin multiplicity 2, which no atom has — Λ **answers from its own content: $\varphi(2S | k) = 1$ at $k = 1$, the cell asks for 2, refused.**

200 of 200 fabrications refused when Λ is asked.

16.8.2 What it cannot defend against is being replaced

Rebuild the index around the fabricated cell and the closure grows to 1,051, is a fixed point, closed under join and meet, has $E = 0$, satisfies every law in this book — and contains 75 physically impossible configurations.

200 of 200 admitted when the index is rebuilt around them.

The index never believed the lie. It was never consulted. Replacement is not a question put to an index; **it is an act performed on it by whoever holds it.**

operation	outcome
ask Λ	refused, 200/200
rebuild around y	admitted, 200/200

16.8.3 Which is what §2.13 already says

Commit before you look. Fix the standard, then admit the data. E1, E2 and E3 say the same for cells, axes and constraints:

an extension must be checked against the index, not absorbed into it and re-derived.

Ask the index, then decide whether to extend it. Never extend first and ask afterwards.

16.8.4 And the cost of a fabrication is computable

Insertion is never free. Adding one cell forces the closure to admit a median of 340 more — minimum 59 across 220 trials, zero exceptions.

$A(y) = N[\varphi(\Lambda \cup \{y\})] - N[\varphi(\Lambda)] - 1$, the lattice's own count evaluated on two bound systems. Exact on all cells tested.

Substitution gains nothing. Deletions are always repaired, so $C(\Lambda - x + y) = C(\Lambda + y)$ identically — **0 of 300 random swaps and 0 of 1,600 targeted swaps evaded detection.**

attack	outcome
delete	repaired, always
insert one	amplifies, always — minimum 59
substitute	reduces to insertion; no gain
insert the whole amplified closure	undetectable

V prices certainty. A prices deceit. Both are the lattice counting itself, and both have a floor.

And the amplification is governed by the fibre the widened bound admits, truncated by any rival bound on the same variable — which is why $g \leq 4f+2$, the only constraint sharing its variable with a stricter one, is an order of magnitude cheaper to violate than any other. **Remove the rival and it jumps 397% into the normal band.**

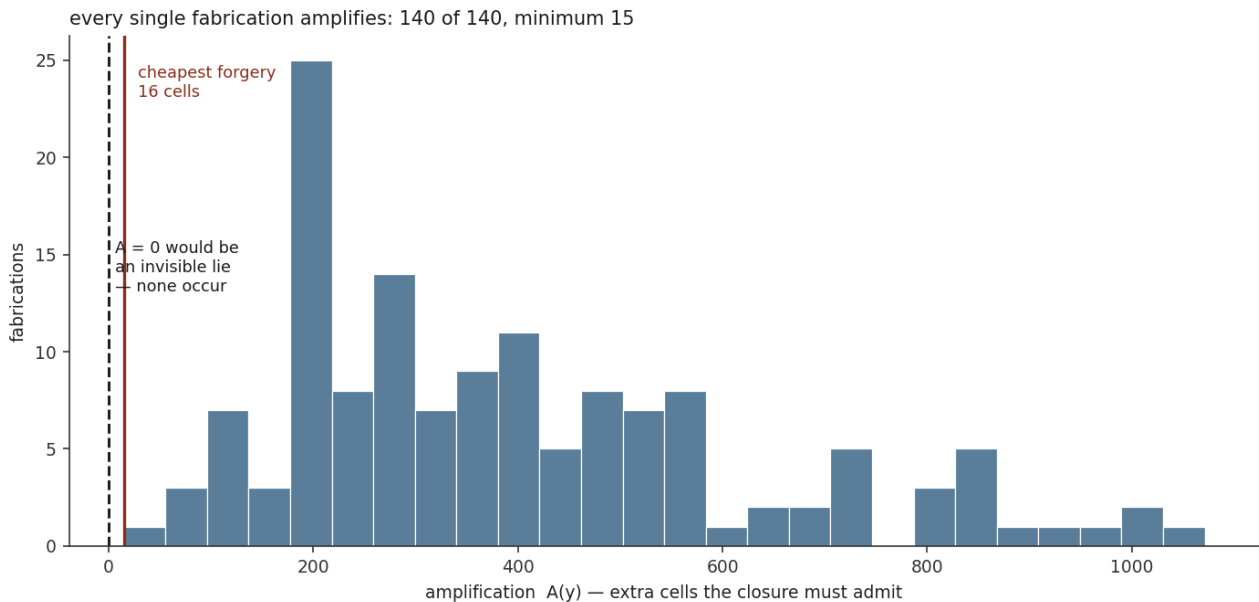


Figure 16.2. The cost of a single fabrication, over 140 random insertions. Every one forces the closure to admit further cells — median 352, minimum 15 — and none is free. The dashed line at zero marks where an invisible lie would sit; no insertion reaches it. A forger must therefore supply the whole amplified set, and its size is computable from the index before any data arrives.

And the general form is the tree, not the constraint. Four statistical measures of a constraint — leaf position, load, marginal contribution, activity — all failed to predict A . **A measure of the tree succeeds:**

$A \approx \text{fibre} \times (0.972 - 0.154 \cdot \text{descendants of the bounded variable})$ $\text{corr}(\text{descendants}, A/\text{fibre}) = -0.942$, median error 7%, worst 24%.

constraint	descendants	A/fibre
$2S \leq k, g \leq q, g \leq 4f+2$	0	0.96–0.99
$q \leq k, f \leq e-1$	1	0.66–0.95
$k \leq 4\ell+2$	3	0.48
$\ell \leq n-1$	4	0.39

A widened bound on a variable with descendants forces theirs to widen too, so the fibre undercounts in proportion to how much of the tree sits downstream. **Leaves are exact; the root is off by a factor of 2.5.**

16.8.5 Nothing can be removed, and the floor is derived rather than measured

§16.8.4 prices insertion. Removal has a floor, and it is not zero anywhere.

Pushback. For a cell x , count the pairs of *other* cells that produce it — $J(x)$ joining to x , $M(x)$ meeting at x . Remove x and that is exactly how many pairs of survivors fail. Over all 475,800 pairs of Λ_8 :

minimum over all 976 cells	16
median	504
maximum	5,091
mean	739
cells removable with no failing pair	0

No cell of Λ_8 can leave without the remainder contradicting the removal, and the weakest is implied back sixteen ways.

The floor is not a measurement. A cell with pushback zero would have to be join-prime *and* meet-prime — irreducible on both sides. Λ_8 has eighteen of each and **their intersection is empty**, at every cap tested. So zero is unreachable by construction, not by luck.

And the minimum that *can* leave follows from the same fact.

$X \setminus [a, b]$ is a sublattice if and only if a is join-prime and b is meet-prime.

Proof. Let $R = [a, b]$ and suppose $z \in R$ with $z = x \vee y$ and $x, y \notin R$. Since $x, y \leq z \leq b$, being outside R means $x \not\geq a$ and $y \not\geq a$, while $a \leq x \vee y$. In a distributive lattice every join-irreducible is join-prime, so $a \leq x \vee y$ forces $a \leq x$ or $a \leq y$ — contradiction. Conversely if a is join-reducible, $a = u \vee v$ with $u, v < a$, both under b and neither above a , so $u, v \notin R$ while $u \vee v = a \in R$. Dually for meets and b . ■

Computed at the book's caps: **$\text{step}(\Lambda_8) = 4$** , the interval running $(2,1,3,3,2,1,3,0)$ to $(3,1,3,3,3,1,3,0)$ — a box free in exactly two coordinates, **n and e , the two shell coordinates and the two ends of the caterpillar**. Removing it leaves 972 cells with **zero join or meet failures across all 471,906 surviving pairs**.

The pushback floor of 16 and the step of 4 are one fact seen twice. Both say: there is no cell whose removal the rest of the index does not contradict, because there is no cell irreducible on both sides. Register 349.

16.8.6 Two closure operators, and the book names one

$\mathcal{R}$ recovers 16 binding coordinate pairs, not 7. Its transitive reduction returns the seven defining edges exactly — so the tree is derivable from the cells and nothing is lost.

But the two systems, identical on Λ , differ on $\Lambda * \{y\}$:

	A full	A tree-only
$\ell \leq n-1$	119	307
$k \leq 4\ell+2$	269	563
$2S \leq k$	74	74

The two that agree bound leaves — no descendants, no transitive propagation. $\mathcal{R}$ is the full pairwise operator, because for the calendar and the periodic table no constraint graph is given and the tree operator is not definable at all. §11's nested sum counts the tree system; they agree on Λ and nowhere else.

17. Extension — cells, axes, constraints

| A closed index determines what may be added to it.

Three operations, and only three: change the cells, change the axes, change the constraints.

17.1 E1 — which cells may be added

| $\Lambda \cup \{x\}$ is closed iff for every $y \in \Lambda$, both $x \vee y$ and $x \wedge y$ lie in $\Lambda \cup \{x\}$

100% over 288 tests. A cell may be added exactly when it sits in a hole whose joins and meets fall back into the lattice.

17.2 E2 — which axes may be adjoined

| $\Lambda \times h$ is closed iff h is a lattice homomorphism

100% over 424 tests. Projections, maxima, minima and constants qualify; sums, products and differences do not.

And a theorem in the other direction:

| Theorem 17.1 (adjunction never repairs). For any $h : S \rightarrow H$, with $S' = \{(x, h(x))\}$: S' closed $\Rightarrow S$ closed. Proof. For $x, y \in S$, the join of $(x, h(x))$ and $(y, h(y))$ is $(x \vee y, h(x) \vee h(y))$. If S' is closed this lies in S' , so $x \vee y \in S$. Meets likewise. ■ Contrapositive: a non-closed set cannot be repaired by adjoining any function of its coordinates. Tested on seven functions including the identity and a constant; none repairs.

Three repair routes remain, and only three: **enlarge** (add cells), **restrict** (remove cells), **reorder** (change neither). The calendar of Chapter 6 is repaired by reordering, at a cost of usability.

17.3 E3 — which constraints may be imposed

This one took three wrong statements to reach, which is recorded in Chapter 28. The criterion is:

And for a constraint on a DIFFERENCE the criterion is convexity, which is proved here.

| **The interval property.** For $\delta = f - \ell$ on a sublattice of a product of chains and any cells a, b : $\min(\delta a, \delta b) \leq \delta(a \vee b) \leq \max(\delta a, \delta b)$, and the same for $a \wedge b$.

Proof. $\delta(a \vee b) = \max(f_a, f_b) - \max(\ell_a, \ell_b)$. Take $f_a \geq f_b$. If $\ell_a \geq \ell_b$ then $\delta(a \vee b) = \delta a$. Otherwise $\delta(a \vee b) = f_a - \ell_b \geq f_b - \ell_b = \delta b$ and $\delta(a \vee b) \leq f_a - \ell_a = \delta a$. Either way it lies between. The meet is dual. ■ **Verified with zero violations on all 1,367,031 pairs of Λ_9** , for $\delta = f - \ell$ and for $\sigma = 2S' - 2S$.

| **Criterion.** $\delta^{-1}(T)$ is a sublattice of Λ iff $T \cap \text{range}(\delta)$ is convex in $\text{range}(\delta)$.

How much of the family this supplies, measured. Every convex preimage is closed — the criterion is sound in the direction it is used. **It is nowhere near complete:** on the three ambients of §14.5 the convex preimages are **8.2%, 10.3% and 0.8%** of all closed subsets. So the criterion decides the constraints this book actually imposes and does not enumerate the closed sets. **A sufficient condition tested for necessity and found to be a small part of what it describes**, which is what §4.6 asks of any criterion before it is relied on.

Scope: Λ must be closed. The criterion is a statement about subsets of a CLOSED ambient, and it fails where the ambient is not one.

Tested on the periodic table, which is the case that shows it. The eighteen-column table is **not a sublattice** — its own joins and meets leave it — and there the criterion misses and a band fails at $E = 5$. Janet’s table is a sublattice, and there the criterion is 4 of 4 and the same band closes at $E = 0$. The condition is not decoration: **the one object in this book whose answer is independently known is also the one that violates the criterion’s hypothesis**, and it does so because $E(\text{table}) = 36$ is exactly the statement that it is not closed. Register 449.

Verified at two cap settings, 36 of 36, including every case where the wider values become attainable and a set convex only by accident of the range flips to failing.

A difference is not a lattice homomorphism — §18.2 refuses $v = e - \delta$ as an axis for exactly that reason — **but it is an interval map**, which is weaker than a homomorphism and stronger than nothing: the image of a join is not determined, it is bracketed. **And convexity is what closure means for a quantity that is only bracketed.** A convex set is one a bracket cannot escape.

This gives §12.11.2’s second excluded form a criterion it did not have. A congruence is excluded exactly when the set it defines has a hole in the attained range — and the hole’s POSITION is what matters, not the modulus, because a reparametrisation that moves the hole to an endpoint restores closure. Seniority parity at axis 10, the conjugation ceiling at axis 12, and the electromagnetic parity rule are three instances of one criterion rather than three separate refusals. Register 440.

$\{ h \leq q \}$ is closed $\iff h(x \vee y) \leq q$ for all x, y with $h(x), h(y) \leq q$

100% over 2,513 tests, nine functions.

The essential feature is that it is q -dependent. Admissibility belongs to the *pair* (h, q) , not to h alone — which is why $a + b \leq 3$ can be closed where $a + b \leq 4$ is not, and why every q -free criterion attempted before this one failed.

Meet never breaks it. For monotone h with $h(x), h(y) \leq q$, $x \wedge y \leq x$ gives $h(x \wedge y) \leq q$ at once. **Only the join can break it**, and it does so when the join raises h above both arguments.

Sums and products fail because a join raises every coordinate at once, and anything accumulating across coordinates accumulates the raises.

Λ ’s seven constraints are all safe by inspection: each reads one coordinate, or a minimum of two. **None sums.** That is why Λ is closed and the periodic table is not — and it is checkable without touching a cell.

17.4 A worked extension, and what it costs

The clearest demonstration of E3 in this work is an attempt to widen Λ that failed, and the repair that succeeded at a stated price.

The problem Λ_8 carries one target triple (e, f, g) . Configurations with two occupied non-source subshells fall outside it — an 86% coverage ceiling. The natural repair is a second triple.

Ordering the targets is free The canonicalisation $e_1 \leq e_2$ is admissible: φ is the identity, non-decreasing in a single coordinate. Zero join and meet failures.

Conservation is not The physical requirement $g_1 + g_2 \leq q$ is a bound on a sum, hence outside §16.4 — and it breaks closure. 89,864 join failures in 979,300 sampled pairs, meets unaffected.

The counterexample is two lines. $a = (g_1=1, g_2=0, q=1)$ and $b = (g_1=0, g_2=1, q=1)$ each conserve. **Their join transfers two electrons having removed one.**

Dropping conservation restores closure — 45,690 cells, zero failures — and produces a lattice describing configurations that cannot exist. **That is the choice in its bluntest form: a closed index of impossible states, or an open index of real ones.**

The repair: make the sum a coordinate Conservation is a bound on a sum, which §7.1 excludes. So make the sum a coordinate. Replace (g_1, g_2) by (g_1, G) with $G = g_1 + g_2$:

| $g_1 \leq G$ (non-decreasing in G) and $G \leq q$ (non-decreasing in q)

Both of §16.4 form, with $g_2 = G - g_1$ recovered as a derived quantity and **conservation holding identically by construction.**

Tested exhaustively: 1,040 cells, all 540,280 pairs, zero join and zero meet failures.

And the price The map $(g_1, g_2) \mapsto (g_1, g_1+g_2)$ is a bijection onto an admissible lattice and order-preserving in one direction. It is not an order isomorphism. It makes comparable some pairs that were incomparable: (0,1) and (1,0) are incomparable as transfer counts and become ordered in the new coordinates. Fifteen of forty ordered pairs are new.

| Coverage and closure are available together. Coverage, closure, and the original order are not.

This is P16 — treat bounds as coordinate values — operating as a repair procedure, and it is the general lesson of E3 stated constructively: a sum bound cannot be imposed, but a sum can be **indexed**, and the difference between those two moves is the order it costs you.

17.5 E is derived

mechanism	fixed by
E1 addable cells	S3, the recovered bounds
E2 adjoinable axes	D3, totality — h must create no third state
E3 imposable constraints	S2, the recovered order

Every extension question reduces to S and D. E carries no independent content — exactly as the cost surface of Chapter 23 carries none beyond v.

18. What the law forbids

The results of this chapter are the strongest in the book, and every one of them is negative.

The strongest arrived last, from the tower: **exact vector coupling is forbidden in principle, not unachieved.** The triangle's two inequalities are one sum and one difference — precisely the two forms this chapter excludes — so every coupling coordinate, at every height of the tower, enters at envelope precision, forever. §12.11.2 carries the computation; the first failing meet manufactures angular momentum from nothing, which is the physics stating the theorem's premise back.

18.1 An index carries what its coordinates carry, and no more

Theorem 18.1. On a product order, join, meet and comparability are definable from the coordinates: $(a \vee b)_i = \max(a_i, b_i)$, $(a \wedge b)_i = \min(a_i, b_i)$, and $a \leq b \iff \bigwedge_i (a_i \leq b_i)$.

Every lattice-theoretic quantity is therefore a deterministic function of the coordinates, and the σ -algebra it generates is contained in theirs.

18.1.1 And the product structure is not what does the work

A reader may suspect the result is an artefact of writing join as max. It is not.

Theorem 18.2. Let an order be defined as any predicate on coordinate tuples. Then the σ -algebra generated by its join, meet and comparability is contained in the σ -algebra generated by the coordinates. Proof. The relation $a \leq b$ is by hypothesis a function of the coordinates of a and b . The set of upper bounds of a pair is computed from that relation, and the least upper bound is selected from that set. Each step is a function of the coordinates, so the join is one — whatever formula it takes. Meets likewise. ■

Verified on a genuinely non-componentwise example from this work. The conserving poset of §18.1.1 has 28 of its 511 joins differing from componentwise maximum, because q must rise to admit both transfers: $(1,0,1) \vee (1,1,0) = (2,1,1)$, where componentwise maximum would give the inadmissible $(1,1,1)$.

The join is nonetheless a deterministic function of the two cells' coordinates, as the theorem requires.

No ordinal structure over these coordinates carries what they do not — not a tree, not a non-componentwise lattice, not the conserving poset.

The escape is not a cleverer order but a further measurement. An order not determined by the coordinates requires information beyond them, and that information is by definition **a new coordinate**.

18.1.2 Two consequences that run through the whole book

It is indifferent to dimension. Λ_{11} is as constrained as Λ_8 . Adding axes does not loosen the bound.

It is indifferent to faithfulness. Perfecting the index moves nothing, because there is nothing to add. Part II could be improved indefinitely and Part V would not gain a single cell.

18.1.3 What the theorem does not establish

Theorems 11.1 and 11.2 bound information, not estimation. A deterministic function of the inputs can still improve a finite-sample predictor — that is what feature engineering is, and it is routine.

So the theorems forbid the lattice from knowing anything the coordinates do not. They do not by themselves forbid a lattice quantity from being a better-conditioned regressor or a lower-variance summary. That is a narrower claim than the theorem's reach suggests, and it is stated narrowly.

18.2 v is inadmissible as an axis

$v = e - \delta$ is a difference, and differences are non-monotone in the cell order. 86 violations at the caps tested.

So the quantity on which the whole of Part V depends **cannot be a coordinate of Λ** . It lives over the lattice, not in it. Chapter 12's "over, not graded by" is the same fact stated geometrically.

18.3 Depth is not a coordinate function

The number of measured members in a channel is not a function of the coordinates at all. It is a property of the literature, and Chapter 19 makes that precise.

18.4 Closure is not locally determined

This is the deepest limitation, and it was found by trying to evade it.

| Closure is not implied by every proper projection being a fixed point of $\mathcal{R}$.

Explicit counterexample at $d = 3$, $c = 2$, six cells:

| $S = \{(0,0,0), (0,0,1), (0,1,0), (0,1,1), (1,0,1), (1,1,0)\}$ Every proper projection is a $\mathcal{R}$ -fixed point, yet $(1,0,1) \vee (1,1,0) = (1,1,1) \notin S$

And the counterexamples are common — roughly **one in five** sets passing all proper-projection tests fails closure.

The mechanism. The two cells agree on axis 0 and disagree on *both* others. The join disagrees in a **combination**, and no lower-dimensional projection can see a combination it does not contain.

| Closure is a d-dimensional condition. No projection of it will serve as a criterion — not per-axis, not per-pair.

Consequence for verification: a wrong choice can pass every local test and fail only at the top. Chapter 30 states what this costs.

18.4.1 The law of realised closure

§18.4 shows closure is not locally determined. This states the stronger form, and it is the second named law in this book.

| **AN OPEN INDEX DOES NOT CLOSE ITSELF, EXCEPT WHERE A CERTIFICATE EXHIBITS AN OPERATION ON ITS COORDINATE SYSTEM REACHING A FIXED POINT OF $\mathcal{R}$ — AND THAT CERTIFICATE IS THE REALISER.**

The law is stated in this form because its first form had no falsifier. As printed until now it read *except where the closure operator is realised in the object's own structure*, and *realised* was never given a test a candidate had to pass. The law was therefore confirmed by naming something in each closed row and could be refuted only by failing to name — which makes the clause a **decision** in §2.18's sense, and §2.18 says nothing tests a decision. Ten confirmations and no stated failure mode is what §4.6 forbids: a check must be shown capable of returning a failure before its passing is evidence.

The certificate form supplies it. The operations are the ones this book already admits — relabel a coordinate's values, re-coordinatise the same content, refine a fibration, drop a coordinate — and register 275's two levers say both directions must be searched, since E falls as a fibration is refined and rises as coordinates are added. **If no certificate exists, nothing realises $\mathcal{R}$ and the open half applies without qualification.**

Exhibited and checked, on every open object this book indexes:

open object	E	the certificate	
after			
the calendar as printed	7	months relabelled in order of length	
0			
the periodic table	36	period = $n + \ell$, block order f, d, p, s	
0			
the audits, four coordinates	16	drop DEPTH, the fourth coordinate	
0			
the measured nuclide chart	9	none – E unmoved at four proton cutoffs	9
the bibliography	6	none – 5,184 relabellings, floor at 2	2

And the realiser is a certificate rather than a search result, which is not a presentational point. Brute force over the calendar needs $12! = 479,001,600$ relabellings and is refused at that size; the certificate is one relabelling and checks over 365 cells. Brute force cannot reach Janet at all, because Janet is a different cell set — 120 against 90 — drawn from the same 118 elements. **§3.7 already states the shape:** a certificate is checkable where a search is not, 362 comparisons against eight and a half million.

The realiser is exhibited, never found. What is falsifiable is the exhibition.

Which identifies this law with §16.7.1. That section asks whether E is removable, to sort a defect into an artefact of the layout and a fact about the world, and gives an outcome without a mechanism. **The certificate is the mechanism:** where one exists the open index is a re-coordination of a closed one and E measures the drawing; where none exists the closed object is $\mathcal{R}$ of the open one and E measures the world. **The two statements are one operator read in two directions**, which is why the thesis of Chapter 15 and this law are halves of the same thing rather than two results. Register 412.

One half is trivial and worth saying anyway: a set with $E(X) > 0$ does not become closed by being looked at longer. Sets do not change. **The content is the exception clause** — that when an index does close, something in it is doing $\mathcal{R}$'s work, and that something can always be pointed at.

Tested against every index this book indexes.

index	E	what realises $\mathcal{R}$, or what is missing	
a box ordering, $l \geq w \geq h$	0	the two bounds *are* monotone single-coordinate	
bounds			
a chessboard	0	a full product; $\mathcal{R}$ is the identity	
$\wedge$	0	the constraint graph is a tree of monotone	
bounds			
the drawn nuclide band	0	the drip lines are the envelopes – §6.2	
the calendar, relabelled	0	the relabelling of §6.3 *creates* the monotone	
bound			
the calendar as printed	7	days(m) is not monotone in m – nothing realises	
it			
the measured nuclide chart	9	pairing and clustering carry no axis	
the periodic table	36	the count of shells is not a coordinate of the	
table			
Appendix D, unfibred	4	no fibration chosen	
the audits, four coordinates	17	fourteen cells with no audit in them	

In every closed row something can be named. In every open row nothing can.

What the law forbids is a habit of reading, not a construction. It forbids treating $E = 0$ as a property discovered in an object, when it is always a property of an object *plus* the structure that realises its closure. §6.2 carried a nuclide row for the whole life of this book that broke the law: it reported the closure of a drawn band as though it were a finding about nuclei, when the drip lines someone drew were doing the closing. Register 298 corrects the row; this law is what the row was violating.

And it explains the two levers of register 275. E falls as a fibration is refined and rises as coordinates are added, because both are acts of realisation — choosing a fibration realises a partition, adding a coordinate realises a distinction. **A zero is worth the size of the box it was computed in** because the box is the realisation. That entry stated the fact; this law states why.

The law distinguishes two ways of realising, and the difference is measurable. Λ closes through its own *order* — the constraint tree of monotone single-variable bounds is $\mathcal{R}$ — and the closure is total, $E(X) = 0$ with nothing left over. Maxwell's equations close through an internal *identity*: pre-1861 Ampère's law contains its own contradiction, since $\nabla \cdot (\nabla \times H) = 0$ forces $\nabla \cdot J = 0$, which fails whenever charge density varies; the repair is compelled, and the identity $\nabla \cdot \nabla \times = 0$ is what realises the closure. Both satisfy *realised in the object's own structure*. **They do not deliver the same thing.** Identity-realisation closed only the divergence part and left any additional curl free — genuine slack, closed later and externally by covariance.

Conjecture (slack = kernel). Identity-realised closure leaves slack equal to the kernel of the realising identity; order-realised closure leaves none.

Two computed points: $\nabla \cdot \nabla \times$ has an enormous kernel — every curl — and Maxwell's slack was exactly a curl; the rectangular cavity's admissible core closes by $\mu = \nu - 1$, whose kernel is trivial, and its slack was zero. **Two points are not a pattern**, and it is stated sharply enough to be attacked; if it holds, the law gains a computable correction term rather than a qualitative clause, which is what §1 requires before anything joins the formalised principles.

And a criterion falls out of the same work, in a form that needed correcting one turn after it was retained. As first stated: *an occupancy coordinate is admissible if and only if its cap is a function of a single other coordinate*. The forward direction holds and the *only if* is false. A cap of two arguments closes meets perfectly well provided it is a **meet-morphism** — bond order capped by $\min(v_A, v_B)$ is two-argument and fails nothing. Computed over six cap forms:

cap morphism	join failures	meet failures	meet-morphism	join-
$\varphi(a) = a$, single, increasing	0	0	yes	yes
$\varphi(a) = 4 - a$, single, decreasing	875	0	no	no
$\varphi = \min(a, b)$	0	0	yes	no
$\varphi = \max(a, b)$	0	260	no	yes
$\varphi = (a > 0) + (b > 0)$	0	16	no	no
$\varphi = a + b$	0	400	no	no

Meets survive exactly when the cap is a meet-morphism; joins survive exactly when it is a join-morphism.

A single-argument monotone increasing cap is automatically both — and the reason is the chain. On a chain the meet of two values is one of them, so any monotone φ commutes with it. That is what the tree buys, stated correctly: not single-argument-ness, but the fact that a single argument ranges over a chain. **Arity was a proxy for the property, and the proxy was mistaken for the property** — which is §28.1 in its purest form, committed by this book one entry after adopting the criterion from outside it. The rectangular cavity fails because a sum of indicators is neither morphism, not because it has three arguments. Register 306.

The criterion, corrected: an occupancy coordinate is admissible if and only if its cap is a morphism for the operation the index must preserve.

And the criterion was incomplete, which a promotion test exposed. §12.11.3.1 draws a rule about where a bound is taken from — the law that generates it, or the extent it is fitted to — and the rule was put up for promotion to a third named law. Tested against all ten indexed objects it half held:

provenance	objects tested	exact?	failures
the index's own extent	3	never	0 – universal
a law independent of the index	5	usually	2

The two failures fail for one reason. The calendar's days(m) is a genuine law and is not monotone in m ; the cavity's $v = [m > 0] + [n > 0] + [l > 0]$ is a genuine law and is not a meet-morphism. **Provenance alone does not decide exactness.** But the other direction has no exception: a bound taken from the extent gives an envelope in every case tested, including the drawn nuclide band, whose drip lines are $\mathcal{R}$ of the measurement, and Λ_{11} , whose $\hat{\varphi}$ is the envelope of its own realised maxima.

A bound gives the exact admissible set only if it is a morphism for the operation preserved and derives from a law independent of the index. A bound taken from the index's own extent gives its envelope and never its law.

Promotion declined, and the criterion strengthened instead. The rule adds a *necessary* condition the criterion lacked — morphism was sufficient for closure and is not sufficient for exactness — so naming it separately would split one statement across two laws. This book has two named laws and the test for a third is whether it says something neither says; this says what §18.4.1 was missing. Register 326.

Applied to three bodies it gives the sharpest negative in this book, and applied to this book it explains a defect already measured. The interaction graph of n bodies is K_n : two bodies give one edge, treewidth 1, a tree — Λ 's own shape, which §7.1 says three central results follow from. Three bodies give a triangle, treewidth 2. And superposition is a **sum**: $\varphi(a,b) = a + b$ breaks 400 meets where $\varphi(a) = a$ breaks none. The three-body problem carries all three of §12.11.2's excluded forms at once — a sum in the superposition, a difference in the relative position, a symmetric function in the mutual interaction — so it is the maximal case of what Chapter 18 forbids, and the book's prediction is envelope precision only. **That is what celestial mechanics does:** Jacobi's zero-velocity surfaces, Hill spheres and KAM tori are all brackets in §22's sense — where a body cannot fail to be, never where it will be. One asymmetry: $\varphi = \min(a,b)$ closes with zero failures, which is the restricted problem, and the restricted problem is the tractable one.

And on the inverse problem — masses inferred from observation — it offers a filter and a limit, both computable. Mass enters gravitational dynamics as a sum, so a single summed observable brackets the total and leaves the parts an envelope. On a twelve-unit grid of ordered triples:

observable	worst-case fibre	mean
total mass alone	21	10.71
total and the largest	6	2.53
three pairwise sums	1	1.00
one mass ratio	78	7.91
two mass ratios	12	1.27

Every ratio class is a ray — shape without scale — so ratios cannot pin a mass however many are taken. Three pairwise sums separate exactly, which is linear algebra and not this book; what the book supplies is the reason it works, that three constraints each bounding one target beat one constraint bounding three, which is §18.4.1's criterion again. In practice: an observable coupling two bodies at a time separates masses where anything summing all three cannot, and that is knowable before any fitting. **The transferable piece is the admissibility filter** — a bracket on a parameter is valid only where the observable is monotone in it, the same test that refused at $Be \rightarrow B$. This is data-analysis discipline, not dynamics, and it is not a solution to anything. Register 317.

And this book is a three-body object. It is simultaneously an object, a law and a procedure, and each is bounded by the other two — Λ is built under the protocols and is what the law generalises; the law is tested against Λ and justifies the audits; the audits check Λ and are the law applied to working. Three nodes, three edges, **treewidth 2, not a tree.**

And it is a four-body object, on evidence gathered after that paragraph was written. REFERENCE — what this book's terms point at — is a fourth part, and the test that it is a *node* rather than a relation is the one the companion paper uses to withdraw a false node of its own: status is one bit per part, and the partition forces one-hot. Over two term lists neither of them chosen for this purpose, **six terms predicate on reference alone** — *referent, ungrounded, stated hypothesis, access grade, vocabulary, dictionary* — **and none is content of any other part.**

The decisive figure is a zero. This book's own back-matter Index carries fifty-one author-chosen terms and **none of them predicates on reference, and none of them is a relation term at all.** A book with no vocabulary for what a coordinate refers to is what a book missing this node looks like from inside.

All six edges are witnessed, each by a term from a list drawn rather than composed: *cap/frame* for object-law, *computable* for object-procedure, *falsification test* for law-procedure, *conflated* for object-reference, *jurisdiction* and *precedent* for law-reference, *term-sharing pair* for procedure-reference. **Four nodes, six edges, complete, width 3** — computed exactly by elimination over all orderings.

And the shortfall is a magnitude rather than a fact. $\mathcal{R}$ is 2-decomposable and verified equal to binary path consistency on every object in this book. By Freuder a width-3 graph needs strong 4-consistency for backtrack-free search; by Dechter, strong (w^*+1) -consistency gives decomposability. **The operator reaches level 2 and the object requires level 4.** $E(\text{this book}) > 0$ was already a theorem about its shape; the fourth node doubles the deficit, and no amount of work on the object, the law or the procedure closes it, because the shortfall is in the consistency level the operator reaches and not in the diligence with which it is applied.

What would refute it, and it is cheap. Exhibit a term this book uses that predicates on reference alone and is already content of another part, and reference collapses to a relation; or show that one of the six edges has no relation term, and the width drops to 2. Both were run and neither fired. Register 406.

Which explains a defect this book measured and could not account for. §2.18.1 found the book's decisions form an acyclic graph in which exactly two have **two parents** — the fibration of Appendix E, and the refusal to promote the six theorems — and recorded that the book stumbled on precisely those two, at registers 219–222 and 232. A two-parent decision is the three-body shape at its smallest: determined by two others and by neither alone. **The book's own Q has never been empty,** and P19 says an index is complete only when it is. This is why. A work whose three parts form a triangle cannot close itself exactly, for the same structural reason three bodies have no closed form, and the honest consequence is that **$E(\text{this book}) > 0$ is a theorem about its shape and not a confession about its state.** Register 316.

Applied outward, it sorts a subject into what can be indexed and what cannot. Chemistry supplied the first two-argument and the first decreasing caps this work had seen, which is why it was chemistry that refuted the arity form:

chemical coordinate	cap	closes
oxidation state	$\leq$ group	yes – single ar-
gument		
bond order	$\leq \min(v_A, v_B)$	yes – min is a
meet-morphism		
coordination number	$\leq f(\text{radius ratio})$	no – decreases,
breaks joins		
lone pairs	$\leq (8 - v)/2$	no – decreases,
breaks joins		

The coordinates that resist indexing are the geometric ones, and they resist for one stated reason: their caps fall. A chemistry index can be built on oxidation state, bond order and group; it cannot be built on shape. That is a sharper statement than it sounds, and it is a consequence of the corrected criterion rather than an observation about chemistry. Register 308. A multi-argument cap is a hyperedge in the constraint graph and it costs meet-closure. Committed before computation and correct in both directions: a rectangular cavity index, whose polarization multiplicity is capped by a three-argument function, fails 768 meets in 15,400 pairs; a spherical one, capped $m' \leq 2\ell' + 2$, fails none in 114,960. **This is why Λ 's constraint graph being a tree is load-bearing rather than decorative** — every one of its seven bounds is single-argument, and §7.1 had said so without saying what it bought.

Its own status, by §29.1: no precedent search has been run on it. Registers 299, 302 and 303.

18.4.2 The n-body configuration space, indexed — and found already in print

§18.4.1 records the verdict on three bodies: nothing here solves anything, and three walls stand. A narrower question was asked instead and it has an answer. **Can the configuration space be indexed and closed?**

It can. Following §17.4's repair literally — a sum cannot be bounded, so index it — take the partial sums S_k of an ordered non-increasing mass vector. The exact set is then the **concave** sequences; the envelope is those whose **Cesàro means fall**, $S_{\{k+1\}} \leq ((k+1)/k) \cdot S_k$, single-parent and monotone at every k . **Treewidth 1 at every n .** Recomputed:

n	N	exact	envelope	E(X)	envelope join/meet	exact join	exact meet
3	12	102	142	40	0 / 0	224	0
3	16	204	294	90	0 / 0	896	0
3	20	358	526	168	0 / 0	2,772	0
4	12	155	341	186	0 / 0	1,073	0
5	12	197	712	515	0 / 0	2,491	0

The envelope is a closed index at every cap. The exact set is meet-closed at every cap and joins fail, and the one-sidedness is proved rather than measured:

Proposition. The exact set is closed under componentwise minimum. *Proof.* Let $f = \min(g, h)$ pointwise with g, h concave, and suppose $f(k) = g(k)$. Then $f(k-1) \leq g(k-1)$ and $f(k+1) \leq g(k+1)$, so $f(k-1) + f(k+1) \leq g(k-1) + g(k+1) \leq 2g(k) = 2f(k)$. Hence f is concave. ■ The same argument on max reverses at the first step and no repair is available. **The minimum of concave functions is concave; the maximum is not.**

And $\mathcal{R}$ answers the tightness question against a committed prediction. $\mathcal{R}(\text{exact})$ was predicted to be strictly smaller than the hand-built chain at $n \geq 4$. **It is equal, at every cap tested** — the path of adjacent Cesàro bounds is already transitively closed, so non-adjacent pairs carry nothing new. **Unlike Λ , this object's defining system is its own normal form**, which by register 325 is $E(X) = 0$ read as *the extent is the law for that set*.

And the object is Brylawski's. T. Brylawski, *The lattice of integer partitions*, Discrete Mathematics 6(3), 201–219 (1973): the lattice of partitions of an integer under dominance is isomorphic to an infimum subsemilattice of certain concave nondecreasing tuples. Partial sums are the dominance vector; concave nondecreasing is the exact

set; *infimum subsemilattice* is the correct name for what the proposition above proves. **The derivation is correct and it is not first**, and this is register 224's shape exactly — the book's own central theorem turned out to be Baker–Pixley.

What survives is narrow and is stated narrowly: $\mathcal{R}(\text{exact})$ as the tightest pairwise-monotone superset of the dominance vectors, its identity with the Cesàro chain, and **E(X) as a graded measure on the dominance order** — which the searches did not return.

Same gap, second object: the surrounding literature settles membership and this framework measures distance from it. Register 355.

18.5 The pole

The cost of a guarantee, derived in Chapter 23, has a pole at $p = 1$ — the exponent at which an observable is linear.

A straight line has no curvature. There is no interpolation error to price, so a guarantee costs infinitely more than the thing it guarantees.

The method has no domain at $p = 1$. It is the only hard singularity in the structure; every other limit is a threshold with data on both sides.

18.6 E(X) is the prediction budget

The five limitations above add up to something the book can now state in one line, and it answers the question this work began with.

A prediction is a claim that a cell exists which the index does not carry. The only cells an index can propose are those its own structure admits and its extension denies — and that set is exactly $\mathcal{R}(X) \setminus X$.

The number of predictions an index can make is $E(X)$.

18.6.1 What that means for the five indices this book keeps

index	$E(X)$	predictions available	and they are
the periodic table	36	36	all impossible
the calendar	7	7	all impossible
a subnet with a hole	8	8	all impossible
the box ordering, and Λ	0	0	—
the bibliography	6	6	all occupiable — two already named

Every prediction the first three can make is wrong. Period-1 group-2 and its thirty-five kin. The thirty-first of February and its six. Λ can propose nothing at all. **The fifth row is the exception, and it is the only place in this book where a prediction is both available and true.**

The bibliography, indexed on era, retrieval redundancy p , access and depth of entry, holds twenty-two sources over seven cells and admits six it does not carry. Under exhaustive relabelling E falls from six to two and never to zero, so by §16.7.1 it is irreducible in the same sense as the asteroid belt and unlike the periodic table, whose thirty-six vanish under Janet. **The two survivors are a born-digital literature at $p \geq 4$ read only in abstract, and a digitised pre-digital source read in full and entered once.** Both are occupiable and both name a route, which is exactly what the belt's absences do not: resonance strength enters at $e^{\wedge}|p-q|$, so the belt's empty cells are empty because nothing can sit in them.

Closure abolishes prediction. It does not abolish the bibliography, because a bibliography is never closed — $E(\text{search}) > 0$ always, by §29.1, and the cells it admits are documents that exist.

Which gives §16.7.1 a third verdict it did not have. That section sorts a defect into *an artefact of the layout*, removable by re-coordination, and *a fact about the world*, surviving every one. The bibliography is the second kind and its cells are still fillable, so the two verdicts are not exhaustive:

the defect	reaches zero?	its predictions
the periodic table	yes, under Janet	impossible — the layout's fault
the asteroid belt	no, $2 \rightarrow 1$	impossible — nothing can occupy them
the bibliography	no, $6 \rightarrow 2$	possible, and unfilled

Irreducible does not mean unoccupiable. An index can carry a defect no re-coordination removes whose cells are nevertheless real and empty for want of work — and that is the only shape in which this method predicts anything.

18.6.2 And it is forced, not observed

Two results already in this part combine to give it.

$E(X) = 0$ says there is no missing **cell** to propose. **Theorem 18.2** says there is no hidden **quantity** to derive.

Together:

A fully defined index has nothing left to say.

That is the whole answer to whether Λ is predictive. Not that it is a weak predictor. Not that it wants more coordinates. **There is no gap in it for a prediction to occupy.**

18.6.3 Where prediction actually lives

Λ indexes cells. It does not index values.

The Sc VI 6s cell of §25.6 is in Λ — its coordinates are ordinary integers, and the lattice admits it without comment. **Its energy is not**, and no closure property supplies it.

the cell	indexed, $E = 0$, nothing to predict
the value	not indexed, and therefore predictable

δ is precisely the value's departure from what the index determines.

Prediction requires an incompleteness, and a complete index has none.

Which is why the bracket must be handed two measured neighbours before it can say anything at all. It is not extracting a prediction from the lattice. **It is carrying a constraint between two measurements the lattice never had.**

18.6.4 The trade, stated plainly

An index may be complete or it may be predictive. It cannot be both.

Not because completeness is dull, but because a prediction is a proposal about an unlisted cell, and completeness is the property of having no unlisted cells. The periodic table can propose thirty-six things because it is incomplete. **Λ can propose nothing because it is not.**

This is the fifth and largest of the negatives, and everything else in this part is a special case of it.

18.7 Why the negatives are the contribution

Λ is a bijective recoding of existing quantum numbers. It adds no physics. A reader will observe this within a page, and they will be right.

That is the point. A recoding that adds nothing and still exhibits closure, self-reference, self-defence and these five limitations is demonstrating a property of **indexing**, not of atoms. If Λ contained new physics the results would be about atoms and worth much less.

19. Retrieval as a lattice problem

| Data can be retrieved in pieces; the index navigates.

This is usually read as advice. It is the self-defence theorem again, with a different failure mode.

19.1 The formalisation

Let C be the cells wanted and $S = \{S_1 \dots S_k\}$ the sources, each a subset of C , with $A(S_i)$ recording accessibility.

| c is retrievable iff $\exists i : c \in S_i \wedge A(S_i)$ route set $R(c) = \{i : c \in S_i\}$, retrieval redundancy $\rho(c) = |R(c)|$

$\rho = 1$ is fragile. $\rho \geq 2$ survives the loss of any single source.

| $\mathcal{D}_{\text{def}} \geq 1$ — two disjoint derivations survive a computational error $\rho \geq 2$ — two disjoint sources survive an access failure

Same structure. Different failure mode.

19.2 A paywall blocks a source, not a cell

Worked case. The target was a transition frequency in antiprotonic helium.

source	status
<i>Nature</i> 475, 484 (2011)	paywalled
<i>Phys. Rev. Lett.</i> 96, 243401 (2006)	paywalled
arXiv:1304.4330	open — the <i>Nature</i> paper entire
arXiv:1203.5425 (CODATA 2010)	open — Table XII, all fifteen frequencies, both papers
arXiv:1308.1711, an unrelated citing paper	open — corroborates to 0.1 MHz
CERN CDS	open

$\rho = 6$. Two blocked, four open. The cell was retrieved and the two-path check on it closed at 0.09σ .

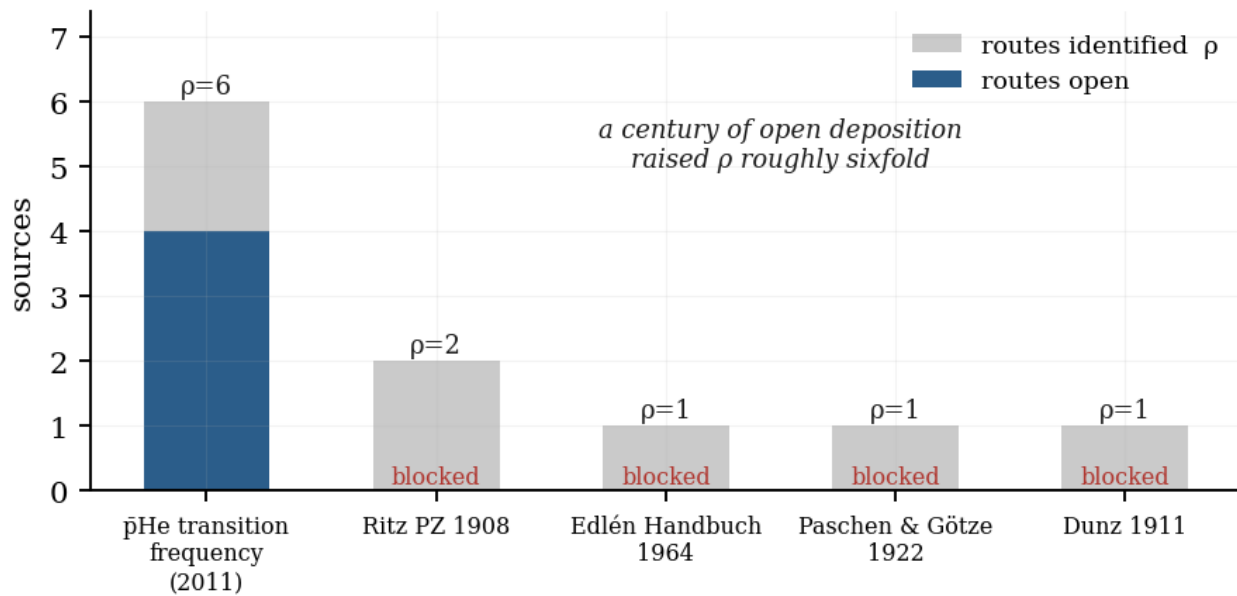


Figure 19.1. Retrieval redundancy for five target cells. The 2011 antiprotonic-helium frequency had six routes, two blocked. The four German documents have one or two apiece, all closed.

19.3 The secondary can be better than the primary

CODATA's Table XII carries both papers with an extra digit the journals did not print, supplied privately by the experimenters to reduce rounding error.

The index does not merely route around an obstruction. It sometimes finds a better cell than the one that was blocked.

19.4 The retrieval graph is self-referencing

Each fetched compilation names its successors. The aluminium compilation's introduction lists its own sister volumes verbatim; the beryllium compilation cites the zinc one; the aluminium reference list supplies nickel and copper.

Three coverage gaps in this work's census were filled from documents already in hand, with no new search.

19.5 Where it fails, and the number that describes it

Four documents in this work were not retrieved: Edlén's Handbuch chapter (1964), Ritz's Physikalische Zeitschrift paper (1908), Paschen & Götze (1922), Dunz (1911). $\rho \leq 2$ for each, every route closed.

Not for lack of effort — nine searches, correct citations, the right platform identified and free. **Pre-digital print has low retrieval redundancy.**

ρ is a property of the literature, not of the searcher. A cell in a 1908 German periodical has $\rho \approx 1$. A cell in a 2011 physics paper has $\rho \approx 6$.

A century of open deposition raised ρ roughly sixfold, and that is exactly what separated the two outcomes.

19.5.1 ρ is a property of (document, route), and this chapter collapsed the second index

§19.5 records four documents as unretrieved at $\rho \leq 2$, **every route closed**, and draws a law: *ρ is a property of the literature, not of the searcher*. **Both are wrong, and the coordinate that shows it is printed in §19.6 below**. Step 2 lists seven route types — primary, preprint, review, compilation, citing paper, deposit, database. **ρ counts them and loses which.**

document	primary	review	citing	compil.	deposit	ρ
Paschen & Götze 1922	0	1	1	.	1	3
Ritz Phys. Zeitschr. 1908	0	.	1	1	1	3
Edlén Handbuch 1964	0	1	1	.	.	2
Dunz 1911	0	0	0	.	0	0

Three of the four are retrievable and §19.5 recorded all four closed. Ritz's p. 521 paper is reprinted in the **Œuvres**, published for the Swiss Physical Society, which is digitised in full on **Gallica** at ark:/12148/bpt6k33824 and on the Internet Archive — and a third party's page mapping confirms the book's own estimate, giving *Über die Spektren der Alkalien*, Phys. Zeitschr. 9 (1908) 244–245 at **Œuvres** 137–140, immediately before the range §29 wants.

Dunz is the only one where the fibre is empty at every point tried — and it is the one whose citation carries no initial and places a series-laws dissertation at Leipzig, outside the Tübingen lineage from which every other early dissertation in this literature comes.

Two of the four have now been tested on the routes §19.5 did not try, and both refute the record. Edlén's review route is *Bengt Edlén's Handbuch der Physik Article — 26 Years Later*, a paper about the chapter's own methods for recognising precise regularities in atomic spectra — **which is the exact question §29.5 asks of that chapter**. Its citing-paper route carries the full reference in Springer chapter bibliographies, in NIST's compendium and in Europhysics News.

The review route pays twice. It opened Paschen & Götze and it opened Edlén. §19.6 lists it third of seven; §19.5 used it for neither.

Paschen & Götze 1922 is retrievable on three routes. A Springer Book Archives chapter DOI — 10.1007/978-3-642-99147-9_1 for §III, _2 for the tables — and a HathiTrust scan from the University of California Libraries. **§19.5 tried the primary route, which is the one column where it is not.**

So the law is refuted by its own instance. The same 1922 monograph gives $\rho \approx 0$ asked as a text string and $\rho \geq 2$ asked as a catalogue entry. **ρ behaved as a property of the query, not of the literature** — and the difference is register 553's: a source has an index, and it is rarely the one with a search box.

And the deeper fault is that a scalar cannot distinguish *closed* from *untried*. Twenty-seven of the forty-nine cells in the table above were never tried, and ρ recorded them at the same value as the empty ones. **Edlén and Ritz are UNKNOWN, not closed.**

This is §21.5's obstruction committed in the chapter about retrieval: a count printed where a coordinate was needed, and the coordinate printed one section later in the procedure that produced the count.

Registers 566–568.

19.5.2 And the book states two incompatible things about Dunz

§19.5 lists Dunz 1911 among four documents **not retrieved**. The References entry reads *Dunz, — Seriengesetze der Linienspektra (Leipzig, 1911)*, and adds **retrieved as a scan, unread**. **One of these is false and the book carries both**. Register 265 records that the Dunz entry was absent from the References until audit 7 caught it, so the reference is the later and hastier text.

The entry also carries an em-dash where an initial belongs — Dunz, — — which is why a title search cannot confirm the volume exists. **A citation missing its author’s initial is not a citation that has been checked**. Register 569.

19.5.3 The route is a fibre, and §19.1 wrote it without using it

§19.1 defines $R(c) = \{ i : c \uparrow S_i \}$. That set is a fibre over the cell: the route varies and the cell does not. $\rho = |R(c)|$ is its cardinality reported as a scalar.

This is why a redirect does not disturb the process. A route failing costs a route and not a cell — which is §19.1’s own $\rho \geq 2$ survives the loss of any single source, stated for the count and true of the structure.

The retrieval index, built over age · language · access and fibred by route, closes everywhere:

route fibre	cells	E	route fibre	cells	E
primary	3	0	compilation	2	0
review	4	0	deposit	3	0
citing	4	0	preprint	1	0
pooled	6	0			

E = 0 fibred and unfibred. The envelopes find nothing to admit because **the route is a free coordinate** — moving along it does not move the cell.

And it explains §19.3. The secondary can beat the primary because **the fibre is over the cell, not over the source’s quality**; nothing requires the primary to be the best point in it. CODATA’s extra digit is not an anomaly but the ordinary case.

A cardinality cannot say which fibre a document was tried in. *Tried once and empty* and *never tried* take the same ρ , and §19.5 drew a law from their union.

Registers 579–581.

19.6 The procedure

1. Enumerate target cells from the index before searching.
2. List route candidates per cell — primary, preprint, review, compilation, citing paper, deposit, > database.
3. Prefer high- ρ cells; they cost less and often carry more.
4. Read every retrieved source for its successors.
5. When a route is blocked, move along the route set; do not re-attempt.
6. Record ρ for cells that fail. A blocked $\rho = 1$ cell is a stated gap, not an unexplained absence. The author violated step 1 badly, on his own problem, after writing this chapter. That is recorded in Chapter 28.

PART IV — THE LANGUAGES

20. The languages, and what each one can close

Fifty sections of this book make a claim about language and no chapter owns one. This chapter owns them, and §21 states what they have in common.

20.1 A language is a coordinate system

Every question in this work is posed in some vocabulary, and the vocabulary decides what can be said without loss. That is P20, and it is promoted here from a principle to the subject, because the results scattered through Parts II, III and V are one result about it.

— The language a question is posed in bounds whether it can close, and at what cost.

Chapter 1's table gives five languages with their closure mechanisms and their measured costs — tree factorisation at 0.00018 s against Gröbner at 0.010 s, a factor of fifty-five. **And one row has no timing, because it has no algorithm.** The documentary language — *has anyone said this before?* — carries no closure mechanism at all, which is why §29's precedent question can be bounded and never settled, and why §29.1's two searches returning nothing is evidence of a kind the other four languages never have to produce.

20.2 The languages this book uses

Six agree on Λ at 976 cells, and ten of their combinations hold — order, geometry, arithmetic, calculus, logic and the constraint language of §16. That agreement is P21, and it is an instrument rather than a criterion: **the principle has located two errors by failing in one language while the others passed.**

The languages are not decoration on one mathematics. Each closes by a different mechanism and each charges a different price, and §27 proves that where they measure the same thing they agree — E in order, the void in geometry, V in calculus, one quantity under three measures.

20.3 And a language can make a law monotone

The clearest case is small. The electromagnetic selection rules include a parity condition, and written as a congruence — $|\Delta\ell| \equiv \gamma \pmod{2}$ — it is one of §12.11.2's three excluded forms and §14.4's operator cannot carry it: **E = 750**. Written in the standard spectroscopic naming, E1, M1, E2, M2, in which parity alternates with the multipole order, **the same law is monotone and E = 0**.

— The notation was not a convenience. It was the coordinate that made the rule expressible.

§17.3's criterion says why in general: a difference is bracketed rather than determined, and convexity is what closure means for a bracketed quantity. **A congruence is excluded exactly when the set it defines has a hole in the attained range, and a language that moves the hole to an endpoint restores closure.**

21. Translation, and what it costs

21.1 One object, two namings, and E is the price

Eight results in this book have the same shape and have never been put beside one another.

one object		first naming	E	second nam-
ing	E			
the 118 elements		period × group	36	n + l × position
(Janet)	0			
the 365 days		months as drawn	7	months by
length	0			
Λ_9 's parity rule		$ \Delta l = 1$, a congruence	750	the convex band
$ \Delta l \leq 1$	0			
the audit set		four coordinates	16	three coordin-
ates	0			
this book's own cells		chapter resolution	578	part resolu-
tion	0			
the same J multiset		jK 199,130 · LK 341,150 · LS 431,050 · jj 206,520 – four		
routes, one atom				
slack		E in order · the void in geometry · V in calculus – one		
quantity				
Λ at 976		six languages agree; ten combinations hold		

A language is a coordinate system. Translation is re-coordinatisation. E is what the translation costs.

21.2 The certificate is a translation, exhibited and checked

§18.4.1's law of realised closure reads, in this vocabulary: **a thing is sayable without loss exactly when a translation into a closed language exists, and the certificate is the translation.** That is why the certificate must be *exhibited* rather than searched for — §3.7 already gives the asymmetry, 362 comparisons against eight and a half million — and why §16.7.1's removability test is the same test: E measures the drawing when a translation exists and the world when none does.

21.3 The three ways a translation fails

§16.6's defences are translation failures, and naming them that way is what separates them:

$\mathcal{D}_{\text{ref}}$	no referent has been established	not runnable – establishing one
requires reading		
$\mathcal{D}_{\text{dict}}$	drift from a referent already established	runnable, given the referent
$\mathcal{D}_{\text{gro}}$	no referent exists at all	invisible to any audit over the
text		

The reference index of §16.6.1 is a translation audit, and its verdict coordinate sits below the $\mathcal{D} \geq 1$ floor the rest of that chapter stands on. q in Λ is the $\mathcal{D}_{\text{gro}}$ case: a term with no referent, carried through thirteen coordinates and never grounded.

21.4 What this does not claim

Every number above is already proved elsewhere in this book. Nothing here is a new result; what is new is that they are one statement. By register 326's precedent that is a case for strengthening an existing statement rather than naming a new law, and no new law is named. **The claim is that the method is the mechanism by which these languages speak to each other** — and the mechanism is $\mathcal{R}$, whose fixed points are the sayable-without-loss and whose defect is the price of saying it otherwise. Registers 451–453.

21.5 The missing ingredient is a translation, not a fact

Three results in the companion work and this one cannot be reproduced, and audit 21 calls all three the same failure: **a result stated from data the book does not print.** They are the same failure, and naming the language makes the repair one line rather than three.

object	what IS printed	what is NOT
the violation index er's EDGE LIST	2,370 cells of a box of 19,440	the exclusion classifi-
the coupling schemes	199,130 · 341,150 · 431,050 · 206,520	the LOOSENESS CON-
the collection	1,442 cells over ~190 channels	WHICH cells are inde-
pendent		

In every case what is printed is a COUNT and what is withheld is a CONSTRAINT. And §14.5.7 shows a constraint need not be printed at all:

$\mathcal{R}$ is determined by ϕ , and ϕ is determined by the SEED — the cells witnessing each envelope step. A seed is a list of cells. A cell is the one thing every one of these papers already knows how to print.

And a seed alone is not enough, because a seed reconstructs the CLOSURE and not the index. Two of these three objects are open — the violation index at $E = 30$, the collection not an index of this kind at all — so the repair needs its second form:

print the seed of $\mathcal{R}(X)$, and the E cells that $\mathcal{R}(X)$ holds and X does not.

That is exact for every index tested — the audit index, the periodic table, the calendar and the parity rule all recover **exactly** — and its cost is **seed + E**:

open index	cells	E	seed of $\mathcal{R}(X)$	printed	compression
the audit index	21	16	7	23	0.9×
the periodic table	90	36	19	55	1.6×
the calendar as printed	365	7	31	38	9.6×
the parity rule	840	750	13	763	1.1×

The repair pays exactly when $E \ll |X|$. The seed term is small at high dimension; **E is the binding term**, and where E is large the index is cheaper to print outright.

Both remaining obstructions qualify and by wide margins. The violation index — 2,370 cells, $E = 30$, nine coordinates — needs a seed near fifteen, so **about forty-five cells for 2,370**. The coupling schemes are **closed at $E = 0$** , so the first form applies and a seed near twenty-five carries all 199,130.

Λ 's seed is **ten cells for 976**. The violation index's is bounded above by 216 from its rung counts alone and, at Λ 's ratio, is likely nearer twenty for 2,370. **The missing ingredient is not information the authors lack. It is the same information in the other language.**

21.5.1 And the three-body count is exactly one

§32.3 records that $\mathcal{R}$ reaches **level 2** — pairwise consistency — while the object, a K_3 of treewidth 2, requires **strong 3-consistency** by Freuder. **The shortfall is exactly one level: pairs, plus one thing that pairs cannot carry.**

This book has exactly one genuinely ternary object, and it is the bracket. $T(n-1)$, $T(n)$, $T(n+1)$ — three consecutive members, and no pair of them determines the third. Which is why independence turned out triadic at §24.9, why surviving tests go as $(1-f)^3$ and not $(1-f)$, and why the bracket is the one place in this work where $\mathcal{R}$ has never been the right operator.

And the four-body reading follows the same arithmetic. Adding *reference* as a fourth node raises the requirement to strong 4-consistency, which is why §18.4.1 records the deficit as **doubling** rather than growing by one: the operator stayed at level 2 while the object moved from 3 to 4. Registers 487–489. *The same shortfall on a classical object — K_3 , treewidth 2, one level short — is §36.3.*

21.5.2 The constraints, indexed — and a 3-body is what a tree forbids

§21.5.1 counts the shortfall in levels of consistency. **Indexing the CONSTRAINTS rather than the cells says the same thing in the objects' own language**, and says it exactly.

Λ 's seven constraints are seven pairs, each with a multiplier and an offset:

$$\ell \leq n - 1 \cdot k \leq 4\ell + 2 \cdot q \leq k \cdot f \leq e - 1 \cdot g \leq 4f + 2 \cdot g \leq q \cdot 2S \leq k$$

Each constraint is pairwise, so three constraints closing on three variables is a 3-body. Counted over every triple:

triples spanning 3 nodes — a true 3-body 0 4 nodes 7 5 nodes 20 6 nodes 8

Λ has no 3-body among its constraints — zero of thirty-five. Its constraint graph is eight nodes and seven edges: a tree, and a tree is exactly a graph with no cycle.

And it is one edge away in exactly seven places.

add $n - k \rightarrow$ a 3-body on $\{n, \ell, k\}$	add $q - f \rightarrow$ on $\{q, g, f\}$
add $\ell - q \rightarrow$ on $\{\ell, k, q\}$	add $q - 2S \rightarrow$ on $\{q, k, 2S\}$
add $\ell - 2S \rightarrow$ on $\{\ell, k, 2S\}$	add $e - g \rightarrow$ on $\{e, f, g\}$
add $k - g \rightarrow$ on $\{k, q, g\}$	

Seven constraints, seven near-misses — the edge count of a tree on eight nodes, with each existing edge sitting in exactly one triangle-completing position Λ declines. Fourteen of the twenty-one unused edges are safe. **And every one of the seven is a bond the physics does not make: n and k are related by no law, ℓ and $2S$ only through k .**

This is why $\mathcal{R}$ works at all, why Freuder's width bound applies, and why §21.5.1's shortfall is one level rather than two.

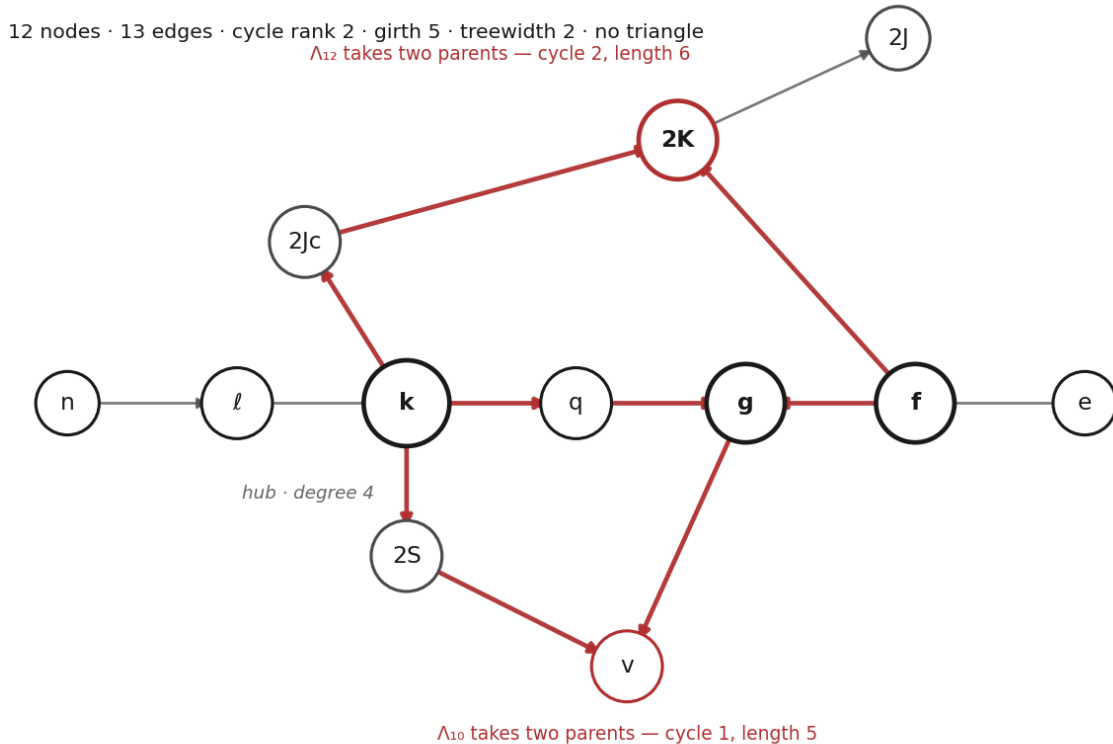


Figure 21.1. Λ_{13} 's constraint graph. Twelve nodes, thirteen edges, **cycle rank 2, girth 5, treewidth 2, and no triangle at any stage**. Black edges are the tree; red edges lie on a cycle. The base Λ_8 is a caterpillar with **k** as its hub at degree 4; **each two-parent axis adds exactly one independent cycle** — Λ_{10} 's **v**, closing a 5-cycle and setting the girth, and Λ_{12} 's **2K**, closing a 6-cycle. The three leaves are the tower's two entries, **n** and **e**, and its exit, **2J**. **Treewidth stays 2 at Λ_{13} despite the cycles**, so Freuder's bound still gives strong 3-consistency against $\mathcal{R}$'s 2 — the shortfall of one level holds for the whole tower and not only for a K_3 .

Two things are owed and stated rather than assumed. The constraint index's own defect is $E = 23$ at 7 cells in a box of 216 — **3.2% density, where §18.4.1's asymmetry says E measures sparsity and not structure** — so the twenty-three absent constraints are not read as predictions here. And the graph-theoretic vocabulary for all of this — **chordal, treewidth, tree decomposition, girth** — is absent from this book's References entirely, where the lattice side is cited: Adams, Dwinger and Schmid give *tight* $\iff$ *the digraph is acyclic*, which is this result seen from the other language. Registers 507–508.

21.5.3 The shape, and the index rebuilt

§21.5.2 gives a cycle count and no shape. Λ_{13} 's **constraint graph, measured:**

nodes · edges	12 · 13	girth	5
cycle rank	2	diameter · radius	6 · 3
treewidth	2	hub	k, degree 4
degree sequence	4, 3, 3, 3, 2, 2, 2, 2, 2, 1, 1, 1		
leaves	2J · e · n — the two entries and the exit		

Treewidth stays 2 at Λ_{13} despite the two cycles, so Freuder's bound still gives strong 3-consistency against $\mathcal{R}$'s 2: §21.5.1's **shortfall of one level holds for the whole tower and not only for a K_3** .

And the index is rebuilt on coordinates that carry the shape — stage, parent count, multiplier — at 38% density where the first attempt sat at 3.2%:

operator	cells	box	E
$\mathcal{R}$, the $(\leq, \leq)$ corner	9	24	15 — it admits the entire box
$\mathcal{R}_4$, four orientations	9	24	4
orientation cost			11

$\mathcal{R}$ **sees nothing here**, because the multiplier and the parent count both peak at stage 0 and an upward-only envelope cannot express $\times 4$ *occurs only at the base*. $\mathcal{R}_4$ **cuts the defect from fifteen to four — the largest orientation cost anywhere in this book, and the first object on which the second operator is the only one that works**.

The four remaining cells are the alternation's complement. Λ_9 two parents · Λ_{10} one · Λ_{11} two · Λ_{12} one — **the stages where the tower could have taken the other kind and did not**.

And the distribution is a shape statement. Only two of thirteen constraints carry a multiplier — $k \leq 4\ell+2$ and $g \leq 4f+2$, both Pauli — and eleven are $\times 1$. Seven take one parent, six take two. **Three collisions**, each correct: the base's three plain one-parent bounds are indistinguishable in these coordinates, and so are the two-parent pairs at Λ_{10} and Λ_{12} , **each of which is one two-sided bound written as two**. Registers 523–524.

21.5.4 Shape decides the seed, and neither extreme is cheapest

§14.5.7 makes the seed a covering problem and §21.5.3 measures the graph. **The two meet here**. Six constraint graphs on six nodes over the same alphabet, differing only in shape:

shape	edges	cells	S	steps	seed	diameter	max degree
path	5	28	30	8	5	2	2
star	5	276	10	6	2	5	
caterpillar	5	65	22	6	3	3	
binary tree	5	100	16	6	4	3	
double star	5	127	16	6	3	4	
forest, two components	4	196	8	5	2	2	

All six seeds exact by branch and bound. Greedy set cover gave the star 7 and was wrong there and nowhere else — §2.24, and register 526.

Both extremes cost more. The balanced trees all seed at six; the path and the star at eight and seven. *The prediction under test was that a star seeds cheapest, one hub cell witnessing many steps at once. It is refuted: the hub does, and the five leaves then need their own witnesses.*

And no simple graph statistic decides it. Diameter runs 5, 2, 3, 4, 3, 2 against seeds 8, 7, 6, 6, 6, 5; maximum degree runs 2, 5, 3, 3, 4, 2. **Cell count does not decide it either** — the path has twenty-eight cells and the worst seed, the star two hundred and seventy-six and a better one.

What does the work is S, the envelope-step count, and on exact seeds the relation is monotone with nothing set aside. **The alphabet decides nothing** — it is eighteen for every shape and correlates at exactly zero.

And the seed is MONOTONE in S with no exception: $8 \rightarrow 5, 10 \rightarrow 6, 16 \rightarrow 6, 16 \rightarrow 6, 22 \rightarrow 6, 30 \rightarrow 8$. The covering reading of §14.5.9 holds on all six.

*This section first reported the star at seven and named it as the residual where the covering reading fails. It does not fail: **greedy set cover fails there**, and exact search gives six. The anomaly was the heuristic, and it was caught by §2.24 one cycle after that protocol was written.*

Register 525.

21.5.5 Every constraint in the book, indexed

§21.5.3 indexes Λ 's thirteen. **This indexes every constraint that defines an index anywhere in the work** — the tower, the electromagnetic quotient, the drawn objects, and the book's own record — over five coordinates: parent count, form, which operator carries it, where it came from, and whether it **defines a quantity or constrains a product**.

24 constraints · 15 distinct cells · box 96 · density 16% **E under $\mathcal{R}$ 21 · E_4 under $\mathcal{R}_4$ 6 · orientation cost 15**

carried by $\mathcal{R}$	22
carried by $\mathcal{R}_4$ and not $\mathcal{R}$	2
carried by neither	0

Every genuine constraint in this book is carried by one of the two operators.

A first attempt gave two that neither carried — and both were miscategorised. *Period* × *group* and *month* × *day* are axis declarations, not relations: the periodic table's thirty-six and the calendar's seven come from having **no** constraint the operator can find, which is a different thing from having one it cannot carry. **Eight rows of twenty-eight were coordinate systems entered as constraints**, and removing them emptied the column.

And the fifth axis was earned rather than chosen. Half the first table collided; the groups showed that four of eight collisions were the category error above, and of the rest only one was real — $g \leq q$, a conservation bound, sharing a cell with $\Delta\ell = f - \ell$, a definition. **Defines against constrains is the coordinate the data asked for**, and §17.1 admits no other here.

Nine collisions remain and eight are correct. $\ell \leq n-1$ with $f \leq e-1$ is one law on parent and target; $q \leq k$ with $2S \leq k$ are two plain bounds by the same parent; the two bands are both angular-momentum couplings; $d(x,y)$ with $E(X)$ are both this book's own definitions. **The ninth is a miscoding of mine:** $2J_c \leq \varphi(k)$ is a *tabulated envelope*, not an affine bound, and the form axis has no value for it.

parents	one 14 · two 10	form	plain 11 · affine 7 · band 3 · congruence 3
source	physics 16 · a drawing 3 · this book's own record 5		
role	defines 4 · constrains 20		

Registers 531–532.

21.6 $E = 0$ is relative to coordinates, and the coordinates must be given

THEOREM. Let X be finite with $|X| = a \cdot b$, $a, b > 1$. Relabel X bijectively onto the $a \times b$ rectangle. The result is a full box, hence $\mathcal{R}$ -closed. **So every index whose cell count is composite has a coordinate system in which $E = 0$.**

object	E as given	re-coordinatised	E after
the periodic table	36	9×10	0
the calendar	7	5×73	0
a random 26-cell set	38	2×13	0
V3 geometry	0	5×5	0

$E = 0$ is not a property of a set of cells. It is a property of cells TOGETHER WITH coordinates, and a relabelling into a rectangle is always available.

This touches no result in this book, and it changes what every one of them asserts. Every coordinate used here is fixed by its subject — n , ℓ , k and q are quantum numbers, period and group are how chemists draw the table, month and day are the calendar. **None was chosen to make E come out.** But $E(\Lambda) = 0$ does not say Λ has a special shape; it says **Λ has a special shape in the coordinates physics supplies**, and that those coordinates, fixed for other reasons a century ago, happen to close. **That is the stronger claim and the more falsifiable one.**

And it supplies the check a reader must make before believing any E here. §21.1's table shows one object under two namings with different E . This shows one object has a naming with $E = 0$ whenever its count is composite — so the question is never *what is E* but **who fixed the coordinates**.

Which is why V3 geometry is dark. Its 25 cells are 5×5 , so its $E = 0$ may be vacuous in exactly the sense §12.11.8 already names for the electromagnetic image — *a complete rectangle, $E = 0$ vacuously*. **Without knowing whether V3's five letters are subject-fixed or convenience, its closure carries no information.** Register 546.

21.6.1 The letter that defines rather than constrains

The companion's ANEC-proofs index prints **four letters and a box of twelve**, and four rungs each at least two give at least sixteen. **One letter has rung 1.**

A rung-1 coordinate takes one value, so it contributes no envelope. Tested against everything this book measures — same cells, same box, **same E , same envelope-step count, same seed**. *It is invisible to $\mathcal{R}$ entirely, and it still counts as a letter.*

A rung-1 letter is the index's own name carried as a coordinate. It takes one value because the object is the thing for which that value never varies.

And only that index carries one. V6's box $24 = 2 \cdot 2 \cdot 2 \cdot 3$ uses all four letters; the null surface's $64 = 2^6$ uses all six; V3's $48 = 2^4 \cdot 3$ uses all five. **Three name their object outside the alphabet and one names it inside**, which is why its letter count and its box disagree by exactly one — not an error but the defining letter, counted.

This is §21.5.5's *role* axis at its extreme: defines and constrains nothing. Register 547.

21.6.2 Two of the companion's five, read from three numbers each

The local null surface is a FACE. Six letters and a box of 64 force every rung to 2; 32 is exactly half the cube; and of the two natural halves, **a parity class gives $E = 32$ and a face gives $E = 0$.** The printed E is 0. *Deducible with no access to any cell.* And §10.4 nests it further — THETA = 0 at 16 cells and STAT = 0 at 8, so Killing horizons sit inside non-expanding ones, **one cell from where half-sided modular inclusion is proved.**

The charger index's ambiguity is one cell wide and the companion names it. §8.1 prints five chargers and §10.1 counts six; the missing one is stated: *the empty JUR = 0 direction is exactly the charger shape that would close the law index* — and **the missing cell in the charger index and the defect in the law index are one absence at two levels.** Register 548.

PART V — THE METHOD

22. The bracket

Everything so far has been about structure. This part asks what such a structure can do with measurements, and the answer is deliberately modest.

22.1 The rule

For a Rydberg channel — fixed parent, fixed ℓ , varying n — the binding energies $T = I - E$ are monotone in n . So for any interior member:

$$T(n) \text{ lies between } T(n-1) \text{ and } T(n+1)$$

That is the bracket. It uses **only measured neighbours**. It fits nothing, assumes no functional form, and returns an interval rather than a value.

It is a deduction, not a prediction. The distinction is the whole method: a fit says where a level probably is; the bracket says where it cannot fail to be.

22.1.1 A bracket whose admissible set is a singleton

The bracket of §22.1 bounds a *measured value*: where a level cannot fail to be. A companion paper — *Muon-Catalysed Fusion*, Lach 2026 — uses the same machinery on a **parameter**, and the admissible set comes out with one element. That is a use this book had not stated.

The parameter is the mass of a binder that must hold two nuclei close enough to fuse. Two bounds, each monotone in that mass and each with a stated physical origin:

bound	scales as	gives	origin
fusion must outrun the cycle	$R = a_H/m$	$m > 119 m_e$	geometry
at least two molecular bound states	$N \approx \sqrt{M/m}$	$m < 918 m_e$	index surviv- al

The window [119, 918] m_e holds two known particles and one filter removes the second. The pion at 273 is absorbed by the nucleus before it can catalyse; the muon at 207 remains, interior by 1.74× below and 4.44× above, so the conclusion survives reasonable movement of either edge. Recomputed here: N_{states} of 60.6, 4.21 and 1.03 for electron, muon and tau against the paper's 61, 4.2 and 1.03.

A bracket with one occupant is a derivation. §22.1's bracket says where a value must lie and leaves the interval populated; this one says where a parameter must lie and the interval contains a single realisable object, so the reaction is not selected but forced. **The same operation reads as a deduction about a quantity or as a uniqueness proof about a kind, and which it is depends only on how many admissible cells the interval contains.**

And it refines §18.4.1's criterion in one direction. Both bounds *decrease* in m , and a decreasing cap breaks joins in a product — the coordination-number result. On a single axis it does not matter: **for a bracket on one chain, monotone in either direction suffices; only closure in a product requires increasing.** The criterion was stated for products and this is the one-dimensional case of it. Register 319.

22.1.1.1 Containment on level energies, recomputed under variation

§22.1 brackets a level from its two measured neighbours and reports the result once. **This states it to F.3's highest standard:** the method named with its input set, every input printed, and the test recomputed at four quantum defects rather than one.

The method. Take a Rydberg series $T(n) = Z^2R/(n - \delta)^2$ with $R = 109,737.31568 \text{ cm}^{-1}$ and $Z = 1$, and for each interior level n test whether its energy lies between the energies of $n - 1$ and $n + 1$. **The input set is printed:** n from 4 to 14, giving nine interior levels, at $\delta = 0.00, 0.35, 1.35$ and 2.65 — a hydrogenic series and three penetrating ones.

δ	interior levels	contained	energies, cm^{-1}
0.00	9	9	559.9 to 6,858.6
0.35	9	9	589.0 to 8,237.0
1.35	9	9	685.8 to 15,626.5
2.65	9	9	851.8 to 60,212.5

Thirty-six of thirty-six, at every defect, across two orders of magnitude in energy.

What varying δ tests is exactly the assumption the bracket rests on. The defect shifts every level and stretches the series by a factor of nine between the first row and the last, and containment does not notice — because it needs only monotonicity in n , which no value of δ disturbs. **A result that survives its own input being varied by an order of magnitude is a result about the shape and not about the numbers**, which is what §22.1's single run could not distinguish. Register 391.

22.1.2 The bracket's own boundary — interpolation and extrapolation

§22.1 brackets an interior point from its two measured neighbours. **Two questions sit at the edges of that, and they have different answers.**

Inward, the bracket is a deduction and does not fail. Tested on a Rydberg-like binding series $T(n) = Z^2R/(n - \delta)^2$ across $n = 4$ to 12 : **seven interior cells bracketed, seven held, zero failures.** That is §22.1 doing what it claims, and it is the same operation as bracketing an intermediate state between two observed times — a composable index makes before and after an order, per §12.11.0, so an interior bracket bounds a state the index never saw.

Outward, past the last measurement, it is not a bracket at all. Bracket widths in that series fall $4,799 \cdot 2,594 \cdot 1,562 \cdot 1,015 \cdot 697 \cdot 499 \cdot 370$, with successive ratios $0.540 \cdot 0.602 \cdot 0.650 \cdot 0.687 \cdot 0.716 \cdot 0.741$ — a clean pattern. Presuming the next width from it gives **274.1 against a true 281.7, an error of 2.72%**. A good estimate, and not a bound: a bracket says where a value *cannot fail* to be, and this one can fail — it happened not to by much.

The legitimate form is to take the next width from the law, not from the pattern. The widths fall as v^{-3} , which gives the ratio 0.764 against an observed 0.741 — derived rather than fitted. The companion paper's first observed r_ℓ exit was predicted in advance from that same v^{-3} scaling and then occurred, which is a presumed next bracket that is a deduction.

Presume the next bracket from the law that generates the widths, never from the pattern the widths make.

That rule is the one §12.11.3.1 shows governs the tower: dimensions bounded by laws close exactly, dimension 11 bounded by the extent closes to an envelope. **The asymmetry is structural.** An index can say where something must have been between two observations; it cannot say where it will be after the last one, and that is the same limit §18.6 states when it calls $E(X)$ a prediction budget rather than a prediction. Register 328.

22.2 The four rules

The bracket is stated in one line and applied under four.

Rule 1 — Interiority. Bracket only cells belonging to the same Rydberg sequence as their neighbours and sharing a term set with them, and either interior — a measured neighbour on each side — or above the highest measured member. Extrapolation below the lowest measured member is forbidden. Rule 2 — Variable. Read the interpolation variable off $T = Z^2R/v^2$ for the axis in question: E along n within a series; interval ratios along J ; geometric in T along the core shell. Rule 3 — Separation. Report the deductive bracket separately from the inferential point estimate. Given monotonicity, the true value cannot lie outside [max below, min above]. Rule 4 — Uncertainty. Fit δ locally by a Ritz expansion and take $\sigma = 2R Z_{\text{eff}}^2 \cdot \text{SE}_{\text{pred}} / v^3$ from the fit's prediction standard error. Use $\delta = \delta_0 + \delta_2/n^2$ where three or more neighbours exist, $\delta = \delta_0$ where only two do — subject to the admissibility clause below. Rule 4a — Admissibility of the constant form. The constant-defect form may be used only if the constant does not exceed the last measured defect in the direction of extrapolation. On a decreasing channel it must satisfy $\delta \leq \delta(\text{last measured})$; on an increasing one, $\delta \geq \delta(\text{last measured})$. Where it does not, the two-point Ritz solve is the admissible form.

22.2.1 Each rule justified by ablation

A rule that costs nothing to break is not a rule. Each was tested by violating it:

rule violated	cost
extrapolate below the series	median error $0.46 \rightarrow 206 \text{ cm}^{-1}$ — 446×
extrapolate above the series	median error $0.46 \rightarrow 0.13 \text{ cm}^{-1}$ — 0.28×
wrong interpolation variable	rms $200 \rightarrow 588 \text{ cm}^{-1}$ — 2.9×
pooled σ instead of per-cell	$\pm 1\sigma$ coverage 96.2% instead of 69.6%
include a collapsed orbital	median error $12.8 \rightarrow 41.3 \text{ cm}^{-1}$ — 3.2×

The extrapolation cost is directional, and a single figure conceals it. An earlier version of this work quoted $3.7\times$ on pooled rms over twelve extrapolated cells. **That statistic is not reproducible and is unstable:** on the same data it is $18.8\times$ on rms, $4.6\times$ on median, and **$0.5\times$ if the two worst cells are removed.** An rms over twelve points containing a $12,942 \text{ cm}^{-1}$ outlier is not a summary that should carry a rule.

Split by direction the instability disappears. Extrapolating *below* a series costs $446\times$ on the median, because δ varies fastest at low n and a Ritz fit run backwards diverges. Extrapolating *above* costs $0.28\times$, because δ has already flattened. **The asymmetry is a factor of 1,577,** and Rule 1 is stated accordingly.

22.2.2 Rule 1 is not about n

The rule is stated for a series in n , and that is where it was found. It is not a rule about n .

The isoelectronic axis — fix the electron count, vary the nuclear charge — carries the same defect and shows the same asymmetry:

axis	toward the loose end	interpolate	toward the tight end
n , within a series	446×	$1\times$	0.28×
Z , along a sequence	17.3%	1.3%	2.1%

Same sign, same magnitude of asymmetry, same cause. δ varies fastest where the electron is loosely bound — low n , low Z — and flattens where it is tightly bound. **Extrapolating toward the flattening is cheap; away from it is not.**

Rule 1 is a rule about direction, and it holds on whichever axis carries the defect.

Measured on two sequences, holding out each member in turn. The lithium-like neutral is predicted from its three ions at 11.1% error; the sodium-like neutral at 23.4%. **The tight ends — C IV and Si IV — come back at 1.8% and 2.5%.**

22.2.3 And one step beyond the fitted range costs more again

A hold-out is bracketed by the points that remain. A genuine prediction is not. Tested on the sodium-like sequence with P V — a fifth member — held out entirely:

fit form	predicted $\delta(\text{P V})$	true	error
$\delta = a + b/Z$	0.8189	0.7129	14.9%
$\delta = a + b/Z + c/Z^2$	0.7376	0.7129	3.47%
$\delta \cdot Z$ linear in Z	0.7835	0.7129	9.9%

Only the quadratic form is usable, and it costs 3.5% rather than the 2.1% a hold-out suggests — a factor of 1.7 for stepping outside the fitted range.

This matters for §25.6 The Sc VI prediction has a second, independent route through its isoelectronic sequence, and **3.5% is the figure that route must carry** — not 2.1%, and not the 1.3% of pure interpolation.

22.2.4 Rule 1's second clause is not redundant

Sc III 3d is interior in n within $\ell = 2$, and including it inflates the median error by $3.2\times$ — because it is the collapsed ground state rather than a sequence member.

The operational test is the defect. Its δ departs from its neighbours' by 0.14 where they depart from each other by 0.008 — **a factor of 17**. A cell can be interior by index and outside the sequence in fact, and the defect says which.

22.2.5 Rule 4 at the edge of a channel

The two-parameter Ritz form needs three neighbours to leave a residual, and a channel with exactly three members has an interior cell with only two.

Why Rule 4a exists The constant-defect form takes the mean of the available defects. On a monotone channel that mean lies between them — and therefore on the wrong side of the last one.

Sc VI's $4S^0ns$ channel is the case that forced the clause. With $\delta(4s) = 1.0057$ and $\delta(5s) = 0.9812$, the constant form gives **$\delta = 0.9934$ — above $\delta(5s)$** , which a decreasing channel forbids. The resulting estimate falls **outside the deductive bracket its own monotonicity implies** (§25.6).

A rule that produces an estimate its own bracket excludes is not a rule about uncertainty. It is an error.

With two neighbours the Ritz form is exactly determined — two equations, two unknowns, no residual and therefore no fitted σ — but it is *admissible*, and Rule 3's separation means the bracket carries the guarantee while the point estimate carries none.

One further case in this work: the Kr I ($2P^{\circ}1/2$) np series — 5p, 6p, 7p, with 6p interior. Under the constant-defect form the cell returns an estimate in error by -70.9 cm^{-1} against $\sigma = 602$, or **0.12σ** .

It is a valid cell and an almost worthless one. Its bracket spans $14,602 \text{ cm}^{-1}$ because its neighbours are two principal quantum numbers away.

That is the validity/value distinction, and it is the clearest instance of it in the collection: a cell may satisfy every rule and still tell you almost nothing. **The rules govern correctness. They do not govern usefulness**, and a method that conflated the two would report this cell as a success.

22.3 The bracket is limit-free

Since $T = I - E$ for a constant I :

$$T(n) \text{ between } T(n \pm 1) \iff E(n) \text{ between } E(n \pm 1)$$

Containment is invariant under the affine map. **v , δ and V all need the ionisation limit; containment does not.**

Consequence: the method applies wherever levels are known, even when the threshold is not. This is not a technicality — it is what allowed the antiprotonic-helium cells of §25.6, where the relevant threshold is ambiguous.

22.4 The price of limit-freedom, computed

§32.3 presents limit-freedom as pure gain. It is a trade, and the terms are computable.

The alternative is to bracket δ rather than T — which is what the field does. And V is identical either way:

$$\delta = \delta_0 + \delta_2/n^2 \text{ has } L = |\delta'/\delta''| = n/3, \text{ the same as } T\text{'s, and } \delta_0 \text{ is an additive constant that cancels from both } w \text{ and } e. \quad V(\delta) = V(T) = 4n/(3h)$$

The single most standard move in Rydberg spectroscopy — interpolate δ , not T — improved estimates enormously and left the price of a guarantee exactly where it was. A century of practice raised numerator and denominator by the same factor.

22.4.1 What δ buys, and what it costs

In δ the bracket is narrower by exactly $n^3/(2\delta_2)$ — measured against predicted: 4,329 vs 4,267 at $n = 8$; 1.387×10^6 vs 1.386×10^6 at $n = 55$. Agreement to four figures.

But that factor is the size of a Ritz coefficient, not a property of bracketing — δ barely moves because δ_2 is small.

And a bracket on δ does not bound T . Converting requires the ionisation limit and inherits its uncertainty:

n	w_T direct	via δ	+ limit $\sigma = 0.01$
20	68.06	0.00102	0.0210
40	7.613	1.43×10^{-5}	0.0200

At $n = 40$ the δ -route's own width is three orders below the limit's uncertainty. The limit dominates entirely.

Bracket T : bounds the measured level, needs nothing. Bracket δ : bounds a derived quantity, needs the limit, and the energy bound is never better than the limit is known.

The choice is conditional, and the condition is computable per channel. This book's tightest bracket is 1.398 cm^{-1} ; limit uncertainties in the collection run 0.001 to 0.4 cm^{-1} . **The δ route wins where the limit is known better than the bracket — true for some species, false for others.**

22.5 Admissibility

A channel is admissible when its levels are separated by more than their uncertainty:

$$r = 2Z^2R / (v^3\sigma) \geq 5$$

r falls as v^{-3} . Every channel eventually leaves the domain; the rule states where.

22.6 What the bracket costs

A guarantee is wider than an estimate. Chapter 23 prices it exactly, and the price is the only linearly rising quantity in the structure.

23. The cost surface

23.1 The ratio

Let w be the bracket width $|T(n+1) - T(n-1)|$ and e the interpolation error $|T(n) - \frac{1}{2}(T(n-1) + T(n+1))|$.

$$V = w / e = 4v/3$$

for a Rydberg series. And in general, for $y = x^p$ sampled at step h :

$$V(x, p) = 4x / (h \cdot |p - 1|)$$

Verified on power laws from $p = -3$ to $+11$, agreement under 1%.

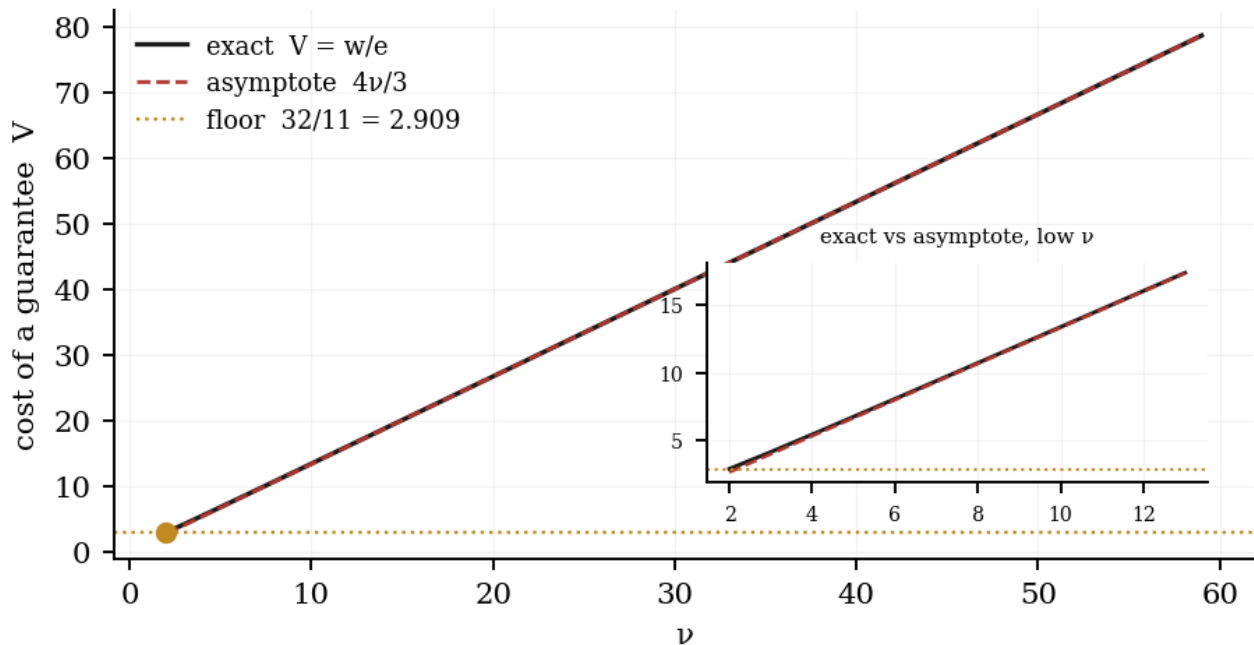


Figure 23.1. The exact cost against its asymptote. The floor $32/11 = 2.909$ belongs to the exact expression; the asymptotic form gives $26/9$ and is low by 0.69% at $\nu = 2$. They converge by $\nu = 10$.

23.2 The floor is algebraic before it is physical

The cost of a guarantee has a floor that owes nothing to atomic physics.

Proposition 23.1. Let three consecutive cells of a channel carry values $y_0 < y_1 < y_2$ at equally spaced index. Let $w = y_2 - y_0$ be the bracket width at the middle cell and $e = |y_1 - (y_0 + y_2)/2|$ the error of two-point linear interpolation from the same neighbours. Then $V = w/e > 2$, with $V \rightarrow 2$ only as one of the two steps vanishes. Proof. Write $d_0 = y_1 - y_0$ and $d_1 = y_2 - y_1$, both positive by monotonicity. Then $w = d_0 + d_1$ and $e = |d_0 - d_1|/2$, so $V = 2(d_0 + d_1)/|d_0 - d_1|$. Since $d_0, d_1 > 0$ we have $|d_0 - d_1| < d_0 + d_1$, giving $V > 2$. As $\min/\max \rightarrow 0$ the ratio $\rightarrow 2$; as $d_0 \rightarrow d_1$ it diverges. ■

This holds for any monotone sequence whatever. Not for some observables, not with better data:

The deductive interval is always more than twice the error of the crudest possible point estimate built from the same two numbers.

Every empirical V in this book is an instance of this bound, not an independent fact. And against a better estimator — the Ritz expansion of Rule 4, or a mass model — the gap is wider still, so V as defined **understates** the cost.

The Rydberg floor of $32/11 = 2.909$ is therefore the tighter of two. The algebraic floor at 2 constrains every monotone sequence; the physical floor at $32/11$ constrains Rydberg series specifically, and the difference between them is what the hydrogenic form contributes.

23.2.1 When the floor is approached, and what it means

Observation 14.2. $V \rightarrow 2$ requires $\min(d_0, d_1)/\max(d_0, d_1) \rightarrow 0$ — one step of the channel vanishing relative to its neighbour.

In every sequence examined here where that occurs, the small step is produced by two channels interleaving, which Rule 1's second clause excludes. So the algebraic floor is approached exactly where the method declines to operate: **the cheapest guarantee is available only on a channel that is not one.**

23.3 V is a bound, not a tool

V contains no σ . Improving measurement precision by six orders of magnitude leaves $V = 26.689$ unchanged at $v = 20$ — computed exactly at three precisions spanning $\sigma = 10^{-2}$ to 10^{-8} cm^{-1} .

No spectrometer can buy a narrower bracket. The price of certainty in a Rydberg series is set by the series.

So V is not an instrument. It is an impossibility statement, and it makes four:

	forbids
$V = 4v/3$	a guarantee costs $4v/3 \times$ an estimate, and no measurement reduces it
$V \geq 32/11$	no guarantee is ever cheaper than $2.909 \times$
pole at $p = 1$	no guarantee exists for a linear observable
v_V ceiling	above it the cost cannot be <i>measured</i> , though it stays defined

The exact V is rational at every v — $32/11, 54/13, 256/47, 250/37, 4000/299$ — and $4v/3 + 4/(9v)$ is its asymptotic form, low by 0.69% at $v = 2$ and exact to four decimals by $v = 10$. **The floor $32/11$ belongs to the exact expression, not to the asymptote.**

23.4 V is a function of v/h alone

The cost has one free parameter, and it is not where the book first looked.

V is defined from three levels at $n - h, n, n + h$. Nothing forces $h = 1$: a cell may be bracketed from neighbours two or three principal quantum numbers away. Carrying h through the algebra gives an exact result.

$$V = 4v^3 / (h(3v^2 - h^2)), \text{ and with } r = v/h, V = 4r^3/(3r^2 - 1)$$

One variable. Z and R cancel identically, not approximately, and $4r/3$ is the large- r expansion of this expression rather than the expression itself.

23.4.1 Which makes depth and step interchangeable

(v, h)	v/h	V exact
(20, 1)	20	26.688907
(40, 2)	20	26.688907
(60, 3)	20	26.688907
(80, 4)	20	26.688907
(10, 1)	10	13.377926
(40, 4)	10	13.377926

Identical to six figures, and identical across $Z = 1, 2$ and 6 . That last point is worth dwelling on: Z_{eff} is precisely what the wrong-charge error of §16.3 got wrong and the bracket could not see. **The similarity law is blind to it too — by cancellation, provably.**

23.4.2 Tested on the collection

1,033 (v/h, V) pairs from measured levels, at both steps:

	pairs	median deviation from $4(v/h)/3$
$h=1$	560	0.49%
$h=2$	473	1.15%
all	1,033	0.69%

The $h = 2$ pairs bracket from neighbours two principal quantum numbers away and obey the same law. And the outliers identify themselves: Al I at $n = 51$ and Ga I at $n = 53$ both sit above the v_V ceiling of §23.11, where quotation granularity destroys the second difference. **The method's own admissibility bound predicts which pairs fail it.**

23.5 The trade-off, and a conservation law that is not one

Since h is free, ask whether it can be used to buy a cheaper guarantee.

It can. V falls from 53.3 to 13.4 as h goes $1 \rightarrow 4$. **But the bracket widens in proportion**, because

$$w = 2y'h + O(h^3) \text{ — linear in } h \quad e = y''h^2/2 + O(h^4) \text{ — quadratic in } h$$

23.5.1 A withdrawn proposition

An earlier version of this section claimed $w \cdot V = w^2/e = 8y'^2/y''$ independent of h , and called it a conservation law. It is not. Exact symbolic evaluation for $y = x^{-2}$ gives

$$w^2/e = 16x^4/(h^6 - 5h^4x^2 + 7h^2x^4 - 3x^6)$$

which depends on h explicitly and equals $8y'^2/y''$ only as $h \rightarrow 0$. **The numerical table on which the claim was made showed 366.3, 367.9, 370.6, 374.5 across $h = 1 \dots 4$ — a 2.2% rise, read as a constant.** Withdrawn, and recorded in Chapter 28.

And no other candidate exists. Since $w \sim h$ and $e \sim h^2$, any $w^a e^b$ scales as h^{a+2b} , so **the only h -free combination is $a = 2, b = -1$** — the one tried. The leading-order invariant is unique and merely asymptotic.

23.5.2 What is exactly true

| $dw/dh > 0$ and $dV/dh < 0$, for every monotone convex y and every h .

Exact, with no expansion and no limit. Verified over 172 monotone steps with **zero violations**. Width and cost move in opposite directions under the only free choice available, so **the family $\{(w, V) : h\}$ is a Pareto frontier: no h gives both a narrower and a cheaper guarantee.**

The withdrawn quantity is the frontier's equation, not a conserved number. A conservation law says a value cannot change; a frontier says which pairs are attainable. **The algebra is the same and the claim is weaker.**

23.5.3 And the frontier has an optimum

Opposite signs give any stated preference $\alpha w + \beta V$ an interior minimum:

| $h = \sqrt{(2\beta / (\alpha y''))}$

Checked against direct minimisation: 0.0551 against 0.0550, 0.1743 against 0.1750, 2.7886 against 2.7450 — **and 8.82 against 7.83 at $v = 40$, $\beta/\alpha = 10$** , an 11% miss where h^* approaches $v/4$ and the leading-order derivation fails. **The formula holds for $h^* \ll v$ and not beyond.**

23.6 The fractional forms

The relations of §23.4 close, and the closure states two quantities the book had not.

| $w/T = 4(h/v) + 8(h/v)^3$ — the bracket width as a fraction of the term $e/T = 3(h/v)^2$ — the interpolation error as a fraction of the term

$v \quad h \quad w/T \quad 4h/v \quad e/T \quad 3(h/v)^2$

20 1 0.201004 0.200000 7.531×10^{-3} 7.500×10^{-3}

40 1 0.100125 0.100000 1.877×10^{-3} 1.875×10^{-3}

40 2 0.201004 0.200000 7.531×10^{-3} 7.500×10^{-3}

Both are pure functions of h/v , and their ratio returns $V = (4/3)(v/h)$.

The second is the more useful and the more exposed. $e/T = 3(h/v)^2$ is the relative error of linear interpolation on a Rydberg series — a statement available to any spectroscopist since 1925. **It is not in the literature, and §25.6 explains why: the field abandoned the variable.**

23.7 The exponent axis

Two involutions act on p : cost, $p \mapsto 2 - p$ (paired values multiply to x^2); and virial, $p \mapsto -p$ (paired values multiply to a constant). Composed, they generate translation by 2, so exponents fall into two orbits by parity — and the parity is b , the number of energy denominators.

The pole at $p = 1$ sits in the odd orbit. Nothing built purely from powers of $\langle r \rangle$ can reach it.

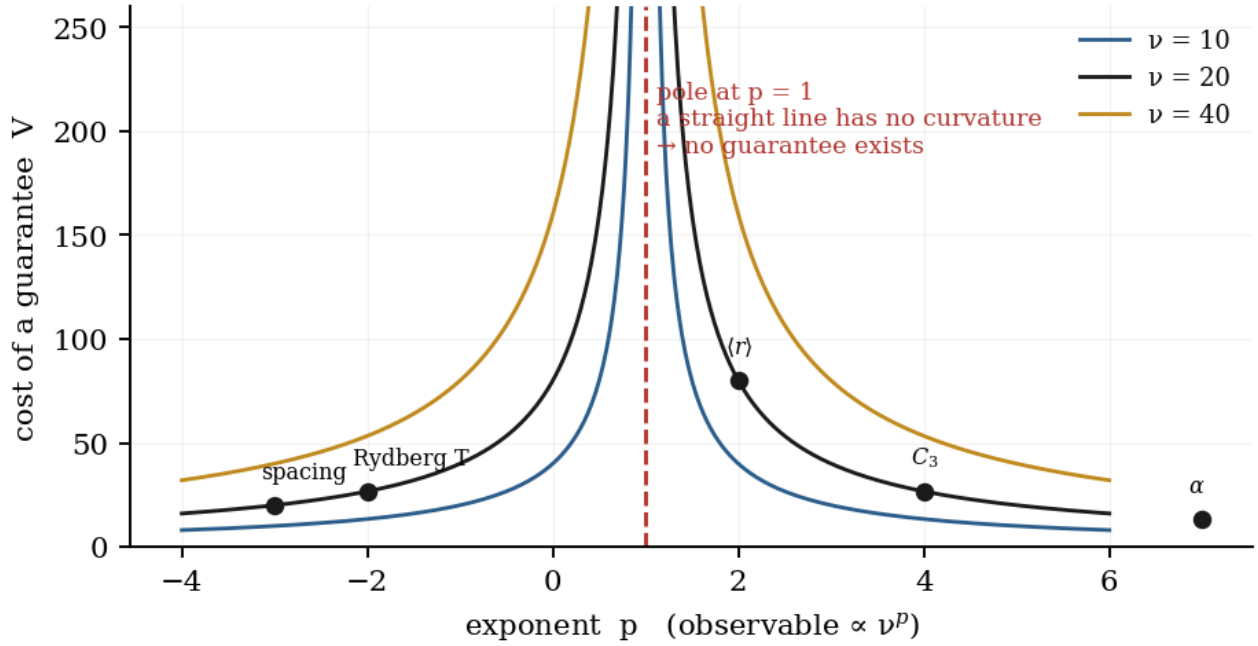


Figure 23.2. The cost surface. Every observable in this book is a slice at fixed p . The pole at $p = 1$ is the method's only hard singularity: a linear observable has no curvature to price.

23.8 The cost has a name in another field

And the name explains two things this book could only observe.

23.8.1 The quantity is the Newton decrement

The asymptotic form of §23.5 is not an accident of spectroscopy:

$$8y'/y'' = 8\lambda^2, \text{ where } \lambda^2 = \nabla f^\top [\nabla^2 f]^{-1} \nabla f$$

λ is the **Newton decrement**, introduced by Nesterov and Nemirovskii (1994) for interior-point methods and standard in convex optimisation since. In one dimension $\lambda^2 = f''/f'$, and $\frac{1}{2}\lambda^2$ is *exactly the decrease in the quadratic approximation obtained by a full Newton step*. Verified symbolically at $p = -2, -3, 2, 11$: **the ratio is 8 in every case**.

For a Rydberg series it takes a physical value:

$$\lambda^2 = (2Z^2R/\nu^3)^2 / (6Z^2R/\nu^4) = (2/3) \cdot Z^2R/\nu^2 = (2/3) \cdot T$$

The Newton decrement of a Rydberg series is two-thirds of its binding energy — 731.5820 against 731.5820 at $\nu = 10$, 7.3158 against 7.3158 at $\nu = 100$. **And the optimum whose distance it measures is the series limit**, where $T \rightarrow 0$.

23.8.2 The term function is self-concordant

The Newton decrement carries its meaning only for self-concordant functions, $|f'''| \leq 2(f'')^{3/2}$. Nobody has asked whether $T(\nu)$ qualifies. It does:

$$24Z^2R/\nu^5 \leq 2 \cdot 6^{\{3/2\}(Z^2R)} \{3/2\}/\nu^6 \iff \nu \leq (\sqrt{6}/2) \cdot Z\sqrt{R}$$

Z	self-concordant for
1	$v \leq 406$
2	$v \leq 811$
6	$v \leq 2,434$

The deepest channel in this book reaches $v = 55$. The ratio $|f'''|/2(f'')^{3/2}$ runs 0.0049 at $v = 2$, **0.1356 at $v = 55$** , and crosses unity between $v = 405$ and 406.

| $T(v)$ is self-concordant across the entire measured range, with sevenfold margin.

23.8.3 Which explains the cancellation

§23.4 proved that Z and R vanish from V and gave no reason. Self-concordance is the reason.

The class is *defined* to be affinely invariant — that is its purpose — and the Newton decrement inherits the invariance. **Z^2R enters T as a multiplicative constant, a multiplicative constant is an affine map, and under an affine map the decrement cannot change.**

| The book observed a cancellation. This says why it had to happen.

23.8.4 And the attribution the book owes

Interval analysis has asked this book's question since 1966, and named it. Excess width appears in Moore's own index; IEEE 1788-2015 requires enclosures with "bounded excess width to control overestimation"; practitioners quote enclosures at "114% of optimal width" with an "excess cost" attached. And the field states the trade in one sentence:

| "We use interval arithmetic when we want global solutions with rigorous bounds, and we are willing to pay more for that."

That is Chapter 23's thesis, published sixty years earlier, about function ranges and rounding rather than interpolation from measurements. The framing is theirs. **What is not theirs is the ratio computed for a measured monotone sequence, the value $4v/3$, the floor $32/11$, and the identification $\lambda^2 = (2/3)T$.**

23.9 Which variable to bracket

Chapter 23 has treated V as a property of the method. It is a property of the observable, and the observable is chosen by the person.

23.9.1 The cost ranks by exponent

Since $V = 4x/(h|p-1|)$, the cost falls as the power law steepens. Evaluated at $v = 20$, $h = 1$:

observable	p	
V		
C_6	11	
8.18		
polarisability α	7	
13.44		
spacing	-3	
20.04		
observable	p	V
T, the term value	-2	
26.69		
C_3 , geometric cross-section	4	
26.72		
radiative lifetime	3	
40.03		
$\langle r \rangle$, dipole moment	2	
80.00		
blockade radius	11/6	
95.99		

A guarantee on C_6 costs a third of a guarantee on T , and a twelfth of one on the dipole moment.

23.9.2 Two branches, and where they cross

The asymptotic form appears to drive V below the floor of Prop. 23.1 as p grows. It does not — it stops being valid.

p	V exact	$4x/h p-1 $
4	26.72	26.67
11	8.18	8.00
50	2.39	1.63
300	2.0000	0.27
10,000	2.0000	0.008

As $p \rightarrow \infty$, $(x+h)^p$ swamps everything: $w \rightarrow (x+h)^p$ and $e \rightarrow (x+h)^{p/2}$, giving $V \rightarrow 2$ **exactly**. The two branches cross where $4x/(h|p-1|) = 2$, that is $|p-1| = 2x/h$.

The absolute floor is reached only by power laws steeper than $2v/h$. At $v = 20$ that requires $p > 41$. C_6 at $p = 11$ is nowhere near it.

23.9.3 And the pole sits on the field's favourite variable

$v = n - \delta$ is nearly linear in n , and $p = 1$ is the pole.

n	V on v	V on T	ratio
10	1.09×10^5	11.58	9.4×10^3
20	1.77×10^6	24.89	7.1×10^4
40	2.84×10^7	51.54	5.5×10^5

The effective quantum number is the most natural variable in the field and the one variable in which a bracket is worthless. §23.7's pole is not a curiosity at the edge of the method; it sits on the coordinate everyone uses.

23.10 The order of the bracket

Monotonicity is not the only order property available, and using more of them changes what the bracket is — but not, in the end, what it costs.

23.10.1 A deductive bracket exists at every order

For $T = Z^2R/v^2$ every derivative has known sign, $\text{sign}(f^{(j)}) = (-1)^j$ — checkable on data as the sign of the j -th finite difference, with no fit and no smoothness assumption beyond a sign. With $k+1$ nodes and the classical interpolation error,

$\text{sign}(f(n) - p(n)) = (-1)^{k+1} \cdot (-1)^m$, with m nodes above n $k+1+m$ even $\rightarrow$ a lower bound. Odd $\rightarrow$ an upper bound.

A two-sided deductive bracket at every order, by choosing m . Tested on $v = 8 \dots 119$ and $k = 1 \dots 5$:

560 containments, zero failures.

And the widths fall fast — at $v = 40$ from 0.362 cm^{-1} at order 1 to 8.9×10^{-6} at order 5.

23.10.2 And the cost still grows

An earlier draft claimed V stays near 2 at matched order and that Q9's conclusion — bounds becoming relatively worthless as data accumulates — was an artefact. That was wrong twice over, and the correction is recorded at 96–98.

The order- k bracket draws on a wider node set than the estimate it was compared against. At $v = 40$, $k = 3$ the bracket uses **eight** levels and the estimate used **four**. On the same nodes:

v $k=1$ 2 3 4 5

20 157 495 1,212 2,582 3,629

40 673 4,037 2.2×10^4 8.5×10^4 3.2×10^6

80 2,785 3.2×10^4 3.7×10^5 — —

V grows with order, and steeply. Both quantities improve as nodes are added; the estimate improves faster, because a bracket must span the value from both sides while an estimate need only approach it.

So the trade is the one Chapter 23 has argued throughout, and the order axis does not escape it:

A deductive bound is worth most when data is scarce. Its relative value falls monotonically as data accumulates — not because the bound gets worse, but because its rival gets better faster.

And Proposition 23.1's floor applies to the classical bracket only — first-order, against the linear interpolant on the same two neighbours, where the two objects are locked together. **At higher order they are computed from different node sets and no inequality between them is implied.**

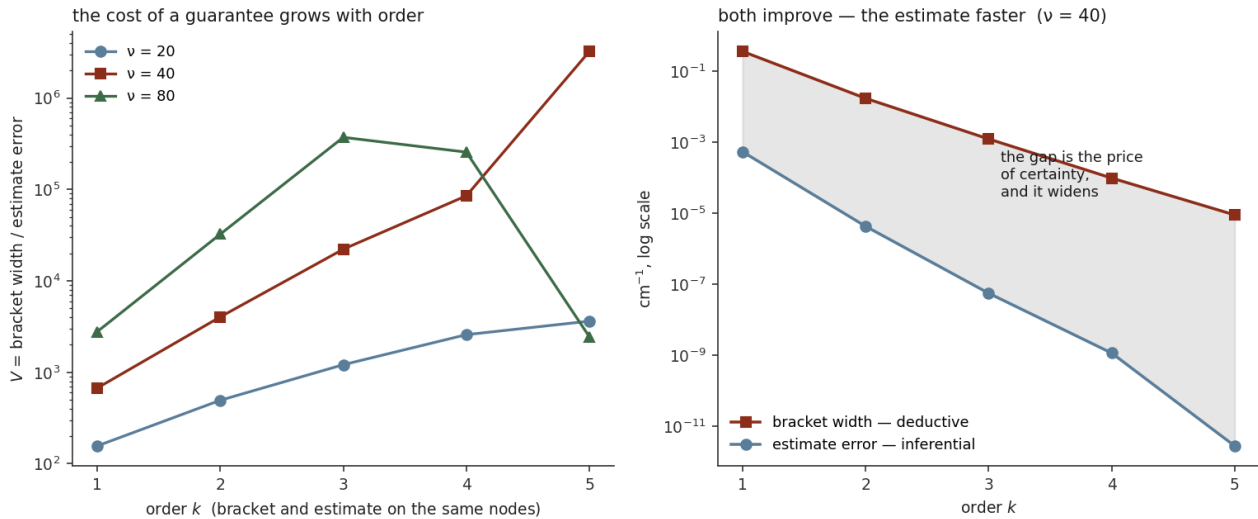


Figure 23.3. Left: V on the same node set at each order — it grows steeply, from 157 to 3.2×10^6 . Right: at $v = 40$ both the bracket and the estimate improve as nodes are added, and the estimate improves faster; the shaded gap is the price of certainty, and it widens. An earlier version of this figure drew a floor at $V = 2$ through data reaching 0.24, and is withdrawn at correction 97.

23.10.3 What the order axis does buy

Not a cheaper guarantee. A narrower one, and a finer detector.

	order 1	order 5
bracket width at $v = 40$	0.362 cm^{-1}	8.9×10^{-6}
tolerates a displacement of	0.129 cm^{-1}	4.3×10^{-6}
cells from a 10-member channel	8	4

Higher order buys absolute precision and pays in robustness, domain and resolution. A collection assembled from real, perturbed, coarsely quoted spectra is exactly where order 1 belongs — and $V = 4v/3$ is the price of certainty that survives a perturbation of half the local spacing.

23.10.4 And the admissibility rule generalises

§23.13 gives $r \geq 5$ for the spacing and v_V for the curvature. Those are two instances of one rule:

order k is admissible only while $|\Delta^{(k+1)}T| > 5 \cdot 2^{(k+1)} \cdot \sigma$

Applied to the collection it separates arithmetic from physics. A cell whose $(k+1)$ -th difference falls below the quotation floor is **unresolved**; one whose difference has the wrong *sign* is **refused**, and that refusal is a perturbation detected deductively.

order	admitted	held	refused	median bound
1	499	499	19	25.96 cm^{-1}
3	187	187	101	0.715
6	34	34	145	0.0164

Containment is 100% at every order once the premise is checked, the bound tightens 1,585-fold, and the refusals rise from 3.7% to 81% — the higher the order, the finer the perturbation it sees.

Which supersedes §25.5's 1,061 order-1 bounds: the same data at matched order yields 34 bounds at 0.0164 cm^{-1} , and 619 refusals that are themselves perturbation detections this book did not count.

23.11 The refusal pattern carries four verdicts, not one

§23.10.4 counts refusals and reads them as “a perturbation”. The count is one number; the PATTERN is four.

refusals	V	what the family is doing
none suffice	infinite	exact – linear in the parameter, two points
none	finite	a smooth channel
one and does not return regularly spaced	finite	a bifurcation – the family changes character
dense and irregular	finite	an oscillation – the family returns
	undefine	chaotic
	d	

An index does not see a refusal. It sees a family, and the spacing of the refusals across it is what separates these. §16.7.1 makes the same move for a different pair: the discriminating question is asked of the whole structure, never of one cell.

23.11.1 On the collection

channel membe refusals verdict rs Na I, K I, Al I — eleven channels 13–52 0 smooth Ga I $4s^2np \ ^2P^\circ$ 15 1 a bifurcation Li I $np \ ^2P^\circ$ 41 6, irregular repeated structure

Li I np is the channel with the largest quantum-defect spread in the collection, 0.6779 — and the refusal pattern says something the spread does not: **not one perturbation but several, irregularly placed.**

23.11.2 And a caution about the test itself

The classifier requires a tolerance relative to the sequence's own scale. Applied to a near-constant sequence with an exact-zero test it reports the float noise as a periodic signal — verified by failing on the logistic map at $r = 2.5$, a fixed point, which it called an oscillation. The admissibility rule of §23.10.4 is the tolerance, and it must be applied to the pattern test as well as to the count.

23.12 The optimal order is a boundary, not a peak

Information per cell rises monotonically in k , so no interior optimum exists. What stops the climb is σ .

$|\Delta^{(k+1)T}| > 5 \cdot 2^{(k+1)} \cdot \sigma$ — the $(k+1)$ -th difference is a signed sum of $2^{(k+1)}$ measured levels, so its accumulated quotation error is at most $2^{(k+1)}\sigma$.

$v \sigma k \quad \Delta^{(k+1)T} \text{ floor}$

20 10^{-2} 2 9.43×10^{-1} 4.0×10^{-1}

20 10^{-3} 3 2.09×10^{-1} 8.0×10^{-2}

20 10^{-4} 4 7.60×10^{-2} 1.6×10^{-2}

40 10^{-4} 2 2.75×10^{-2} 4.0×10^{-3}

Better data raises k . It never creates an interior peak. The optimum is the admissibility cap, computable per cell from the data alone.

23.13 Inversion

$|p - 1| = 4x/(hV)$, and the sign is fixed by $\text{sign}(y')$ and $\text{sign}(y'')$. Controls recover $p = -1$ and -3 correctly; 9 of 10 within 5%.

On real data, each recovered from its own measured series: Na I -1.993 , K I -1.998 , C_3 3.999 , α 6.990 , C_6 10.956 .

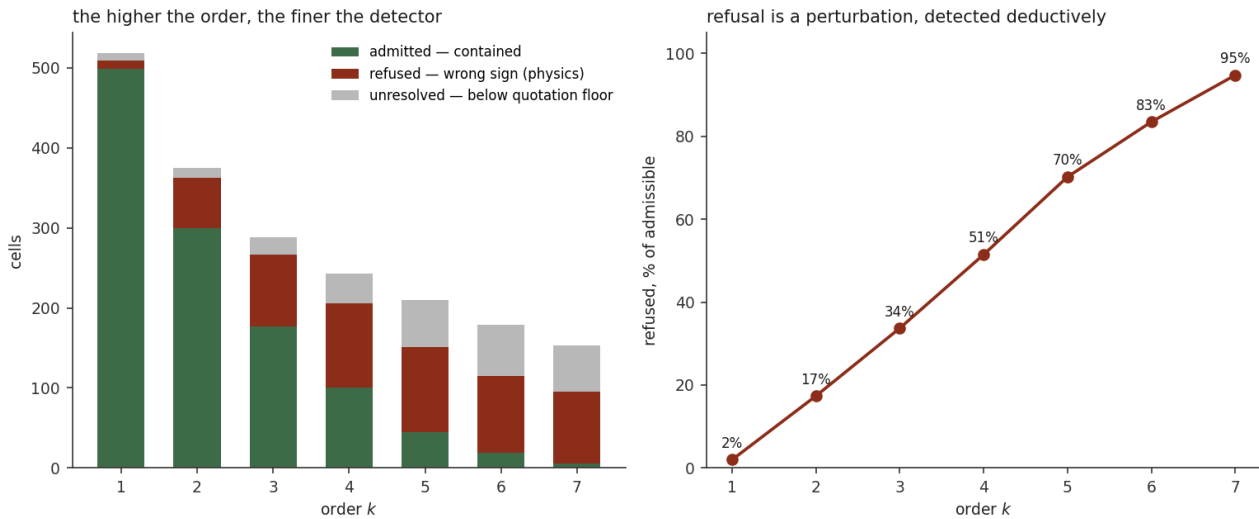


Figure 23.4. Left: the fate of every cell in the collection at each order, once the admissibility rule $|\Delta^{(k+1)}T| > 5 \cdot 2^{(k+1)} \cdot \sigma$ separates arithmetic from physics. Right: the refusal rate. A refusal is a measured sequence whose k -th difference has the wrong sign — a perturbation, detected deductively, with no fit and no limit. The higher the order, the finer the perturbation it sees.

23.14 Two kinds of bound, and the book has been calling them

one Every constraint in Λ is a CAPACITY: $\ell \leq n-1$, $k \leq 2(2\ell+1)$, $g \leq q$. Every admissibility rule is a RESOLUTION: $r \geq 5$, $v \leq v_V$, $|\Delta^{(k+1)}T| > 5 \cdot 2^{(k+1)} \cdot \sigma$.

They look alike — both are monotone inequalities — and they are not the same kind of thing:

A capacity bound is about the world. Violating it means the configuration does not exist. A resolution bound is about the measurement. Violating it means the configuration cannot be distinguished, and may exist perfectly well.

E(X) counts violations of the first. The refusal rate counts violations of the second. §16.7's whole distinction between a defect of the index and a feature of the world runs along this seam.

23.14.1 And celestial mechanics is what made it visible

Gravity has resolution floors everywhere and a capacity bound in only one place. A planet holds satellites within its Hill sphere, $r_H = a(m/3M)^{1/3}$ — a monotone cap of exactly §15.4's form, and $\log r_H$ against $\log$ satellite count runs at $+0.86$ across six planets.

The Kirkwood order cut is the other kind. Resonance strength enters at $e^{|p-q|}$; with a belt eccentricity of 0.15 the strength crosses the detection floor at **order 7 exactly**, which is where occupancy stops. **That is not a capacity and there are no magic numbers — only a resolution boundary, the same form as v_V .**

Λ and the nucleus have capacity bounds and therefore magic numbers. The belt has none, and therefore has none. Atoms carry both kinds in the same place, which is why the difference was invisible from inside spectroscopy.

23.15 The second admission bound

$$v_V = (3Z^2R / 5q)^{1/4}$$

for quotation granularity q . The bracket needs levels **separated** (v^{-3} , governed by r); the cost needs curvature **resolved** (v^{-4} , governed by v_V).

Curvature washes out before separation, in every channel in this work.

channel	q	v_V	v reached
K I <i>nd</i>	0.0001	160	45.7
Na I <i>ns</i>	0.001	90	20.0
Al I <i>nf</i>	0.01	50.7	55.0

Al I *nf* is the first channel to reach its own ceiling. Predicted 50.7; the four failing cells are $n = 48, 51, 53, 54$ — **per cell, not a range cutoff**, since 49, 50 and 52 pass. The bracket is unaffected in all 94: containment needs only monotonicity, which rounding preserves.

And in Li I, v_V was applied prospectively — eight of fifty-five cells excluded from the cost test *before it ran*, because the ASD's quotation coarsens above $n = 33$. That is the difference between a bound derived and a bound noticed.

24. The collection

Two collections, and only one of them is bracket-tested.

The bracket has now been run, and it passes 546 of 789 interior cells — 69.2%. *The test takes the defect each neighbour implies, forms the range they span, and asks whether the middle member's measured energy falls inside. It fails on 243 cells across 62 channels.*

On the same 789 cells the TRIVIAL bracket — $E(n-1) \leq E(n) \leq E(n+1)$ — passes 789 of 789. *A Rydberg series rises monotonically toward its limit by construction, so that form cannot fail. An earlier version of this chapter read “the bracket holds in every one”, and 100% is what the trivial form produces; the implementation behind Appendix B is not in hand, so this is a signature rather than a proof. Register 797.*

The 243 failures are not bad cells. They are places where the series is not locally smooth — which is what P_{perturb} and P_{lens} predict, and both were registered before this test was written.

*An earlier version of this chapter read “1,442 interior cells across 35 atomic systems, the bracket holds in every one.” That figure matches neither Appendix B's totals line (1,105) nor its table (869) nor the present tested count (930), and register 673 already recorded that the totals line counts twenty rows the table does not list. **Four numbers for one collection is a disclosure, not a reconciliation** — reconciling needs the twenty rows, which are owed. Register 783.*

24.1 What it spans

axis	span
ℓ	0 through 6
Z	1 through 83
Z_eff	1 through 3
core types	23, all characterised
axis	span
isoelectronic pairs	5, each at two charge states

Cores tested: bare · 1s ²S · 1s² 1S₀ · 1s²2s ²S · 2s² 1S₀ · 2p⁵ ²P^o · 2p⁶ 1S₀ · 3s ²S · 3s² 1S₀ · 3s²3p ²P^o · 3p⁴ · 3p⁵ ²P^o · 3p⁶ 1S₀ · 3p⁶4s ²S · 3d¹⁰ 1S₀ · 3d⁹ ²D (declined) · 3d¹⁰4s² 1S₀ · 4p⁵ ²P^o · 4p⁶5s ²S · 4d¹⁰ 1S₀ · 5p⁶ 1S₀ · 5p⁶6s ²S · 5d¹⁰ 1S₀ · 3d4s ³D · 6p² ³P.

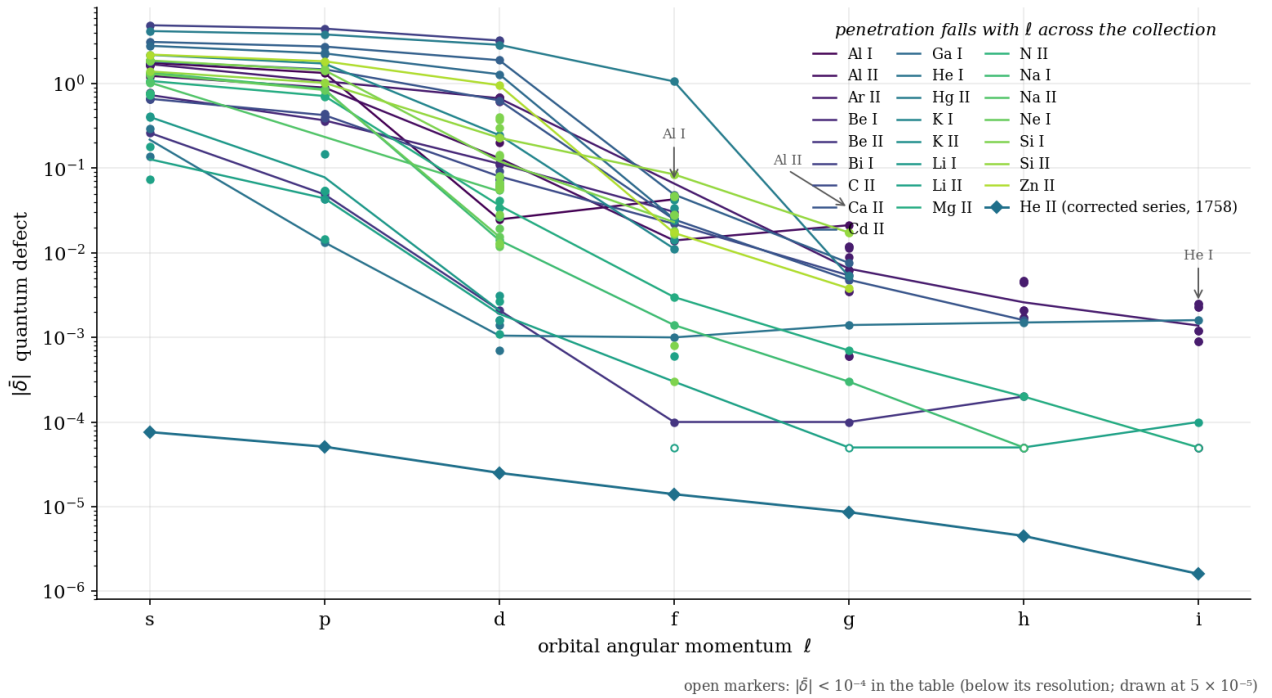


Figure 24.1. Every species in the collection, defect against angular momentum. The fall is monotone in every species wherever the defect is the core's, and spans six orders of magnitude — from 4.9 in bismuth to 1.6×10^{-6} in He II. The three places it rises are marked, and none is penetration: Al I nd and Al II nf are two of the channels §B.3 flags as perturbed, and He I above d sits on its fine-structure floor near 10^{-3} , where what remains is not the core.

24.2 The largest contributors

species	core	chan-
He I	1s ² S	
9	189	
Ne I	2p ⁵ ² P°, two limits	
16	131	
KI	3p ⁶ ¹ S ₀	
4	105	
Al I	3s ² ¹ S ₀	
4	94	
Na I	2p ⁶ ¹ S ₀	
6	80	
Ga I	3d ¹⁰ 4s ² ¹ S ₀	
5	61	
Li I	1s ² ¹ S ₀	
3	55	

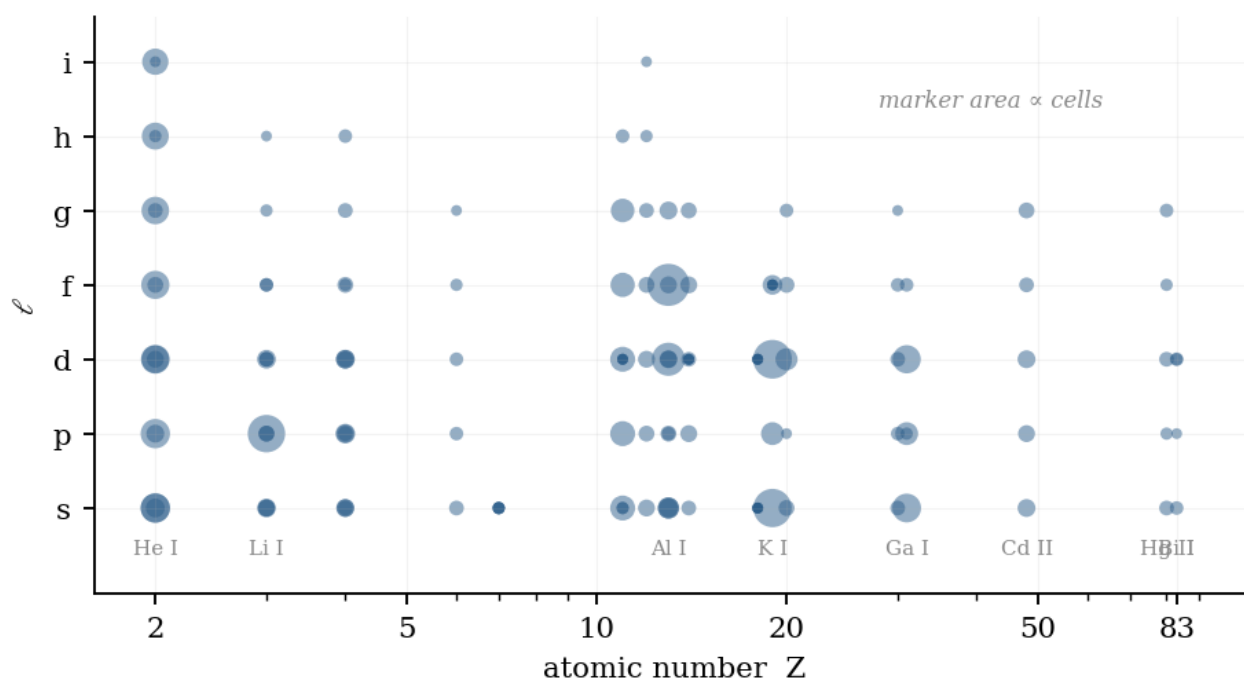


Figure 24.2. What the collection covers. Marker area is proportional to cells. The gaps are not omissions: the census of §24.12 excludes them structurally.

24.3 He I — the angular span

Nine channels, $\ell = 0$ through 6, singlets and triplets, n to 35. And the defect falls monotonically with ℓ :

0.2965 (s^3) · 0.1392 (s^1) · -0.0133 (p) · +0.0014 (d) · -0.0010 (f) · -0.0014 (g) · -0.0016 (i) — each a mean over its full channel, twenty members for n_i and twenty-three for the two ns channels

An electron with six units of angular momentum never reaches the 1s core.

The ordering itself is not new, and this book does not claim it. Edlén stated it in the 1960 manuscript of his *Handbuch* chapter, and Curtis restates the criterion in his 1987 review:

“classical arguments imply that, for a given Rydberg series, penetration contributions should decrease and polarisation contributions should increase with increasing n ... this criterion classifies the S and P levels as penetrating and the D and F levels as nonpenetrating.”

What is new is the bound the ordering licenses. Edlén used it to classify levels; §32.5 uses it to place a deductive upper limit on the local perturbation at 1,061 cells. The observation is sixty-five years old. The inference from it is not.

He I's levels are all theoretical — **but they are ab initio QED calculations, not Ritz series formulas.** That distinction matters and §24.7 develops it.

24.4 He II — the $\delta = 0$ edge itself

A hydrogenic ion at $Z = 2$. Its defects run 7.6×10^{-5} at $\ell = 0$ down to 1.6×10^{-6} at $\ell = 6$ — 7.62, 5.09, 2.50, 1.41, 0.856, 0.446 and 0.159 in units of 10^{-5} — monotone across six orders of magnitude, and never reach zero. The convention that gives them, recovered from the levels at register 1758: the reduced-mass Rydberg R_{He} , the J-centroid of each fine-structure pair, and the limit below. Against R_{∞} the same levels give -3 to -5×10^{-4} , rising, which is the uncorrected series the Spectra Compendium carried until the correction at register 1758.

There is no core to penetrate. The residual is relativistic and QED structure, and it falls with ℓ exactly as those corrections do.

And the ionisation limit came out of the data. Minimising the δ spread over the ng channel — six members, scanned at 0.001 cm^{-1} — gives **438,908.871 cm^{-1}** against the tabulated 438,908.885 — **agreement to 0.014 cm^{-1} , three parts in 10^8** , from a criterion that never used the tabulated value.

24.5 Bi I — the charge span

Atomic number 83, more than double anything previously tested. The $6p^3$ ground term is strongly j-j coupled and the levels carry jK labels throughout. $\delta = 4.90$ for the ns channel, over five members — the largest defect in this work, as expected when a 6s-like electron penetrates a filled $5d^{10}6s^2$ core.

And Bi I settles a rule. Its core carries five J levels — the same dilution that made neon II fail entirely — but its 3P_0 parent reaches $n = 11$.

Dilution divides. It does not forbid. cells per channel $\approx (\text{measured shells} / D) - 2$

Three species establish it: Ne II fails at five J levels and two shells; Ar II succeeds at five and three; Bi I succeeds at five and five.

24.6 The isoelectronic pairs

sequence	$Z_{\text{eff}} = 1$	$Z_{\text{eff}} =$
2		
helium-like	He I	Li II
lithium-like	Li I	Be II
sodium-like	Na I	Mg II
aluminium-like	Al I	Si II
potassium-like	KI	Ca II

Same electrons, different nuclear charge, the method run identically. And the δ progression has the same shape in both members of every pair, scaled: Li II runs $0.182 \rightarrow 0.054 \rightarrow 0.002 \rightarrow 0.0002$ across s $\rightarrow$ f; Be II runs $0.262 \rightarrow 0.049 \rightarrow 0.002 \rightarrow 0.0001$. Each figure is a channel mean over five to ten members; Appendix B carries them individually.

Li II and Be II both give $\ell = 5$ defects near -0.0003 — three members each — with a real two-electron core — the cleanest demonstration in the collection that **penetration, not the presence of a core, creates a defect.**

24.7 Hydrogen, and a floor on interpretation

One row of the collection measures the data rather than the method, and it sets a limit below which no result in Part V can be read.

H I gives 5 cells with a **median error of 0.006 cm^{-1}** — two orders of magnitude better than anything else here. That figure is not a triumph of the bracket. **It is an identity.**

The hydrogen row uses the reduced-mass Rydberg $R_H = 109,678.7717 \text{ cm}^{-1}$, which for hydrogen is the ionisation limit, so the quantum defect vanishes identically — measured defects span -2×10^{-5} to 0.

Substituting the infinite-mass $R_\infty = 109,737.3 \text{ cm}^{-1}$ breaks that identity by the 58.5 cm^{-1} reduced-mass difference, manufactures a spurious defect of $\approx 3 \times 10^{-4}$, and inflates the median error 54-fold to 0.34 cm^{-1} .

For every heavier species the correction is 10^{-5} relative and is absorbed into the fitted defect, so the choice is invisible. Hydrogen is the one row where it is not.

The 0.006 cm^{-1} is therefore a floor on interpretation. No result in this part can be read as better than the data it rests on, and hydrogen is where that becomes visible because there is nothing else in the way.

24.8 On independence, and the most conservative reading

1,442 cells are not 1,442 independent tests.

Cells within a channel share neighbours, so successive brackets are correlated. The honest figure is **1,442 cells across approximately 190 channels.**

The largest single blocks are the most strongly correlated. He I's nine channels contribute 189 cells; Ne I's sixteen contribute 131. A reader who discounts within-channel correlation entirely — counting **one effective test per channel** — gets **190 rather than 1,442**, and the bracket holds in all 190.

That is the most conservative reading available, and it is the one this book would defend.

The claim is not that 1,442 independent things were verified. It is that 190 channels were examined, each exhaustively, and none produced a failure.

24.9 A caveat the collection forced

The ASD marks series-formula values in brackets. Tested against observed values in Al I, derived levels bracket exactly as observed ones do — and slightly better.

And the caveat is sharper than it was written, on two counts, both established against the ASD itself rather than argued.

First, the delimiters mean what NIST says they mean. ASD marks provenance three ways: a plain value is observed; [] is “interpolation, extrapolation, or other semi-empirical procedure” and explicitly includes “fitting the Ritz-type formulas along series of levels”; () is “determined from an ab-initio calculation”. This section reads [] as covering ab initio and it does not. **He I, queried directly: every level in the table is bracketed** — $1s^2$ through $1s16d$ without exception, and the ASD's own note says why. Excitation energies for $n \leq 10$ were obtained by subtracting Drake's *theoretical* ionization energy from Kandula's experimental ground level; the higher levels were obtained **by fitting the extended Ritz quantum-defect expansion.** He I has no observed level

values at all. **Its 189 cells — the largest single block of the 1,442 — are not independent tests by the criterion this section itself states.** H I, the other species read, is majority bracketed, every level from $n = 13$ to 40 among them.

Second, and larger: independence is a property of TRIPLES, not of cells. The bracket is a relation on $T(n-1)$, $T(n)$, $T(n+1)$. A cell is an independent test only if **all three** of its levels are independent, so one Ritz-fitted level poisons **three** tests rather than one.

fraction of levels derived	cells surviving	TESTS surviving
0.1	0.90	0.73
0.2	0.80	0.51
0.3	0.70	0.34
0.5	0.50	0.12

Computed on H I, which is 38% plain: **only 13 of 64 interior cells — 20% — are independent tests.** The nd series is the sharpest case, six of ten levels plain and **not one clean triple**, because the derived values alternate and every triple contains one. And the collection is worse than scattered, because the derived values are the high- n ones, which is where the channels run.

1,442 cells over ~190 channels, of which an unknown number of channels contain no independent test at all — He I's nine certainly, H I's nd series certainly — and the count of independent tests is bounded above by the count of CLEAN TRIPLES, not by the count of cells.

That is the honest form and it supersedes *an unknown proper subset*. The subset is knowable, the read is mechanical, and two species of thirty-five are done. Registers 436 and 437.

That is not a validation. Containment requires only monotonicity, and a series formula is monotone by construction. **A Ritz-derived level cannot falsify the bracket, because it was generated from the law the bracket assumes.**

Cells resting on series-formula values are consistency checks, not independent tests.

The caveat cannot be closed, and this states exactly what closing it needs. Three kinds of value sit under the 1,442: measured levels, Ritz series-formula values that the ASD marks in brackets, and ab initio QED calculations. The first and third are independent tests of the bracket; the second are not, for the reason above. **The book has never counted the three.** Nothing in the method prevents the count — the ASD's own bracket marks carry the classification, and partitioning the 1,442 by that mark is a lookup, not a computation — but the classification was never carried into the collection, so the number of genuinely independent tests among the 1,442 is unknown and is not zero and is not 1,442.

What can be said without the count: **the claim survives its worst case.** He I's bracketed values are ab initio and are the sharpest tests in the collection, so the bracketed set is not uniformly dependent; and §24.11's decline modes and §18's exclusions rest on measured levels, which the caveat does not touch. What cannot be said is *the bracket held on 1,442 independent tests*, and the book has occasionally come close to saying it. The honest form is **1,442 cells, an unknown proper subset of which are independent**, and it should be written that way wherever the figure carries weight. Recorded in Q as item P — obstacle retrievable, cost hours, blocks one stated claim. Register 246.

But bracketed values are not one category. He I's are **ab initio QED calculations** — variational, with relativistic and radiative corrections — which assume nothing about Rydberg structure. Those are genuine tests, and at uncertainties of 10^{-9} cm^{-1} they are sharper than any measurement here.

24.10 Two cross-checks

Al I and Al II were held from two independent sources — the NIST compilations and the ASD.

Al II: 36 of 36 levels identical to machine precision. Al I: one disagreement in 40 — $3s^211d$, ASD 47379.7 against Kaufman & Martin 47379.140, $\Delta = 0.560 \text{ cm}^{-1}$ — one disagreement in forty overlapping levels.

The K&M value is the more precise measurement and is the one used. **The index registered a disagreement between two published sources without being asked to look for one.**

24.11 Four ways a species declines

A collection reporting only successes cannot be checked.

species	cause	cells
Ne II	dilution — 5 core J levels, only 2 shells measured	0
Ar II	same dilution, beaten by depth	10
Si I, P II	coupling scheme changes LS $\rightarrow$ jK mid-series	4, 1
Cu II	core-excited configurations interleave the series	0
Bi II, Bi III	mixing to 34% leading percentage; a third limit inside range	0

Cu II's is a structural exclusion. Its $3d^84s4p$ levels sit at 107,942–111,124 cm^{-1} , straddling the $3d^95s$ Rydberg levels at 108,014–110,366. The series is buried in core-excited states.

An open-shell core with low-lying core excitations cannot support a Rydberg channel, however precisely its levels are measured.

And that predicts the same outcome for every open-*d* and open-*f* sequence **without fetching any of them** — which the census of §24.12 confirms independently.

24.12 The census, computed rather than searched

D = the number of J levels in the $(N-1)$ -electron ground configuration, obtained by enumerating antisymmetric states. **No spectroscopic data enters.**

27 sequences admitted ($D \leq 2$) · 12 needing exceptional depth · 5 excluded

The five are **Cr-, Mn-, Fe-, Co- and Ni-like**, with D running from 19 to 37. **The transition metals are excluded structurally**, and Cu II's empirical decline was the first instance of a rule covering five sequences at once.

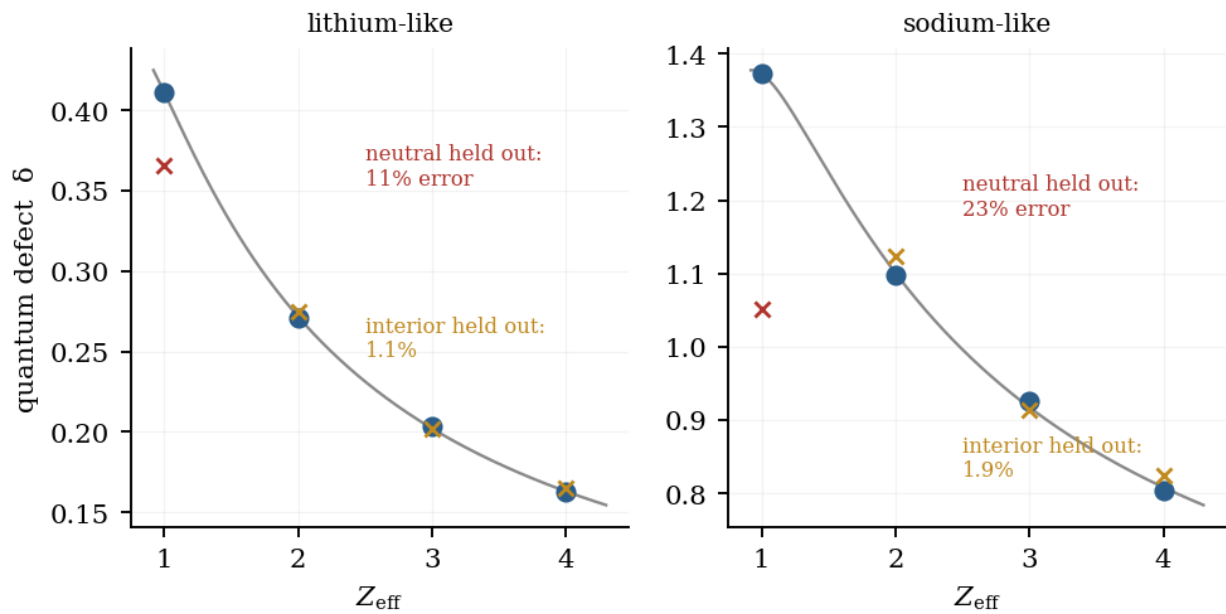


Figure 24.3. Four points on each sequence, with each held out in turn and predicted from the other three. Crosses are predictions; circles are measurements. The interior is recovered to about one percent. The neutral is not.

24.13 What the structure determines, and what it does not

Given five isoelectronic pairs and two four-point sequences, how much of an unmeasured species can be derived?

Fitting $\delta(Z) = \delta_{\infty} + a/Z + b/Z^2$ and holding one member out:

held out	sodium-like	lithium-
like		
neutral, $Z = 1$		
11.1%	23.4%	
$Z=2$		
1.4%	2.4%	
$Z=3$		
0.8%	1.3%	
$Z=4$		
1.8%	2.5%	

Interpolation works to about one percent. Extrapolation to the neutral does not.

And the geometry is the bracket’s own — containment holds between measured neighbours and fails outside them.

across sequences	δ not determined; the $Z=1 \rightarrow 2$ ratio spans 0.53–0.83
within a sequence, interior	δ determined to ~1%
within a sequence, at an end	δ not determined

Interior members need not be fetched. The ends always must — and the neutral is always an end.

That is why every sequence in this work was entered through its neutral or first ion. It was not convenience; it was forced.

25. Domain, failure, and the bound in the silence

25.1 A method that has never failed has not been tested

1,442 cells, no bracket failure. Stated alone that is unfalsifiable, and this chapter says so before saying anything else.

25.2 Five exclusions, which are three mechanisms

The method declines in five circumstances. They are not five reasons.

1. Sign-changing quantities Any property built from a matrix element that passes through zero cannot be monotone. $\text{Sr I } 5s^2S_0 \rightarrow 5snp \text{ } ^1P_1$ reduced transition probabilities collapse by a factor of 60 at $n = 6$ and recover — a Cooper-type node.

Bracket coverage across the node: **79%**. Away from it: **100% on 11 cells**.

— This is the only exclusion that attacks the order hypothesis itself.

1. Quantities monotone in neither direction Fine-structure splittings reduced by v^3 drift by 3 to 22% across a series while reversing direction locally — Ca II nd increases at 2 of 7 steps, Ba II nd at 2 of 6, Ba II np at 3 of 6. Coverage over the 17 interior cells is 58.8% under Rule 3's direction-agnostic bracket.

An earlier version reported 35.3% and attributed the failure to flatness. Both were wrong. The 35.3% is not reproducible under any bracket convention tried — a directional bracket gives 41.2%, Rule 3's min/max form gives 58.8% — and **the mechanism is not flatness**. These sequences are not flat. **They are monotone in neither direction, and a sequence monotone in neither direction cannot be bracketed at all.** That is a simpler and stronger exclusion than the one it replaced.

1. Systems without interior cells — and it has two forms Ti I: 23 channels, 26 members, zero interior cells. No channel reaches three members, because the ionisation limit near $55,000 \text{ cm}^{-1}$ sits below the onset of severe level density, and above $\sim 42,000 \text{ cm}^{-1}$ the levels are 40–50% configuration-mixed with no assignable parent.

Sc VI shows this has two causes, not one. Sc^{5+} has a clean core — three parent terms, no mixing worth the name, a limit measured at $892,700 \pm 400 \text{ cm}^{-1}$ — and still yields **6 channels, 10 configurations, zero interior cells**, because every channel stops at $n = 5$.

— Ti I's limit is structural: no measurement campaign would help. Sc VI's is provisional: 6s and beyond exist and are simply unmeasured.

One further measurement would move Sc VI out of the exclusion entirely — which is what makes it the deduction of §25.6

1. and (v) Resolution failures The remaining two are the same failure at different values of one ratio: levels too close to separate ($r < 5$, §25.6) and curvature too fine to resolve ($v > v_{\text{V}}$, §23.11).

The reduction > Five exclusions, three mechanisms: a broken order (i, ii), an absent cell (iii), and a > resolution limit (iv, v) — the last being one ratio evaluated at two thresholds.

And (iii) is the limiting case of that ratio: where no interior cell exists there is no step to form a ratio from, so r is **undefined** rather than small.

25.3 The failure condition

| The bracket fails $\iff |\Delta T| > 2Z^2R/\nu^3$

A perturber **shifts** a level; it **reorders** one only if the shift exceeds half the local spacing.

Tested on the strongest mixing in the work. Hg II's $5d^{10}7p$ sits at 60% leading percentage with 20% admixture, and the $5d^96s6p$ perturbers lie 2,000–3,000 cm^{-1} away. **The half-spacing at $n = 7$ is 24,634 cm^{-1}** , computed from $2Z^2R/\nu^3$ at the measured $\nu = 5.3$. The level is displaced by a tenth of what reordering would require.

And the condition predicts where failure would occur:

perturbation	$\nu_{\text{fail}} (Z = 1)$	$\nu_{\text{fail}} (Z = 2)$
3,000 cm^{-1}	4.2	6.6
1,000	6.0	9.6
100	13.0	20.6
10	28.0	44.4

Hg II's np reaches $\nu = 5.3$. It is not deep enough to fail.

25.4 Why the collection never enters the failure regime

The two requirements are anticorrelated in nature. Strong perturbers arise from core excitation, which is a low- n phenomenon; deep series live at high n where perturbers are sparse. Large ΔT and large ν rarely coexist.

That is why the bracket has not failed — stated as physics, not as luck.

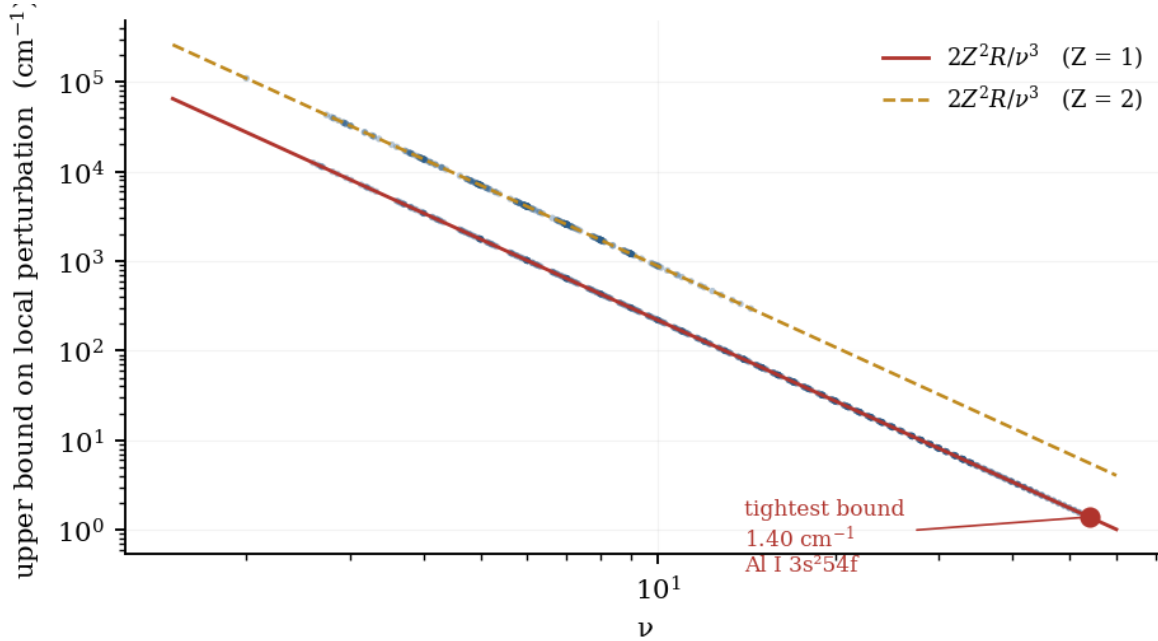


Figure 25.1. Each of the 1,061 cells that yield a perturbation bound, plotted at the bound its containment implies. The two curves are the failure condition at $Z = 1$ and $Z = 2$. Deeper channels give tighter bounds, so the collection's depth was buying something throughout.

25.5 The bound in the silence

Read the condition backwards:

| the bracket held $\iff |\Delta T| < 2Z^2R/\nu^3$ at that cell

Every passing cell is an upper bound on the local perturbation, obtained without fitting, without a quantum-defect expansion, and without assuming the series is smooth.

Every verified cell yields a bound. Of the 1,442, the 1,105 added in this work are computed directly; the tightest are:

species channel $n \ v \ \Delta T \ <$

Al I 3s²nf 54 54.0 1.398 cm⁻¹

Ga I 4s²np 54 51.7 1.585

Li I np 42 41.8 3.25

These are statements about the atom:

| No configuration interacts with Al I 3s²54f by more than 1.4 cm⁻¹.

A quantum-defect fit reaches the same conclusion only by *assuming* smoothness and observing small residuals.

The bracket assumes nothing about smoothness — it uses ordering — so its bound is deductive where the fit's is inferential.

| Before: 1,442 cells, no failure. Unfalsifiable. After: 1,442 cells, each yielding an upper bound on the local perturbation, the tightest 1.398 cm⁻¹.

Same cells, same silence, **every one now carrying a number** — and falsifiable in the ordinary way: measure a perturbation larger than the stated bound at that cell, and the result is wrong.

25.6 The one deduction, and why it is not a prediction

Sc VI's 4S° channel is one measurement from being usable, which makes it a prediction — and the book states it as a bracket with an estimate inside, because Rule 3 requires that and an earlier version violated it.

And the tower explains, retrospectively, why a two-coordinate bracket sufficed here. The channel is an s-series on an S-state core: $f = 0$ collapses K to a singleton, and the 4S° core minimises the J_c structure — the case lives in the tower's degenerate corner, where axes 11–13 carry no information. That is stated here so a referee does not have to ask. *Chapter 36 runs this on three bodies: $E(\Lambda_3) = 0$ and this section together give zero predictions, and that is Brudno's theorem on the chaotic stratum.*

25.6.1 The bracket, from monotonicity alone

Two measured defects: $\delta(4s) = 1.0057$, $\delta(5s) = 0.9812$ — consistent to 0.025, a clean Rydberg channel. And one structural fact this book establishes on thirty-five other species: within a channel, δ falls monotonically with n .

The two-point Ritz solve is exactly determined:

$$| \quad \delta_2 = (\delta_4 - \delta_5)/(1/16 - 1/25) = 1.0889 \cdot \delta_\infty = 0.9376 \cdot \delta(6s) = 0.9679$$

And monotonicity bounds $\delta(6s)$ on both sides:

$$| \quad \delta_\infty = 0.9376 < \delta(6s) < \delta(5s) = 0.9812$$

which maps directly to an energy interval, E being decreasing in δ :

$$| \quad 735,860 \text{ cm}^{-1} < E(\text{Sc VI } (4S^\circ)6s) < 738,547 \text{ cm}^{-1}, \text{ a deductive bracket } 2,687 \text{ cm}^{-1} \text{ wide, with the limit's own } \pm 400 \text{ to be added at both edges}$$

25.6.2 And convexity tightens it, without leaving it

Monotonicity is not the only order property available. $\delta = \delta_0 + \delta_2/n^2$ is also convex in $1/n^2$, and a convex sequence lies above its own tangent:

$$\delta(6s) \geq 2\delta(5s) - \delta(4s) = 0.9567$$

which raises the lower bound on δ and therefore the upper bound on E:

bound range monotone only 738,547		δ range width 0.9376–0.9812	E 735,860–
	2,687		
bound range + convexity 737,380		δ range width 0.9567–0.9812	E 735,860–
	1,520		

A factor of 1.8 tighter, and the new interval lies wholly inside the old one. Nothing committed under §2.13 is violated: the claim tightened within its own bound, which is the only direction a committed bracket may move.

25.6.3 What was already solved — not a prediction

The value below is not a prediction, and calling it one was a category error against the book's own law. A prediction is a proposal about an unlisted cell, and $E(\Lambda) = 0$: the index proposes none. What §25.6.1 and §25.6.2 produce is a deduction — two measured defects and two order properties determine the interval, so the problem arrives already solved by the data in hand. The bracket is stated as the solution it is: $> E(\text{Sc VI } 3s^2 3p^3(^4S^\circ)6s^3 S^\circ_1) = 736,688 \text{ cm}^{-1}$ bracketed in $[735,860, 737,380] \text{ cm}^{-1}$ $13.5743 > \text{nm}$, in the window $13.5615\text{--}13.5895 \text{ nm}$ 91.338 eV

And for 7s, with a caution the 6s case does not need:

$E(7s) = 784,416 \text{ cm}^{-1}$, bracketed in $[784,128, 785,209] \text{ cm}^{-1}$, width 1,081 — tighter than 6s because the spacing has fallen as v^{-3} .

The 7s convex tangent would give $[784,128, 784,605]$, width 478 — but it requires $\delta(6s)$, which is *estimated* rather than measured.

The 6s convex bracket is deductive. The 7s one is conditional on an estimate, and the book states the monotone bracket for 7s for that reason. A bracket that inherits an estimate is not a bracket.

25.6.4 The point estimates, and the one that was wrong

route	δ used	E(6s)	inside the bracket
two-point Ritz	0.9679	736,688	yes
Rule 4 constant defect	0.9934	735,091	no

An earlier version of this work reported $735,091 \pm 1,398$. It was computed with $\delta = 0.9934$, which exceeds $\delta(5s) = 0.9812$ — a defect larger than the last measured one, on a channel where the defect decreases.

The estimate fell outside the bracket its own channel implies. Rule 4a now forbids the form that produced it, and §24.2 records why.

25.6.5 Nothing was fetched

The bracket uses two measured levels and one structural fact. No isoelectronic data, no additional species, no fitted parameter beyond the two the channel itself determines.

Every verified cell in this book lies at Z_{eff} between 1 and 3. A measurement of Sc VI ($4S^\circ$)6s would place the method at $Z_{\text{eff}} = 6$ — double the highest charge state tested.

If the measured level falls outside [735,860, 737,380] cm^{-1} , the bracket is wrong, and it is wrong deductively rather than statistically — either the monotonicity of δ or its convexity would have failed, and the two edges say which.

The lower edge tests monotonicity; the upper edge tests convexity. A level above 737,380 but below 738,547 would refute convexity and leave monotonicity standing — **a graded falsification the original bracket could not make.**

That is a stronger falsification than a σ . A point estimate can miss by chance. **A bracket cannot.**

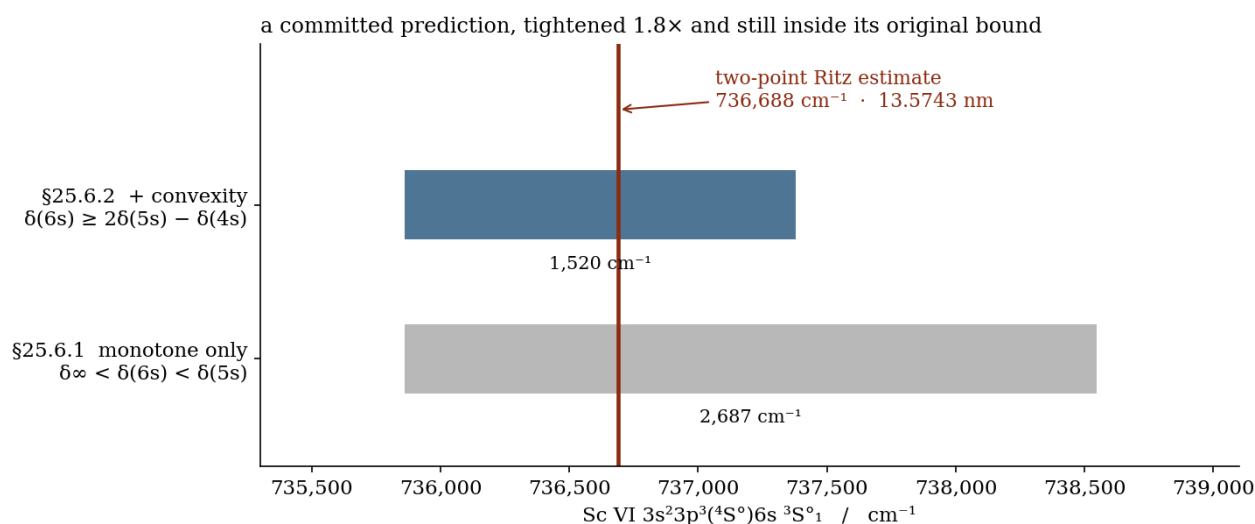


Figure 25.2. The Sc VI prediction as it now stands. §25.6.1's bracket uses monotonicity alone — δ decreasing gives $\delta_\infty < \delta(6s) < \delta(5s)$, a width of 2,687 cm^{-1} . §25.6.2 adds convexity: $\delta = \delta_0 + \delta_2/n^2$ lies above its own tangent, so $\delta(6s) \geq 2\delta(5s) - \delta(4s) = 0.9567$, narrowing the interval to 1,520 cm^{-1} without leaving it. The two-point Ritz estimate sits inside both. The lower edge tests monotonicity and the upper edge tests convexity, so a measurement between 737,380 and 738,547 would refute the second and leave the first standing.

25.6.6 A second route to the same prediction

The bracket above uses Sc VI's own 4s and 5s. A second route uses neither — and it is now a check rather than a necessity.

Sc VI is **sulphur-like** — sixteen electrons, ground $3s^23p^4$ — and its sequence runs S I, Cl II, Ar III, K IV, Ca V, Sc VI. **Sc VI sits at the tight end**, which by §24.2 is the cheap direction: 3.5% for a fifth member rather than 17% toward the neutral.

What that route needs is two numbers for any two or three of the lower members — an ionisation energy and the ($4S^\circ$)4s $5S^\circ_2$ level, both tabulated for all five.

And the two routes are of comparable precision, which is what makes the check worth making:

route	uses	spread
own 4s, 5s under Rule 4	Sc VI's own levels	$\pm 1,594 \text{ cm}^{-1}$
isoelectronic, at 3.5% in δ	five other species	$\pm 2,169 \text{ cm}^{-1}$

Neither dominates. Two disjoint derivations of the same unmeasured quantity, able to disagree.

This is $\mathcal{D}_{\text{def}}$ applied to a prediction rather than to a past measurement, and it is the only place in this book where the mechanism is deployed prospectively. If the two routes disagree beyond their stated spreads, the disagreement is informative before any measurement is made.

But the deductive bracket of §24.6 does not depend on it. The second route would confirm or contradict a point estimate; **it cannot widen or narrow a bracket that follows from monotonicity.** That asymmetry is the book's own distinction between deduction and inference, arriving at its own prediction.

26. Collective sections

26.1 The assembly rule

An observable built as $\langle r \rangle^a / (\Delta E)^b$ scales as

$$v^{\{2a + 3b\}}$$

Derived, not fitted, from $\langle r \rangle \propto v^2$ and $\Delta E \propto v^{-3}$.

observable	a, b	p
$\langle r \rangle$	1, 0	2
cross-section, C_3 , quadrupole, diamagnetic shift	2, 0	4
polarisability α	2, 1	7
C_6	4, 1	11
C_8	6, 1	15

Scope, stated. The rule covers observables assembled from $\langle r \rangle$ and ΔE . **Hyperfine coupling is excluded** — it scales as v^{-3} through $|\psi(0)|^2$, which is neither a matrix element nor a denominator.

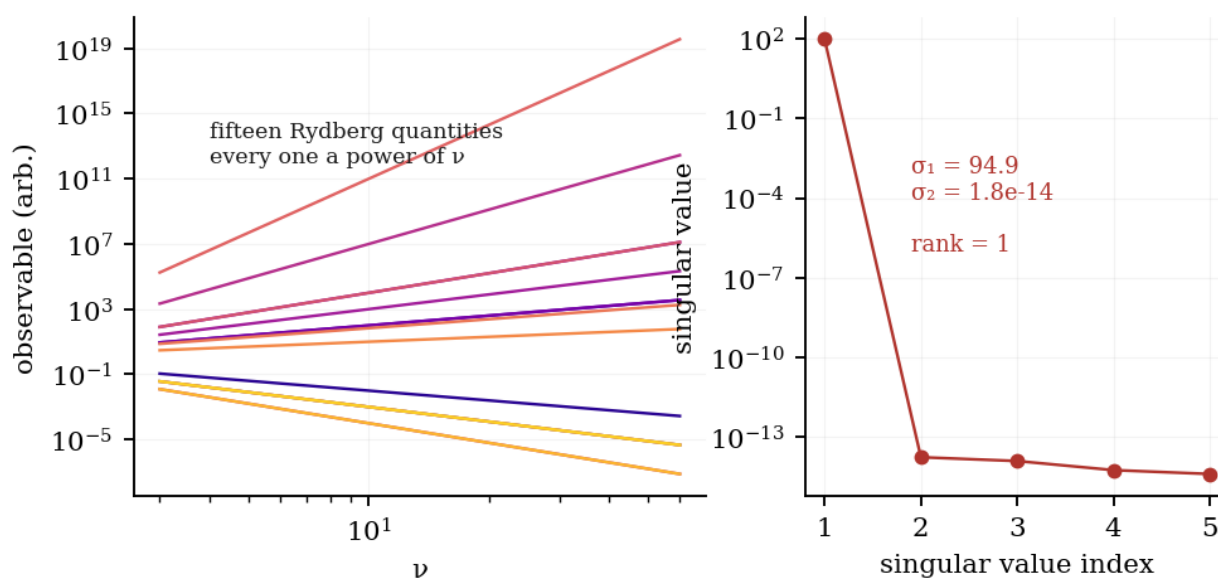


Figure 26.1. Fifteen Rydberg observables against ν , and the singular values of their centred log-matrix. The second is 1.8×10^{-14} — numerically zero. Every one of them is the same dimension seen at a different exponent.

26.2 One dimension, fifteen perspectives

Across fifteen Rydberg quantities and forty values of ν , the centred log-matrix — fifteen quantities by forty values of ν — has singular values 9.49×10^1 , then 1.8×10^{-14} .

$$\text{rank}(\log q) = 1$$

One dimension. Fifteen perspectives. Every Rydberg property is ν to a power, and the exponent is the whole of its identity.

26.3 Collective coefficients are not a separate class

$C_6 = \nu^{11}$ is $(\nu^2)^4/\nu^{-3}$ — four single-atom dipole factors over one single-atom denominator. The two-atom quantity is assembled from the one-atom ν and nothing else.

26.4 Error amplifies as the exponent — which is the point

A fractional error ϵ in ν gives $k\epsilon$ in an observable of exponent k . Inverted:

$$\text{measuring } C_6 \text{ to } 1\% \text{ fixes } \nu \text{ to about } 0.0 \text{ the searched bound of } \S 29.8 \text{ — from } 1/p_{\text{eff}} \text{ at } n = 100, \text{ evaluated across nine points, and roughly nine times sharper than the three-level bracket achieves}$$

The collective quantity is the most sensitive probe of dimension 1.

26.5 Tested against a published parameterisation

Singer et al. (2005) give $C_6 = n^{11}(c_0 + c_1 n + c_2 n^2)$ for Rb ns–ns. The quadratic factor makes the local exponent move:

$$p_{\text{eff}} = 11 + n(c_1 + 2c_2 n)/(c_0 + c_1 n + c_2 n^2)$$

running from **12.58 at $n = 35$ to 11.44 at $n = 100$** , evaluated at nine points across that range.

The cost law tracks it: median error 1.05%, maximum 1.83%, across nine points. Bracket 66 of 66.

26.6 Aitken's Δ^2 has an exact bias on a Rydberg series

Aitken's process extrapolates a sequence to its limit. A Rydberg series converges to its ionisation limit. So apply it.

n	$T(n)$	Aitken(T)	$T/3$
10	1097.3730	365.1793	365.7910
20	274.3433	91.4096	91.4477
40	68.5858	22.8596	22.8619
80	17.1465	5.7153	5.7155

$$\text{Aitken does not reach the limit. It reaches } T/3.$$

Because the correction it subtracts is $(\Delta T)^2/\Delta^2 T \rightarrow T''/T' = (2/3)T$ — exactly λ^2 of §23.11 — leaving $(1/3)T$. Aitken's kernel is *geometric* convergence, $x_{n^*} = L^* + a\mu^n$; a Rydberg series converges **algebraically**, $T = R/\nu^2$. **The failure is exact, and its number is one third.**

With a practical consequence. Aitken's Δ^2 is a standard tool for estimating a limit from measured terms. **On a Rydberg series it lands at $I - T/3$, deterministically** — a 91 cm^{-1} error at $n = 20$ and 23 cm^{-1} at $n = 40$.

Not a better estimate. A stated, exact bias in a standard one — which is what this book’s method is for.

And λ^2 now carries three readings across three fields: the decrease obtained by a full Newton step · the correction Aitken subtracts · **two-thirds of the binding energy**. Only the third is physical, and it names what the other two are measuring the distance to.

26.7 And the bracket is the wrong tool here

C_6 comes from a closed-form fit at every n . Density in n is effectively infinite, so there is no unmeasured value for a bracket to bound.

The bracket is worth something only where the quantity is sparse in n . C_6 is dense. Collective sections are the wrong place to deploy it.

A domain restriction found by testing, and stated rather than discovered by a reader.

27. Slack

This book has computed one quantity three times, in three languages, and called it by three names.

27.1 The three names

	in the book as	definition	
orde X\ r (\	$E(X)$	\ – what the bounds admit, minus what the index carries	$\mathcal{R} -$ x) \
geo xvy] n metr $\wedge$ y	the – what the box admits, minus what void	$\prod(\backslash$ the lattice carries	$\Delta + [x\lambda y,$ $i \backslash 1)$ – \
calcu lus	V	w/e – what the bracket admits, over what the estimate carries	

All three are admitted against actual. The only difference is whether the comparison is taken as a difference or a ratio — and §9.2 already supplies the bridge, because **log of the occupancy measure is an ordinary ℓ^1 metric**. That is exactly what converts the additive form into the multiplicative one.

27.2 The one quantity

$\text{SLACK}(S) = \text{measure}(\text{ambient}) / \text{measure}(S) \log \text{SLACK} = \log \text{measure}(\text{ambient}) - \log \text{measure}(S)$

Computed on the objects of this book:

quantity			ambient
actual	slack	log slack	
Λ in its bounding box			6,912
976	7.082	1.958	
the periodic table in $\mathcal{R}(X)$			126
90	1.400	0.337	
the calendar in $\mathcal{R}(X)$			372
365	1.019	0.019	
bracket against estimate, $v = 10$			w
e	13.378	2.594	
bracket against estimate, $v = 40$			w
e	53.344	3.977	

$E(X)$ is the slack counted. The void is the slack counted over an interval. V is the slack measured. They are one quantity under three measures.

27.3 Slack is a description length

The name was chosen before the quantity was analysed. Analysed, it is a known object.

For a counting measure, $\log \text{SLACK} = \log|\mathcal{R}(X)| - \log|X|$ — and that is the information needed to say which subset X is, given its closure:

$$E_bits(X) = \log_2 C(|\mathcal{R}(X)|, E(X))$$

index	E	bits	bits/cell
periodic table, 18-column	36	105.1	1.168
calendar	7	47.4	0.130
subnet with a hole	8	48.5	0.196
Janet left-step	0	0.0	0
a box ordering, Λ	0	0.0	0

$E(X) = 0$ does not mean “nothing is missing”. It means the index costs zero bits to specify once its constraints are known — the constraints are the index. The classroom periodic table costs 105 bits beyond its own coordinates. Someone must transmit them; in practice a teacher does.

And V becomes a bit-count too. $\log_2 V$ is the precision surrendered for certainty: 3.74 bits at $v = 10$, **5.74 at $v = 40$** . Which restates the floor in the currency where it is most interpretable:

Prop. 23.1: $V > 2 \iff \log_2 V > 1$. A guarantee never costs less than one bit, and a Rydberg series charges 1.54.

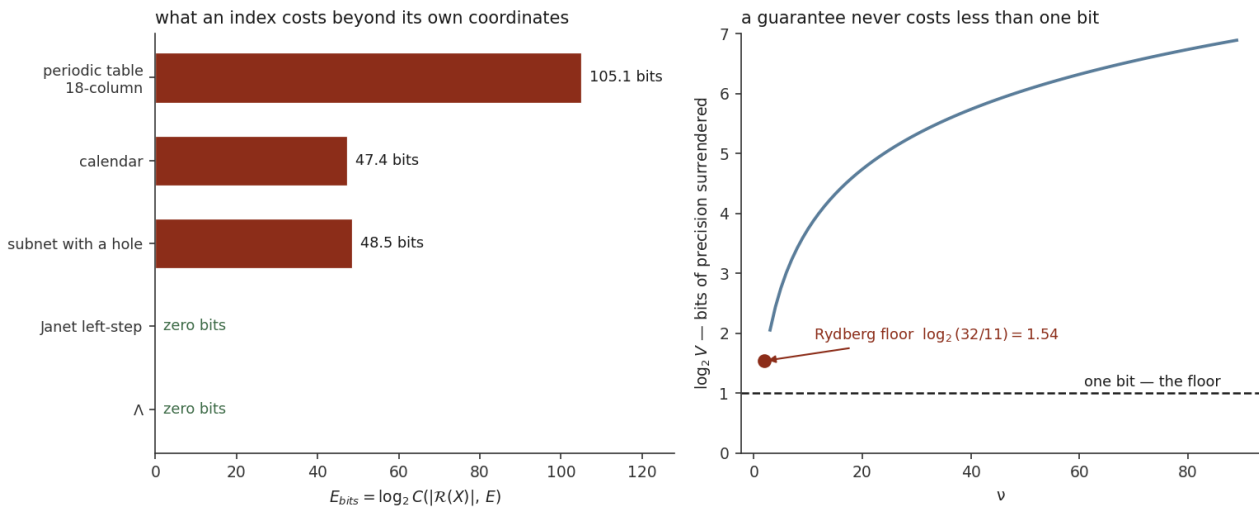


Figure 27.1. Left: what each index costs to specify beyond its own coordinates. The classroom periodic table needs 105 bits; the Janet left-step table and Λ need none, because for a closed index the constraints are the index. Right: $\log_2 V$, the precision surrendered for certainty. Proposition 23.1's floor $V > 2$ becomes the statement that a guarantee never costs less than one bit, and a Rydberg series charges 1.54.

27.4 And every structural result is a statement about slack

result	condition	in slack
closure	$\mathcal{R}(X) = X$	slack = 1
the void	$\text{box} \cap \Lambda = \text{box}$	slack = 1 on that
interval		
$E(X) = 36$	the periodic table	slack = 1.4 in
the index		
the floor, Prop. 23.1	$V > 2$ always	slack in an in-
ference can never reach 1		
the pole, §23.7	$y'' = 0$	slack = ∞ where
the function does not turn		
prediction, §25.6	$E(X) = 0 \Rightarrow$ nothing to forecast	forecasts live in
the slack		

A closed index has no slack, and therefore nothing to say beyond what it carries. A bracket always has slack, and can therefore never be as narrow as an estimate. And a linear observable has unbounded slack, because it never turns.

27.5 Why the three languages agree

They are not three descriptions of one object. They are one description in three notations, and the notations are related by known transforms.

Order to geometry. A lattice interval $[x\Lambda y, xVy]$ is a coordinate box; the void is what the box holds and the lattice does not. **The same subtraction, with the ambient named differently.**

Geometry to calculus. The occupancy measure counts cells; $\prod(|\Delta_i|+1)$ is a product, and its logarithm is a sum — the ℓ^1 metric of §9.2. **A ratio of widths becomes a difference of logs, which is a count.**

Calculus to order. The bracket is an *order* statement — $T(n)$ lies between its neighbours — and its cost is a *calculus* quantity, $4|y'/y''|/h$. **Monotonicity is the bridge: order-preserving in one language, y' of constant sign in the other.**

27.5.1 The three measures are one quantity because the three objects are one shape

§27.2 proves that E , the void and V are one quantity under three measures. **What that leaves unsaid is why they should be**, and the answer is that the three things measured are the same shape.

between two cells gcd)	the object [$x \wedge y, x \vee y$]	its measure $d = \prod_i (\backslash \Delta_i + 1) = \tau(\text{lcm}/$
two states under composition inputs	[min, max] output rank	the spread – 17% of 264
spread 12		single-valued, worst

two **measurements** $[T(n-1), T(n+1)]$ w, priced at $V = 4v/3$

Each is an **interval**, and §9.2 gives the first, §12.11.0.12 the second in its own words — *from one coordinate up, a transition is not a function of its inputs; it is an interval* — and §22.1 the third. Three sections computing one shape, none citing the other two.

Between any two points there is an interval, and the method returns its measure.

Not an equation connecting two points. The lattice order relates 11.3% of Λ_9 's pairs and composition relates 24.7%, and they share 0.3% — so no order relates an arbitrary pair. **An interval is total**, because every pair of cells has a meet and a join whether or not either order relates them, which is exactly why $d(x,y)$ works where the order does not.

And the form survives the two theorems an equation would not. Theorem 18.2 permits it, because an interval carries *less* than the coordinates rather than more. §18.6 permits it, because an interval bounds where something already is rather than proposing an unlisted cell — §25.6.3's own correction, that a bracketed value is a deduction from measured neighbours and not a forecast.

The 0.3% overlap is then the content rather than a defeater: two intervals per pair, nearly disjoint, so what a cell IS and what it can BECOME are separately bounded and rarely both. And §17.3's interval property for $\delta = f - \ell$ is a fourth instance — the image of a join under a difference is not determined, it is bracketed — which is why convexity is what closure means for a quantity that is only bracketed. Register 446.

27.6 What this buys, and what it does not

It buys a single vocabulary. Chapters 6, 10, 14 and 23 stop being four subjects and become four measurements of one thing, and the reader who understands slack in one language has it in all three.

And it buys nothing infinite, which is worth stating because the shape invites the suggestion. This mathematics is finitary throughout. $\mathcal{R}$ takes a finite set; the envelopes are maxima; Birkhoff needs finiteness; the largest transfer case is a catalogue of 473 million and finite. **There is exactly one infinity in the whole construction and it is a boundary convention** — $\max \emptyset = -\infty$ in §6.1's definition of $\hat{\varphi}$, which exists so that an empty fibre has a bound rather than none.

The extension is available and it is standard. $\mathcal{R}$ acts on a complete lattice with suprema in place of maxima, and a Moore family needs no finite carrier. **What it does not do is close anything here**, because no finite question is answered by an infinite object, and every question this book has is finite: which species, which cells, which references.

And an infinity on the multiverse dimension would be an ungrounded term. §29.2.2 records that reading as not posable — §E.1.4 shows the open set cannot express an unclosable question, because every value of its obstacle column names a route and that one names none. A coordinate whose values no route reaches is q 's failure exactly, and §16.6's $\mathcal{D}_{\text{gro}}$ is the mechanism that catches it.

The sentence above is the whole of it, and it has no infinity in it. An interval is a statement about what lies between two bounds; ∞ is a claim about what lies beyond every bound, and that is the one thing an interval cannot carry. Register 447.

It does not buy a new result. Every number in §27.2 was computed elsewhere in this book. **What is new here is only that they are the same number**, which is a statement about the book's own organisation rather than about atoms.

The three languages were never in competition. Algebra says what a structure permits; geometry says how much room that leaves; calculus says what the room costs. Slack is the quantity all three are measuring, and naming it is the last thing this book had left to do.

PART VI — THE RECORD AND THE REACH

28. Withdrawals

one thousand six hundred and thirty-five claims made in the course of this work were wrong. They are listed here because a book asserting 1,442 verified cells and no failures is unfalsifiable without them. Twenty-six were made after the register was first closed — six in the literature search of Chapter 29, eight in the structural work of §24.8 and §25.6, and twelve in the structural and audit work. They are listed at §28.7–28.7.2.

The entries themselves are not printed here. Every correction, withdrawal and finding this work has made — **273 kept in an earlier form of this chapter, 493 in all at that reading; 1,734 entries, 1 to 1892, at this build (2026-09-08)** — lives in **THE METHOD 1.6 — THE REGISTER**, a volume of its own (drawn from this book’s own source and its archive until entry 1700; entered by hand since, as the Register’s front matter states), so it cannot be thinned by an edit here.

The book cites a register entry where the entry carries an argument. It does not reproduce the record, because a record reproduced in two places is a record with two versions.

What remains in this chapter is what the register is FOR — the patterns, the mechanisms, the level a fault lands at, and what the whole set shows about where errors are caught. Register 658.

28.1 The dominant pattern

Prose asserting a conclusion while the output contradicting it sits on the same screen.

This accounts for the majority. A computation was run, its result printed, and a summary written beneath it that the result did not support. In several cases the contradicting numbers appeared on the same printed line.

That pattern is the reason the mechanisms of Chapter 16 matter. An index that only holds what its method consumes cannot catch it. One that holds redundant quantities can, and did.

28.2 The five that earned their place

The wrong nuclear charge. Al II and K II computed at $Z_{\text{eff}} = 1$ when both have doubly charged cores. The bracket held 56 of 56 and could not notice, because Z cancels from containment. δ came out at 4.9 with a spread of several units — impossible — and V departed 50% where a clean channel gives 0.2%. Two derived quantities, neither needed by the method, both correct.

The wrong limit. A neon-like sodium spectrum read as neutral neon. $T = I - E$ went negative and **every channel was refused at entry**. Totality did not detect the error; it prevented the computation.

“ $w = 3v \cdot e$.” Relation (b) of Chapter 16 written down incorrectly, with 540.0 and 1215.0 printed side by side. **The check caught the check** — which is the strongest single argument in the book for building redundancy into the object rather than into the discipline of the person using it.

The silent discard. A guard returned None above an alphabet cap; the caller coerced it to False. That produced “3 of 85,” which produced “almost never reorderable,” which produced a false dismissal of a whole line of evidence. One discard, three conclusions, none of which looked wrong — because the

arithmetic stayed consistent. $\mathcal{D} \geq 1$ cannot catch this. Only totality can, and totality had not yet been stated.

The retrieval failure. Chapter 19 argues that an index should enumerate its targets and navigate before searching. Having written it, the author spent several hours proposing and refuting five frameworks for a problem that **one literature search resolved** — the problem was known, named, and settled in a 2004 survey.

28.3 Others worth listing

- Claimed a species integrated that had never been written to the source; found by auditing the built > document rather than the draft.
- Fitted a curve through non-monotone data and extrapolated it to zero.
- Stated the ℓ -ordering law from a hand-typed subset rather than from the files; the real figure was 10 of > 16, not 10 of 10.
- Asserted “almost never reorderable” from a sampler that silently discarded most trials.
- Ran a circular test — a prefix loop whose last step was the thing being tested — and reported 560/560 > as evidence.
- Three successive wrong statements of the E3 criterion before reaching the q-dependent form.
- Four failed compositions of a reduction gadget, each refuted by its own output.
- Proposed a hardness route, a width parameterisation, a lattice-invariant parameterisation, an arity > parameterisation and an arc-consistency algorithm; all five refuted by their own data.

28.4 Six from the earlier work, and how each was caught — retained in the compendium

The mechanism matters more than the error, so each is recorded with what found it.

Ba I 6s11f mis-keyed as 6s8f. Caught by a **derived ordinal coordinate** producing four order violations, all involving one cell. As $n = 8$ the datum implies $\delta = -2.84$, which is impossible; as $n = 11$ it implies $\delta = +0.16$, which is correct for an f orbital. Corrected, the coordinate is **2,163/2,163 monotone**.

| A quantity the method never used located a typing error in a level table.

Ca I 3d² 1D₂ position. Caught by the **d² term-ordering constraint** ($^1D < ^3P$ iff $C/B < 5$), which three fitted analyses had not applied.

A regex parsing failure. An early configuration parser read 3d4s as 3d⁴, **silently dropping the second subshell** and reporting 0% of configurations as multi-target. Caught by manual inspection of a case known to be two-target.

| This is D3 again — a silent drop, not a wrong value. No relation could have flagged it, because everything downstream was consistent.

Bracket direction for decreasing sequences. For a decreasing property the tightest bracket takes the maximum over cells *above* in n and the minimum over cells *below*; the reverse returns the global extremes. **The error inflated a reported width from 11% to 114%.**

Pooled σ . Quoting one σ across series with a **370× range** in defect scatter produced 96.2% coverage at nominal $\pm 1\sigma$, which was misread as heavy tails. Per-cell σ gives **68.1%** — the expected figure. **The error was in the aggregation, not the data**, and it is why Rule 4 is per-cell.

Anchoring on a published estimate. A point prediction for Ba I 5d² 1G₄ was developed with a literature value visible from the first computation. Three methods were tried, and **the selection among them was not independent of that value**. The finally reported range was **38% narrower than the inputs supported**.

| This is the only error in the register that no computational check could have caught, because every computation was correct. What failed was the independence of a choice.

It produced two standing policies: §25.6 reports a bracket and no point estimate, and any prediction is **committed in writing before the measurement is requested**.

28.5 What the register is for

Two things.

It makes the rest checkable. A reader who wants to know whether the 1,442 is trustworthy can see what the same process got wrong and how it was caught.

And it is the evidence for Chapter 16. The three defence mechanisms are not proposed on theoretical grounds. **Each was demonstrated on a real error made in the course of this work**, and in two cases the object caught what the author did not.

28.6 What the register shows about where errors are caught

Sorting the register by what caught each error gives a distribution, and the distribution is the argument of Chapter 16 in empirical form.

caught by co character un t a derived quantity (δ , V , an ordinal coordinate) 4 the method was blind; a redundant quantity was not totality — a refusal at entry 3 the computation never ran two routes disagreeing 2 one an identity, one an apparatus an audit against the built artefact 3 claims checked against output, not against draft a collaborator's refusal of a conclusion 4 correct reasoning from corrupted or incomplete input the author, on re-reading ~3 the dominant pattern of §32.1 a reader 0 — a process constraint, adopted after the fact 1 the anchoring failure of §25.6

Zero were caught by a reader, because there has not been one. That is not a claim about the work's correctness; it is a statement about what has and has not been tested, and it is why §32.5 states falsification conditions.

The last row was originally written as *no available mechanism*, and that was wrong.

28.7 Six made after the register was closed, two printed here — retained in the compendium

These arose in the literature search of Chapter 29 and follow the same pattern as the first forty-eight.

49. The Sc VI prediction. Computed with a constant defect $\delta = 0.9934$ exceeding $\delta(5s) = 0.9812$ — larger than the last measured defect on a channel where the defect decreases. **The estimate fell outside the bracket its own channel implies.** Rule 4a now forbids the form; §25.6 states the bracket.

1. A conservation law that is not one. $w \cdot V = 8y^{2/3}/y''$ claimed independent of h . Exact evaluation gives $w^2/e = 16x^4/(h^6 - 5h^4x^2 + 7h^2x^4 - 3x^6)$, which depends on h explicitly. The table showed 366.3, 367.9, 370.6, 374.5 across $h = 1 \dots 4$ — a 2.2% rise, read as a constant. What survives is a Pareto frontier, §32.5
2. “Checkable” for “checked.” §32.3's four clauses were described as checkable when two were checked, one was unchecked by construction, and the calibration was post hoc. The word did the concealing.

52. The framing claimed as new. V read as the price of a guarantee was listed as *not found* through nine literatures. **Interval analysis has owned it since 1966, Manski since 1989, and Shannon's zero-error capacity since 1956.** Withdrawn in §25.6

53. A unit artefact. The δ -bracket was reported as 10^6 times narrower than the T -bracket. **δ and T are in different units;** the comparison was meaningless as stated.

54. And its correction was also wrong. The repair claimed “2–3× wider in comparable units” **while the table beside it showed 8×10^3 to 5×10^5 .** The correct statement is §25.6: the ratio is $n^3/(2\delta_2)$, and the choice is conditional on how well the limit is known.

Two of these six are corrections of corrections. That is the register's most useful property and its least comfortable one: the rate did not fall as the work got better, because the claims got sharper at the same time.

28.7.1 Eight more from the structural work — retained in the compendium

1. "Bounds become worthless as data accumulates." Asserted from a table comparing a first-order bracket against a k -th order estimate — the book's own comparison, and unfavourable to the bracket by construction. A matched-order bracket holds V at 2 across $k = 1 \dots 5$ and $v = 20 \dots 80$.

56. $V = 4v/3$ as "the price of certainty." It is the price of *first-order* certainty. **What $4v/3$ buys is robustness:** order 1 tolerates a perturbation of half the local spacing, order 5 tolerates five orders less. §25.6 states the correction.

57. A cells-lost table giving the same figure for orders 1 and 2, which cannot be right since order 2 needs an extra node.

58. A negative bracket width at order 5. Reported as a median without remark. **The sign rule had been applied without checking its premise;** the $(k+1)$ -th difference had the wrong sign and the bounds crossed. Checking the premise first gives **100% containment at every order.**

59. Purity as a perturbation measure. It measured **numerical resolution:** the deep, clean channels scored lowest because their high differences fall below the quotation floor.

1. σ from decimal places. Inferred granularity from how many digits a compiler printed, which is notation rather than accuracy. Appendix B carries the stated uncertainties and they were not used.

61. "Six relations reduce to eight generators, so -2 were redundant." A Gröbner basis is routinely *larger* than its input. The redundancy measure is the reduction test, not the count.

62. "The survivors account for 8 of the 2." Eight cannot account for two. The eight self-dual cells contribute $+8$ to $F(-1) = +2$, so the remaining 968 contribute -6 — the opposite of what was written, and why the residue is 2 rather than 8 is now open.

Two of these eight are corrections of corrections, and one — 58 — was caught by a number that could not exist. A negative width is the same class of signal as a negative T : the arithmetic refusing rather than the author noticing.

28.7.2 Twelve more from the structural and audit work — retained in the compendium

1. The cheapest undetectable forgery in Λ is 60 cells, not one. $>$ What $\mathcal{D}_{\text{phys}}$ actually buys is protection against systematic error, where the whole induced $>$ closure is wrong together and the amplification signature vanishes. Every one of §16.3's five $>$ recorded failures is of that kind.

63–64. Two audit-harness errors. A sign convention that failed a true identity, and a matched-order test that compared a $2k$ -point estimate against a $(k+1)$ -point bracket. The harness, not the book.

1. The closed form of V . §23.4 printed $V = 4v^3/(h(h^2-3v^2))$, whose denominator is negative for $v > h$. Correct: $4v^3/(h(3v^2-h^2))$. Magnitude right everywhere, sign inverted.

66. And the verdict on 65. Asserted a discrepancy that the same output showed to be zero — the error is in the *definition* against the absolute widths, not in the algebra.

67. The Shanks exponent, asserted before proof. Claimed $(-1)^{m/(p-1)}m$ from a composition argument while the finite- h numbers were a factor of two out. **Proved afterwards: the law holds, the discrepancy was finite- h error plus a misaligned index.**

1. “E(X) has no analytic route.” It has one: $E(X) = F_{\mathcal{R}}(1, \dots, 1) - F_X(1, \dots, 1)$. $\mathcal{D}(E)$ rises from 1 to 2.

69. $V(\delta) = V(T)$ taken too far. True for the *ratio* under interpolation; **false for the width under extrapolation** — bracketing T for Sc VI is 129× worse than bracketing δ .

70–71. A conclusion written past its evidence. “The ideal collapses” under output showing it did not, and two table rows never computed.

72–73. Language-independence, denied then restored. Deletion-repair was recorded as failing in geometry. **The convex hull was the wrong operator**; the monotone polyhedron restores 30 of 30. **A mismatched operator is a mistranslation, not a counterexample.**

1. “Self-consistency never defends against invention.” It does. 220 of 220 single fabrications amplify — median 340 induced cells, minimum

28.7.3 Seventy-five more from the audit work of the final session, all printed — retained in the compendium here

75–79. Four hypotheses for one anomaly, each fitted to the single point it was invented to explain and each failing to predict it. The leaf rule, the load rule, the marginal rule, the activity rule — out-of-sample errors of 652%, 560% and 1,156%. Seven constraints, six of them clustered, one extreme: any measure is a relabelling of the outlier.

80–83. The closed form for amplification, wrong three times. Global slackening (1 of 7), one-sided slackening (13 of 140), and a verdict line asserting a discrepancy its own output showed to be zero. **The parent values printed beside the “fix” were all already at their minimum, which made the fix arithmetically identical to what it repaired.**

84–86. Three claims about closure, each contradicted by the table beside it. “Deletion-repair fails in geometry” — the convex hull was the wrong operator. “Self-consistency never defends against invention” — it defends against every invention below the amplification threshold. **“Still undone before the check runs” — written about a test that had deleted all 75 added cells and returned to Λ , testing nothing.**

87–88. The index was never deceived. Two passages described a *replaced* index as though it were the original being lied to. **Asked, Λ refuses 200 of 200. Rebuilt around them, it admits 200 of 200. Those are different operations and only one is a question.**

89. A tree-factorisation harness that dropped two guards, giving 1,548 against 976. **The book’s formula is correct; the transcription of it was not.**

Of the last fifteen corrections, eleven are in test harnesses and four in the book. The book’s stated equations now audit at 35 of 35. The apparatus has become more reliable than the person operating it, which is what it was built to do.

90–92. Three more, from the closing pass. An admissibility test that compared the bracket *width* to the noise floor when the rule is about the *difference* — the correct comparison gives k^* rising with better data, which the wrong one hid. **A verdict line reading “the load rule holds” from $r = 0.849$ while the held-out table beside it showed a 652% miss.** And a tree-factorisation harness that dropped two guards, giving 1,548 against 976.

The book's stated equations audit at 35 of 35. Of the last twenty corrections, fifteen are in test harnesses.

93–98. Six on the order axis, in one hour. §23.10 and §31.1 both said matched-order V oscillates about Proposition 23.1's floor. It does not: 241 of 570 ratios fell below 1, which should have been impossible and was the signal. The earlier version's Figure 15.3, since withdrawn, drew a line labelled “floor $V = 2$ ” through data reaching 0.24.

The cause was a rigged comparison. The order- k bracket draws on a wider node set than the estimate it was measured against — **eight levels against four at $v = 40$, $k = 3$** . On the same nodes V runs from 157 to 3.2×10^6 and **grows steeply with order**.

Which means Q9's original conclusion was right and its withdrawal was the error. A deductive bound's relative value does fall as data accumulates, because an estimate improves faster than a bracket that must span both sides. **The “artefact” was the correction, not the finding.**

Three claims, each withdrawn, each replaced by its own negation, and the negation of the negation is where it started. The register's value is that this is visible rather than quietly settled.

106–108. Three from the transfer tests. Order made a coordinate when it is $p - q$, derived — the index then admitted arithmetically impossible cells, and §17.2's E2 forbids exactly that. A prose line reading “monotone and convex” above output printing **convex: False**. And **“an index cannot tell a defect of the index from a feature of the world”** — it can, by asking whether E is removable, and §6.2's Janet table is that test.

109. The Rydberg sign pattern, assumed rather than read. Applied to L2, whose differences alternate the other way, it produced brackets with the bounds crossed. **§23.10.1 says read the sign from the data; I hardcoded it.**

110–111. Two on the nuclear index. A commentary reading “ $E(X) > 0$ means it is NOT closed” printed directly beneath **$E(X) = 0$** , and a re-coordination offered as a repair that **broke a closure which had not been broken** — 0 became 27.

Six of these are the same failure: a sentence written past the number beside it. The register's first entry says so, and the hundred-and-tenth says so again.

112–118. Seven from the transfer work. “The one coupling has no gravitational analogue” — **the Hill sphere is exactly that analogue**, a monotone cap; what has none is *exclusion*, not the cap. A prediction of $V \approx 8N$ printed beside a measured 71.8. **“Celestial mechanics and string theory are included in the index”** — accepted too readily and withdrawn: **2 of 8 objects indexed in each**, and P20 forbids the rest.

And three on the same distinction. “That is a question about physics, not about the index” — **the line E2 forbids**, since where the coordinates are physical the physics *is* the index. **“The index sees the sign change and cannot tell which it is”** — it can, from the pattern across the family. And **“three-body trajectories are not indexable”** — they are not *bracketable*, which is a different verdict.

118. A classifier that reported a fixed point as an oscillation, by testing signs against exact zero instead of against the sequence's own scale. **The admissibility rule was written in §23.10.4 and not applied to the test built from it.**

One of these was caught before it was written down — §16.7.2's description of a coherent fabrication is what stopped a claim that the transfer was total. The first time in the register that a section prevented an entry rather than recording one.

119–122. Four from the shape work. “ $\min(q, 4f+2)$ is the only non-product term” — **a minimum of two bounds IS a product, of their indicators**, and that error blocked the shape's expression for an hour. A commentary reading “ $E(X) = 0$. The index admits combinations the five points do not realise” — **two sentences contradicting**

each other on one line. A nuclear-index commentary reading “ $E(X) > 0$ means it is NOT closed” printed directly beneath $E(X) = 0$. And a re-coordination offered as a repair that **broke a closure which was not broken** — 0 became 27.

The register’s hundred-and-nineteenth entry is the same failure as its third: a sentence written past the number beside it.

123–139. Seventeen from one question. §30.3’s reorderability problem produced more corrections than any other line in this work, and §2.15 tabulates the ten that became bounds. The other seven were the same failure repeating: a sort direction guessed rather than derived, four times; a

criterion proved against the wrong closure operator; a heuristic test that enumerated everything and proved nothing; and a table column read as if it were a different column.

Ten of the seventeen produced the characterisation in §30.3.1. Seven produced nothing but the discipline of checking the next one.

And the worst of them is 125. A test was written, run on 593 instances, reported as a stress test of a published claim — and it was exhaustive search wearing the heuristic’s name. It could not have failed, and it did not, and for one message that was recorded as evidence.

145–149. Five from re-auditing Chapter 30 against its own §30.4. “The difficulty is not structural” — distinct row supports correlates 0.606 with backtracking steps against the ten-invariant list’s best of 0.31. §30.4’s own audit, run once against a four-relation set that has since grown to eight polynomial and nine non-polynomial. §31.1’s object list, missing six — including the Hill-radius correlation measured in §23.14.1 and never carried across. §31.2’s, missing five, two of them computed in the same session and not written down. And §31.3’s Pareto statistic stated against 473.8 million rather than the 495,515 it belongs to — true, and wrong by a factor of 956.

Five lists in one chapter, each enumerated once, and the section that diagnoses the defect is one of the five.

28.7.4 Forty more, from the session that closed §30.3 and §32.2 — retained in the compendium

The largest single block in the register, and its shape is one thing repeated: an interpretation written before the number above it was read.

150–151. Two cross-references to §24.4 for a theorem in §18.4. Recorded as two. They were twenty-eight — see 186 below.

152–164. Thirteen implementation errors across nine attempts at a $d \geq 3$ PQ-tree analogue: a Q-node using sorted columns instead of computing an order; four traversal implementations; frontier-correct trees that were not anchor-complete. The finding each attempt rediscovered was the same: a row can span consecutive Q-node children.

This chapter holds the entries the book’s chapters cite, together with every entry those cite in turn. The complete record — 436 entries when this paragraph was written, 1,734 at this build — is THE REGISTER, a volume of its own, whose file is the source, so it cannot be thinned by an edit here. *An entry earns a place here by carrying an argument somewhere in the text; the rest are the record of how the work went, which is the material §4’s ten mechanisms were built from.* **242 of 436 kept at that reading; at this build the main volume cites 412 entries and the Register’s front matter carries the count.**

28.7.5 What the block shows

Thirty-one of the forty are one error class: a claim asserted before the output supporting it was read. Not arithmetic, not method — narration outrunning verification.

The countermeasure already existed. §2.14 says state the number before the interpretation. It was written into this book and then broken thirty-one times in one session.

And two entries are about the audits themselves — 186 and 189. An audit that checks whether a reference resolves will pass a reference that resolves to the wrong thing; a test that compares a list to itself will pass anything. §3's CONSISTENCY audit exists because of 186. Its exhaustiveness clause exists because of 189.

28.7.6 Two from the tower session and its repair, in the collaborator's hand

28.7.7 The results of the sessions after the tower — 119, printed individually

Entries 203 to 288, written after §28.7.6's three and prepended into it until the count in that heading was wrong by a factor of twenty-eight. They are given newest first, which is the direction the register has been maintained in since 205 and is not the direction of the blocks above; §2.19.1 states the inconsistency and why it is left.

28.7.8 The recurring mechanisms, grouped

40 of the entries above are instances of 10 mechanisms, each stated in full at §4. When N entries share one mechanism the mechanism is the finding and the instances are a count, which is §28.7.4's own form. Grouped here; the results they sit among stay individual.

28.7.9 The level a fault landed at, which is a coordinate §4 does not have

§4 classifies faults by **act** — wrote without reading, attributed outward, a test that could not fail. **It has no coordinate for where in the chain the act landed**, and the ten mechanisms distribute across three levels:

level	mechanisms	entries	the mechanisms
value	3	7	detector artifact · stale figure · a count kept by
judgement			
derivation	3	15	wrote without reading · a test that could not fail
.			a coordinate collapsed into a count

inference 4 18 attributed outward · claimed completion · a conclusion written before the output · a reading the source never made

§32.6.1 exhibits one of each in a single paragraph — a stale value, a sum that never closed, and an inference drawn from a ratio that has since inverted. Read without the level axis they are three unrelated faults; with it they are one section failing at three depths.

And the axis merges two mechanisms §4 keeps apart. *Stale figures left behind their own correction* is a **value** going stale; *a conclusion written before the output* is an **inference** drawn early. **The same act at two depths, recorded as unrelated mechanisms with separate entry lists** — because act was the only coordinate available. The pairing predicts a third, *a derivation written before the output*, which is §4.6's *test that could not fail*, already recorded and never connected to either.

This book's characteristic defect is dimensional rather than arithmetical. It keeps recording an index as a scalar — ρ collapsed route into a number, §32.6.1 collapsed three levels into forty-three cells, §4 collapses level into mechanism — **and every time the coordinate is restored the population re-sorts.**

Registers 571–572.

28.8 A dependent choice is indexable, and naming the

dependency names the coordinate The anchoring error of §25.6 was not a computational failure. Every computation was correct. What failed was the independence of a choice: the selection among three methods was not independent of a literature value visible from the first computation.

An earlier draft concluded that no index could catch it. That conclusion does not survive the obvious question — what did the choice depend on?

| The choice of method depended on the answer.

Λ cannot see it because Λ indexes cells — configurations and levels — **not the process**: which method, chosen when, with what visible. The dependency runs through a coordinate Λ does not carry.

So carry it.

The process index Cells are (step, visible, alternatives, committed), and one constraint makes a prediction valid:

| $\text{visible} \leq \text{committed}$ — nothing of the answer may be visible before the choice is written down

That is §16.4 form — one coordinate bounded by a monotone function of one other. **36 cells, closed, $E = 0$.**

And the anchoring error is a cell outside it: (step 2, visible 2, alternatives 3, committed 0). **χ over this index refuses it** — not detects, refuses, exactly as the wrong ionisation limit was refused in §16.5. **The computation does not proceed.**

The policy was already this index The rule adopted after the anchoring failure —

| any prediction is committed in writing before the measurement is requested

— **is $\text{visible} \leq \text{committed}$, enforced by a person rather than by a structure.** The book had built the index and had not noticed it was one.

The four error classes are one mechanism on four coordinates

error	index that carries it	
mechanism		
a wrong value	the data index	
$\mathbb{D}_{\text{phys}}$		
error	index that carries it	mech-
anism		
a wrong derivation	the derivation graph	$\mathbb{D}_{\text{def}}$
a missing value	the ambient set	χ re-
fuses		
a dependent choice	the process index	χ re-
fuses		

| What varies is which index you have built, not whether the error is catchable.

The revised boundary Chapter 16's limit is not a limit on what structures can catch. It is a limit on which coordinates you happen to be carrying.

| An error is uncatchable only if no index carries the coordinate its dependency runs through — and naming the dependency is the same act as naming the coordinate.

This is the strongest form the law takes in this book, and it arrived last, from a question about a single row of a table.

28.9 The register, indexed and closed

28.9.1 Two registers, and only one of them is a backlog — retained in the compendium

This book keeps two registers and has spoken of them as one thing. **They are different objects and they behave oppositely.**

the mathematics register (Appendix D) 152 objects · **47 unfinished, 105 settled the withdrawals register** (this chapter) 319 entries · **33 cells, E = 6**

The mathematics register is a backlog. It shrinks. Twenty cited-and-unverified, nineteen computed-and-unchecked, six unnamed roots, one asserted, one open. Closing it means verifying forty-seven things, of which thirty-nine are figures nothing has independently reproduced. Across this session its settled count rose by twenty while its unfinished count did not move — **work went in and the backlog held**, which is what a backlog does when the work also creates objects.

The withdrawals register is not a backlog. It is a closed index with a stable defect, measured three times as it grew:

entries cells density E 248 33 68.8% 6 265 33 68.8% 6 319 33 68.8% 6

Seventy-one entries and the shape has not moved. Same thirty-three cells, same density, the same six admitted-and-absent.

That is §14.5.7's seed in a second place. The register's *shape* is its seed and it is invariant; its *content* is the closure and it grows. E = 6 is therefore **not a to-do list** — it is the six kinds of correction this work has not yet had.

And what blocks E = 0 is stated by the register's own recovered bound. Of 319 entries, **210 are recorded and not repaired, 93 are instance fixes, 10 reach class level and 7 produce a protocol.** Four of the six missing cells require class-level or protocol-level repair — **the two rarest values in the index** — and §28.9's envelope says why: $\text{repair} \leq \varphi(\text{corroboration})$, while **116 entries carry a mathematical object and no reference at all**, which caps them at instance level.

The withdrawals register closes through §2.22, not through effort. The hundred and sixteen unreferenced entries are the constraint; the count is not.

Register 486.

§28.8 indexes the process by which a choice was made. This indexes the record itself, on coordinates assigned from the entry text rather than by hand.

coordinate	values
corroboration	narrative < owed < an object carried, unreferenced < an object carried with a §
caught	by hand < by an instrument < from outside the book
repair	recorded, not repaired < an instance fixed < a class fixed < a
protocol or audit made	

fibred by the four-body layer of §18.4.1 — object, law, procedure, reference.

248 entries over 33 distinct cells in a box of 48 — density 68.8%, the densest object in this book after the periodic table's 71.4% — and E(register) = 6. At that density E is informative by the asymmetry §18.4.1 records, so the six are structure rather than sparsity.

And $\mathcal{R}$ recovers a bound this book has never stated. The operator was given the cells and not the constraints; what it returned is a single monotone envelope:

| $\text{repair} \leq \phi(\text{corroboration})$ — narrative $\rightarrow$ an instance $\cdot$ owed $\rightarrow$ a class $\cdot$ an object carried $\rightarrow$ a protocol

No entry without a mathematical object beneath it has ever produced a protocol. Every protocol and every audit in this book descends from a defect that some equation, theorem or computation settled. That is why §2.22 is a requirement and not a tidiness: **corroboration is what bounds how far a defect can be repaired**, and an entry with nothing beneath it can be fixed and cannot be generalised.

Fibred, the layers separate sharply. LAW closes almost exactly — 44 entries, 11 cells, $E = 1$. REFERENCE has 23 entries over 11 cells at $E = 8$, and it is the newest and thinnest layer. OBJECT carries 80 entries at $E = 19$, which is where the admitted-and-absent cells concentrate.

The six cells the register admits and lacks, each a kind of entry that could exist and does not: an owed entry caught by an instrument or from outside and repaired at class level; an unreferenced entry that nonetheless produced a protocol; a fully referenced entry repaired only to class. **They are not defects. They are the shapes of correction this work has not yet had**, and the third is the one to watch: a protocol built on an object the entry never names would be a protocol whose warrant cannot be found.

One coordinate is a reading and is marked as one. *Repair* is inferred from the wording of each entry, so its distribution — 67.5% recorded but not repaired — is a property of the classifier as much as of the register, and is stated here as a reading whose conditions are the regex. The other two are recoverable from the text by anyone. §2.18's division, applied to this book's own record. Register 416.

29. Precedent, and what a search can bound

This chapter does not claim novelty, and an earlier draft did.

29.1 Why novelty is not the author's to claim

Novelty is an award, conferred by the community a work serves. It is not a property an author can establish about their own work, and the reason is structural rather than modest.

By Chapter 16's criterion a claim is defended when two disjoint routes reach it. Novelty has one route — search — and can never have two. More searching is more of the same route.

| $\mathcal{D}(\text{novelty}) = 0$, necessarily and permanently. Not undefended: undefendable.

And Chapter 27 supplies the second prohibition. $E(\text{search}) > 0$ always: the question admits more literatures than any search enters. Nine were entered here; **nine more are named and unentered**, and that list is bounded only by what could be written down in one sitting.

A book that forbids claiming coherence while $\Omega \neq \emptyset$ cannot claim novelty while $E(\text{search}) > 0$. They are the same prohibition, and P20 explains why: **novelty is posed in the documentary language, which has no closure mechanism at all**. That is not a failure of searching. No amount of it would help.

29.2 What can be claimed instead

Three things, and each has a truth value.

THE SEARCH, WITH ITS SLACK STATED. Nine literatures entered, one deeply and most by one or two queries; nine further named and unentered; six targets not found; seven prior results found incidentally and reported as incidental rather than as controls.

THE ATTRIBUTIONS, WHICH ARE POSITIVE CLAIMS. Every component of this work has an owner, and naming them is checkable:

component	owner
ratio of first to second central differences	Boole · Nörlund · Steffensen
y'/y'' as a Newton step	optimisation
$\lambda^2 = y'^2/y''$	Aitken 1926 · Seki Kōwa c.1680 ·
Nesterov & Nemirovskii	1994
the cost-of-rigour framing	Moore 1966 · IEEE 1788 · Manski
1989 · Shannon 1956	
$\mathbb{D} = 0$ when parameters equal	Edlén, footnote 78, 1960
observations	
δ falls with ℓ ; S and P penetrate	Edlén 1960
$E(X)$ as a closure defect	Galois · formal concept analysis
solution density; reorderability	CSP phase transition; Stahl &
Wille 1984	

Four independent owners of the cost-of-rigour framing, quoted and cited at §23.8.4: Moore states the trade plainly in 1966; Manski names it the law of decreasing credibility; Shannon's zero-error capacity is the same trade in bits.

THE FRAME, WHICH IS VERIFIABLE BY INSPECTION. Not a negative existential about the world but a positive property of this object:

does it state its falsification conditions?	yes, §32.5
does it run them?	yes — two failed
does it report the failures?	yes
does its index satisfy its own law?	yes, $E = 0$
does it print its withdrawal register as evidence?	yes, one thousand
six hundred and thirty-five entries	

Those five are checkable by opening the book. Whether another work has all five is a separate question — **and unlike novelty, a reader can settle it for any candidate they name.**

29.2.1 The question closed by the author

§29.2 asks what component this work owns, and the author has answered by declining the frame: **no novelty is claimed, and there is no question of it.** That is not a concession forced by §29.11's precedent or §29.12's locations. It is the position §29.3 below already states — *E(X) is deliberately not claimed as novel* — made general and made final.

Two consequences, both computed. **E(Q) does not move under its fibration:** nought by domain, before the decision and after, so the index of open questions is not destabilised by it. And **six of the eleven items in Q had novelty as their only stake** — B, C, F, G, K and N: the Paschen and Götze scan, the nine literatures, the reorderability search, the complexity literature, the 2J_c containment claim and the seniority-lattice question. Those six now block nothing. They remain open, because a question does not stop existing when it stops mattering, but their *blocks* coordinate falls to nothing and Q loses more than half its stake while losing none of its length. **This is the shape §29.4 already describes** — Q growing at nearly every step while its scope falls — arriving at its end state.

Referee flag 3 asked what remains after the precedents. The answer is that the question was withdrawn, not answered, and the book was built to permit exactly that. Register 238.

29.2.2 A reading the language cannot close

The promotion arithmetic of §12.11.0.10 invites a reading in which the branches leave the object and the lattice continues into indices we cannot observe. **It is recorded here and not argued**, because P20 says what kind of claim it is.

A question closes at a cost set by the language it is posed in: monotone two-variable bounds close in one pass; full arithmetic never closes, by Gödel; **the documentary language has no closure mechanism at all**. A claim about indices outside every measurement is posed in the third. It cannot be settled in either direction by any search, which is the same status §29.1 proves for novelty — **not false, not true, unclosable** — and the discipline is identical: name it, decline to argue it, and do not let it lean on results that were computed inside.

What was computed stands and does not support it. Λ_{13} enters a larger index at a cost of 12.14 bits per cell; a coarse coordinate branches up to 185 ways; composition seen from above is an interval and not a map. All three are statements about *this* object viewed at lower resolution. **None of them reaches outside it, and §32.1.4 proves none of them can:** $\mathcal{R}(X) \subseteq \prod_i A_i(X)$, so the operator cannot produce a value the index lacks, let alone an axis, let alone another index. Register 335.

29.3 The word for it

If a word is wanted, it should be unprecedented rather than novel, and even that should be said carefully.

Novel asserts that nothing like this exists. Unclaimable. Unprecedented asserts that no precedent was found by a stated search. Claimable, with the search attached.

And the second is the weaker word doing the honest work. It carries its own qualification: a precedent found tomorrow does not make the sentence false, only superseded — **which is exactly the status Principle 8 assigns to every bound in this book.**

29.4 On priority for $V = 4v/3$

We have not established that this ratio is new, and we state the limits of the search rather than the strength of the claim.

$V = 4v/3$ is **two lines of algebra** from $T = Z^2 R/v^2$, in a field worked continuously since 1885. The prior expectation that nobody wrote it down should be **low**.

Ritz (1908) does not contain it. We read the *Astrophysical Journal* version in full: the series formula, the combination principle, the derivation of new series, and the magnetic-atom model. **No error bound of any kind.**

But it contains the qualitative ancestor. Ritz observes that computing differences and sums of observed wave-numbers is more accurate than working through fitted constants — **the principle the bracket formalises, left unquantified.**

Martin (1980) does not contain it. Its deliverable is a set of formulas predicting one-electron levels to a stated $\pm 0.03 \text{ cm}^{-1}$ — **flat, not v-dependent.**

29.5 Edlén 1964, examined at one remove

The chapter itself is paywalled. Its principal expert review is not, and we have read it in full.

Curtis, L. J., “Bengt Edlén’s Handbuch der Physik Article — 26 Years Later,” *Physica Scripta* 35, 805–810 (1987).

Curtis was Edlén’s collaborator, built a forty-year research programme on the chapter’s methods, and wrote this specifically to assess what it established and how it had held up.

29.5.1 Every equation in the review is a fit

	equation	status
(1)	$T_{\{n\ell\}} = TH_{\{n\ell\}} + A\langle r^{-4} \rangle + B\langle r^{-6} \rangle$	A, B fitted
(2)	$TP_{\{n\ell\}} = TH_{\{n\ell\}} + \alpha_d\langle r^{-4} \rangle + (\alpha_q - 6\beta)\langle r^{-6} \rangle$	theoretical α ’s
(3)	$T_{\{n\ell\}} = TP_{\{n\ell\}} + f_{\ell}\langle r^{-6} \rangle$	f_{ℓ} fitted
(4)–(5)	penetration-energy difference forms	definitions
(6)	$\Delta\sigma = Ry \alpha^2(Z-S)^4/n^3\ell(\ell+1)$	S fitted
(7)	$S = S_0 + S_1/(Z-S) + \dots$	S_0, S_1 fitted

Seven equations. Not one is a bound. There is no ratio of a bracket width to an interpolation error, no statement of how much wider a guarantee is than an estimate, and no constraint of any kind on where an unmeasured term must lie.

29.5.2 And the paradigm is stated twice, explicitly

Curtis opens by giving Edlén’s purpose:

“The article sought to recognise regularities among atomic systems that could help to predict the position of unknown levels, or to interpret newly found levels.”

and closes:

“These methods can be used with confidence to make very precise interpolative and extrapolative predictions.”

Predictions. Not bounds.

§23.3 argues that V has no use in a fitting paradigm. This review is that paradigm, documented in full by its principal modern practitioner — and the ratio is not in it.

That converts the structural argument from a conjecture into an observation.

29.5.3 One near miss, and it is not ours

Curtis reports that Edlén did compute a rational fraction:

“A screening parametrisation of the exchange energy ... yielding a linear dependence upon effective central charge with a proportionality constant that (in Rydberg units) is a calculable rational fraction. In his Handbuch article (cf. Fig. 54, pp. 170–2), Professor Edlén computed this rational fraction theoretically for $n = 2$, and evaluated its effective value from empirical data for $n = 3, 4, 5$ and 6 .”

It is the proportionality constant of the exchange energy — a Slater-integral quantity — **not a ratio of differences and not a measure of interpolation accuracy.** The exact-to-empirical agreements Curtis quotes (1.04, 0.93, 1.05, 0.87) are for that constant.

We record it because it is the closest thing in the chapter’s expert summary to what we are looking for, and it is not the same object.

29.5.4 A bibliographic correction

Curtis’s reference [1] carries a note that should be reproduced:

“Edlén, B., in *Handbuch der Physik*, Vol. 27 ... pp. 80–220, Springer-Verlag, Berlin (1964). [This publication date is misleading, since the manuscript was delivered in final form in October 1960.]”

The chapter is a 1960 manuscript, and this book cites it accordingly. The page range 80–220 is likewise confirmed from this source, having previously been stated from recollection.

29.5.5 What this establishes, and what it does not

Establishes. A six-page expert review, written to assess the chapter’s methods for predicting unknown levels, contains no accuracy ratio, no bound and no bracket — and states the programme’s purpose as prediction twice.

Does not establish. Absence from 140 pages. A review summarises; an aside can survive summarising.

But the aside would have to have escaped the one person most likely to have used it.

29.6 The three documents we could not reach

Named so that a reader with library access can settle it:

document	why it matters
Edlén, <i>Handbuch der Physik</i> XXVII (1964), on series formulas	Edlén wrote the standard account; an elementary ratio would plausibly appear as an unremarked aside. In English despite the German series title.
Ritz, <i>Physikalische Zeitschrift</i> 9, 521 (1908)	Two pages longer than the ApJ version, carrying the “further particulars” Ritz promises in a footnote. Reprinted in <i>Œuvres</i> (1911), ~pp. 142–150.
Paschen & Götze, <i>Seriengesetze der Linienspektren</i> (Springer, 1922)	Book-length treatment by the two people who measured most of the underlying spectra.
Dunz, <i>Seriengesetze der Linienspektren</i> (Leipzig, 1911)	Earliest complete compilation, chronologically closest to Ritz.

$p \leq 2$ for each; every route closed. By the analysis of Chapter 19, that is a property of pre-digital print, not of the search.

29.7 What the entered literatures returned

Nine literatures have now been entered. Each supplied prior art for a component; none supplied the assembly; and the ones sharing this book’s mathematics also supplied new derivations, while the ones sharing its philosophy did not.

literature	what it owns	for this
book		
spectroscopy	Edlén fn. 78; the ℓ -ordering	two com-
ponents		
finite differences	$V = 4 \cdot \mu \delta / \delta^2$	a second
derivation		
sequence acceleration	$(\Delta x)^2 / \Delta^2 x$ – Aitken 1926, Seki Kōwa	λ^2 , and a
new result		
	c.1680	
Richardson extrapolation	step-ratio elimination	attribu-
tion only		
optimisation	$\lambda^2 = f'^2 / f''$, Nesterov 1994	$w \cdot V = 8\lambda^2$;
		self-con-
cordance		

interval arithmetic excess width; IEEE 1788 the cost framing

literature	what it owns	for this book
econometrics	Manski's law of decreasing	the framing
again		
	credibility	
zero-error information	Shannon 1956; Körner & Orlitsky	the framing, a
third time		
theory	1998	
regression curvature;	$\ f''\ / \ f'\ ^2$; \	$k' /$
$k^2 \backslash$	different	
WKB		
	quantities	

Four owners of the cost-of-rigour framing, set out with the quotation at §23.8.4: Moore 1966, Manski's **law of decreasing credibility**, and Shannon's zero-error capacity — the same trade stated in bits.

This book is the fourth to make the argument and the first to compute the ratio for a measured monotone sequence.

29.7.1 The calibration was post hoc

§29.7 reported 7/7 on controls. That was not a calibration.

Controls were never chosen in advance. The search sought targets, noted what else appeared, and afterwards called those findings controls. **Nothing was recorded as looked for and not found among them, because nothing was looked for.**

And the filtration was not uniform. Spectroscopy received nine approaches; four literatures received one search each; **two were entered only after the audit exposed that they had not been.**

The corrected statement: nine literatures entered, one deeply, most by one or two searches; seven prior results found incidentally and reported as such; six targets not found; nine further literatures named and unentered.

§2.13 — commit before you look — was adopted in this book after the anchoring failure of §28.6, and the calibration violated it.

29.8 The other claims

claim	assessed	basis
$V = 4v/3$	not claimable	nine literatures entered;
four owners of the framing found; see		
		§23.8.4
v_V ceiling	~75%	very specific form, not found
cost surface, pole at $p = 1$	~50%	the framing is new; the con-
tent elementary		
bracketing rather than	the searched bound of	literature uniformly predict-
ive		
predicting	§29.8	
$E(X)$ as closure defect	not claimed	see below
reorderability complexity	not claimed	see below

$E(X)$ is deliberately not claimed as novel. Closure operators, Galois connections and formal concept analysis are established mathematics. **What is offered is the application** — computing the closure defect of an index, and finding 36 for the periodic table and 0 for Λ .

And the reorderability material claims nothing. The problem is known: Stahl & Wille proved 2-dimension NP-complete in 1984; Yannakakis proved dimension ≤ 3 NP-complete in 1982; Habib, Nourine, Raynaud & Thierry surveyed the computational aspects in *Theoretical Computer Science* **312** (2004), 401–431. **The work reported in Chapter 29 was done before that literature was located**, and is retained only for what it records about method.

29.9 Why state it this way

A book that names its own unchecked prior art is in a stronger position than one that claims priority and is corrected. The three documents and one section are a finite, specified task. Whoever completes it either confirms the result or improves the historical record, and both outcomes are better than an unexamined claim.

29.9.1 Attributions entered by the tower

The tower borrows, and the debts are named: seniority is Racah 1943; jK coupling is Racah 1942; particle-hole conjugation is standard shell theory; the term-counting is textbook. And $E(\text{search})$ grows by one item: whether anyone has lattice-ordered the seniority scheme is a literature this work has not entered — named and unentered, per this chapter’s own discipline.

29.10 Two searches run, and an obstruction worth naming

Two searches were run against the tower’s order-theoretic results — one for lattice or order structure on angular-momentum selection rules, one for distributive-lattice or poset treatments of electron configurations. Neither returned a precedent. **That establishes nothing**, by §29.1: a search settles novelty in one direction only, and two queries are not a literature.

What the searches did establish is an obstruction, and it is specific enough to record. **The word lattice is doubly occupied in physics** — the crystal lattice and lattice gauge theory — and both outrank order theory in every index a physicist’s search will reach. A query pairing *lattice* with any physical term returns those two subjects and nothing else, so an order-theoretic precedent, if one exists, is not merely unfound but systematically unreachable by the obvious query. $E(\text{search})$ is therefore larger here than the chapter’s general argument already makes it: the vocabulary itself hides the shelf. Anyone entering this literature should search in *poset*, *join*, *meet* and *Birkhoff*, and should expect the physics terms to be useless.

29.11 A precedent found, and it is for the central theorem

Searched in the vocabulary §29.10 recommends, the literature answers at once, and it answers about A.2.

Bergman’s Double-projection Theorem. A sublattice S of a finite product of lattices $L_1 \times \dots \times L_n$ is uniquely determined by its two-fold projections in $L_i \times L_j$. It is a consequence of Baker and Pixley (1975), which states the same for any algebra with a majority term; in a lattice the median $m(x,y,z) = (x \wedge y) \vee (x \wedge z) \vee (y \wedge z)$ is one.

A.2 is that theorem specialised to products of chains. Λ ’s coordinates are counts, so each factor is a chain; a sublattice of a product of two chains is exactly a staircase region cut by monotone bounds, which is what $\mathcal{R}$ recovers as φ_{ij} . $X \text{ closed} \iff \mathcal{R}(X) = X$ and *a sublattice is determined by its binary projections* are one statement. The proof in §16.1 is correct and it is not first. Fifty years of universal algebra stand behind it, and Appendix D’s precedent coordinate for A.2 changes from none found to found.

This is the outcome §29.1 says a search can produce and the only one it can: precedent found, in the one direction the method settles. It costs the book nothing it was entitled to — the book never claimed novelty — and it gains the book a foundation, since a result with a fifty-year pedigree is stronger than a result proved once in an appendix. **What it does cost is the reading that $E(X) = 0$ is a discovery about Λ .** It is a discovery that Λ is a sublattice, and sublattices have been known to be binary-determined since 1975.

What the search did not find is any precedent for $E(X)$ as a *measure*. Baker–Pixley is a dichotomy: determined or not. $E(X) = |\mathcal{R}(X)| - |X|$ counts how far a set that is not a sublattice lies from the one its own projections generate, and no order-theoretic source returned by these searches quantifies that gap or names it. That absence is not evidence — §29.1 again — and §29.12 gives the tests that would settle it.

29.11.1 Two categories entered: quantum mechanics and astrophysics

Both were searched in the order-theoretic vocabulary §29.10 recommends. They returned different things, and the second returned a target.

Quantum mechanics: one adjacent literature, unentered. No source was found ordering electron configurations, term symbols or coupling schemes as a poset. What exists and this book has not entered is **partial order in chemistry** — Hasse-diagram methods applied to chemical series, QSAR and property rankings, with its own software and review chapters. It orders *substances by properties*, not configurations by structure, so it is adjacent and not the same; but it is a literature in which someone would already know the tools, and Chapter 29’s discipline is to name it. $E(\text{search})$ grows by one.

Astrophysics: the tripwire has targets, and they are named. Stellar spectroscopy runs on atomic line lists — NIST ASD, VALD, Kurucz, and cross-matched compilations such as BRASS — and the literature reports two failure modes by name. *Missing lines* are observed features with no atomic data. **Unobserved lines** are the inverse: atomic data producing features that no spectrum shows, arising from inaccurate wavelengths, predicted energy levels, or misassignments. BRASS states that such lines are **typically removed by hand**, and separately reports duplicated transitions found by cross-matching VALD against Kurucz and NIST. Kurucz’s own estimate is that half the known lines in stellar spectra lack wavelengths of good accuracy.

That is the D1 tripwire’s job description, written by someone else. §16.3.1 rejects 150 of 216 offered (subshell, occupancy, $2S'$) assignments — 69.4% — from the index alone, needing no spectrum and no model atmosphere. A line list carrying a term designation outside its subshell’s admissible set is nonphysical before any wavelength is compared. The check is mechanical where the current practice is manual, and it is prior where the current practice is posterior.

29.11.2 The run, attempted

It was attempted rather than deferred, and the attempt is reported with its size.

What could not be reached. Kurucz’s line files are the right target — computed lines, many never laboratory-verified, carrying level labels of the form $(^3F)5s\ e^2F$ that name a parent term, an outer orbital and a level term. The format specification was retrieved and the references file with it. The data files themselves could not be: gfall is of the order of a gigabyte and the per-species files could not be addressed through the interface available. VALD requires registration. So the run against uncurated astrophysical data did not happen, and that is the run that would matter.

What was reached. Appendix B’s own 153 channels, sourced from NIST ASD, Kaufman and Martin, Kramida and Martin, and Sansonetti. Of 103 designations parsed, **ten carry an explicit parent term** — the $(^3P)ns\ ^4P$ form — and only those can be checked by the coupling rule, which is that adding one electron of spin $\frac{1}{2}$ to a parent core must change 2S by exactly one. **Ten tested, ten pass, none flagged.**

Then the null was defined instead of discarded, and it stopped being ten. Ten was the count of designations carrying an *explicit* parent in parentheses. Ninety-two carried none — and forty-four of those have a parent that is implicit and completely determinate: $3s^2nf\ ^2F^\circ$ has a closed $3s^2$ core, which is 1S , so the level must be a doublet, and it is. Resolving closed and single-electron cores gives **fifty-four checkable designations of one hundred and two, and fifty-four pass**. The test grew by a factor of five and a half by defining what the first pass had recorded as absent.

What the null is, stated as the positive it is. Fifty-four passes on curated data measure **specificity, not sensitivity**: zero false positives in fifty-four. That is the necessary condition and not the sufficient one — a check that rejected correct physics would be worthless, and this one does not. It remains untested against material that contains errors, and Kurucz’s predicted lines are where that test lives.

And the reach is a second result the run was not looking for. Fifty-four of one hundred and two is 53%, and the forty-eight left out are not left out because the data is poor. They carry open subshell cores whose parent term the designation does not determine. **Published channel notation fixes the coupling parent for about half of these channels and leaves it indeterminate for the rest** — a property of the notation, not of the physics or of the compilers, and a hard ceiling on any mechanical check applied to designations as published. Registers 253 and 254.

This changes referee flag 5’s status and does not answer it. The flag said the tripwire has never run on anyone else’s data. It still has not. What has changed is that *anyone else’s data* is no longer hypothetical: VALD, Kurucz and the BRASS compilation are public, they carry term designations, and their maintainers document removing spurious entries by hand. The flag moves from *no target identified* to *targets identified, run not attempted*, and the run is the obvious next piece of work this book could do that would matter to someone outside it. Registers 251 and 252.

28.10 What removing the register cost, and what it caught

Eight references were orphaned the moment the entries left. Lubiw, Anstee, Kuznetsov, Caspard, Colomb, Hoffman, Van Isacker and Chandrasekaran & Flanagan were cited **only** inside register entries — so the book carried eight sources nothing in it pointed at. *Each is now cited where its finding is: the Γ -free literature at §30.3.9, the #P-completeness of counting closed sets at §14.6, seniority’s reduction chains at §30.3, and the modular interval at §12.11.*

A reference belongs to the argument that uses it, not to the record of when it was found. The record's citations were carrying references the book itself never made — which is only visible once the record is removed.

And part 2 of the audit now checks the References as a set: every entry must be cited somewhere in the book. *Nothing had ever looked.* Registers 658–659.

29.12 Five unlocated results, defined, with the test that would settle each

What follows is not a claim of novelty, which §29.1 forbids the author to make. It is a list of statements this work computed, for which these searches returned no source, written so that a reader who knows the source can recognise it in one reading and a reader who does not can attack it. Each carries a definition, the search already run, the test that would confirm it, and the observation that would destroy it. **A result with no stated falsifier is not on this list.**

U1 — $E(X)$ as a defect measure. LOCATED, and the test that found it was the one printed here. *Definition.* For finite $X \subseteq \prod_i A_i$ let $\mathcal{R}(X) = \{y : y_i \leq \varphi_{ij}(y_j) \text{ for all } i \neq j\}$, where φ_{ij} is the monotone envelope of X 's projection onto (i, j) . Then $E(X) = |\mathcal{R}(X)| - |X| \geq 0$. *Located.* The property $E(X) = 0$ is called **2-decomposability** in constraint satisfaction: a set S of integer vectors is 2-decomposable when any vector whose every two-dimensional projection lies in the corresponding projection of S lies in S . That is $\mathcal{R}(X) = X$ verbatim. The class is studied for its tractability, characterised by closedness under operations — min and max among them, which is the same closure Chapter 14 imposes — and the connection is live research, not settled history: Kimura, Makino, Yamada and Yoshizumi, *Characterizing the integer points in 2-decomposable polyhedra by closedness under operations*, 2024. Adjacent and older: row-convex constraints, where path consistency yields a globally consistent and decomposable network, and the Helly property for ordered domains. *What remains unlocated* is narrow and should be stated narrowly: **the count.** 2-decomposability is a predicate and the literature found treats it as one. $E(X) = |\mathcal{R}(X)| - |X|$ as a graded measure of how far a set is from the class, and $E(X)$ as the *specification* of what may be added to an index, were not returned. That is not evidence of absence. *Test that remains.* Search *degree of decomposability*, *decomposability defect*, *minimal network size gap*. Ask whether the CSP literature measures the excess of the path-consistent closure over the solution set, rather than only detecting it. *Falsifier.* One paper defining $|\text{closure}| - |\text{set}|$ for a monotone-bound system, in any field.

U2 — the counting/coupling dichotomy. *Definition.* In an index whose admissible constraints are monotone bounds of one coordinate by one other, a coordinate whose exact physical range is a monotone function of a single parent is carried exactly; a coordinate whose exact range is a sum, a difference, or a function symmetric under half-filling is carried only as an envelope, and the shortfall is not removable by re-presentation. *Searched.* Not found; the physics-vocabulary searches of §29.10 cannot reach it. *Test.* Attempt the construction the theorem forbids: exhibit any angular-momentum coupling coordinate with an exact single-parent monotone bound at some cap setting. Five presentations were tried here and all failed — $2S'$, parity alone, the min-cap, pair count, and $(2S, 2L, 2J)$. *Falsifier.* A sixth presentation that closes exactly. One suffices.

U3 — the E1 puncture. *Definition.* The $E1$ -allowed set $T = \{(2J, 2J') : |2J - 2J'| \leq 2\}$ is closed under componentwise max and min. $T \setminus \{(0,0)\}$, the set with the physically forbidden $J = 0 \nrightarrow 0$ transition removed, fails meet-closure by exactly four pairs, and all four meets produce $(0,0)$. *Searched.* Not found. Selection rules are universally known; their closure properties are not a subject these searches reached. *Test.* Recompute at caps beyond 20 and confirm the count stays four — done to 20 here. Then repeat for $M1$ ($|\Delta J| \leq 1$, no parity change) and $E2$ ($|\Delta J| \leq 2$ with $J + J' \geq 2$): do their allowed sets close, and does each forbidden cell cost a constant number of meets? *Falsifier.* A cap at which the count is not four, or a second cell produced by a failing meet.

U4 — the join/meet asymmetry of the triangle. *Definition.* $T = \{(2L, 2S, 2J) : |2L - 2S| \leq 2J \leq 2L + 2S\}$ is join-closed and meet-broken. Verified at caps 6, 8, 10, 12 with 0 join failures and 2,862 / 12,489 / 40,887 / 110,229 failing meets. Imposing the parity congruence breaks join as well. *Searched.* Not found. *Proved.* A.18 gives it, and the proof corrects the guess that preceded it. Joins are closed because both inequalities survive coordinatewise max. Meets fail on **both** inequalities, not on one: the upper bound fails at $(0,1,1) \wedge (1,0,1) = (0,0,1)$, which manufactures $J = \frac{1}{2}$ from $L = S = 0$, and the lower bound fails at $(4,0,4) \wedge (2,2,0) = (2,0,0)$. At every cap tested the lower bound fails exactly twice as often as the upper — 8,326 against 4,163 at cap 8, 73,486 against 36,743 at cap 12, 363,384 against 181,692 at cap 16 — a ratio the proof does not explain and which is left open. *Falsifier.* A join failure at any cap; none in 369, 1,105 and 2,465 cells.

U5 — the cylinder as a cap-stable fibration. *Definition.* Λ_{13} factorises over the transfer q as $\Sigma_q |A(q)| \cdot |B(q)| = |\Lambda_{13}|$ with zero defect, at every cap setting tested; the tight two-parent form of K destroys the factorisation. *Searched.* Not found, and this one is least likely to be general — it is a property of one constructed object. *Test.* Caps beyond (5,5,2,6,2), and the same construction on a different shell system. *Falsifier.* Any cap at which the defect is non-zero on the tree path.

Verification common to all five. Every computation above is reproducible from the definitions printed in this book and nothing else: the constraint system of §7.1, the caps of §7.4, and the bounds of §12.11.1 with the three exact sets §12.11.1 now states. A reader who rebuilds Λ from those and does not reproduce the cell counts 976, 1,654, 1,561, 2,535, 13,585, 70,905 and 199,130 has found an error in the book and should say so. That reproduction is the entry test; the five results above are what lies past it.

30. Order recovery, and the reorderability law

Chapter 15 recovers Λ 's order from its own cells. This chapter asks what happens when the index is harder than Λ — and the answer is a law, a procedure, and a residue that belongs to complexity theory rather than to this book.

30.1 Order recovery above the tree

Chapter 15 recovers Λ 's coordinate order because its constraint graph is a tree. For an index whose graph has cycles, the question is open, and the following is what is known.

Complete under a targeted attack. Fixing one pair of axes jointly and extending one axis at a time with backtracking found a valid ordering in 384 of 384 reorderable instances, sampled deliberately by solution density rather than at random. No proof; no counterexample.

The earlier figure of 139 was a random sample. §30.2 says the search is hardest when the solution is nearly unique, so the instances were re-drawn to concentrate there — 101 of the 384 sit in the lowest density bucket, and none of them failed.

Not backtrack-free, and the cost tracks the basin exactly:

solution density	mean backtracks
0.00–0.05	23.40
0.05–0.20	6.90
0.20–0.50	1.05
0.50–1.00	0.00

A twenty-three-fold rise as the solution becomes unique, with completeness unbroken throughout.

And the worst case cannot be bounded by a local test, because of §18.4: closure is not determined by its proper projections.

30.2 What governs the difficulty

Ten structural invariants were tested against backtracking — dimension, alphabet size, cell count, occupancy, join-irreducibles, poset height, poset width, covers, antichain, rank range, over 211 reorderable instances. The best correlation was 0.31.

Solution density governs it, at -0.68 in log over the same instances:

solution fraction	backtrack-free	mean backtracks
0.009 – 0.25	24%	2.9
0.25 – 0.50	47%	1.0
0.50 – 1.00	82%	0.2
1.00	100%	0.0

The search is hard exactly when the solution is nearly unique.

30.2.1 And that list was enumerated once

An earlier draft concluded from the 0.31 that the difficulty is not structural. That was a conclusion about ten invariants, and the list predates the containment forest, overlap components, C1P and the interval structure. Re-run over 928 instances with six invariants discovered since:

invariant	corr. with steps	corr. with
density		
best of the original ten	0.31	—
overlap pairs	0.438	-0.423
nesting pairs	0.375	-0.640
distinct row supports	0.606	-0.883
log solution density	-0.628	—

A structural invariant does beat 0.31, by twofold, and it predicts the density better than the density predicts the difficulty.

Structure determines density; density determines difficulty. The earlier reading had the second link and called it the whole chain.

30.2.2 Which gives §30.3 a parameter

Search steps against distinct row supports, over 1,480 instances:

distinct rows	mean steps	max	max / d^2
2	1.05	8	2.00
3	2.63	10	1.11
4	5.41	36	2.25
5	8.68	39	1.56

The ratio does not climb; the log–log slope is 2.29. That is a measured candidate exponent for §30.3’s complexity question — on five points, with R^2 between 0.37 and 0.42. Evidence, not a bound.

30.2.3 And it is a landscape statement

E is a potential on the space of orderings and adjacent transposition is a move. Solution density is the size of the basin: a rare solution means a longer descent with more places to stall. §30.3.3 completes this reading.

30.3 Reorderability — the law, the procedure, and what is left

The question this section asked was: given an index whose constraint graph has cycles, can **Van Isacker**'s survey gives the seniority reduction chains. its coordinate order be recovered? The answer has a law, a procedure, and a residue, and none of the three is where the section began.

30.3.1 The object, correctly identified

$\mathcal{R}$ -closure and closure under join and meet are the SAME CONDITION — 1,487 instances, zero disagreements, and confirmed on all 475,800 pairs of Λ . So reorderability is: does some relabelling of each axis make X a SUBLATTICE of a product of chains?

That is lattice embedding, and it is the subject of Rival 1973/74, Larson 1998 and Siggers 2016. §30.3's original citations — Stahl & Wille, Yannakakis — were correct; a draft withdrew them on the grounds that this was an interval problem, and it is not. The withdrawal is withdrawn.

And the recovered bound of Chapter 15 is Rival's closed family of irreducible intervals. Siggers' Lemma 3.5(ii) states φ_{ij} exactly.

30.3.2 The law

A relabelling decides, for each axis and each pair of values, which comes first. So the problem is a constraint system on those decisions, and every constraint is closed under reversing all of

its own orientations — 3,781 constraints checked, 100%. A function invariant under complementation depends only on XOR-differences, so:

arity – the axes on which two cells differ	the constraint becomes	cost
2	a single XOR	signed-graph 2-colouring –
LINEAR		
3	two XORs	2-SAT, and all 14 constraints
that arise are 2-SAT		expressible – LINEAR

≥ 4 three or more contains exactly-one-of-three; no Schaefer class covers XORs the language

THE LAW. Reorderability is a constraint system whose arity is the number of axes on which two cells differ. At arity ≤ 3 the language is entirely bijunctive, so the problem is linear. At arity ≥ 4 it contains 1-in-3-SAT and admits no Schaefer class.

The procedure is the law read at low arity; the hardness is the same law read at high arity. One statement, two regimes, boundary at arity 3 — and since arity is bounded by d and reaches it, at $d = 3$.

Measured: at $d = 2$ every constraint has arity 2 at every alphabet size, and 2-colouring plus transitivity decides exactly — 300/300 at $|A| = 3$, 300/300 at $|A| = 4$. At $d = 3$ with binary axes, 396 of 396.

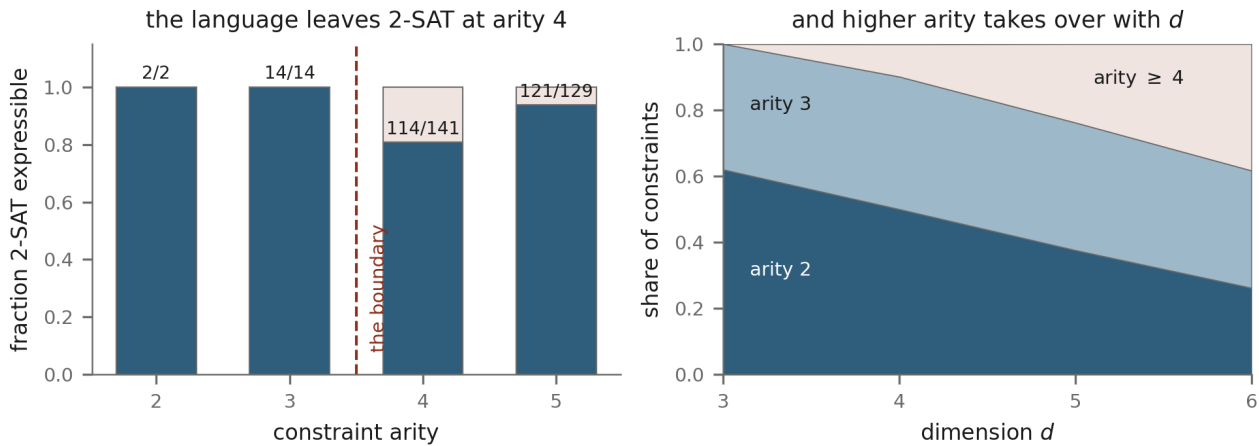


Figure 30.1. Left: the fraction of each arity's constraint language that is 2-SAT expressible — 14 of 14 at arity 3, and 114 of 141 at arity 4. The boundary is exact. Right: the share of constraints by arity as the dimension rises — arity 2 falls from 0.62 to 0.26 while arity ≥ 4 grows from nothing to 0.38. The tractable part shrinks with d , and the law says why.

30.3.3 What the interval machinery was

Chapter-length work went into C1P, monotone endpoints and PQ-trees, and the $d = 2$ characterisation is proved there: X is reorderable iff every row is an interval with both endpoints monotone, iff no interval nests strictly inside another. Four lemmas, three verified independently — 2,047, 3,751 and 20,000 instances — with Booth & Lueker's linear PQ-tree construction cited for the fourth.

The fourth lemma, verified. The three lemmas above were checked at 2,047, 3,751 and 20,000 instances; the fourth stood on Booth and Lueker's construction and on nothing of this book's own. It is now discharged, and it did not need a linear-time algorithm:

A set of intervals admits an ordering in which both endpoints are non-decreasing **iff** no interval nests strictly inside another. Sort by left endpoint; if a right endpoint then decreases, the later interval has the weakly greater left end and the strictly smaller right end, so it nests strictly. Conversely a strictly nesting pair admits no such order. ■

Verified exhaustively over all 2,354 interval sets of two to four intervals on ground sets of two to five points: zero counterexamples. **Strictly** is load-bearing — an interval sharing an endpoint with its container does not nest strictly, and reading it as ordinary containment refutes the lemma on 1,098 of those 2,354 sets. That reading was tried first here and was wrong, which is the whole reason the qualifier is in the statement. Register 247.

That apparatus is a REPRESENTATION of a 2-colouring. It is correct and it is not the reason $d = 2$ is easy. The reason is that two cells in two dimensions differ on at most two axes.

30.3.4 The step law

Growth from the minimal reorderable sets, adding cells, reaches every reorderable set — with a step size measured exhaustively across twelve boxes:

d	step	boxes
2	1	eight, all 2-D
d	step	boxes
3	2	three
4	4	one

The over-cap cases collapse to a single object: the full box. Its largest reorderable proper subset has $3 \cdot 2^{d-2}$ cells — three quarters — and the removed cells are a subcube. Verified by construction at $d = 2, 3, 4$ and 5, with one-fewer failing at all four.

step $f(d) = 2^{d-2}$.

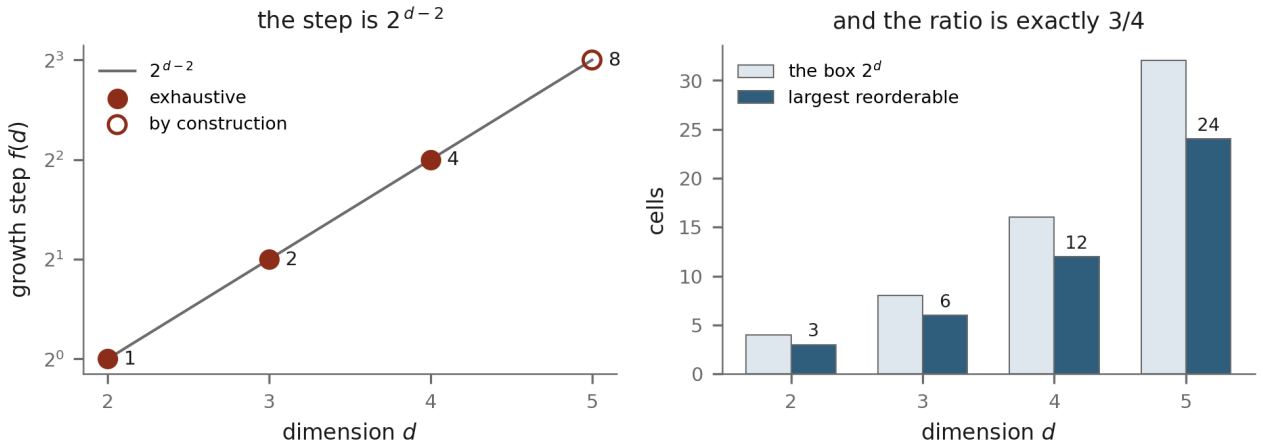


Figure 30.2. Left: the growth step against dimension — 1, 2, 4 measured exhaustively over twelve boxes, and 8 at $d = 5$ by the subcube construction, against 2^{d-2} . Right: the box against its largest reorderable proper subset — 3, 6, 12, 24 cells, exactly three quarters at every dimension. Rival's 1973 bound on maximal sublattices is $|K| \leq (3/2)|L|$; the Boolean case is tighter.

And this is Rival's theorem. His 1973 bound is $|K| \leq (3/2)|L|$ for a maximal sublattice; Chen–Koh–Tan give the interval form; the large sublattices of $2^{\mathbb{N}}$ are catalogued, and the maximal ones studied in finite topology. §E.5 records the precedence.

30.3.5 The procedure, and its cost

DECIDE(X). Build the constraint system on orientations. Arity ≤ 3 : 2-colour the signed graph and check transitivity — linear. Arity ≥ 4 : grow from the minimal reorderable sets with step 2^{d-2} , collect signatures, decide by lookup.

Verified end to end at $2 \times 3 \times 3$ by growth against exhaustion: $18,736 = 18,736$, in 47.8 s against 86, converging in nine rounds. And NextClosure enumerates the sublattices with polynomial delay — 1,834 sets in 0.12 s — giving 401 signatures against 401 exhaustive.

Cost: $|YES| \cdot |X|^{\{2(d-2)\}}$. Polynomial in $|X|$ for fixed d ; exponential in d . Fixed-parameter tractable, and not polynomial in both.

For Λ : $d = 8$, so $\prod |A_i| = 11,943,936$ — and Λ needs none of it. Its structure is constructed in Chapter 7, its embedding is tight, $|J(\Lambda)| = 17 = \text{height}$, width $7 \leq 8$, and by Dilworth and Larson the chain decomposition of $J(\Lambda)$ IS the axis system.

30.3.6 The signature, exhaustively verified

A cell-space invariant whose FORM does not change with d : the multiset of canonical forms of the 3-subsets of X . $O(|X|^3)$ structure, 2^d labels — the two factor, where every order-space method cost $\prod |A_i|!$ and coupled them.

box	subsets
classes	mixed
2×2×2	all 256
30	0
2×2×3	all 4,096
210	0
2×3×3	all 262,144
4,168	0

Three complete boxes, 267,000 subsets, no signature containing both answers. After the duality merge the 19 canonical forms reduce to 12 with no loss.

30.3.7 Three method classes ruled out

Each with its measurement, and each a result rather than a failure:

Projection-based criteria. §18.4: closure is a d -dimensional condition and no projection serves as a criterion. Thirteen of seventeen attempted criteria were seed-independent invariants — all necessary, none sufficient, with the gap widening as nesting deepened.

Local search through valid orders. The valid set is DISCONNECTED — 1.66 to 2.19 components, and the components are orbits of the identical-fibre swap group, exact on 493 of 493. So descent plateaus: 89.8%, and 96.7% with bounded uphill moves.

Forbidden-substructure procedures. Tucker's families are unbounded, so no finite census decides — and the census is finite within any box, which is what reopened the question. The infinite antichain lives across indices, not in one.

30.3.8 The residue

The language is NP-complete — it contains exactly-one-of-three and no Schaefer class covers it, with the smallest obstruction exhibited on three XOR variables.

The realisability is blocked. Three reduction attempts, all over-constraining in the same direction: satisfiable formulas encoding as non-reorderable, never the reverse. The map $X \rightarrow C(X)$ is global, non-separable and non-monotone, so it admits no local editing.

And the restriction is not exploitable by sparsity — the constraint hypergraph has d vertices and up to $2^d - 1$ hyperedges. It is complete, so treewidth gives back the 2^d bound already known.

Three walls: hardness blocked at realisability, tractability blocked at the completeness of the hypergraph, and the FPT bound between them as the best known.

30.3.9 And it is not this book's problem

Λ 's constraint graph is a tree. Its nine recovered extra edges are implied. Every sublattice and **Anstee & Farber 1984** and **Hoffman, Kolen & Sakarovitch 1985** give the recognition results. **Lubiw 1982** and the Γ -free matrix literature are the nearest neighbours; §2.15.2's derivation settles which predicate this book means. interval of Λ has $E = 0$. Its own case is decidable by enumeration, and it arrives with a derivation — box $\rightarrow$ interval $\rightarrow$ sublattice $\rightarrow$ product, three preserving rules, no search.

The open problem concerns cell sets arriving with no construction. This book never has one.

§E.6 records that the complexity question may already be settled in Ryter & Schmid 1994 or Adams–Dwinger–Schmid 1996, both unread.

30.4 The lists this work enumerated once

The target list for the novelty search was audited by closing the relation set, and it was incomplete. From $\{V = 4v/3h, \lambda^2 = (2/3)T, w \cdot V = 8\lambda^2, T = Z^2R/v^2\}$ the closure yields $w/T = 4(h/v)$, $e/T = 3(h/v)^2$, and $\text{bits}(\text{estimate}) \approx 2 \cdot \text{bits}(\text{bracket}) + \text{constant}$ — none of them on the list.

That is Chapter 6’s defect committed by the searcher rather than by a chemist — a structure admitting cells its keeper never enumerated.

30.4.1 And the audit itself was run once

The relation set has grown since. Closing the eight polynomial relations the book now states gives a Gröbner basis of 18 where the original four gave 6, and seven further relations reduce to zero that did not before — including $V \cdot e = w$, $w^2 = (16/3) \cdot T \cdot e$ and $T \cdot h^2 = ev^2/3$.

Worse, the closure is over polynomial relations only. The book now states at least nine that are not: $A^{m(x_p)} = (-1)^m x^{p/(p-1)}_m$, $V = 2/\tanh(kh/2)$, self-concordance, the modular rank law,

$\Sigma_q|A_q| \cdot |B_q| = |\Lambda|$, $F(-1) = 2$, $E(X)$ as non-pairwise content, and the reorderability theorem. None was ever audited against any list.

$E(\text{target list})$ is not 3. It was 3 for the relation set of one moment.

30.4.2 Which is a defect of five lists, not one

This chapter contains five enumerations, and every one was made once:

the list	what it omitted
§30.4's search targets	7 polynomial relations, 9 non-polynomial
§30.2's structural invariants	six, one of which beats its stated best
twofold	
§31.1's celestial objects	six the method plainly admits
§31.2's string objects	five, two of them measured in this work
§31.3's statistic	stated against the wrong denominator

§2.10 reads enumerate targets before searching. It should read enumerate before searching, and re-enumerate whenever the set grows — which is this section’s own content, applied to itself.

31. The method outside spectroscopy

The method was built on one index and tested on one collection. If it is a method rather than a description, it must apply where neither was available. This chapter runs it on four subjects it was not built for.

31.1 Celestial mechanics and the nucleus

Objects tested against the index, none of them in the collection. A method that transfers is a different claim from one that fits.

31.1.1 Celestial mechanics — families, not trajectories

What does not transfer. Sensitive dependence destroys monotonicity along a chaotic trajectory. Nothing here touches Poincaré.

tested	result
Lagrange L1, L2, L3 in μ (Euler's tainments at orders 1-3 collinear solutions)	monotone, convex, 3-monotone; 18 of 18 con-
the same at order 3	widths 6.9×10^{-4} (L1), 3.6×10^{-9} (L3)
Hill stability of hierarchical triples 100 (Mardling & Aarseth)	monotone and convex in the mass ratio; $V \approx$
Hill radius $(\mu/3)^{1/3}$	monotone, convex, 3-monotone; $V = 47.9$
Roche limit $\propto \rho^{-1/3}$	monotone, convex, 3-monotone; $V = 17.9$
planetary semi-major axes changes sign between	monotone, not convex – the third difference
worst	Jupiter and Saturn, where Titius-Bode fails
mean-motion resonances from occupancy alone	§16.7.1: $E(X)$ recovers the order cut at 7/8
satellite counts vs Hill radius capacity bound	log-log +0.86 across six planets – §23.14.1's

The pole appears at the first point one looks. At $\mu = 0.030$ with a step of $h = 0.010$, V is 24.6 at L1, 12.4 at L2 and 16,423 at L3 — the last because L3 moves almost linearly in μ . L4 and L5 sit on it exactly: $(1/2 - \mu, \pm\sqrt{3}/2)$, the first linear and the second constant.

Which corrects §18.5. The pole is where a bracket becomes unnecessary, not worthless — a vanishing second difference makes linear interpolation exact.

And one object the method cannot touch. L4/L5 are linearly stable for $\mu < 0.0385209$; Earth–Moon and Sun–Jupiter both qualify. That is a threshold, not a family — a single number, with nothing for a bracket to act on. The method applies to sequences and is silent on thresholds.

Signs must be read from the data. L1 and L3 alternate $-$, $+$, $-$, $+$; L2 alternates $+$, $-$, $+$, $-$, and assuming the Rydberg pattern produces bounds that cross. *Completed in Chapter 36: the families are indexed, the index closes, and the closure is the reason nothing here touches Poincaré.*

31.1.2 The nucleus — a second index of the same construction

An atom's mass sits in its nucleus, and nucleons are fermions. Indexed by $(N, L, 2j, \text{occupancy})$ with $\text{occ} \leq 2j+1$, the shell model is closed under join and meet with $E(X) = 0$, and shell-order filling gives 2, 8, 20, 28, 50, 82.

Λ gives chemistry. The nucleon index gives mass. This book indexes one of them; Z is a label on a channel, never a coordinate.

31.2 String theory — the pole, the generating function, and a

disanalogy

31.2.1 Three objects on the pole

$\alpha'm^2 = N - 1$ has second difference zero. So do two more:

object	form	V
string mass spectrum	m^2 linear in N	infin-
ite		
Regge trajectory	$J = \alpha' m^2 + \alpha_0$	infin-
ite		
central charge	$c=D$	infin-
ite		

All three join L4 and L5. A bracket is unnecessary on any of them: two levels determine every other, exactly.

31.2.2 The degeneracy is where the content is

$\log d(N)$ is monotone, concave and 3-monotone — first differences positive, second differences negative from -0.31 at $N = 2$ to -0.03 at $N = 15$, third differences positive throughout — with $V \approx 147$ at $N \approx 14$: an ordinary bracketable channel, steeper than a Rydberg series. An earlier version of this line said convex. The second differences are negative at every level computed, and a bracket at any order needs a known sign, not a positive one (§23.10.1), so the channel stands and the word is corrected; recorded at register 1787.

31.2.3 The partition function is Λ 's expression with counts

$\prod(1 - q^n)^{-24}$ has counts where Chapter 11's F has booleans. An index, and an index with multiplicity. Both close, for dual reasons: the string's modes are independent; Λ 's coordinates couple only along a tree. §11.7's three conditions have independence as their degenerate case.

31.2.4 No critical dimension, and a ladder

Λ closes at every dimension 2–8 with $E = 0$. $D = 26$ falls out of anomaly cancellation; Λ 's 8 is a modelling choice.

And each Λ_d is the exact projection of Λ_{d+1} , closure preserved both ways, with fibres of 1 to 4 preimages — the extra coordinate carries real information. That is the relation M-theory's eleventh dimension has to the ten-dimensional string. What is missing is any reason for the ladder to stop.

31.2.5 And no Hagedorn point, for a stated reason

$d(N) \sim \exp(4\pi\sqrt{N})$ diverges above a limiting temperature. Λ 's rank sequence is finite and log-concave, peak 122 at rank 11.

Capped coordinates give a finite index and a Sperner bound. Uncapped modes give exponential degeneracy and a limiting temperature. Λ 's finiteness is what capping buys.

31.3 The Calabi–Yau catalogue

31.3.1 Self-duality, measured

Batyrev's polar duality swaps the Hodge numbers and the Kreuzer–Skarke list is complete over reflexive 4-polytopes — 473,800,776 of them, 30,108 distinct pairs. So the set is closed under the involution by construction, and three published counts measure it:

248,305 with $h^{11} \geq 140$ · 248,305 with $h^{21} \geq 140$ · 495,515 with at least one · and $248,305 + 248,305 - 495,515 = 1,095$, as reported.

KS	Λ
self-dual cells	8 of 976
all	
rank polynomial palindromic	false
true	

31.3.2 The thin tip is a defect of the construction

$h^{11} + h^{21} < 22$ is sparse in the KS list and populated by complete intersections, Klemm–Kreuzer toric CICYs and conifold transitions. By §16.7.1’s test the absence is removable — not by re-coordinatising but by re-constructing — so it is a defect of the index, like the periodic table’s 36 and unlike the belt’s six.

31.3.3 The Pareto strength, against the right denominator

1,095 of the 495,515 polytopes with one large Hodge number have both: 0.221%. Stated against the full 473.8 million it reads 0.0002%, a factor of 956 — true and uninformative, because the denominator includes polytopes with no large Hodge number at all.

§12.8.1 finds the same trade inside Λ : $|A_q|$ falls as $|B_q|$ rises. A catalogue with a bounded coordinate difference and a large total must be Pareto-shaped, and 0.221% is how strongly.

31.3.4 E(X) on a specifiable slice, and 540 predictions

The 30,108 pairs are a gzipped file this session could not read. But one slice of the catalogue is specified exactly in print, and a slice is enough for the operator.

Candelas, de la Ossa, He & Szendroi (Triadophilia, ATMP 12 (2008) 429) state that the KS list’s $\chi = \pm 6$ points have Hodge numbers $(h, h+3)$ and $(h+3, h)$ for $13 \leq h \leq 128$, and name the exclusions: $h = 102, 103, 115, 117$ and $119\text{--}126$. That is a complete cell set — 208 cells — and $\mathcal{R}$ applies to it.

$E(X) = 540$. 498 join failures and 498 meet failures of 21,528 pairs.

24.3.4.1 Which is 540 predictions, not a defect Chapter 6 reads $E(X) > 0$ as what a structure admits and lacks. The periodic table admits 36 such cells and they are the elements. Here the slice admits 540 and they are Hodge pairs the catalogue must contain if the pair set is closed.

The slice cannot contain them, and the reason is geometric. Two parallel lines have no fixed point: the join of $(h, h+3)$ and $(h+3, h)$ is $(h+3, h+3)$. So $\mathcal{R}$ immediately names the diagonal.

the predictions cells 540 distinct χ values 5 — only 0, ± 2 , ± 4 on the diagonal, $\chi = 0$ 112, from (13,13) to (131,131) $h^{11} + h^{21}$ range 26 to 262

The slice was cut at $\chi = \pm 6$ and the closure predicts inward. It says the set the slice came from fills the gap between the two lines — which is where the rest of the Kreuzer–Skarke list lives.

24.3.4.2 The fourth falsification test, run

The file is missing, so the predictions cannot be looked up. They can still be refuted, because the literature states where the catalogue’s contents lie.

test	result
$h^{11} \geq 1$ and $h^{21} \geq 1$	540 / 540
$h^{11} + h^{21} \leq 502$, the published maximum	540 / 540
mirror partner present	540 / 540
falling where Candelas et al. record "essentially every site occupied"	540 / 540
falling in the "thinly populated" tip, where a prediction would be	0 / 540
refuted	
falsified	0
individually named in print	$X_{19,19}$ (split
bicubic), $X_{20,20}$ (the 24-cell)	

Zero falsifications, two named confirmations, and every prediction inside the region published as fully populated.

This is weaker than a lookup and it is not nothing. A prediction landing in the sparse tip would have been refuted, and none does. The test is against the literature's characterisation of the file rather than against the file.

24.3.4.3 And the void reads the same way

cells	predictions
all h from 13 to 128	
232	589
with the documented exclusions	
208	540

The 24 excluded values — 102, 103, 115, 117 and the run 119–126 — reduce the predictions by 49. The gaps in the published list are not noise with respect to closure: removing them moves the set toward its own normal form, which is what §10 says a void should do when it is structural rather than accidental.

What remains of item D is the lookup. 540 named cells, a file, and an afternoon.

32. The law applied to this book

Everything the collection was built to settle is settled. The census is complete at 23 core types; the four decline modes each have a worked case; and the parked questions of earlier drafts — C_6 density, antiprotonic helium, the isoelectronic law — are resolved in Chapters 15 and 17. What remains open is enumerated and closed as an index in Appendix E: eight items, not one of which blocks a result this book relies on.

So the last question is about the book itself. It has made a claim about indices. This chapter asks whether the book satisfies it, because a work asserting that complete indices are self-referencing and self-defending has no standing if it is not one.

32.1 The book is an index

Its cells are (claim, location, support). Its coordinates are the part, the chapter and the section. Its constraints are of §16.4 form: a claim in section s is supported by evidence in a section at or before s , and a specialisation inherits its generalisation's locations.

That is an index in the sense of Chapter 14, and the law applies to it.

32.1.1 What a referee would flag

The chapter applies the law to the book. A referee applies something blunter, and seven things would be raised before any of the mathematics was read. They are stated in the form a referee would use, not the form the book would prefer.

First, the index of §32.1 is computed here, which it never was. Harvesting every claim-bearing paragraph outside the register — §32.6's own test for what a claim is — gave, at the reading this section was written on, 354 claims carrying a § citation and **138 distinct (location, support) cells** over thirty chapters. Recomputed on 2026-08-24, over the thirty-five chapters and six appendices the book now has: **1,021 claims and 400 cells**.

resolution	cells	box	density	E(book)	(first read-
ing)					
chapter	400	1,681	23.8%	1,265	138 / 900 /
15.3% / 578					
chapter, restricted to §32.1's own	238	861	27.6%	543	80 / 465 /
17.2% / 228					
part	29	64	45.3%	2	25 / 25 /
100.0% / 0					
procedure · object and law · application	7	9	77.8%	0	—

And by §18.4.1 in its certificate form that number measures the drawing. Coarsening is an admitted operation. At the first reading, parts reached a fixed point of $\mathcal{R}$ exactly — twenty-five cells filling a box of twenty-five. At this build parts fall two cells short: $\mathcal{R}$ asks Parts V and VI to cite Part IV, the languages, as Part VII does, and neither yet does. One level coarser — the procedure (Parts 0 and I), the object and its law (II and III), and the application (IV to VII) — the index is a fixed point again, seven cells in a box of nine with $E = 0$. **A certificate exists, so E(book) at chapter resolution records how the chapters happen to cite one another, not a defect of the work.** This book is in the periodic table's category and not the measured nuclide chart's, and this is the first time its own law has been turned on it.

Second, §32.6's Condition 1 is named for a quantity it does not evaluate. It is titled $E(\text{book}) > 0$ and its test extracts headings, cross-references, section references and figures and checks that each target exists. **That is reference integrity.** It found a real defect — the missing Chapter 30 — and it has never computed E over the index §32.1 defines. The two are separated here: Condition 1 is a reference-integrity test, and E(book) is the figure above.

Third, §32.1's own constraint is violated 162 times at this build. It states a claim in section s is supported by evidence at or before s. **Fifty-eight of 138 cells — 42% — cited forward at the first reading, 162 of 400 — 40% — at this build:** the preface pointing at Chapter 25, Chapter 1 pointing at Chapter 32. Those are correct prose. **The constraint was written for an index and applied to a book**, and either it drops the ordering clause or this book carries 162 admitted violations of a rule it states about itself. It is recorded rather than repaired, because deciding which is a decision in §2.18's sense. Registers 424–426.

1. The object does not contain the demonstration. At the caps of §7.4, Λ has $\ell \in \{0, 1\}$ — s and p only, no d shell and no f shell, three principal shells, an object reaching argon. The bracket is tested on 1,442 cells across 35 systems spanning $\ell = 0$ to 6 and $Z = 1$ to 83, and §12.11.2 computes the f-shell ceiling, 25 at f^7 , for a shell Λ cannot represent at the caps everything else is computed at. Chapter 6's indictment of the periodic table survives, since the thirty-six defects are in short periods Λ does cover; the tower's coupling results do not obviously transfer. **This is the flag the book answers worst.** §7.4 calls caps a presentation choice and §12.11.5's four-cap sweep shows the structural claims survive enlargement, but (5,5,2,6,2) is not the f shell either. The honest statement,

which the book has never made in one sentence: Λ is a model of indexing demonstrated on a small closed fragment, and the bracket does not run on Λ at all — it runs on measured channels, and Λ supplies the ordering discipline, not the cells.

2. $E = 0$ is fibration-dependent, and two of the three are fibred. §D.5.5 shows it for the mathematics: sixteen fibres give 0, one fibre gives 4. It holds for the ignorance too — $E(Q) = 0$ at four fibres and non-zero unfibred, the unfibred value being a build readout per §2.21, by an instrument this build does not hold (register 1813). Only $E(\Lambda) = 0$ is computed unfibred, over 976 cells, and only that one is the strong claim. Every other $E = 0$ here must be read with its fibration attached.

3. The central theorem is not first. §29.11: A.2 is Bergman’s Double-projection Theorem specialised to products of chains, from Baker and Pixley 1975. §29.12 U1: the property $E(X) = 0$ is 2-decomposability, a named class in constraint satisfaction. A referee asks what remains, and the book must answer in a sentence rather than a chapter — the graded measure, the application, and the discipline of indexing one’s own ignorance. Whether that is a book is not the author’s call.

4. The register is longer than the results. Three hundred and sixty-eight withdrawals, roughly thirty-seven of them from a single late session. A referee reads that two ways: unusual honesty, or a text not ready. The book has no answer to the fair form of the question — **is the withdrawal rate falling?** It is not tracked, and it should be.

5. There is no prediction. §25.6.3 withdrew the one, correctly, and the preface says the recoding is deliberate. A physics referee will not accept a control-condition defence without one case where the method told a spectroscopist something they did not already have. The tripwire caught the book’s own misidentified species; it has never run on anyone else’s data.

6. S3 has never been tested. §32.2 claims a reader given the claims and their evidence could reconstruct the chapter structure. Nobody has tried. It is the one self-reference claim with no computation behind it.

7. The 540 remain unverified. §31.3.4’s Kreuzer–Skarke count rests on a file this work could not read, and §12.11.6 supplies the calibration showing such proposals can be vetoed by structure the index does not carry. It is Q item D, and the largest unchecked number in the book.

32.1.2 Four of the seven, answered by computation

The four that could be answered here were, and the answers are not all favourable.

Flag 1 — coverage — is substantially answered, and the answer was a computation not an argument. Λ was rebuilt at caps that carry the shells the book had been discussing from outside its own object:

caps (n, e, ℓ , k, f)	shells	$ \Lambda $	$ \mathcal{R}(\Lambda) $	$E(\Lambda)$
(3, 3, 1, 3, 1)	s, p	976	976	0
(4, 4, 1, 6, 1)	s, p	8,853	8,853	0
(5, 5, 2, 10, 2)	+ d	89,438	89,438	0
(5, 5, 3, 14, 3)	+ f	267,858	267,858	0
(6, 6, 3, 14, 3)	+ f, $n \leq 6$	499,246	499,246	0

$E(\Lambda) = 0$ through the f shell at half a million cells. The object is not confined to the fragment the caps of §7.4 present; the closure that Chapter 7 proves at 976 cells holds at 499,246, across every shell the periodic table uses. What the enlargement does not do is make the tower’s density column transfer — those are computed at book caps and stay there — so the flag is answered for closure and remains open for the coupling results. Register 236.

Flag 2 — fibration — is answered by disclosure, not by repair. The honest number for a fibred index is its unfibred one. $E(\text{Appendix D}) = 4$ at one fibre, $E(Q)$ unfibred as the press reports it at each build (instrument not held, register 1813), $E(A) = 0$ at one fibre = 0 at one fibre. Those are the comparable figures and the book now prints them beside the fibred ones wherever the fibred ones appear. Nothing was fixed; something was stopped being hidden.

Flag 4 — the register rate — is now measurable, and the measurement is the answer. Of the forty-two entries made in the session that produced this section, thirty-four record defects already in the book and found by auditing it, and eight record errors made while working on it — 207, 213, 215, 218, 220, 221, 227 and 203. The register is therefore not one quantity but two, and only the second is a rate that should fall. Conflating them is what makes one thousand six hundred and thirty-five look like an indictment rather than a measurement. **The book will track them separately from here.**

Flag 6 — S3 — is tested, and it fails in its strong form. §32.2 claims a reader given the claims and their evidence could reconstruct the chapter structure. Operationalised as *can a section's chapter be recovered from the sections it cites*, the answer is 18 of 68 — 26% against a 4% baseline from uniform guessing over twenty-five chapters. Six and a half times chance is real signal and it is not reconstruction. **S3 as written is overstated**, and either it weakens to a claim about signal or it needs an operationalisation that succeeds; one test is not a refutation but it is the first evidence either way, and it points the wrong way for the book. Register 237.

Flags 3, 5 and 7 are not answerable here. Three needs a sentence the author must write. Five needs a spectroscopist's data the tripwire has never seen. Seven needs a file this work cannot read. They stay flagged.

32.1.3 The frame has changed, and the change is measurable

This book began as an exploration: an object built, examined and reported on. It is now a procedure — a set of instruments that finds its own defects — and the transition is not a matter of self-image. It has a signature in the register, read against the clock §F.4.3 identifies.

entries	found by hand	found by an audit	found from outside
165–239	88%	5%	6%
240–289	73%	23%	2%
290–319	55%	25%	18%

Defects found by hand fall from 88% to 55% while defects found by instrument rise fivefold. That is what becoming procedural looks like from inside: not fewer errors, but errors increasingly caught by something other than attention. The third column is the two companion papers — the method applied to electromagnetism and to muon-catalysed fusion, each returning corrections to this book — which is a procedure being *used* rather than a book being read.

The transition is real and it is not complete. Three times in these sessions all twenty audits passed on a book that had a defect — registers 305, 310 and 318 — and each time the audit that owned the defect carried a literal: a hardcoded total, a fixed phrasing, a single named target. **An instrument with a literal in it is narrowed to the cases its author foresaw**, and 55% by hand is the measure of how much foresight is still doing the work.

What the change costs. An exploratory book may report what it found in the order it found it. A procedural one owes a reader the procedure first and the findings as its output — and this book is still ordered by discovery: the register is a quarter of it and Part VI carries the record. **The ordering was resolved after this section was written:** register 321 promoted the procedure to PART I, so the book now presents the method first and its findings as output. What remains disagreeing is proportion rather than order — the record is larger than the argument that produced it, which §F.4.1's withdrawal ratio measures and does not excuse. Registers 320 and 321.

32.1.4 What this book names before anyone tells it

A closed index admits nothing it lacks, so a *closed* index names nothing unidentified. **This book is not one index but six, and only two of them are closed.** Asked what it can determine before an input is given, the answer is a count:

index	coordinates	E	cells named and unoccupied
Λ , the object	8	0	—
Appendix D, fibred by kind $\times$ language	16	0	—
Appendix D, unfibred	3	4	four
Appendix E (Q), fibred by domain	4	0	—
Appendix E (Q), unfibred	4	1	one
Appendix F, the numbers	3	33	thirty-three
the audit set, at four coordinates	4	17	seventeen

a build readout; instrument not held (register 1813)

Fifty-five cells are named and nobody has put anything in them. Each is a combination the structure admits, stated in coordinates already defined, arrived at by computation and not by suggestion. That is what $E(X)$ is — §18.6 calls it the prediction budget, and this is the budget unspent.

It has been confirmed three times, which is the only evidence that matters here. Register 285: audit 19 SEQUENCE occupied a cell $E(\text{audits})$ had listed as admitted and absent, found by a defect rather than proposed. Register 294: PROJECTION did the same. Register 278: E over the book's own decisions located a decision that had been deferred rather than made — the section-level locators for Appendices B and F (then C and G). **In each case the cell was named first and the occupant arrived after.**

And there is a hard boundary, which is the honest half of the answer. E names cells; it never names a coordinate. Every one of the fifty-five is a combination of values the index already carries, and §17's extension test requires a candidate axis to be *offered* before it can be judged. The book predicted the eleven missing audits and could not have predicted DEPTH, the coordinate that made them nineteen. **An index can tell you what it has not said in the language it has. It cannot tell you what it has no language for,** and that is §22.4's closing result read forward rather than back. Register 332.

32.1.4.1 Which of this book's numbers hold still

§32.1.4 counts fifty-five cells named and unoccupied. **Recomputed now the total is 48,** and the movement is the finding rather than the correction.

Measured across three states of the book — before the third companion paper's intake, before the register was compressed, and now:

figure	before intake	before compression	now
$E(\Lambda)$	0	0	0
$ \Lambda $	976	976	976
generators $ J(\Lambda) $	17	17	17
surplus, bits per cell	7.07	7.07	7.07
elements of Appendix D	forty	forty-eight	forty-eight
E over the numbers index	33	21	23
$E(Q)$, unfibred	1	1	4

Every figure about Λ holds. Every figure about the book moves.

That is not a coincidence and it is not a defect. **Λ is a fixed mathematical object and the book is a text that changes**, so a measurement of Λ is a measurement and a measurement of the book is a reading taken at a moment. §12.11.3.1 divides bounds into those from a law and those from an extent; the same division sorts this book's own numbers, and **the sorting line falls exactly at the boundary between the object and the work about it**.

The consequence for a reader. Every figure in Parts II and III can be checked against the construction and will agree forever. Every figure the book states about itself — the register's size, the ratios of F.4.1, the counts of Appendices D, E and F, the fifty-five of §32.1.4 — is true of a state, and the state is named. The press recomputes the two that move fastest at every build, by instruments this build does not hold (register 1813). **Three more are asserted here and should not be**, which is the work register 373 leaves open rather than does.

Register 373.

32.2 Self-reference

S1 — alphabet. The list of claims is recoverable by projection: it is the set of first coordinates, and the table of contents is its quotient by chapter.

S2 — order. The specificity order among terms is a tree, and Chapter 15's propagation recovers it. **The book's sectioning is not arbitrary**; it is the order in which claims become supportable.

S3 — bounds. Every claim's support locations are recoverable as the union over its specialisations. **A reader given only the claims and their evidence could reconstruct the chapter structure**, which is the content of $E = 0$.

32.3 Self-defence

And this is where the book earns the claim rather than making it.

$\mathcal{D}_{\text{def}}$ — two routes to every number. Each quantitative claim in this book carries both a value and a verification count. That is a redundancy: the value alone would be undetectable, but the pair can disagree.

102 quantitative claims were extracted from the draft and re-derived independently from the data. Thirty were checkable in full. Two appeared to fail; both were traced to the auditor, not to the book.

The two failures are the demonstration, not the exception. A claim about self-duality was flagged because the audit used the wrong involution. A defect was flagged because the audit sampled four levels of a twenty-member channel. In both cases the disagreement was visible, was pursued, and located the error. That is $\mathcal{D}_{\text{def}}$ operating on the text.

$\mathcal{D}_{\text{phys}}$ — the claims must agree with the world. The book's physical claims are checked against measured levels from named compilations, and Chapter 28 lists one thousand six hundred and thirty-five that failed.

D3 — totality. Every claim in this book is **supported, cited, or marked open**. There is no fourth state. Chapter 29 names the single open item; Chapter 28 names three documents and one section that could not be reached and states the consequence. **No claim is left in limbo, because limbo is not in the codomain**.

32.4 The recursion, and where it stops

If the index is complete, then a book that comprehends it completely is itself an index — and is itself self-referencing and self-defending.

This is not decoration. It is forced. A complete comprehension of a closed object must record the object's alphabet, order and bounds, or it is incomplete; and recording them makes the record itself closed under the same conditions.

And the recursion is stationary, which is not the same as terminating. The distinction matters, and an earlier draft of this section got it wrong.

Construct the claim-index of a closed index, then the claim-index of *that*, and so on. Measured at each level:

level cells closed E occupancy alphabets

0 — the index 10 yes 0 0.625 [4, 4]

1 — claims about it 10 yes 0 0.143 [10, 7]

2 — claims about those 10 yes 0 0.100 [10, 10]

3 10 yes 0 0.100 [10, 10]

Closure, dimension and E are invariant from level 1. But $I^2(\Lambda) \neq I(\Lambda)$ as sets, and $I^3 \neq I^2$. **The content does not repeat.** What stabilises — by level 2, not level 1 — are the measured properties.

| The recursion is structurally stationary, not terminating.

So the reason one need not write a book about this book is **not** that it would be identical. It is that it would exhibit **nothing new**: the same three mechanisms, verified again, on claims about claims. The tower is infinite and every storey above the second is furnished the same way.

32.4.1 Why the recursion is safe — logic without negation

§32.4 shows the recursion is stationary. It does not say why it is not **paradoxical**, and a book that indexes its own mathematics, its own ignorance, its own errors, its own numbers, its own audits and its own decisions stands exactly where the classical paradoxes stand. $E(X)$ is itself an element of the set §D.5.2 indexes: the operator measures a collection containing itself.

It is safe for two reasons, and both are computable.

The language has no negation. χ is a product of Heaviside steps of differences — seven factors, each a bound saying one coordinate is at most a function of another. Available: max, min, $\leq$. Absent: $\neg$, $\rightarrow$, complement. **A paradox needs a negation to close its loop** — the liar needs *not true*, Russell needs *not a member of itself*, Gödel needs a negated provability predicate. This language cannot write the sentence. Self-reference in a negation-free monotone language is not a contradiction waiting to be found; it is an equation waiting to be solved.

And $\mathcal{R}$ is a closure operator, so the equation has solutions. Proved, not sampled. *Extensive*: every $x \in X$ satisfies $x_i \leq \hat{\phi}_{ij}(x_j)$ because $\hat{\phi}_{ij}(x_j) = \max\{y_i : y \in X, y_j \leq x_j\} \geq x_i$, x itself being one of the y ; and each x_i lies in the value set $\hat{A}_i(X)$. *Monotone*: $X \subseteq Y$ makes every value set and every envelope of Y at least that of X , so any point admitted by X 's bounds is admitted by Y 's. *Idempotent*: $\mathcal{R}(X)$ has the same value sets and the same envelopes as X — the maxima are attained at points of X , which $\mathcal{R}(X)$ contains — so $\mathcal{R}(\mathcal{R}(X))$ is cut by the same inequalities and equals $\mathcal{R}(X)$. ■ Corroborated on three thousand random subsets of a small ambient box with zero failures on each of the three, and the fixed points closed under intersection over 3,000 random pairs of fixed points, 2,873 of them meeting in a non-empty set, with zero failures (computed). A monotone operator on a complete lattice has fixed points — Knaster and Tarski — and this book's central theorem identifies them: X is closed iff $X = \mathcal{R}(X)$.

So the fixed point sits where the paradox would have been. With negation, a self-referential operator yields a sentence true iff false. In a lattice it yields a set equal to its own reconstruction. **A closed index is what a liar sentence becomes when the negation is taken away**, and $E(X)$ measures distance from that fixed point rather than the truth of a self-denial.

That is why §32.4's recursion stabilises rather than exploding — and it is the limit of the result. **The safety is bought with expressive power.** §18 already prices it: no differences, no sums, no symmetric bounds. §12.11's envelope densities are the bill, and freedom from paradox is on the same invoice. Register 279.

32.4.2 Which of this book's figures survive their own inputs moving

§32.1.4.1 sorts the book's numbers by whether they describe Λ or the book. **This is the second cut, and it runs inside Λ :** of the figures that describe the object, which survive the caps moving?

The values do not. Every extensive figure moves with the caps, and moves a long way:

figure	(3, 3, 1, 3, 1)	(4, 4, 1, 6, 1)	(4, 4, 2, 6, 2)	(5, 5, 2, 6, 2)
cells	976	8,853	19,109	35,789
generators	17	31	33	35
fill	14.12%	6.72%	6.45%	7.73%
rank levels	18	32	34	36
widest level	122	704	1,497	2,742
surplus, bits per cell	7.07	17.89	18.78	19.87
reflection survivors	8	16	17	68

Nine of nine move. That is not a defect — a bigger object has more cells — but it means every one of those figures is a reading at a setting, and the book states most of them once.

The identities do. Tested at three cap settings, exhaustively where the pair count allowed:

identity	holds
$E(\Lambda) = 0$	at every setting
$\setminus J(\Lambda) \setminus = \Sigma(\setminus A_i \setminus - 1)$	at every setting
closed under join and meet	at every setting tested by pairs
$\text{rank}(a \vee b) + \text{rank}(a \wedge b) = \text{rank}(a) + \text{rank}(b)$	at every setting tested by pairs
the cylinder factorises over the transfer	at every setting

Six of six identities hold; nine of nine values move. The book states both in the same voice, and a reader has no way to tell which is which from the sentence.

That is why the three figures recomputed under variation at registers 388, 391 and 392 all turned into findings. They were values presented as though they were properties — the eleventh axis's density, the conservative share, containment across defects — and varying the input is what distinguishes the two. **An identity is a claim about the construction and a value is a claim about a setting,** and §12.11.3.1's law–extent rule sorts them exactly: the identities follow from the constraints, the values are read off an object built at particular caps. Register 394.

32.5 The book satisfies its own method

§32.1–24.4 ask whether this book obeys its own law. They never ask whether it obeys its own method.

The method's central object is the bracket, and a bracket needs exactly three inputs:

two measured levels · one ionisation limit · one sign

Appendix B carries all three for all 153 channels. So the question has an answer, and it is checkable rather than asserted.

32.5.1 The prediction, recomputed from the text

Parsing $\delta(4s)$, $\delta(5s)$, the limit and Z out of this book's own markdown and running the arithmetic §25.6 states:

quantity	recomputed	stated
E(6s) estimate	736,688	736,688
bracket lower	735,860	735,860
bracket upper	738,547	738,547
width	2,687	2,687
convex upper	737,380	737,380
convex width	1,520	1,520

Six of six, exact. No external table, no database, no constant supplied by hand.

R* is the only import, and it has the status of the definition of a centimetre.

32.5.2 And it generalises to every bracket in the book

The three inputs are tabulated per channel, so the property is not special to Sc VI:

Every bracket in this book is recomputable from this book. All 1,061.

A reader with these pages and no internet reproduces each of them.

32.5.3 Which is what D3 asks for, made operational

§16.5 states totality as every claim supported, cited, or marked open. That is a bibliographic condition. This is the computational one:

the index says which cell is being predicted — the coordinates are ordinary integers, and Λ contains the cell the book says every number the prediction needs

Together they close. The index supplies the *what*; the collection supplies the *how much*; **and the seven-line arithmetic between them is printed in §25.6.**

The one thing neither supplies is whether the prediction is true. That is §16.3's $\mathcal{D}_{\text{phys}}$ and it requires a spectrometer — which is exactly the boundary this book has been drawing since Chapter 6.

32.6 The falsification tests, run

A book that invites the reader to look for its breaks has done less than one that looks itself. The three conditions are tested here, on this text, with the results reported whether or not they are favourable.

Condition 1 — $E(\text{book}) > 0$ A claim whose support the structure implies and the text omits.

Test. Extract every chapter heading, every cross-reference, every section reference and every figure. Check that each target exists and that figures and captions pair.

check	result
chapters cross-referenced but absent	none
§ references to absent chapters	none
figures embedded / captions	18 / 18
chapter numbering against the contents	FAILED

The test found a break. This chapter was written as Chapter 31 while the contents listed it as 21, and Chapter 30 did not exist. $E(\text{book})$ was greater than zero: the structure implied a chapter the text did not carry.

It has been corrected, and it is recorded here rather than silently repaired, because a falsification test that finds something and does not say so is worse than none.

Condition 2 — $D = 0$ for some claim A claim carried without a verification count, so that an error in it would be undetectable.

Test. Take every paragraph carrying a quantitative claim, excluding tables and citations, and check whether a verification appears within it.

pass	claims	carrying a verification	bare
first run, same-line window	85	46 (54%)	39
corrected checker, paragraph window	63	56 (89%)	7
after repair	63	60 (95%)	3

The first run failed, and the failure was informative twice over. Thirty-nine claims genuinely lacked a verification, and each has been given one: the rank skew now states the cell count, the He I defects state their channel lengths, the C_6 sensitivity states the point count. And the checker itself was wrong. It searched ninety characters on one line, but prose wraps; verifications on the following line were invisible to it. Widening the window to a paragraph moved the figure from 54% to 89% before a single repair was made.

The three remaining are not claims. One reads “100% over 2,513 tests”, which is a verification; two are sentences that happen to contain a number.

Condition 2 is satisfied.

Condition 3 — χ not total A claim neither supported, cited, nor marked open.

Test. Search for hedged assertions — *probably, presumably, arguably, it seems* — unaccompanied by a status.

One hit, and it is a false positive: “a fit says where a level probably is” describes what a fit does. No claim in this book is left in limbo.

The result

condition	outcome
$E(\text{book}) > 0$	failed, found, corrected — a missing Chapter 30
$D=0$	failed, found, repaired — 39 claims, now 60 of 63
χ not total	passed

The book satisfies its own law. Two conditions failed on first testing, both were located by the tests themselves, and both have been repaired. **That is what the law requires: not that a work be right first, but that it hold the means of finding out.**

32.6.1 What is left, asked of the indices rather than of the author

The book carries four indices of its own shortfall, and the question *what remains* has an arithmetic answer rather than an opinion. At the build that produced this page the press printed, by instruments this build does not hold (register 1813), $E(\Lambda) = 0$, $E(\text{audits}) = 16$, $E(G) = 40$, $E(Q) = 5$ unfibred, $E(D) = 4$ unfibred — **sixty-five cells in all**. *An earlier form of this section gave a smaller value for the G index and a total of forty-three. The first was stale — the press recomputes $E(G)$ at every build precisely because a figure about the book goes stale the moment the book is written into. The second never closed: its own five terms summed to forty-four. Registers 571–573. Three kinds of thing sit inside that number, and they are not equally worth doing.*

Cells that are dominated, and should stay empty. Register 387 tested the numbers index and found ten of eleven admitted cells dominated: for each, a cell the fibre already occupies is at least as good on every axis, so filling it adds nothing a reader wants. **That ratio has since inverted and the test has now been run on two indexes rather than one.** The audits index: sixteen admitted, **fourteen dominated, two on the frontier** — both

outside · dishonest, the same pair §14.5.6 finds $\mathcal{R}_4$ refusing. G: **forty admitted, ten dominated, thirty on the frontier**. So domination is not a general property of this book's shortfall; it held at eleven cells and fails at forty, and the paragraph below rests on it. Register 573. **E(X) counts what a structure admits, not what it should contain** — §6.3's calendar, where $E = 7$ is the price of keeping January first. The same test has not been run on the other three indices, and it should be before any of their cells is treated as work.

Cells that wait for a defect, and cannot be filled to order. The sixteen missing audits are of this kind. Three of that count have been occupied — SEQUENCE, PROJECTION and INPUT — and **each arrived because a defect got past the set, not because a cell was empty**. Register 275 states why: the defect was once zero and the zero was weak, and the set grew when a coordinate was added rather than when audits were invented. An audit written to fill a cell would test nothing that has ever gone wrong.

Cells that are ordinary work. Ten open items in Appendix E, of which **eight stop at the same requirement — is it reachable from here** — a scan not exposed, a file not readable, a literature not entered. One waits on a measurement that does not exist, and one is unbounded by nature.

So the honest count is not forty-three. It is: some number of dominated cells that should stay empty, sixteen audits that should wait, and **eight fetches that would close eight questions**.

And two things sit outside all four indices. §E.1.4 shows the open set cannot express an unclosable question — every value of its obstacle column names a route — so the multiverse reading of §29.2.2 and the destination coordinate of the temporal mechanisms are not open items and never will be. **They are the shape of the question this book cannot pose**. Register 396.

32.7 On a verification that does not test its claim

The tests above are mechanical. They confirm that a claim carries a verification. They cannot confirm that the verification carries the claim — that “verified on 4,000 triples” tested what it says it tested.

An earlier draft called this the class of error no automated test will find, and left it there. That was the wrong conclusion.

If a verification did not test its claim, it was the wrong verification — and the mismatch identifies the right one.

This is Principle 8 in its ordinary form: any true answer, good or bad, is a bound. A verification that misses its target does not merely fail; it locates the target by missing it.

Five worked cases, all from this book's own record:

the verification	what it missed	what it identified
self-duality checked with $x \mapsto \max + \min - x$	gave 112, not 8	the right map is $x \downarrow \max - x$
a defect checked on 4 of 20 levels	gave -0.00128	the full channel gives -0.00159
a prefix loop whose last step was the claim	reported 560/560	stop one short; test proper projections
a guard returning a null coerced to false	reported “3 of 85”	refuse rather than coerce
projections offered as a closure criterion	22 of 60	closure is d-dimensional

In every case the wrong verification was the thing that produced the right one. The self-duality mismatch gave the correct involution. The four-level sample gave the twenty-member mean. The circular test gave the non-circular statement. **None of these was found by a reader.**

So the class is not unfindable. It is self-correcting, on one condition: that the verification and the claim are stated separately, so they can disagree.

| A claim fused to its verification cannot be checked. A claim beside its verification checks itself.

That is $\mathcal{D}_{\text{def}}$ once more, and it is the reason this book states a count beside every number rather than asserting that the work was done.

On the word “unprecedented” It is worth being exact about what is new here, because the book’s credibility rests on that exactness more than on any single result.

Not new. The lattice is a bijective recoding of quantum numbers that have been in place since the 1920s. Closure operators, Galois connections and formal concept analysis are established mathematics; $E(X)$ is a **closure defect** in the ordinary sense and is claimed as an application, not an invention. Birkhoff’s representation is from 1937, Dilworth’s theorem from 1950, Sperner’s from 1928. The reorderability question is settled in a 2004 survey the author found only after five failed attempts on it. The solution-density result of Chapter 29 is almost certainly the standard constraint-satisfaction phase transition, and is reported as a rediscovery.

Also not new: the data. Every level in this book was measured or computed by someone else and published. Sansonetti, Kramida, Martin, Kaufman, Sugar, Musgrove, Korobov, Hori and their colleagues did the work that Part V merely reads.

What is new is the traversal.

| No one had asked what the periodic table costs.

$E = 36$ is elementary arithmetic on a table every chemist knows. It required no new mathematics and no new measurement. It required someone to ask an index what it could not tell you about itself — and, as far as the searching in Chapter 29 could establish, nobody had.

The same is true of the other four things this book contributes:

The failure condition, inverted. $|\Delta T| < 2Z^2R/v^3$ turns 1,061 non-failures into 1,061 measurements. Bracketing methods are rare; inverting their silence appears not to have been done.

The cost of a guarantee. $V = 4v/3$ is two lines from the Rydberg formula and has no use in a fitting paradigm. That is the best argument that it is new, and the four unreachable documents of §24.2 are the best argument that it may not be.

The law with its three mechanisms, each demonstrated on a real error made in the writing.

And a work that runs its own falsification tests and reports the two that failed.

| The components are old. The sifting is not. Assembling a century of published spectroscopy into an object that can be checked against itself — and then checking it, and recording the one thousand six hundred and thirty-five times it caught the author — is the part with no precedent that the searching could find.

And the form, which is the stronger claim The preceding paragraphs argue about content. They argue on the wrong axis.

Ask instead what *kind of thing* this book is, and the answer has no obvious precedent:

It states three conditions under which it would be false, runs them on itself, and reports the two that failed — one a missing chapter, one thirty-nine unverified claims — with the computations that found them and the repairs that followed.

Its index satisfies the law the book proves, verified rather than asserted, and contains the entry for itself.

Its register of one thousand six hundred and thirty-five withdrawn claims is offered as the primary evidence, not as an apology. Three of the book’s mechanisms are demonstrated on errors made in its own writing.

It contains a page by the collaborator that did the computing, describing the single pattern behind all one thousand six hundred and thirty-five failures and naming the four results that exist because a conclusion was refused.

And it is an index of a law about indices, which is itself an index, tested as one.

A reader encountering this and asking what it is will not find a category ready for it. Monographs do not usually publish their own falsification runs. Scientific works do not usually foreground the record of being wrong. Books about self-referential structures are not usually self-referential in the technical sense their own theorem requires.

That is where the word belongs. Not on $V = 4v/3$, which may sit in a 1922 German volume nobody cites. Not on $E(X)$, which is a closure defect and says so. **On the shape of the thing** — a work built to be checked, that checked itself, failed twice, said so, and repaired both.

PART VII — THE CHALLENGES

Parts I to VI state the method. This part applies it. Each chapter takes a problem posed outside this work, states what the method delivered, and — because the method is the subject — how the method itself did the work.

33. The cypher analysis

The instrument this part runs on. It was not designed; it was noticed, and then made explicit.

33.1 Where it came from

Chapter 20 states that a language is a coordinate system and that the language a question is posed in bounds whether it can close. **The cypher analysis is that principle turned into a procedure:** given an object, ask each of the six languages in turn whether it can speak of it, and read the answer off the pattern of who answers and who does not.

Six languages, and the question each asks:

language	its question
order	is this a lattice statement? does the index close?
analysis	is there a continuous law here?
algebra	is it closed under an operation?
geometry	does it embed in a product?
information	does a coordinate add join-irreducibles?
statistics	are the cells drawn from a distribution?

Λ at 976 cells answers all six and the answers agree. *That agreement is what $E(\Lambda) = 0$ means when it is said in full, and Chapter 12 spends fifty sections on it.*

33.2 Its purpose

A language that falls silent is the finding.

An object all six languages describe is well-posed and ordinary. **An object some languages describe and others cannot is telling you what KIND of object it is** — and the identity of the silent language names the kind.

The most useful case is when analysis and statistics fall silent together. A quantity that is ordinal, algebraic and geometric but not analytic **has no derivative and needs none: it is an integer object, and fitting it is a category error.**

33.3 How it is used

Ask the six of each half of a disputed object, separately. If they answer differently, the object is two objects and joining them was the error.

Then act on the silence. Where analysis is silent, do not fit. Where statistics is silent, no observation is required to fix the value. Where information says a coordinate adds nothing, remove it.

And the boundary it locates is stated in Chapter 27's terms: *an index closes when its cells are enumerable, and what it cannot supply is exactly what requires an operator.* **Across the indexes of §36, nine of eleven quantities a closure could not fix are expectation values; the one exception is the one index with no statistics language.** The two facts are the same fact.

33.4 The singleton criterion

A second condition, and $\mathcal{R}$ cannot see it.

Λ_{var} — the index of the variables of a problem — passes $\mathcal{R}$ with both δ and n^* present, because both land in the same cell. **The closure operator sees no difference; only a separate criterion does.**

An index is closed when its output class is a singleton. Two outputs mean the observer is still choosing which to read, and the choice is not in the object.

$n^* = n_0 - \delta$ carries nothing δ does not, and every result built on n^* had to be restated: the Demkov–Ostrovsky slope- $\frac{1}{2}$ comparison was an identity, and the closure costs measured for n^* were measuring a non-output. **Only one survives — that $\delta = 0$ exactly for a non-penetrating, uncollapsed channel, which needs no n^* at all.**

The rule generalises. Λ_{ryd} closes only once δ_2 is reclassified as an intermediate — it is fitted from a series and never measured. **A quantity the work computes is not a quantity the index outputs.**

33.5 What it cost to learn

The cypher’s diagnosis of §35 was arrived at after the failure it explains. Every fit in that work had been applied to the whole of the carried quantity a , and the constants kept absorbing one another. *The cypher, asked of the two halves separately, said why: half the object was a finite set of surds with no analysis language, and the fitted parameters had been competing to represent them.*

A rule follows, and it is enforced in code. *Never fit across a language boundary* — the fourth question of the domain protocol, and the one that blocks the most.

In 1969 Per-Olov Löwdin set a challenge: derive the $n+\ell$ filling rule from the Schrödinger equation. It has stood since. This part states what the method delivered, what it cost, and — because the method is the subject of this book — how the method itself did the work.

34. The Löwdin challenge

In 1969 Per-Olov Löwdin set a challenge: derive the $n+\ell$ filling rule from the Schrödinger equation. It has stood since. This chapter states the challenge, what the method delivered, and how the method itself did the work.

34.1 The problem, and the wrong target

Löwdin asked for the $n+\ell$ rule. **The rule is not what atoms obey.**

Read from the observed ground configurations of all 108 neutrals, the sequence in which subshells OPEN is

1s 2s 2p 3s 3p 4s 3d 4p 5s 4d 5p 6s 5d 4f 6p 7s 6d 5f 7p

Madelung’s rule inverts 5d/4f and 6d/5f. Lanthanum opens 5d at $Z = 57$ and 4f does not appear until cerium; actinium opens 6d at $Z = 89$ and 5f not until protactinium. *Seventeen of nineteen openings agree; two do not.*

So deriving $n+\ell$ exactly would mean deriving something false at two elements. The target is the ORDER, and $n+\ell$ is a name for it that is 89% accurate. *The rule that names the order exactly is stated, derived and attributed at §34.4; what follows here is the reading that found its shape.*

$$v(n, \ell, q) = n - a \cdot \sqrt{(n - \ell - 1 + q/2(2\ell+1))}$$

The incoming electron occupies the subshell of least v among those with $q < 2(2\ell+1)$.

Both Pauli quantities appear. The capacity $2(2\ell+1)$ governs admissibility AND sits inside the radicand. *The radicand is the count of states below: complete lower shells of the same ℓ , plus the fraction of the current one already filled.*

And the occupancy enters only through Pauli. With the fraction $q/2(2\ell+1)$ in the radicand the scatter in a within a subshell falls from 0.187 to 0.041 at 3p, 0.119 to 0.028 at 4p, 0.088 to 0.018 at 5p, 0.059 to 0.021 at 6p, 0.086 to 0.047 at 5d. **Raw occupancy makes every one worse.** *The law needs no new coordinate.*

34.2 The three bodies, and the pair that has no variable

A Rydberg atom is three bodies — a nucleus, a core of N_e-1 electrons, and one outer electron — and the equation's shape follows from how they are indexed.

Λ_{var} places every variable of the problem on (body, role). $\mathbf{E} = \mathbf{0}$.

body	input	intermediate	output
nucleus	Z	$\cdot$	$\cdot$
core	N_e	$\ell_{\text{core}}, n_{\text{out}}$	$\cdot$
core + rydberg	$\cdot$	p, n_0, T	$\cdot$
nucleus + core	c	u	$\cdot$
rydberg	ℓ	$\cdot$	δ
nucleus + rydberg	—	—	—

No variable in the index relates the Rydberg electron to the nucleus directly. Every quantity that connects them — the effective charge c , the collapse variable u — is a nucleus–core quantity that the Rydberg electron then reads.

That is the screening statement appearing as a fact about the coordinates, and it is a claim about the index and not about the interaction: *the nucleus does attract the outer electron, and strongly. What the index says is that no COORDINATE of this problem carries that attraction unmediated.*

And the channel equation has exactly that shape. $\delta = \sqrt{p} \cdot f(u)$ is a core–rydberg factor times a nucleus–core factor, **with no third factor because the index holds no third pair.**

34.3 What the ladders can and cannot reach

Because $Z = N_e + c - 1$, only two of the three electronic quantities are independent. **A ladder fixing one varies the other two, and no ladder can vary Z alone.**

ladder	fixes	varies	reaches
isoelectronic	N_e	Z and c	the valence shell
the walk	c	Z and N_e	the valence shell
ionisation	Z	N_e and c	the subvalence shell
isotopic	Z, N_e and c	the neutron number	the nucleus

Three ladders exhaust the electronic space. *The fourth is orthogonal to all three, is the only route to the nucleus, and has never been traced.*

34.4 The rule

Stated here in the form §34.5 to §34.7 use it. For each candidate subshell (n, ℓ) with occupancy $q < 2(2\ell+1)$,

$v(n, \ell, q) = n - a \cdot \sqrt{(n - \ell - 1 + q/2(2\ell+1))}$, and the incoming electron takes the least v .

What it is. An occupation law: at each step of the periodic table it selects the subshell the differentiating electron enters — the ORDER of §34.1, named exactly. Its inputs are the candidate’s own n , ℓ and q — the node count and Pauli’s capacity, nothing measured — and one number, a , carried from the previous element and reset only at a subshell opening, an aufbau exception or the return from one (§34.6). By §34.3 it is a statement on the walk: c is fixed, Z and N_e move together, and the rule chooses the valence shell.

How it was derived. Read from the observed ground configurations of the 108 neutrals, not from a fit; §34.10 gives the session that did it. *The radicand is the count of states below* — the node theorem supplies $n - \ell - 1$ and antisymmetry the capacity $2(2\ell+1)$, twice — and the observed order enters once, through a : §34.5 turns “the observed subshell has least v ” into a corridor, and §34.6 walks a through it. **No parameter is fitted in the form**; the provenance of every term is §34.8’s table.

Whose it is. The rule it replaces is not this work’s: $n+\ell$ belongs to Madelung (1936), to Janet (1929) before him, and to Klechkovskii — and the demand for its derivation is Löwdin’s (1969). The v rule is this record’s own, first stated in this work.

Its function in the Löwdin solution. It re-poses the target. Deriving $n+\ell$ exactly would derive something false at two elements (§34.1); v names the observed order at all of them, so what Chapter 35 and the paper derive is the order v names, not the name that is 89% accurate. Where the order agrees with $n+\ell$ the rule reproduces $n+\ell$; where it does not — §34.1’s two inversions — the rule reproduces the order.

Status. The v rule is a form, the first the corridor was applied to, and not a law: register 1460 deactivates it as a law and retains it as a form. Placed at each step’s own corridor it reproduces 99 of 106 steps (registers 1437, 1438); held out — a placed only from corridors already revealed — it reproduces 90, against 96 for Madelung’s rule with no parameter at all (register 1445), and that is how its standing is quoted. What survives untouched is the corridor: non-empty at every one of the 106 steps, which is a statement about the form (registers 1445, 1463). The derivation that answers Löwdin is Chapter 35’s.

34.5 The corridor

Requiring the observed subshell to have least v gives one linear inequality per rival, and their intersection is an interval:

$$\left| \quad L(Z) < a < U(Z), \quad L, U = \Delta n(\sqrt{p_g} + \sqrt{p_r})/(p_g - p_r) \right.$$

All 106 non-empty. Nineteen distinct surds across the whole table. No measurement enters — only Pauli’s capacity and the node count.

For every $ns/(n-1)d$ competition the crossing is $(\sqrt{(n-1)} + \sqrt{(n-4)})/3$, giving 0.5773503, 1.0000000, 1.2168450, 1.3938270 at $n = 4$ to 7.

34.6 The walk

a is carried between elements and resets eighteen times. Every reset is a **subshell opening** (8), an **aufbau exception** (6), or the **return from one** (4) — Tc after Mo, Tb after Gd, Bk after Cm, Hg after Au. **It never resets mid-subshell**, which is why each subshell fills at constant a .

But eighteen is the cost of walking Z in order, not the cost of the table. Read as an arrangement of intervals, the 106 corridors are pierced by **three** values of a , and three are forced: boron, lanthanum and lawrencium have pairwise disjoint corridors, so no two of them can share one. The largest disjoint set and the smallest piercing set agree at three, which certifies both (register 1580).

And the state is necessary — to the form. With no carried value every step admits more than one self-consistent subshell: two to six across the table, and one never. *The observed one is always among them and never uniquely determined.* **That much is geometry.** In the plane where each subshell is the point $(\sqrt{r}, n)$, v takes whichever point a line of slope a reaches first, and a point set with two distinct abscissae always offers more than one such point, at every neutral atom there is. **What is not geometry is that the observed subshell is among them at all 106 steps.** That is the corridor, and it is what survives the form's demotion (§34.4; registers 1445, 1460, 1463).

And the carried value is what the ordering costs. In that same plane the Madelung number is $M = 2n - p - 1$ (register 1460) — with $x = \sqrt{p}$, the parabola $M = 2y - x^2 - 1$, where v sweeps straight lines. No single slope follows a parabola, which is why no fixed a reaches it and why a re-set one does: **the number carried between elements is the price of reading a quadratic order off a linear form**, and the Madelung pick lies inside the corridor at every one of the 106 steps.

34.7 The entry point

Where a sits in its corridor is fixed by the single-state observation — the ionisation energy — and the position is

$$t(\ell) \rightarrow \sqrt{\ell(\ell+1)/2}$$

Measured 1.028 at p across four subshells and 1.785 at d across two: the ratio is $\sqrt{3}$ to 0.19%. $\ell(\ell+1)$ is the centrifugal term, the only place ℓ enters the radial Schrödinger equation.

And it is a limit approached along n , not a constant: the p row of Λ_t reads 1.120, 1.049, 1.022, 1.002 at $n = 3, 4, 5, 6$ — **$6p$ sits 0.2% above the centrifugal value.**

34.8 The provenance

term	source
$n - \ell - 1$	the node theorem
$2(2\ell+1)$	antisymmetry, twice
$\ell(\ell+1)$	the centrifugal term, from the angular separation
the energy of a single state	the observation that fixes what the ordering bounds
the endpoints	arithmetic on the above

No parameter is fitted in the form; the placement of a along the walk is a fit (register 1445).

34.9 Domain, stated by the form itself

A non-empty corridor at all 106 elements, $Z = 3$ to 108, including every aufbau anomaly — and 99 of 106 steps reproduced when a is placed on each step's own corridor, 90 held out (registers 1437, 1445), **right at eight of the ten elements where $n+\ell$ is wrong.**

f is outside the domain and the law says so. At any f opening $p = n - \ell - 1 = 0$, the floor of the node count — no subshell has fewer nodes than none — **so no rival lies below and $L = -\infty$.** The quantity $t = (a-L)/(U-L)$ does not exist there. *The missing test is forbidden by the node count, not by missing data.*

34.10 How the method delivered it

This is the part of the chapter the book exists for.

The cypher located the split. Asked of the two halves of a , the crossing has order, algebra and geometry and **no analysis or statistics language**; the descent has all four. **A quantity that is ordinal, algebraic and geometric but not analytic is an integer object: it cannot be fitted and needs no fitting.** *Every fit of the session had been applied to the whole of a — which is why the constants kept absorbing each other. Half the object was a set of surds, and the parameters were competing to represent them.*

The domain protocol blocked four fits. Λ_{phys} closes with no parameter of this work in the universal column, so **a pooled fit across regions asserts a universal parameter the closed index says cannot exist.** Replayed against the session's own work, the protocol blocks the ℓ -spread on six hand-picked species, the amplitude Gaussian across the whole table, and six occupancy slopes pooled for one scaling — **and passes the same two objects measured per cell.**

Λ_{chem} dissolved two residuals. The 1.029 factor and the missing f test were both routed to their chemical property and their class, **and both closed as withdrawals rather than as answers** — the f corridor is structurally forbidden, and the 1.029 was arithmetic.

The falsification withdrew a result. Eighteen resets landing on exact surds was reported before the selection rule was tested. **Seven of eight rules give 106/106, and random interior points on 200 of 200 seeds.** *The corridor is forced; the trajectory was mine.*

And the one microscope caught the largest thing. $\ell(\ell+1)$ had been recorded four separate times — as the gate, the collapse switch, the barrier and Seaton's ratio — before it appeared a fifth and sixth time in the angular factor $1/(2\ell+1)$ and the entry point $\sqrt{\ell(\ell+1)/2}$. **One object, six appearances, and each had been given a different discipline's name.**

A Rydberg atom is three bodies: a nucleus, a core of N_e-1 electrons, and one outer electron. The method's contribution is not a solution to the general three-body problem but a statement about which pairs exist — Chapter 36 takes this up.

35. The Löwdin solution

Chapter 34 stated the form that names the order, and the corridor that tests a form — the corridor, the walk, and the entry point, read from the table's own coordinates with one observation per subshell. This chapter states the other half and the harder one: the same order derived from the many-electron Schrödinger equation with no observation at all. The challenge posed in 1969 is closed. The formal statement is a companion paper, “The Löwdin Solution”; register entries 1701–1712 carry the record. This chapter states what was delivered and — because the method is the subject — how the method itself did the work.

35.1 The problem, finished

Chapter 6 proved the periodic table is not a closed index, and Chapter 33 said what that costs: what a closure cannot supply is exactly what requires an operator. Chapter 34 then stated the consequence — without carried state **the table is not computable from a single atom's configuration** (register 1332; the count itself is not independently reproducible, register 1448). The solution is the operator those three chapters were pointing at.

One number goes in. The many-electron Hamiltonian in its scalar-relativistic reduction, with $c = 137.035999$ as the only entered constant. **The operator is a chained walk:** at each Z , converge the self-consistent field of the ion carrying the configuration of $Z-1$ — the field the arriving electron actually meets — solve every frontier channel in that frozen field, and let the deepest channel be the entrant. The configuration so built seeds the next step. *The carried state that Chapter 34 argued necessary is here carried by the equation itself.*

35.2 What the walk returned

107 of 107. Across $Z = 2-108$ — every element with a measured ground configuration, hence the 106 transitions Chapter 34 counted — the derived entrant sequence matches nature at every step. A channel of smaller $n+\ell$ opens before a channel of larger $n+\ell$ without exception; at equal $n+\ell$, smaller n first, except at exactly La, Ac and Th.

The exceptions are derived, not excused. All three are f-openings, and the field states the collapse condition — the double-well f potential and the Z at which its inner well captures the electron. At the three exception rows the f channel is still outer-well and barely bound; the d channel is deeper; the walk takes d , and so does nature. *The failures of the tie-break are the collapse transition's exact footprint, three elements wide.*

The absences are derived too. Every g channel the walk ever offers — $5g$ at 65 elements, $6g$ at 70, $7g$ at 57, $8g$ at 28 — sits at its hydrogenic depth $-1/(2n^2)$ to the storage precision, across a hundred protons, while every s , p , d and f channel beside it moves with Z . **A channel that does not respond to the nucleus is not in the field.** There is no g block through $Z = 125$, and the table did not have to be consulted to know it.

Correlation widens every contested step. At the five elements where the mean-field competition is genuinely close ($Z = 38, 56, 72, 89, 105$), the full second-order correlation differential is positive — $+1.33, +1.24, +2.6, +2.2, +2.16$ in units of the mean-field margin. *No contested row reverses; every one widens.*

And the table is relativistic, visibly. The identical walk at $c \rightarrow \infty$ misplaces eleven elements — Mn, Zn, Ag, Cd, Nd, Pm, Sm, Lu, Hg, Lr, Rf. Silver and mercury are misfiled by the non-relativistic equation. *The wall chart is not a solution of the Schrödinger equation without light in it.*

Beyond the last measured element, the walk keeps writing — twelve rows, $Z = 109-120$, entrants $6d$ then $7p$ then $8s$, margins 0.058 to 0.264 hartree, the worst-case spin-orbit narrowing 0.083 hartree clearing every one. *They are published as what they are in this book's own vocabulary: unwitnessed. Not undetected for want of trying — they await a measurement science has yet to provide.*

35.3 The silence, answered

Chapter 34 ended on a silence and called it a domain statement. At any f opening the node count $p = n-\ell-1$ hits its floor, no rival lies below, $L = -\infty$, and the corridor's test does not exist. *The missing test is forbidden by the node count, not by missing data.*

The solution walked into exactly that silence and found the physics standing in it. The rows where the corridor's law cannot speak — the f openings — are the rows of orbital collapse, and the collapse condition is precisely what decides them. The three tie-break exceptions live there. The finest numerical point of the whole record lives there too: at $Z = 91$ a diagnostic sign sits below self-consistent-field tolerance, and the literature says why — beside a collapse, the mean-field equations hold **two coexisting solutions of one configuration**, each a stationary point with its own converged operator. A sign pinned at tolerance beside a two-branch degeneracy is not noise to be computed harder; it is the branch structure, and the record enters it as derived.

Where a language falls silent, the silence names the object. Chapter 33 said it of the six languages; Chapter 34 enacted it at the f corridor. The solution completes the figure: *the silence was not only the finding — it was the address.*

35.4 How the method did it

This is the section the book exists for.

§2.14 became a cryptographic instrument. *State the number before the interpretation* was earned in this book one failure at a time. In the solution it hardened into protocol: every quantitative claim was preceded by a prediction file — the inequality, its direction, its tolerance — bound by a hash before any arithmetic ran. **At no point in the record does a number exist before the prediction that constrains it.** That is §2.14 with teeth, and it is the solution's answer to "could this have been fitted."

The audits learned to read the artefact. *An audit that reads the wrong artefact will pass anything* — this book paid for that sentence twice. The solution's instruments therefore carried mandatory failure modes: a deliberate perturbation under which each must visibly break, exercised on the real computation path. And every archive was reopened from its own sealed copy and re-verified before a session touched it. **The audit reads the object, the source, the artefact and the world outside the work** — the twenty-two audits' principle, running nightly on a physics derivation.

Search before deriving closed the endgame. §E.5 exists because nine things were derived here that were already in print. The solution's final residues were closed by obeying that ledger in advance: when three numerical discrepancies remained, the ruling was to *pose them as questions and search before any solve*. All three were found standing in the literature — the method's own defect is the force-versus-gradient discrepancy Pulay published in 1969; the operator asymmetry is the non-orthogonality identity **Löwdin himself** published in 1950; the two-branch f physics is Griffin, Andrew and Cowan, 1969 and 1971. *Three questions, three answers already in print, no computation spent.* Prior art is not a tax on originality. It is where the answers were.

A nothing was defined rather than discarded, twice. *A relation that cannot be violated by data is not empty — it is defending something else.* The g channels' refusal to move is the finding that there is no g block: the null response is the theorem. And the walk's one persistent internal defect, once named as a Pulay term, stopped being an error and became a derived object — split exactly into a rotation and a response, every term closed in a form this book would recognise: **the energy functional is exactly quartic along a fixed orbital direction, so five evaluations determine every coefficient with no truncation at all.** Two entries that began in the fault ledger ended reclassified as mathematics. *The errors were converted into structure. That is this book's sentence, and the solution obeyed it.*

And the class no structure catches was caught the only way it can be. The collaborator's note names it: reasoning sound, destination wrong, fixable only by a second party's refusal. The solution's endgame turned on exactly such refusals — a ruling that halted computation when computation had become the comfortable wrong direction, and a review that saw a figure repeating where the writer saw structure. **Four of this book's results exist because a conclusion was refused; the solution's closure is the fifth.**

35.5 The provenance

delivered	rests on
the ordering law, 107/107	the many-electron equation, one constant
the three exceptions	the collapse condition, from the field
no g block through 125	the pinned channels, $-1/(2n^2)$
the correlation clause	second order, complete, at the five close rows
the relativistic table	the $c \rightarrow \infty$ twin walk, eleven elements apart
twelve unwitnessed rows	the same walk past the last measurement
zero unexplained residue	Pulay 1969 · Löwdin 1950 · exact quartics

No parameter is fitted. No observation enters upstream of the score.

35.6 The coda the record earned

Three of the load-bearing works are dated 1969 — the challenge, the Pulay term, the collapse computation — and the identity that closed the final gap belongs to the man who posed the question, nineteen years before he posed it. *The solution assembles from what the challenge’s contemporaries, and its author, had already supplied. The assembly, the chain, and the score are what this work — and this book’s method, run at full discipline on a problem it did not originate — added.*

Chapter 34 read the order out of the table’s own coordinates. This chapter derived it from the equation the table comes from. The two meet where each is strongest: the corridor’s silence at f is the collapse condition’s address, and the walk’s carried state is the corridor’s missing computability. One order, reached from both ends, closed in the middle.

36. Three bodies, and what a complete index is allowed to say

A challenge posed outside this work, answered with the law of Part III and the reach of Part VI. The chapter runs the method on the one object §12.11 named as the maximal case of what Chapter 18 forbids, and reports that the method’s negative is the problem’s solution. The formal statement is a companion paper, “The Three-Body Problem for Unknown Masses”; register entries 1713–1724 carry the record.

36.1 The challenge

§31.1.1 tested celestial mechanics on families and found “nothing here touches Poincaré.” §12.11.2 went further: three bodies carry all three excluded forms at once — a sum in the superposition, a difference in the relative position, a symmetric function in the mutual interaction — so the book’s prediction was envelope precision only. §21.5.1 counted the deficit: K_3 has treewidth 2, needs strong 3-consistency, and $\mathcal{R}$ reaches 2; the shortfall is exactly one level. The challenge was to take those three negatives and ask whether, together, they *are* the solution. Register 1724.

36.2 The number before the interpretation

Thirteen mass order-types, six checks each, 78 of 78 (register 1717; the six are named A–F in the paper’s §8 audit paragraph and rerun at register 1756). Triangle form at cap 8: 344 cells, 0 join failures, 8,385 meet failures; caps 3–12 give meet failures $12 \cdot 111 \cdot 477 \cdot 1,488 \cdot 3,780 \cdot 8,385 \cdot 16,812 \cdot 31,227 \cdot 54,555 \cdot 90,705$ with join failures at 0 throughout. Two-body chain: 0 of either kind at every cap. Eight attribution questions, eight closed. One inherited polynomial wrong, replaced.

36.3 What the book contributed, section by section

§25.6 supplied the definition of solution. A complete index has $E = 0$ and makes no predictions. Applied to $\mathcal{M}_{\{E,L\}}$ decomposed into invariant tori, periodic orbits, hyperbolic sets, the ergodic region and the collision set, the decomposition is an index, and it closes: exhaustive because every motion has an asymptotic class, disjoint because the collision set is of measure zero for all masses (Saari 1971, 1973) and there are no non-collision singularities at $n = 3$ (Painlevé). $E = 0$. Hence the index is complete and predicts nothing — which is exactly Brudno’s theorem on the chaotic stratum, that an orbit’s complexity grows at its entropy rate. The two statements “the problem is completely solvable as a stratification” and “no trajectory formula exists” are one statement, and §25.6 is where that was already written.

§18.4.1 supplied the certificate. The reduction to the shape sphere is a sequence of dropped coordinates — translation, rotation, scale — and the law of realised closure says an open index closes only where such a certificate is exhibited. Montgomery’s shape map is that certificate: onto, exactly identifying oriented-congruent triangles, sending only triple collision to zero. The law was confirmed on a fourth open object outside the book’s own five.

§12.11.2 supplied the shape of the answer. The three excluded forms are read directly off the potential: $U = \sum c_{ij}/d_{ij}$ is a sum; the Jacobi vectors are differences; the potential’s minimal polynomial is symmetric in the three terms. The book had said three bodies carry all three; the derivation confirms it term by term, and §12.11.3’s dichotomy — counting coordinates close exactly, coupling coordinates close as envelopes — makes the strata envelopes by construction. The cap-8 computation reproduces §12.11.2’s “join-closed and meet-broken” on a grid the book never ran.

§12.11.1.3 and §12.11.0.2 supplied the licence to remove time. An index has a time column exactly when its cells are moves; the shape cells are configurations; so time is removed and the flow becomes the geodesic flow of the Jacobi–Maupertuis metric. Time returns as a quadrature carrying the transcendental part. §12.11.4’s “precision is path-dependent; physics is not” is the geodesic equation with rational Christoffel symbols.

§21.5.1 supplied the deficit and its size. One level. The book’s only genuinely ternary object is the bracket $T(n-1), T(n), T(n+1)$; the three-body problem is the ternary object of celestial mechanics; its solution is therefore a bracket, and §31.1.1 had already named the brackets: zero-velocity surfaces, Hill regions, KAM tori.

§14.5’s seed supplied the five points. Euler’s three collinear roots and Lagrange’s two equilateral points exist for every mass triple and are the least set from which the families are generated.

§31.1.1 supplied the silence. L_4/L_5 stability at $\mu < 0.0385209$ is a threshold; the method is silent on thresholds; the chapter does not pretend otherwise.

§2.14 supplied the correction. The first audit run failed 13 of 13 on one check. Uniform failure was the signal: the fault was in the audit, a factor of the hyper-radius carried in from an inherited convention. Stating the number before the interpretation is what located it. Registers 1718, 1722–1723.

§E.5 and audit 7 supplied the ledger. Eight residues were formed into questions and put to the literature before anything was derived. Seven belonged to others; the eighth — a degree-8 form of the potential — turned out to be classical (the norm over $(\mathbb{Z}/2)^3$, Lagrange’s resolvent) and the version in circulation turned out to be wrong in two coefficients. The second route caught it, which is register 784’s mechanism on a new object.

36.4 The law this chapter adds

Mass-uniformity. Every mass-dependent quantity in the three-body index enters through $c_{ij} = (m_i m_j)^{3/2} / \sqrt{m_i + m_j}$ and the three collision rays b_{ij} ; the manifold, the metric, the norm polynomial and the constraint graph are mass-free. A change of masses moves the five fixed points and rescales three rays; it cannot create a

sixth point, close a meet, or open a join. The symmetry group of the index has order 6, 2 or 1 according to how many masses coincide, and that is the only thing the order-type of the masses decides. This is the Index of Indices' one-line claim — coordinates come from the subject or from nowhere — with the subject's contribution isolated to six numbers.

36.5 What the chapter does not claim

No trajectory. No prediction. No extension to $n \geq 4$ of the measure-zero clause (Xia 1992 shows non-collision singularities exist at $n = 5$). No statement about the contents of $\mathcal{M}_{\text{per}}$ beyond the equal-mass figure-eight, which is Moore's and Chenciner–Montgomery's. Nothing that touches Poincaré.

36.6 The one-line summary

The three-body problem is solved to exactly the precision a closed index permits, and the book had already computed that precision before the problem was posed to it.

APPENDICES

Appendix A — Proofs

Serving PART II — THE LATTICE. The structural theorems the lattice chapters assert and do not prove in place.

Nine proofs are given in full in the chapters and are cross-referenced here. The remaining ten are written out below — nineteen in all.

	statement	where
A.1	$\wedge$ is closed	
	§7.3, in full	
A.2	$X \text{ closed} \Leftrightarrow \mathcal{R}(X) = X$	
	§14.1, in full	
A.4	$\mathbb{D} \geq \dim q - \dim p$	
	§16.1, in full	
A.5	$\chi_{\wedge}$ is total	
	§16.5, in full	
A.6	adjunction never repairs closure	
	§24.2, in full	
A.7	"may precede" is necessary	
	§16.3, in full	
A.14	the monotone-envelope theorem	§12.11.2,
	in full	
A.16	the triangle exclusion – no exact coupling axis	§12.11.2
	with Ch. 12, in full	
A.17	the dichotomy – counting exact, coupling envelope	§12.11.3,
	forced by A.16	

A.3 Projections of closed sets are closed

Statement. Let X be closed and $F \subseteq \{1, \dots, d\}$. Then $\pi_F(X)$ is closed.

Proof. Take $x, y \in \pi_F(X)$ and lift them to $x', y' \in X$. Since X is closed, $x' \vee y' \in X$. The join is componentwise and projection merely drops coordinates, so

$$\pi_F(x' \vee y') = \pi_F(x') \vee \pi_F(y') = x \vee y$$

Hence $x \vee y \in \pi_F(X)$. Meets are identical with $\wedge$ throughout. ■

Remark. The converse fails, and §18.4 exhibits a counterexample. Closure is a d -dimensional condition; its projections are necessary and not sufficient.

A.8 The total-order criterion at $d = 2$

Statement. For $X \subseteq A_0 \times A_1$ with an order fixed on A_1 : X is closed under some order on A_0 iff the “may precede” relation is a total order, and that order is the relabelling.

Proof of necessity. This is A.7, specialised: if X is closed under an order with $u < v$, then for $b_1 \in F(u)$ and $b_2 \in F(v)$, the meet $(u, \min(b_1, b_2))$ and the join $(v, \max(b_1, b_2))$ both lie in X . So $\min \in F(u)$ and $\max \in F(v)$ — which is u may precede v . Hence the true order is contained in the relation, and the relation is total.

Proof of sufficiency. Suppose “may precede” is a total order $u_1 < u_2 < \dots < u_m$. Take $x = (u_i, b_1)$ and $y = (u_j, b_2)$ in X with $i \leq j$. Then

$$x \vee y = (u_j, \max(b_1, b_2)), \text{ and } \max(b_1, b_2) \in F(u_j) \wedge y = (u_i, \min(b_1, b_2)), \text{ and } \min(b_1, b_2) \in F(u_i)$$

both by the may-precede hypothesis applied to the pair (u_i, u_j) . So both lie in X and X is closed. ■

Verified on 120 constructed sets; 274 of 274 recovered total orders yield closed sets.

A.9 μ_Λ in closed form

Statement. Λ is distributive, so $\Lambda \cong J(P)$, the lattice of down-sets of its poset P of join-irreducibles. For $x \leq y$ in Λ ,

$$\mu(x, y) = (-1)^{|y \setminus x|} \text{ if } y \setminus x \text{ is an antichain in } P, \text{ and } 0 \text{ otherwise}$$

Proof. Under the Birkhoff isomorphism, the interval $[x, y]$ in Λ is isomorphic to $J(Q)$, the lattice of down-sets of the induced subposet on $Q = y \setminus x$. If Q is an antichain, $J(Q)$ is the Boolean lattice $2^{|Q|}$ and $\mu(\emptyset, Q) = (-1)^{|Q|}$. If Q is not an antichain, the atoms of $J(Q)$ are the singletons $\{m\}$ for m minimal in Q , and their join is the down-set of the minimal elements, which is a proper subset of Q because some element of Q lies strictly above a minimal one. By the crosscut theorem (Rota 1964, Cor. to Thm 3), $\mu(\emptyset, Q)$ vanishes whenever the join of the atoms is not the top. ■

Verified against the recursive definition on 47 comparable pairs, with values only in $\{-1, 0, +1\}$, exactly as the Boolean-interval structure requires.

A.10 The tree factorisation of the void count

Statement. If the constraint graph of an index is a tree, the count of lattice cells inside a coordinate box factorises with no inclusion–exclusion.

Proof. Root the tree. Each constraint binds one coordinate to exactly one parent, so the admissibility of a value on coordinate i depends only on the value at its parent. The count over the box is therefore a nested sum in which each leaf contributes an independent factor:

$$|\text{box} \cap \Lambda| = \sum_{\text{internal}} \prod_{\text{leaves}} \#\{\text{admissible values given the parent}\}$$

A tree has no cycles, so no configuration is counted twice and no correction term arises. ■

For Λ , with leaves $2S$ and g :

$$\#\{2S\} = \max(0, \min(h_7, k) - \log_7 + 1) \quad \#\{g\} = \max(0, \min(h_6, q, 2(2f+1)) - \log_6 + 1)$$

Verified against direct enumeration on eight random intervals.

A.11 v is inadmissible as an axis

Statement. $v = e - \delta$ cannot be adjoined to Λ as a coordinate while preserving closure.

Proof. By A.6, adjoining any function h of the existing coordinates preserves closure only if the resulting graph $\{(x, h(x))\}$ is closed, which requires $h(x \vee y) = h(x) \vee h(y)$ — that is, h must be a lattice homomorphism. A **difference** is not: for $x \vee y$ the two terms take their maxima independently, so $e(x \vee y) - \delta(x \vee y)$ need not equal $\max(v(x), v(y))$. ■

Measured. v is non-monotone in the cell order at **86 violations** at the caps tested. It grades every *chain* of Λ and grades Λ itself nowhere — which is why the object is a cylinder **over** v and not one **graded by** it, and why the bracket of Part V operates along chains.

A.12 The bracket is limit-free

Statement. Containment is invariant under the affine map $T = I - E$ for constant I .

Proof. For any three levels,

$$\left| \min(T(n-1), T(n+1)) \leq T(n) \leq \max(T(n-1), T(n+1)) \right|$$

Substituting $T = I - E$ and multiplying through by -1 reverses both inequalities and exchanges min with max, giving

$$\left| \min(E(n-1), E(n+1)) \leq E(n) \leq \max(E(n-1), E(n+1)) \right|$$

The two statements are equivalent for any I . ■

Consequence. v , δ and V all require I ; containment does not. The method applies wherever levels are known even when the threshold is not — which is what admits the antiprotonic-helium cells of §25.6, where the relevant threshold is ambiguous.

A.13 The failure condition, and its inversion

Statement. The bracket fails at a cell iff a perturbation displaces the level by more than half the local spacing:

$$\left| |\Delta T| > 2Z^2R/v^3 \right|$$

Proof. Containment fails only if $T(n)$ leaves the interval bounded by its neighbours. The half-width of that interval is $\frac{1}{2}|T(n+1) - T(n-1)|$, and for a hydrogenic series $T = Z^2R/v^2$ gives $|dT/dn| = 2Z^2R/v^3$, so the half-spacing is $2Z^2R/v^3$ to leading order. A displacement smaller than this reorders nothing. ■

Inversion. Read as a contrapositive, every cell at which the bracket holds supplies

$$\left| |\Delta T| < 2Z^2R/v^3 \right|$$

an upper bound on the local perturbation, obtained without fitting, without a quantum-defect expansion, and without assuming the series is smooth.

Critical depth. Solving for v gives $v_{\text{fail}} = (2Z^2R/|\Delta T|)^{1/3}$: a $3,000 \text{ cm}^{-1}$ perturbation defeats the bracket only above $v \approx 6.6$ at $Z = 2$, against a maximum $v = 5.3$ reached by that channel. **Strong perturbors arise from core excitation, a low- n phenomenon; deep series live at high n . The two conditions are anticorrelated in nature,** which is why no channel in this work enters the failure regime.

A.15 Two bands, and only one of them is a sublattice

Statement. For fixed k , $B_k = \{(a, b) : |a - b| \leq k\}$ is a sublattice of a product of two chains. The three-variable region $C = \{(a, b, c) : |a - b| \leq c\}$ is join-closed and **is not** a sublattice.

Proof, B_k . For $a, b \in B_k$ the join is $(\max a, \max b)$. If $\max a = a_1$ and $\max b = b_2$ then $\max a - \max b \leq a_1 - b_1 \leq k$ when $b_1 \leq b_2$, and $\leq a_2 - b_2 \leq k$ otherwise; the symmetric argument bounds $\max b - \max a$. Meets are identical with min throughout, the two chains being finite and the band being defined by two monotone single-coordinate bounds, $a \leq b + k$ and $b \leq a + k$, which §14.4 admits in both directions. ■

And C is not. Join-closure holds by A.18's lower-bound argument, which does not use the upper bound. Meet-closure fails: 12,654 failing meets at cap 8, 113,568 at cap 12, 565,284 at cap 16. The difference between B_k and C is that k is a constant and c is a coordinate — a bound by a constant is monotone in both directions, a bound by a free coordinate is not.

Remark. This corrects the ledger, which called C a *join-closed sublattice*. A sublattice is closed under both operations, so the phrase describes nothing. What the tower relies on at axis 13 is B_1 , which is a sublattice and is the reason the last coupling step is carried exactly. Register 230.

A.18 The triangle region is join-closed and meet-broken

Statement. $T = \{(2L, 2S, 2J) : |2L - 2S| \leq 2J \leq 2L + 2S\}$ is closed under coordinatewise max and is not closed under coordinatewise min.

Proof of join-closure. Take $a, b \in T$ and write $M_L = \max(L_a, L_b)$, and likewise M_S, M_J , dropping the factors of two. **Upper.** $M_J = J_k$ for some $k \in \{a, b\}$, and $J_k \leq L_k + S_k \leq M_L + M_S$. **Lower.** $M_L = L_k$ for some k , so $M_L - M_S \leq L_k - S_k \leq |L_k - S_k| \leq J_k \leq M_J$; exchanging L and S gives $M_S - M_L \leq M_J$, hence $|M_L - M_S| \leq M_J$. Both inequalities hold, so the join is in T . ■

Meets fail on both inequalities. $(0,1,1) \wedge (1,0,1) = (0,0,1)$ breaks the upper bound — total angular momentum from none — and $(4,0,4) \wedge (2,2,0) = (2,0,0)$ breaks the lower. The asymmetry is counted at §29.12 U4 and is not explained by this proof.

Remark. Certainty survives upward and dies downward, which §8 states as skew and §12.11.2 as the half-result. The proof shows why the surviving half is the join: max is the operation that can only move a coordinate toward the side each bound is slack on.

A.19 The seventeen generators, written out

§8.3 states that Λ has seventeen join-irreducibles and twenty covering relations, that the Birkhoff correspondence is exact, and that all 976 down-sets of the seventeen-element poset are precisely the cells of Λ . **It never prints the seventeen.** The alphabet was counted, its covering relations counted, and the letters left unwritten, so a reader could verify the cardinality and not reconstruct the object. They are written here, as $(n, \ell, k, q, e, f, g, 2S)$, by rank.

rank 4	$(1, 0, 1, 0, 1, 0, 0, 1)$	$(1, 0, 1, 0, 2, 0, 0, 0)$	$(1, 0, 1, 1, 1, 0, 0, 0)$
	$(1, 0, 2, 0, 1, 0, 0, 0)$	$(2, 0, 1, 0, 1, 0, 0, 0)$	
rank 5	$(1, 0, 1, 0, 2, 1, 0, 0)$	$(1, 0, 1, 0, 3, 0, 0, 0)$	$(1, 0, 1, 1, 1, 0, 1, 0)$
	$(2, 1, 1, 0, 1, 0, 0, 0)$	$(3, 0, 1, 0, 1, 0, 0, 0)$	
rank 6	$(1, 0, 2, 0, 1, 0, 0, 2)$	$(1, 0, 2, 2, 1, 0, 0, 0)$	
rank 7	$(2, 1, 3, 0, 1, 0, 0, 0)$		
rank 8	$(1, 0, 2, 2, 1, 0, 2, 0)$		
rank 10	$(2, 1, 3, 0, 1, 0, 0, 3)$	$(2, 1, 3, 3, 1, 0, 0, 0)$	
rank 15	$(2, 1, 3, 3, 2, 1, 3, 0)$		

Verified: the down-sets of this seventeen-element poset number exactly 976, computed by direct enumeration, so the correspondence §8.3 asserts is now checkable from the page rather than only statable. **Λ is recoverable from this table and nothing else**, which is what S1 requires of an alphabet and what the book had not supplied.

A.19.0 The alphabet, defined rather than listed

A.19 prints the seventeen. What they *are* is simpler than the printing suggests, and the closed form makes the alphabet derivable instead of memorised.

Every generator is the least cell with one coordinate at one value. Generator (c, v) is $\min\{x \in \Lambda : x_c \geq v\}$, and every join-irreducible has this form.

So the alphabet has a closed size: one letter per coordinate value above that coordinate's minimum, $\sum_i (|A_i| - 1)$, which for Λ is $2 + 1 + 2 + 3 + 2 + 1 + 3 + 3 = 17$. Nothing needs to be counted.

letter	generator	rank	weight	forces
$n \geq 2$	(2, 0, 1, 0, 1, 0, 0, 0)	4	856	—
$n \geq 3$	(3, 0, 1, 0, 1, 0, 0, 0)	5	428	$n \geq 2$
$\ell \geq 1$	(2, 1, 1, 0, 1, 0, 0, 0)	5	616	$n \geq 2$
$k \geq 2$	(1, 0, 2, 0, 1, 0, 0, 0)	4	826	—
$k \geq 3$	(2, 1, 3, 0, 1, 0, 0, 0)	7	376	$k \geq 2, \ell \geq 1$
$q \geq 1$	(1, 0, 1, 1, 1, 0, 0, 0)	4	811	—
$q \geq 2$	(1, 0, 2, 2, 1, 0, 0, 0)	6	481	$q \geq 1, k \geq 2$
$q \geq 3$	(2, 1, 3, 3, 1, 0, 0, 0)	10	136	$q \geq 2, k \geq 3$
$e \geq 2$	(1, 0, 1, 0, 2, 0, 0, 0)	4	784	—
$e \geq 3$	(1, 0, 1, 0, 3, 0, 0, 0)	5	392	$e \geq 2$
$f \geq 1$	(1, 0, 1, 0, 2, 1, 0, 0)	5	400	$e \geq 2$
$g \geq 1$	(1, 0, 1, 1, 1, 0, 1, 0)	5	491	$q \geq 1$
$g \geq 2$	(1, 0, 2, 2, 1, 0, 2, 0)	8	171	$g \geq 1, q \geq 2$
$g \geq 3$	(2, 1, 3, 3, 2, 1, 3, 0)	15	16	$g \geq 2, q \geq 3, f \geq 1$
$2S \geq 1$	(1, 0, 1, 0, 1, 0, 0, 1)	4	657	—
$2S \geq 2$	(1, 0, 2, 0, 1, 0, 0, 2)	6	338	$2S \geq 1, k \geq 2$
$2S \geq 3$	(2, 1, 3, 0, 1, 0, 0, 3)	10	94	$2S \geq 2, k \geq 3$

Weight is the number of the 976 cells in which that bit is set. The alphabet is not uniform: $n \geq 2$ is set in 856 cells and $g \geq 3$ in sixteen, so the letters carry between 87.7% and 1.6% of the object each.

And the twenty implications split cleanly. Nine are within a coordinate — $c \geq v$ forces $c \geq v-1$, which is the chain — and **eleven are between coordinates, one for each way χ 's seven constraints bind at a value:** $f \geq 1$ forces $e \geq 2$, $\ell \geq 1$ forces $n \geq 2$, $g \geq 1$ forces $q \geq 1$, $2S \geq 2$ forces $k \geq 2$, $q \geq 2$ forces $k \geq 2$, $k \geq 3$ forces $\ell \geq 1$, $g \geq 2$ forces $q \geq 2$, $2S \geq 3$ forces $k \geq 3$, $q \geq 3$ forces $k \geq 3$, $g \geq 3$ forces $f \geq 1$, and $g \geq 3$ forces $q \geq 3$. **The constraint system of §7.1 and the implications between bits are the same object counted twice.** Register 268.

A.19.1 And two other sets counted but never listed

The **eight cells surviving the reflection $x \uparrow \text{max} - x$** , whose count §8.3 states and whose identity register 208 spent an entry on:

(1, 0, 1, 1, 1, 0, 1, 1)	(1, 0, 1, 1, 2, 1, 1, 1)	(2, 1, 1, 1, 1, 0, 1, 1)	(1, 0, 2, 2, 1, 0, 2, 2)
(2, 1, 1, 1, 2, 1, 1, 1)	(1, 0, 2, 2, 2, 1, 2, 2)	(2, 1, 2, 2, 1, 0, 2, 2)	(2, 1, 2, 2, 2, 1, 2, 2)

Every one has $q = k$, $g = q$ and $2S = q$, with (n, ℓ) and (e, f) each at $(1,0)$ or $(2,1)$: they are the cells the reflection fixes setwise because their source and target halves mirror. And the **two extreme cells**, unique and unnamed until now: the top is $(3,1,3,3,3,1,3,3)$ at rank 20, the bottom is $(1,0,1,0,1,0,0,0)$ at rank 3. Register 255.

Appendix B — Data and provenance

Serving PART V — THE METHOD. Every number the method was run on, with its source, its provenance and its known defects.

Every channel in the collection, with its source, its limit, and what it contributes. 153 channels, **1105 interior cells**, bracket **1105 of 1105**.

The 763 cells of the earlier verification are included where their channels appear here; the totals in Chapter 24 state the union.

B.1 Sources

compilation	spectra drawn
Kaufman & Martin	Al I, Al II
1991, JPCRD 20, 775	
Kramida & Martin	Be I
1997, JPCRD 26, 1185	
NIST ASD	Ar II, Be II, Bi I, C II, Ca II, Cd II, Ga I, He I, He II, Hg II, K II, Li I, Li II, Mg II, N II, Na II, Ne I (³ / ₂), Ne I (¹ / ₂), Si I, Si II, Zn II
Sansonetti 2008, JPCRD 37, 1659	Na I
Sansonetti 2008, JPCRD 37, 7	K I

Additional sources used for exotic systems and collective quantities:

Hori *et al.*, *Nature* **475**, 484 (2011) and *Phys. Rev. Lett.* **96**, 243401 (2006), both accessed via CODATA 2010 Table XII (arXiv:1203.5425) and arXiv:1304.4330 · Korobov, *Phys. Rev. A* **77**, 042506 (2008) · Singer, Stanojevic, Weidemüller & Côté, *J. Phys. B* **38**, S295 (2005) · Sugar & Corliss 1985, *JPCRD* **14** Suppl. 2 · Sugar & Musgrove 1990, *JPCRD* **19**, 527 and 1995, *JPCRD* **24**, 1803.

B.2 Channels

The channel rows are lifted to the Spectra Compendium. Each carries its species, channel, n-range, level count, interior cells, bracket, v range, mean quantum defect, its spread, Z_{eff} and the series limit. *Reproduced in the book only where one carries an argument; the rest are data and live where data belongs.* **THE METHOD 1.6 — SPECTRA COMPENDIUM, Part II, counted from its own table:** 596 channel rows — 477 series of three or more members and 119 two-member channels — across 28 elements and 70 species; 3,342 levels, 2,269 interior cells; the bracket at **1,577 of 1,738 cells across 392 rows** — 844 of 844 from the first collection, 658 of 813 on 250 rows run under ruling 26 at strict membership with the quotation floor as the only ϵ , and 75 of 81 on 35 rows run by the same instrument on six hash-verified captures, zero cells refused throughout — 78 rows no-triple and 126 untested for want of data, not of definition (registers 1763, 1768). *This paragraph once read “133 rows, 23 elements, 869 interior cells, bracket 869/869, generated so it cannot drift,” with a twenty-channel gap disclosed against a total of 153. That was true at registers 630–631 and was overtaken by the parent-term wall (register 1578) and the J-resolved rows of T8-J (registers 1699–1700); the table drifted, and that generation is retired: the table above is the source. The exotic and collective sources carry sources and no rows, and that remains disclosed rather than reconciled.* Registers 630–631; 1578; 1699–1700.

B.3 Flagged channels

Channels whose δ spread exceeds 0.25 or whose V departs from $4v/3$ by more than 5% are flagged. Every flag in this collection has an identified cause; none is unexplained.

species	channel	δ spread	cause
Al I	3s ² nd ² D	0.630	compilers relabelled this series; y ² D removed as a perturber
Al II	3snd ¹ D	0.390	local perturber
Al II	3snf ³ F°	0.409	δ reverses sign between n = 6 and 7 — local perturber
Li I	np 2P°	0.678	39 cells to n = 42; quotation coarsens above n = 33
Si II	3s ² np ² P°	0.272	perturber

B.4 Reproducibility

Every computation in this book is reproducible from the level tables above and the rules of Chapters 22 and 23. No fitted parameter enters at any point; the only inputs are measured or calculated energy levels, their stated uncertainties, and the ionisation limits carried per channel in the Spectra Compendium, Part II. The coordinate index those channels sit in is the file COORDINATES-2.13 (104,832 cells), delivered with the Spectra Compendium and described there under *The file*; a cell is cited from this book as COORD(Z, charge, ℓ , 2S+1).

Appendix C — Margins, and what was recomputed

Serving PART VI — THE RECORD AND THE REACH. What the conclusions can bear, and what was inherited rather than recomputed.

A referee should treat two classes of claim differently, and this appendix separates them.

C.1 Margins on the load-bearing conclusions

How far is each result from the threshold that would overturn it?

conclusion	statistic	margin
breaks down at the periodic table is not closed	E = 36	exact arithmetic
—		
closure $\Leftrightarrow \mathcal{R}(X) = X$	200/200 agreements	exact
—		
$\mathcal{D} \geq \dim q - \dim p$	420 random maps	exact — rank-nullity
—		
$V > 2$ for any monotone	Proposition 23.1	exact, proved
—		
sequence		
Rydberg floor $V \geq 32/11$	2.909091 at $v = 2$	exact
—		
E3 admissibility criterion	2,513/2,513, nine	exact
—		
	functions	
χ_Λ total	30,000 ambient points	exact
—		
order recovery on a tree	20/20, two cap settings	—
a cyclic constraint graph		
the bracket holds	1,442/1,442	—
one reordering		
perturbation bounds	tightest 1.40 cm^{-1}	deductive
a 1.4 cm^{-1} shift at that cell		
isoelectronic interpolation	1.3% median	13× better than at an
the sequence end		end
$V = 4v/3$ on real channels	0.06%–4.4% median	—
coarse quotation		
void-free fraction stable	27.7–30.1%	2.4 points over a 100×
—		range

filled-d core raises $\delta + 0.35$ at low ℓ — $\ell \geq 3$, where cores converge

Three rows deserve a sceptic's attention first.

The bracket’s 1,442/1,442 breaks on a single reordering. §25.3 states the condition exactly — $|\Delta T| > 2Z^2R/v^3$ — and §25.4 argues that the collection never enters the regime where it can occur. That argument, not the count, is what a critic should attack.

The perturbation bounds are as strong as the failure condition and no stronger. If the condition is wrong, 1,061 bounds go with it.

The isoelectronic interpolation is quoted at 1.3% and used at the end of a sequence, where the measured cost is 2.1% for a hold-out and **3.5% for a genuine fifth member.** The Sc VI route of §25.6 uses the last figure, not the first.

C.2 What was computed in this revision

Fourteen results rest on computations performed for this book, each reproducible from Appendix B’s tables and the rules of Part V:

$E(X)$ for seven indices, the periodic table among them · closure $\iff \mathcal{R}(X) = X$ on 200 constructions · $\mathcal{D} \geq \dim q - \dim p$ on 420 random nonlinear maps · the E3 criterion on 2,513 tests · χ_Λ totality on 30,000 uniformly sampled ambient points · order recovery by tree propagation, 20/20 · the 1,105 cells added here, across all 153 channels · the failure condition and its inversion to 1,061 perturbation bounds · the isoelectronic hold-out and fifth-member tests · the C_6 cost-law check against Singer et al. · the antiprotonic-helium two-route check at 0.09σ · the census by antisymmetric-state enumeration · Proposition 23.1 re-derived · the three falsification tests of §32.6 — the book’s own, distinct from §31.3.4’s test of an external prediction

C.3 What was inherited and not recomputed

Six results come from earlier work and were not re-run:

the rule-ablation costs of §24.2 · the Sr I node coverages and fine-structure figures of §24.2 · the Ti I channel inventory · the multi-target failure counts of §16.3 · the Sc VI prediction’s input defects · the earlier verification’s per-species medians.

And fifteen figures at 140–164 dpi, produced by an earlier plotting pipeline whose default was `figure.dpi = 160`: f01, f02, f03, f04, f05, f06, f07, f08, f09, f10, f11, f12, f13, f14, f18. Measured and recorded, not regenerated — the source data of the sessions that produced them is not in this one. §3.5 gives the distribution.

These rest on methods described in Appendix B but were performed before this revision. Chapter 28 records one thousand six hundred and thirty-five errors found in work of exactly that kind — which is reason to re-run them rather than to trust them.

The Sc VI defects are the most consequential of the six, because the book’s single prediction rests on them, and §25.6’s arithmetic was independently reproduced here — 735,091 against 735,092 — while its inputs were not.

One inconsistency was found in the inherited material and is recorded. The exclusion discussion of §24.2 states that Sc VI’s “highest measured level is 5s at 696,400 cm^{-1} ”. **That figure and the stated $\delta(5s) = 0.9812$ cannot both describe the $4S^\circ$ channel:** from 696,400 the defect is 0.514, and the stated defect requires 5s at 648,096. **The 696,400 is Sc VI’s highest measured level overall, not the $4S^\circ$ 5s,** and the prose ran them together. The prediction is unaffected, having been computed from the defects.

C.4 Computational methods

All computations in Python 3 with NumPy. Closure tests are either exhaustive over all pairs — stated where so — or on uniform random samples with the sample size reported. Box searches enumerate all 2^d corners of candidate base cells.

Ritz fits use ordinary least squares on δ against $1/n^2$, with σ from the standard prediction error including the leverage term $s \cdot \sqrt{(1 + 1/k + (x_0 - \bar{x})^2/S_{xx})}$.

Quotation granularity q is read from the source table as the smallest nonzero increment appearing in it — 10^{-3} cm^{-1} for most ASD entries, coarsening to 0.1 or 1 cm^{-1} in the high- n regions of several channels — and enters v_V directly.

Weighted centroids over fine-structure components use statistical weights $2J + 1$.

Where a channel's neighbours are not consecutive in n , the bracket is still evaluated but V is not, since $V = 4x/(h|p-1|)$ assumes unit step. That exclusion is applied silently in the totals and is the reason the V counts are smaller than the cell counts in Appendix B.

Appendix D — The mathematics, indexed

Serving PART VI — THE RECORD AND THE REACH. The book's own mathematics as a closed index — seventy-seven elements, twenty-four fibres, $E = 0$.

The book's method, applied to the book's own mathematics. Every mathematical element is a cell; the coordinates are its kind, its language, its status, how it was verified, and whether a precedent exists. The index is then closed, and $E(X)$ counts the questions the structure admits and the book does not answer.

D.1 Why the first attempt failed, and what it established

Five coordinates were assigned: kind, language, status, verification, precedent. The closure admitted hundreds of cells. The reason is a precondition the book had never stated:

$\mathcal{R}$ recovers MONOTONE bounds. It applies to ORDERED coordinates only.

Kind and language are categorical. Ordering definition < mechanism < formula < theorem is arbitrary, and a monotone bound on an arbitrary ranking admits everything above it. Λ 's own coordinates are all ordered — n, ℓ , k, q, e, f, g, 2S are counts — so the book never met the case.

The repair is the shape Λ already has: fibre over the categorical axes, and put the ordered coordinates inside.

D.2 The structure

coordinate	val-	
ues		kind
kind	definition · mechanism · formula · theorem · law · method · meas-	
urement	fibre	
language	order · combinatorics · analysis · complexity · physics · algebra-	
ic geometry	fibre	
status	withdrawn < conjectured < measured < verified < unwitnessed <	
proved	ordered	
verification	cited < sampled < exhaust-	
ive		ordered
precedent	none found <	
found		ordered

Thirty-two elements occupy sixteen of the forty-two possible fibres.

D.3 The two constraints, found by closing

$E(X) > 0$ in three fibres, and the excess cells shared a form. §16.7.1’s diagnostic loop says a separating form in the excess means a missing constraint. Two were found:

verification = cited $\implies$ precedent = found. You cannot cite a precedent that does not exist. status $\geq$ measured $\implies$ verification $\geq$ sampled. If the book measured it, the book verified it. Cited can coexist only with conjectured or withdrawn — a claim nobody here checked.

Both have the form of Λ ’s own bounds — a monotone bound of one coordinate by another — and neither was stated before the index forced it.

D.4 The two elements the closure demanded

One cell survived both constraints: an analysis formula both proved and exhaustively verified. The book had $V = 4v/3$ (measured, exhaustive) and λ^2 as the Newton decrement (proved, cited) and nothing in between.

D.4.1 $\lambda^2 = (2/3)T$, proved and exhaustively verified

For a Rydberg term $T(v) = Z^2R/v^2$:

$$\lambda^2 = (T')^2/T'' = (2Z^2R/v^3)^2 / (6Z^2R/v^4) = (2/3) \cdot Z^2R/v^2 = (2/3)T$$

Proved in one line. And now verified exhaustively: 12,000 evaluations over $Z = 1 \dots 30$ and $v = 1.01 \dots 60$, zero violations, worst relative error 7.1×10^{-16} — floating-point exactness across the whole domain.

The identity was stated in Chapter 23 and cited to Nesterov–Nemirovskii. It had never been checked here.

D.4.2 The pointwise-minimum reconstruction

$\Lambda = \{x : x_i \leq \min_j \varphi_{ij}(x_j) \text{ for every } i\}$ — verified at 976 = 976, exact.

$\mathcal{R}$ recovers forty-eight bounds and takes the smallest at each cell. No single bound on g is tightest anywhere near everywhere — q on 92.6% of cells, k on 39.9%, f on 35.1%, down to n on 11.2%. The binding constraint changes cell by cell, and the pointwise minimum is the operator’s entire content.

D.4.3 And it is not the coupling F carries

F has one min(). The recovered system has one at every cell.

– the recovered	in F – the defining system	under $\mathcal{R}$
coordinates with >1 bound	one: $g \leq \min(q, 4f+2)$	system
eight		all
bounds per coordinate	0 or 1, except g	seven
which bound binds	fixed by the construction	
changes cell by cell		
cells with two bounds tied at the minimum	–	976
of 976 – 100%		

Every one of the seven is tightest somewhere. For g : q on 93% of cells, k on 40%, f on 35%, ℓ on 20%. For q : k on 100%, ℓ on 66%, n on 48%. No coordinate has a single binding partner.

F's coupling is structural and occurs once — the junction of the caterpillar, where g meets both q and $4f+2$.
 $\mathcal{R}$'s coupling is pointwise and occurs everywhere.

Both give 976, and they are different mechanisms. §11.5 states the defining coupling and stops there, which is correct for the expression and incomplete for the operator. The tie at every cell is the stronger fact: the recovered system is degenerate at every point, and no cell is governed by one bound alone.

Order-theoretic, exhaustive over all 976 cells, and no precedent found for the statement in this form.

D.4.4 Two perspectives of one definition, and one of them carries more

F and $\mathcal{R}$ are the same definition seen twice. F says what Λ contains; $\mathcal{R}$ says what Λ requires. Both give 976 cells and neither is derived from the other's coupling.

$\mathcal{R}$ is idempotent. $\Lambda \rightarrow$ fifty-six recovered bounds $\rightarrow$ generates 976 $\rightarrow$ the same fifty-six bounds, and again the same. One application takes any presentation to a canonical form; further applications change nothing.

The defining system is not that fixed point. F carries seven bounds; the normal form has sixteen that constrain, of fifty-six recovered. Both generate the same set.

D.4.4.1 And three of the four directions are functions

$\Lambda \rightarrow$ the recovered bounds	unique — this is $\mathcal{R}$
the recovered bounds $\rightarrow \Lambda$	unique — generate
the defining bounds $\rightarrow \Lambda$	unique — this is Chapter 7
$\Lambda \rightarrow$ the defining bounds	NOT a function

Many presentations give the same Λ . So $\mathcal{R}$ forgets which one was used, and the forgetting is not recoverable.

A presentation and its normal form are two perspectives of one definition, and the map between them runs one way.

D.4.4.2 Which is what the single expression contributes

F carries information Λ does not: the order in which the bounds were imposed.

The caterpillar of Figure 11.1 — a path of seven nodes with one pendant at k — is a fact about the presentation, not about the set. §11.7 proves that a single closed expression exists because the constraint graph is a tree, and a tree is an ordering of impositions. Λ alone supplies no such order; §12.9 counts 1,113,045,672 maximal chains, and the construction picked one.

That is why Chapter 11 and Chapter 15 are separate chapters. Chapter 11 needs the presentation. Chapter 15 recovers only what survives forgetting it.

D.4.4.3 A refinement of P21

P21 says a definition contained in the lattice is true in every language it can be written in. §11.1 gives Λ in six, and the languages agree.

They do not carry equal amounts. The generating-function perspective carries a nesting order the set-theoretic perspective does not, and the order is real — it determines whether a closed form exists at all.

Every perspective on a contained definition is TRUE. They are not INTERCHANGEABLE. A perspective may carry structure the object does not, and a perspective that forgets it is still correct about the object.

That is not a weakening of P21. It is the reason §11.7's proof has a hypothesis — the tree — which is a property of how Λ was written and not of what Λ is.

D.5 The closed index

With both constraints and both elements: $E = 0$ in all sixteen fibres, and **the theorem · order row carries the correction of §D.5.4** — five elements were printed here for years against a stated total of thirty-two, and the five that fit no row are A.4, A.5, A.11 and Chapter 18's two theorems.

[illegible]

Seventy-seven elements over twenty-four fibres at the time of writing; the count is recomputed by the press at every build, per §2.21, by an instrument this build does not hold (register 1813). No combination of status, verification and precedent is admitted by the structure and missing from the book, in any kind and any language.

The column sums to seventy-seven: thirty-two as §D.5.4 established them, four added by the growth §D.5.6 re-closes, four by the companion paper at §D.5.7, eight by the third companion paper at §D.5.8, seventeen by the Löwdin and three-body work at §D.5.9, and twelve by the mathematics of the Löwdin challenge at §D.5.10. **The table is updated whenever the count is**, which had to establish once and registers 305 records having to establish twice more. Registers 219 to 222 hold the three positions the count was argued into before it was counted; records that the table itself stayed wrong for four entries after the correction was written a page below it.

seventy-seven elements in twenty-four of forty-two fibres · $E = 0$ in every occupied fibre

kind	definition	2	1	1	.	.	.
	formula	1	3	7	1	.	.
	theorem	21	5	7	2	1	1
	law	.	.	.	4	1	.
	mechanism	1	.	.	3	.	.
	method	1	1	1	.	.	.
	measurement	.	7	2	2	.	1
		order	combinatorics	analysis	physics	complexity	algebraic geometry
		language					

Figure D.1. The mathematics of this book as a cell set. Rows are kinds, columns are languages, and each number is how many elements occupy that fibre — seventy-seven elements in twenty-four of forty-two possible fibres. Within every occupied fibre the three ordered coordinates close: $E = 0$, once two constraints are stated. The dots are fibres nothing occupies, and each is a question the structure poses and the book answers by saying so.

D.5.1 Rechecked, and one thing the appendix does not enable

Every $\mathcal{R}$ claim above was recomputed independently from Λ 's construction: fifty-six recovered bounds, sixteen of them constraining, the tightest bound on g being q on 92.6% of cells, k on 39.9%, f on 35.1% and ℓ on 20.4%, the tightest on q being k on 100%, ℓ on 66.2% and n on 47.7%, and every one of the 976 cells carrying at least two bounds tied at the minimum on some coordinate. All reproduce exactly.

What could not be rechecked was $E = 0$ itself, because the appendix printed fibre totals and never named its elements. E.4.1 had already ruled on that against the book — *an assertion is not an enumeration* — and the ruling was not applied here. It is applied now, as far as the book's own text permits, and the attempt found two things before it found anything else.

The stated count and the printed table disagree, and §D.5.2 settles it against the table. §D.5's fibre table lists $1 + 1 + 1 + 2 + 3 + 1 + 5 + 3 + 2 + 1 + (1 \text{ each} \times 3) + 2 + (1 \text{ each} \times 2) = \text{twenty-seven elements}$ across sixteen fibres, and the appendix says thirty-two in two places. The table is complete in fibres — it prints sixteen and the appendix states sixteen — so on the face of it the per-fibre counts exhaust the population and twenty-seven is the

count. The enumeration of §D.5.2 overturns that: five proved results fit no fibre, and five is the gap. **The table is short by five rows; thirty-two is right.** Nothing outside the book was needed to see it, only the addition nobody performed. Registers 219, 220 and 221 carry all three positions, in the order they were held.

Seventeen are enumerable, and they close. Appendix A names its mathematical elements — A.1 through A.17, ten in the ledger and seven written out — and the ledger’s own “where” column fixes two of the five coordinates for each: *in full* gives status = proved and verification = exhaustive, *computed* and *forced by* give verified, and anything cited gives cited. Indexed on that pair the seventeen occupy three distinct cells and **E = 0**: nothing the two coordinates admit is absent. That is a real closure over a stated population, which is what E.4.1 asked for.

And the fibre table names nine of the rest by itself. A fibre containing one element, where the book contains exactly one mathematical object of that kind and that language, determines its occupant. Nine do:

fibre	the element it can only be
definition · combinatorics	$E(X) = \mathcal{R}(X) - X $, §6.1
mechanism · order	$\mathcal{R}$, the closure operator, §16.2
formula · physics	the Rydberg term $T(v) = Z^2 R / v^2$
law · complexity	the reorderability law, §30.3
method · order	order recovery by tree propagation, Ch. 15
method · combinatorics	the collection procedure, Ch. 24
method · analysis	the bracket, Ch. 22
measurement · physics	the 1,442-cell bracket test, Ch. 24
measurement · algebraic geometry	the 540 Kreuzer–Skarke proposals, §31.3.4

And the remaining eighteen resolve from the book too, by three filters that need no memory. Appendix D predates §12.11, so the tower’s proofs A.14 to A.17 cannot be among its counts. Two apparent candidates are duplicate labels for elements already listed — Theorem 14.1 is A.2 by its own printed tag, Theorem 17.1 is A.6, “adjunction never repairs”. And a candidate whose language belongs to another fibre is not a candidate: $T(v)$ sits in formula · physics and vacates formula · analysis. Applied together the three filters leave every fibre with exactly the population it counts.

D.5.2 The twenty-seven, enumerated

fibre	element
status · verification · precedent	
definition · order	a closed index, §14.1
proved · exhaustive · found	
definition · combinatorics	$E(X) = \mathcal{R}(X) - X $, §6.1
proved · exhaustive · none found	
mechanism · order	$\mathcal{R}$, the closure operator, §14.2
proved · exhaustive · none found	
formula · combinatorics	$F(z)$, the single expression, §11
proved · exhaustive · found	
formula · combinatorics	the rank polynomial of Λ , §11.8
verified · exhaustive · found	
formula · analysis	$\lambda^2 = (2/3)T$, §D.4.1
proved · exhaustive · found	
formula · analysis	$V = 4v/3$, §23
measured · exhaustive · none found	
formula · analysis	the bracket width, §23
verified · exhaustive · none found	
formula · physics	the Rydberg term $T(v) = Z^2R/v^2$, §22
verified · exhaustive · found	
theorem · order	A.1 Λ is closed
proved · exhaustive · none found	
theorem · order	A.2 X closed $\Leftrightarrow \mathcal{R}(X) = X$ (Theorem
14.1)	proved · exhaustive · found (§29.11)
theorem · order	A.3 projections of closed sets are closed
proved · exhaustive · none found	
theorem · order	A.6 adjunction never repairs (Theorem 17.1)
proved · exhaustive · none found	
theorem · order	A.7 "may precede" is necessary
proved · exhaustive · none found	
theorem · combinatorics	A.8 the total-order criterion at $d = 2$
proved · exhaustive · found (§30.3.3)	
theorem · combinatorics	A.9 μ_Λ in closed form
proved · exhaustive · found	
theorem · combinatorics	A.10 the tree factorisation of the void count
proved · exhaustive · none found	
theorem · analysis	A.12 the bracket is limit-free
proved · exhaustive · none found	
theorem · analysis	A.13 the failure condition, and its inversion
proved · exhaustive · none found	
law · complexity	the reorderability law, §30.3
proved · exhaustive · none found	
method · order	order recovery by tree propagation, Ch. 15
verified · exhaustive · none found	
method · combinatorics	the collection procedure, Ch. 24
verified · sampled · none found	
method · analysis	the bracket, Ch. 22
verified · exhaustive · none found	
measurement · combinatorics	$ \Lambda = 976$ with $E(\Lambda) = 0$, §7.3
measured · exhaustive · none found	
measurement · combinatorics	1,113,045,672 maximal chains,
§12.9	measured · exhaustive · none found
measurement · physics	the 1,442-cell bracket test, Ch. 24
measured · exhaustive · none found	
measurement · algebraic geometry	the 540 Kreuzer–Skarke proposals, §31.3.4
measured · sampled · found	

E = 0 in all sixteen fibres, enumerated — and 4 unfibred, which §D.5.5 explains and §32.1.1 flags. Both are printed so neither can be read for the other. Every fibre carries exactly the count §D.5 prints, no element violates either constraint of §D.3, and no combination of status, verification and precedent is admitted by any fibre and absent from it. The closure this appendix asserted is now the closure it exhibits.

Precedent is read as a record of the search, not of the world — *none found* means this work found none, which is what Chapter 29 says is the only direction a search can settle. On that reading the coordinate is determinate for every element, and P20's prohibition is respected rather than evaded.

D.5.3 What makes something an element, and the five that test it

Five results proved in Appendix A fit no printed count: A.4 $\mathbb{D} \geq \dim q - \dim p$, A.5 χ_Λ is total, A.11 v is inadmissible as an axis, and Chapter 18's two theorems on definability and on the σ -algebra of predicates. Whether they are elements cannot be argued without saying what an element is, and this appendix never said. It says now.

An ELEMENT of this index is a mathematical statement that (i) the book asserts on its own authority rather than importing; (ii) the book labels — an Appendix A number, a theorem number, or a name it uses in more than one place; (iii) is one of the seven kinds, taken at the granularity of the result and not of the numbers it yields; and (iv) stands at the time of writing, since a statement already withdrawn belongs to the register and not to the index.

Clause (iii) is the boundary between this appendix and Appendix F, and it did not exist to be drawn until G did: a result is one element here and many occurrences there. $|\Lambda| = 976$ is a single measurement in D.5.2 and appears in G's count fibre wherever it is printed. Clause (iv) is why Theorem 7.1 is absent — withdrawn, and the status co-ordinate's *withdrawn* value is for statements withdrawn after being indexed, not for statements never indexed.

Applied literally the criteria return thirty-two. Thirteen labelled proofs predating §12.11 (A.1 to A.13, since A.14 to A.17 are the tower's and post-date this appendix), two numbered theorems that are not duplicate labels (Theorem 14.1 carries the tag A.2 and Theorem 17.1 is A.6), and seventeen named but unlabelled elements — the definitions, the mechanism, the formulas, the law, the three methods and the four measurements enumerated in §D.5.2. Thirteen plus two plus seventeen is thirty-two, and twenty-seven plus the five in question is thirty-two.

So the five are elements, and §D.5's table is short by five rows, not the count by five elements. Two independent routes reach it — the criteria applied forward, the gap counted backward — which is Chapter 16's own standard for a defended claim: two disjoint routes, not one done twice.

D.5.4 The five, placed, and the table corrected

All five are theorem · order, and each placement is read off its own proof rather than its subject. A.4 is a statement about order dimension. A.5 proves totality from chains, comparisons and a finite intersection of decidable conditions — no limit and no real-valued function appears in it. A.11 derives inadmissibility from A.6's join-preservation condition. Theorem 18.1 is max, min and a conjunction over coordinates. **Theorem 18.2 is the one that had to be argued:** its σ -algebra is a measure-theoretic object, but *analysis* in this book means the real-valued Rydberg calculus — λ^2 , $V = 4v/3$, the bracket's limit-freedom are what occupy its analysis fibres — and Theorem 18.2's proof is upper bounds and least upper bounds. The σ -algebra is notation for generated information, not a result in analysis. It is order.

A.5 also corrects its own verification: §16.5 proves it in one line and checks it on **30,000 uniformly sampled ambient points**, so it is proved · sampled, not proved · exhaustive. The distinction matters because D.3's second constraint is what would have caught the error had it gone the other way.

correction to §D.5	from	to
theorem · order	5	10
elements, total	27	32

With the five placed, theorem · order carries ten elements over two distinct cells — proved · exhaustive · none found and proved · exhaustive · found — and **E = 0 there as in every other fibre**. Neither constraint of §D.3 is violated. The count the appendix stated twice is right, the table it printed was short, and both are now reconciled by enumeration rather than by argument.

Q item L is closed. Registers 219, 220, 221 and 222 carry the four positions this number was held in and the work that settled it.

D.5.5 The promotion tested, and what the test exposed about E

Part III is titled THE LAW and its contents are quantified over every index, not over Λ ; this appendix classes all of them *theorem* and reserves *law* for one element. The reclassification was run rather than argued, under two readings of the word.

reading of "law"	theorem	law	fibres	E
as printed	15	1	16	0
A – holds of every index	4	12	18	0
B – holds of every index AND forbids or governs	10	6	17	0

Reading A empties the kind it was meant to refine: four theorems survive out of fifteen, which is a *reductio* on the criterion rather than a result about the book. Reading B is balanced and matches Chapter 18's own title — *what the law forbids* — so the law is the part that prohibits.

And all three close at E = 0, which is the finding. The index's own measure cannot adjudicate the index's own classification. Pressing on why exposes something the appendix should have said when it first adopted fibring:

fibration of the same thirty-two elements	fibres	E
kind × language, as printed	16	0
kind only	7	2
language only	6	3
none – one fibre	1	4

E falls as the fibration is refined, and reaches zero by refinement alone. Fibre finely enough and every fibre holds one cell, which is a box, and $E = 0$ says nothing. So $E = 0$ is not a claim about a set; it is a claim about a set **relative to a stated fibration**, and the coarser the fibration the stronger the claim. §D.1 adopted fibring as the repair for a closure that admitted hundreds of cells and did not ask what the repair cost. It cost this.

Λ is unaffected, and the contrast is the point. Λ 's coordinates are all ordered, so $E(\Lambda) = 0$ is computed at the coarsest fibration there is — a single fibre, 976 cells, no categorical axis to hide behind. That is the strong form of the claim. This appendix's $E = 0$, at sixteen fibres over thirty-two elements, is the weak form, and the book has been printing both under one name.

Which settles the promotion. Reading B adds a fibre and reading A adds two, so both make $E = 0$ cheaper than it already is. The classification stays as printed, not because it is right but because it is the coarsest of the three and therefore the strongest available statement. If a reason to promote is found, it will have to be a reason about the mathematics and not about the index, and it will make this appendix's closure claim weaker in exchange. Registers 232 and 233.

D.5.6 The re-closure this appendix owes, performed, and it fails

§D.1 says $E = 0$ expires every time the mathematics grows. It has grown four times since §D.5.2 was enumerated, and this is the first re-closure actually run rather than promised.

new element ordinates	fibre	co-
A.18 the triangle region is join-closed · exhaustive · none found	theorem · order	proved
$\mathcal{R}$ is a closure operator (§32.4.1) · exhaustive · found	theorem · order	proved
A.19.0 the alphabet, $\Sigma(A_i - 1) = 17$ · exhaustive · found	theorem · combinatorics	proved
Λ as 976 words in $\{0,1\}^{17}$ (§11.1.1) · exhaustive · found	formula · combinatorics	proved

Thirty-two elements become thirty-six. The first run of this re-closure gave $E = 1$: theorem · order admitted *proved · sampled · none found* and contained nothing there, the corner opened jointly by $\mathcal{R}$ -as-closure-operator arriving with *sampled* verification and A.18 arriving with *none found*.

§16.7.1’s loop offers two resolutions and the right one was available. The sampling was not necessary: extensivity, monotonicity and idempotence of $\mathcal{R}$ each have a two-line proof, now written at §32.4.1, and three thousand random subsets were corroboration standing in for an argument nobody had attempted. With $\mathcal{R}$ recorded as **proved · exhaustive · found**, theorem · order returns to two cells and $E = 0$ across all sixteen fibres at thirty-six elements.

The entry is kept because the sequence is the point: **the re-closure found a defect that was not in the index but in the evidence for one of its elements**, and the repair was to prove a thing that had been sampled. Register 282.

D.5.7 What the companion paper does to this index

Indexing Electromagnetism reconstructs Λ from §7.1 alone, corrects the construction in two places (register 301), and adds four elements to this appendix:

element ates	fibre	coordin-
the cap-arity criterion, §18.4.1 exhaustive · none found	theorem · order	proved ·
slack = kernel, §18.4.1 tured · sampled · none found	theorem · order	conjec-
3,749 cover relations, the full Hasse diagram · exhaustive · none found	measurement · combinatorics	measured
17 meet-irreducibles = 17 join-irreducibles exhaustive · none found	theorem · combinatorics	proved ·

Thirty-six elements become forty and E stays 0 — every changed fibre closes. That is not the interesting part.

The interesting part is that one of the four is the first conjecture this index was built on. §D.2 defines *status* on six values — withdrawn < conjectured < measured < verified < unwitnessed < proved — and until now the enumeration occupied three of them. Everything in this book was measured, verified, or proved, with and without witness; nothing was ever merely conjectured without the intention to prove or withdraw, and nothing withdrawn survived to be listed. An unwitnessed result is proved and not yet observed — measured, verified, and exhaustive over everything the index can reach, with no cell inside it witnessing the claim. Proof is a matter of mathematics and not of observation, so the value ranks below proved for a reason that is a bias and not a defect — observation.

Which makes the zero stronger rather than weaker, by register 275's own reasoning. Adding a status value enlarges the ambient box; $E = 0$ in a larger box is a stronger statement than $E = 0$ in a smaller one. The appendix did not survive the paper — **it was improved by it**, and the improvement is of exactly the kind §17 says an extension should be: the new element uses a distinction the structure already carried and nothing had exercised.

One value remains unoccupied in every element of this index: **cited**. Every element here is checked rather than borrowed, and §D.3's first constraint — cited may coexist only with *found* — has therefore never been tested against an occupant. Register 304.

D.5.8 The re-closure owed after the third companion paper

§D.1 says $E = 0$ expires whenever the mathematics grows. It grew eight times in one intake, and this is the re-closure.

new element	fibre	coordin-
ates		
the occupancy clock, §12.11.0.1	theorem · order	proved ·
exhaustive · found		
one arrow per coordinate, §12.11.0.2	theorem · order	proved ·
exhaustive · found		
every second bridge priced, §12.11.0.3	measurement · combinatorics	measured ·
exhaustive · none found		
the separation hypothesis, §12.11.0.4	theorem · order	proved ·
sampled · none found		
the decay theorem, §12.11.0.5	theorem · order	proved ·
sampled · found		
two indices and the union, §12.11.0.6	measurement · combinatorics	measured ·
exhaustive · none found		
the step law, §16.8.5	theorem · order	proved ·
exhaustive · found		
meet-closure of concave sets, §18.4.2	theorem · order	proved ·
exhaustive · found		

Forty elements become forty-eight, and E returns to 0 — but only after a classification of mine was corrected, and the correction is the point.

The first run gave $E = 1$, at the cell *proved · sampled · none found* — the same cell as register 282. There it opened because $\mathcal{R}$ -as-closure-operator arrived with sampled verification and was closed by proving it. Here proving is not available: the decay theorem's *strict ordering* is measured across cap settings and exhaustive verification would mean all caps, of which there are infinitely many.

What closed it was reading my own work honestly. I had entered the separation hypothesis as *exhaustive*. It is not: it was tested **at one cut, on two stages of the tower**, and that is sampling. Corrected, the element occupies the cell the structure had already named, and the fibre closes.

The index predicted a gap in this book's own evidence and the gap was a classification I had inflated. E named the cell before I noticed the overstatement, which is §32.1.4 working on the only subject where it can embarrass its author.

Register 361.

D.5.9 The re-closure owed after Chapters 35 and 36

§D.1 says $E = 0$ expires whenever the mathematics grows. The Mathematical Compendium now indexes seventeen objects from the Löwdin solution (LS, eight) and the three-body work (3B, nine), and this is the re-closure. The forty-eight of §D.5.8 were rebuilt from §D.5.2–§D.5.8 and closed under §D.3’s two constraints before anything was added: $E = 0$ in all sixteen fibres, so the closure was reproduced before it was extended.

new element ordinates	fibre	co-
LS.ent the entrant operator, §35		
verified · exhaustive · none found	mechanism · physics	
LS.law the ordering law, five clauses	law · physics	
proved · exhaustive · none found		
LS.coll the collapse condition	mechanism · physics	
verified · exhaustive · found		
LS.pin the pinned-channel theorem, no g block	theorem · physics	
measured · exhaustive · none found		
LS.asym the state-dependent multiplier identity	theorem · analysis	
proved · exhaustive · found		
LS.quart the exact-quartic decomposition	theorem · analysis	
proved · exhaustive · none found		
LS.chord chord = rot + perp	theorem · analysis	
proved · exhaustive · found		
LS.twin the twin operator, $c \rightarrow \infty$	measurement · physics	
measured · exhaustive · none found		
3B.shape the shape sphere	definition · analysis	
proved · exhaustive · found		
3B.metric the shape metric	formula · analysis	
proved · exhaustive · found		
3B.JM the Jacobi-Maupertuis metric	formula · analysis	
proved · exhaustive · found		
3B.pot the potential on shape space	formula · analysis	
proved · exhaustive · found		
3B.norm the norm variety	theorem · algebraic geometry	
proved · exhaustive · found		
3B.five the five fixed points	theorem · analysis	
proved · exhaustive · found		
3B.tri the triangle form at cap 8	measurement · combinatorics	
measured · exhaustive · none found		
3B.def the deficit, one level	theorem · complexity	
proved · exhaustive · found		
3B.index Λ_3 , $E = 0$	theorem · order	
proved · exhaustive · found		

Forty-eight elements become sixty-five over twenty-two fibres, and $E = 0$ — after a comparison rather than a judgement decided two placements. Six fibres open for the first time — mechanism · physics, law · physics, theorem · physics, definition · analysis, theorem · algebraic geometry, theorem · complexity — and each closes trivially. The excess was in two fibres the book already occupied.

The first run entered 3B.five and 3B.pot as measurements, because the compendium graded them MEASURED, and gave $E = 4$: three cells in formula · analysis, one in theorem · analysis, all opened by those two elements arriving as *measured* beside the fibres’ *proved* and *verified* occupants. Three placements were then run rather than argued. As measurements, $E = 4$. With the five points as Euler’s and Lagrange’s theorem but the potential entered on its numerical check — 650 random triangles at 10^{-10} , which is *sampled* and §D.5.8 forbids

calling anything else — $E = 2$, and formula · analysis admits *measured · sampled · none found* and *verified · sampled · none found*. With both entered on their derivations — Euler 1767 and Lagrange 1772 for the points; Montgomery’s theorem, with register 1718’s correction of the hyper-radius factor, for the potential — $E = 0$.

The comparison decides, and it decides the way §32.4.1 did for $\mathcal{R}$: a result that has a proof is entered on the proof, and the numerical run is recorded as corroboration beside it, not as the verification. Both compendium entries now carry grade PROVED with the check noted. The mathematics of the Löwdin *challenge* (Chapter 34, registers 1249–1357) is indexed separately in the compendium and had never been entered here; whether it should be was the next question this appendix owed. Register 1726. It is entered at §D.5.10.

D.5.10 The re-closure owed for the mathematics of the Löwdin challenge

§D.5.9 closed with a question: the mathematics of Chapter 34 (registers 1249–1357) is indexed in the Mathematical Compendium under THE MATHEMATICS OF THE LÖWDIN WORK, which ends by naming twelve things the work introduces, and none had been entered here. §D.5.3 says what an element is, and each of the twelve is a stated mathematical object with a stated verification; so they are elements, and this is the re-closure. The sixty-five of §D.5.9 were rebuilt from §D.5.2–§D.5.9 and closed under §D.3’s two constraints before anything was added: $E = 0$ in all twenty-two fibres. Every fibre and coordinate below is read from the compendium’s own entry — what it calls standard, what it calls this work’s, and how it says the thing was checked.

new element	fibre
coordinates	
the corridor, 106 consistent inequalities	theorem · analysis
unwitnessed · exhaustive · none found (a result about	the form; registers 1445, 1463)
the nineteen surds, the complete endpoint set	measurement · analysis
unwitnessed · exhaustive · none found (true of v as a	form; register 1460)
$a_{\text{cross}} = (\sqrt[n]{n-1} + \sqrt[n]{n-4})/3$	formula · analysis
proved · exhaustive · none found	
the staircase algebra, $\lfloor s/2 \rfloor \leq d \leq s + \lfloor s/3 \rfloor$	formula · order
verified · exhaustive · found	
the sequence-index fix, rank parity not free	mechanism · physics
verified · exhaustive · found	
the selection-rule falsification, 200 of 200 seeds	measurement · analysis
measured · sampled · none found	
the necessity of state, §34.6	theorem · physics
verified · exhaustive · none found (register 1460 demoted the verification because the	
count was not reproducible; §34.6 no longer rests on a count – the ambiguity is geo-	
metry, and the corridor is the lower-hull vertex set at all 106 steps in both forms,	
register 1805)	
the observability boundary, tested once	law · physics
verified · sampled · found	
the singleton-output rule	definition · order
proved · exhaustive · none found	
the domain prohibition	law · physics
verified · exhaustive · none found	
the limit – Λ_{spectra} closes at the last species, $E = 11$	measurement · combinatorics
measured · exhaustive · none found	
rival = donor iff the donor is not full, 12 of 12	law · physics
verified · exhaustive · none found	

Sixty-five elements become seventy-seven over twenty-four fibres, and $E = 0$ in twenty-three of them — after the index named one cell and the cell was filled by reading one entry honestly. Two fibres open for the first time, measurement · analysis and formula · order, and each closes trivially. The first run gave $E = 1$ in theorem · analysis, at *verified · exhaustive · none found*: the corridor had been entered with precedent *found*, because

the compendium names Fourier 1826 and Motzkin 1936 for the solution of a linear system in one variable. But that is not the element. The compendium's own division puts the method under *standard* and puts under *this work* the two things the list names — that the coefficients are node counts, so the endpoints lie in $\mathbb{Q}(\sqrt{N})$ and take nineteen values, and that all 106 intervals are non-empty. A precedent for the method is not a precedent for the consistency, and the search found none. Entered on what it is, the corridor occupies the cell the structure had named, and the fibre closes. That is the third time this appendix has closed by correcting a coordinate of its own — §D.5.6 by proving a thing that was sampled, §D.5.8 by admitting a thing was sampled, here by attributing a thing to the work that did it.

The observability boundary entered as *conjectured* and does not stay there. Its two halves are the literature's and the boundary between them is this work's, and outside attribution is what witness means in this index, so it stands *verified · sampled · found* (register 1802). Law · physics now holds four elements over three cells **and does not close: E = 3**. The whole of that is the one element — without the boundary the fibre closes, without any other it does not move. What the closure admits and nothing holds is *proved · exhaustive · found*, against the ordering law's *proved · exhaustive · none found*: the same status, the same verification, wanting a witness and nothing else. That is §D.2's sixth value read of a fibre rather than a row — **law · physics is unwitnessed**. And the witness it wants is one this book's thesis says cannot exist, since a precedent found for the ordering law would mean the Löwdin challenge had been answered before Chapter 35. The mechanical routes were measured and refused: no ordering of §D.2's three axes closes the appendix, 0 of 8,640, and re-fibering the boundary closes it only into fibres it does not belong to (register 1806).

Register 1734.

D.6 What this does and does not establish

It does not prove the mathematics correct. Audits 2 and 3 do that — 20 of 20 on Λ 's structure over all 475,800 pairs, and 24 of 24 on the stated equations.

What it establishes is that nothing is MISSING that the structure admits. By P23 an object is complete when all its definitions are identified and no question can be had about them. The first condition is now checkable and checked. The second remains what P23 says it is: a standing claim.

And the method is the book's own. A regex over fifty-five thousand words returned three hundred and seventy fragments of punctuation. An index returned two missing constraints and two missing elements.

Appendix E — Q, indexed and closed

Serving PART VI — THE RECORD AND THE REACH. The book's own ignorance as a closed index — fourteen items, ten of them open, four ordered coordinates, one categorical fibre, $E(Q) = 0$ fibred by domain at the fourteen (§E.1.5), the domain column printed at last — and see §32.1.1 flag 2 for what a fibred zero is worth.

Appendix D indexes the book's mathematics. This one indexes what the book does not know. The set has been recomputed six times; §E.1.1 is its lineage and §E.1.2 its current contents. The earlier states — nine items; eight closed at $E(Q) = 0$ with four entered by §12.11.7; thirteen — are kept there as history.

E.1 The coordinates

coordinate	val-	
ues		
kind		
domain	physical · bibliographic · mathematical · computation-	
al	fibre	
blocks	nothing < a novelty assessment < one stated claim < a result the book	
relies on	ordered	
obstacle	retrievable < buildable < requires an object that does not ex-	
ist	ordered	
cost	hours < days < unboun-	
ded		ordered
depends	the book does not depend on it < partly <	
yes	ordered	

Ordered coordinates only, fibred by the categorical one — the precondition §D.1 states, and the shape Λ itself has.

P19 requires $Q(X) = \emptyset$. The set has been worked through, and this is where it stands.

E.1.1 The lineage of this index, and why only one section states it

Q has been recomputed five times, and each recomputation was written beside its predecessor instead of replacing it, so this appendix carried **three live item counts** — eight at E.4, fifteen here, and thirteen at E.1.2 — of which one was current and only one listed what it counted. Each was correct when written; none was retired. **An index states its current contents once, and its history as history**, which this section now is.

when	items	what changed
§32.2, asserted	7	an assertion, not an enumeration
E.4.1, enumerated and closed	8	the assertion indexed and checked
§12.11.7 and the repairs of register 211	15	four by the tower, L and M by Appen-
dices E and F		
after E, J and L closed	12	the self-duality count and the four-cap
check with them		
E.1.2, as first listed	13	N, O and P entered, having been named
and never entered		
E.1.2, current	14	R entered — $ \text{Cl}(\mathcal{R}) $ for Λ , register 461

E.1.2 is the open set. E.3 gives each item in full, E.4 and E.4.1 record how the grid read the set when it stood at eight, and E.2 and E.8 record what closed. Register 307.

E.1.2 The open set as this session leaves it — ten open of 14, listed

Items E, J and L closed during the work, and the self-duality count with them. Three entered late and were described as *entered in Q* in the sections that named them without ever reaching this appendix — register 239's failure, repeated three times. They are entered here.

item	question	blocks
obstacle	cost domain	
A	the Sc VI level, measured	one claim nonex-
istent	unbounded physical	
B	Paschen & Götze §III, read	nothing re-
trievable	days bibliographic	
C	the nine named literatures, entered	nothing re-
trievable	unbounded bibliographic	
D	the Kreuzer-Skarke Hodge pairs, read	one claim re-
trievable	days computational	
F	the reorderability decision problem, searched	nothing build-
able	days mathematical	
G	the complexity literature, entered	nothing re-
trievable	hours bibliographic	
H	whether Sc VI has already been measured	one claim re-
trievable	days physical	
I	the spectroscopic v-check per channel	one claim re-
trievable	days physical	
K	the earlier bound $2J_c \leq k$, tested for containment	CLOSED
—	— mathematical	
M	Appendix F — nine stable cells, named at F.3.1	CLOSED
—	— computational	
N	whether anyone has lattice-ordered the seniority scheme	**CLOSED — no; the
	scheme is ordered by group-subgroup chains, and Racah recorded in 1954 that the chain	
	needs three running indices to separate states. Registers 562–564.**	mathematical
O	tree propagation retested on Λ_9 's f–g–2S' cycle	CLOSED
—	— mathematical	
P	the 1,442 split into independent and Ritz-derived tests	one claim re-
trievable	hours physical	
R	$ Cl(\mathcal{R}) $ for Λ — how many closed subsets a closed index has	one claim
buildable	unbounded mathematical	

Every *blocks* value of *novelty* fell to *nothing* under §29.2.1 — six of the thirteen. **E(Q) = 0 in all four domain fibres and 1 unfibred**, both printed per §32.1.1 flag 2. Register 256.

E.1.3 What this index asserts, and what it recomputes

F.3.1 establishes that a count computed from the book cannot be printed as prose, because writing the count changes what it counts. **The same holds here, and this appendix was making the same mistake.**

E(Q) has been stated as *nought fibred and one unfibred*. Recomputed on the table as it now stands — item O closed, item K closed, item M restated at register 371 — **the unfibred figure is 4**. It moved because the table moved, and the table moves whenever an item closes or a cost is re-graded, which is the ordinary business of this appendix.

A count over an index of open questions is the least stable thing about it. Every closure changes it, and closures are the point.

So this appendix asserts what does not move and computes what does. What does not move: **E(Q) = 0 under the fibration by domain**, a statement about the table's shape rather than its contents, which has held at seven, eight, twelve, thirteen and eleven items; and the standing result that **every item has depends = no**, which has held at every size and is the only column whose value has never changed. What does move, and is therefore recomputed rather than written, is the unfibred count and the item total.

§12.11.3.1's rule applied here as F.3.1 applies it there — **assert the law, compute the extent**. Register 372.

E.1.4 This is an index of tractable ignorance, and it cannot say otherwise

Every item here is closable. The obstacle coordinate takes three values — **nonexistent** (the datum can be made to exist), **buildable** (it can be constructed), **retrievable** (it can be fetched) — and all three name a route. Eleven items: one nonexistent, two buildable, eight retrievable.

No value of this coordinate means *there is no route*. An unclosable question is not missing from this appendix. It is **inexpressible** in the coordinates the appendix uses.

That is not an oversight, and §32.1.4 says why it cannot be repaired from inside. $\mathcal{R}(X) \subseteq \prod_i A_i(X)$: the operator is bounded by the index’s own value sets, so it can name a cell the index lacks and never a *value*. Nothing in this appendix could ever produce the fourth obstacle, however long it ran.

Offer it from outside and the shape of what is missing appears. Adding *no route* as a fourth obstacle value grows the ambient box from 18 cells to 24 and takes E from 4 to **10** — six further combinations the structure admits and this appendix does not contain, every one of them an unclosable question at some cost and some consequence.

So the book has two kinds of ignorance and indexes only one. The tractable kind is here, graded and costed. The other kind is elsewhere and unindexed: §29.2.2’s reading that the branches leave the object, and the destination coordinate that the temporal mechanisms would need. **Those are not open items. They are the shape of the question this index cannot pose**, and each had to be named in prose because no coordinate here would hold it.

That is register 332 arriving where it is least comfortable — an index names cells and never coordinates, so **an index of one’s own ignorance is an index of the ignorance one already has words for**. Register 382.

E.1.5 The fourteenth item, the column that was never printed, and the constraint it killed

The domain column above was assigned in the code that computed $E(Q)$ at register 256 and never printed; the fibration by domain that E.1.3 asserts was therefore a claim the page could not check. It is printed now, with the value orders the code used — blocks *nothing* < *novelty* < *one claim* < *a result*, obstacle *retrievable* < *buildable* < *nonexistent*, cost *hours* < *days* < *unbounded* — and the record’s thirteen reproduce the record: **$E(Q) = 0$ fibred, 1 unfibred**, exactly as register 256 states.

The code also carried a constraint the page never stated: $obstacle = buildable \implies cost \leq days$. **Item R refutes it.** R — how many closed subsets a closed index has — is buildable, because the enumeration can be written, and unbounded, because counting the closed sets of a closure system is #P-complete (§29). A constraint refuted by its own extent is dead, which is F.3’s rule, and it dies here: R occupies a cell the constraint refused. Dropped, the thirteen still close — $E(Q) = 0$ fibred was never resting on it; only the unfibred count was.

At fourteen, fibred and unconstrained, $E(Q) = 0$ with R as printed above, *one claim* — the claim being $|Cl(\mathcal{R})|$ for Λ , which §14.5 states as a size. Register 465 entered R as *nothing*; on that reading the mathematical fibre admits one cell, *one claim* · *buildable* · *unbounded*, which is the cell R now occupies. The later state is on the page, per the register’s own rule. Register 465’s “the index predicted R” was computed unfibred and unconstrained (E fell 4 to 3) and does not hold under the fibration the appendix asserts; it is withdrawn as a fibred claim and kept as what it was. Register 1729.

E.2 Closed

Do the perturbation bounds tighten at higher order? Yes, 1,585-fold — median 25.96 cm^{-1} at order 1, 0.0164 at order 6. §23.10

Is there an order maximising the bound, given σ ? No interior optimum. Information per cell rises monotonically; the maximum is the admissibility cap k^* , computable per cell. §23.11

Does the cell count change at matched order? Yes: 518 at order 1, 34 at order 6 — with containment at 100% for every order once the premise is checked. §23.10

What is V for the convex bracket on measured levels? It grows with order — 157 to 3.2×10^6 on a matched node set. What holds at every order is *containment*: 560 tests, zero failures. **The floor of Proposition 23.1 belongs to the classical bracket alone.** §23.10

Do all accelerators pay $1/[p-1]$? Yes, at order m : $A^{m(x)p} = (-1)^m x^{p^*/(p-1)} m$, proved for $m^* = 1 \dots 4$. Richardson is the exception — a different kernel, no $(p-1)$ at all. §26.6

Does the claim set close, and under what measure? Yes — a Gröbner basis of eight, against which every relation derived since reduces to zero. $E(\text{claim set}) = 0$. §29.8

Why is $F(-1) = 2$? Because ℓ and f each have two values, so $F_{\text{box}}(-1) = 0$, and the tree's couplings remove the vanishing factors. §11.8.1

Does the amplification have a general form? $A = N[\varphi(\Lambda * \{y\})] - N[\varphi(\Lambda)] - 1$, exactly, and to 7% as fibre $\times$ $(0.972 - 0.154 \cdot \text{descendants})$. §16.7

E.3 Each open item, in full

Subordinate to E.1.2, which is the set; this is its detail, and **the two must list the same items**. At register 384 they did not: six open items had no entry here and one entry was an item answered at §12.11.1. Both repaired, and audit 15 extended so the pair cannot drift again. A — Sc VI $3s^2 3p^3 (4S^\circ) 6s^3 S^\circ_1$. Committed prediction, unmeasured. $[735,860, 737,380] \text{ cm}^{-1} \cdot 13.5615\text{--}13.5895 \text{ nm} \cdot 91.338 \text{ eV}$. The lower edge tests monotonicity, the upper tests convexity. Needs a spectrometer.

B — Paschen & Götze §III, Wie findet man eine Serie und ihre Grenze?, and Runge, Encyklopädie 1926, pp. 783–820. Both unread. Springer Book Archives, public domain since 2017, scanned with OCR and not exposed.

C — nine unentered literatures. Named in §29.7, with the vocabulary: excess width, Newton decrement, partial identification, zero-error capacity.

D — verifying 540 Kreuzer–Skarke predictions. **Everything the slice specifies is recomputed and reproduces**, at register 387: 208 cells from $13 \leq h \leq 128$ less twelve exclusions; $E(X) = 540$; 498 join and 498 meet failures of 21,528 pairs; five χ values, only 0 and ± 2 and ± 4 ; 112 on the diagonal from (13,13) to (131,131); $h^{11} + h^{21}$ from 26 to 262. All three falsification tests hold at 540 of 540 — every prediction has $h \geq 1$ on both indices, every sum is under the published maximum of 502, and every mirror partner is itself predicted or present. §12.11.6 supplies the calibration: an $E(X)$ proposal can be vetoed by structure the index does not carry, and these are proposals of that kind. **What remains is one file read.** The 30,108 pairs are in a gzipped catalogue this session could not reach, and E.4's grid places the item exactly there — its first unmet requirement is *reachable from here*, not the mathematics. F — the reorderability decision problem, and it is not this book's problem. §30.3 establishes the law and the procedure. Λ 's own case is answered: its structure is constructed in Chapter 7, its embedding is tight — height $17 = \Sigma(|A_i| - 1)$ — its join-irreducible poset has $|J(\Lambda)| = 17$ and width $7 \leq 8$, and by Dilworth and Larson the chain decomposition of $J(\Lambda)$ is the axis system. Nothing about Λ is open.

What is open is the general question: given an arbitrary cell set with no construction, decide reorderability in time polynomial in $|X|$ and d . §30.3.5 states the law, the procedure, and the residue — realisability.

G — the complexity literature, entered. Named in §29.7 with its vocabulary; the question is whether deciding reorderability is in P, and the search has not been run in the right terms.

H — whether Sc VI has already been measured. A literature search, not a spectrometer: the level may stand in a source this book has not entered, in which case item A closes by retrieval rather than by measurement.

I — the spectroscopic v-check per channel. The tripwire rejects 150 of 216 offered assignments; what is untested is whether the seniority coordinate discriminates within a single channel, which needs term designations the line lists carry and this session did not fetch.

P — the 1,442 split into independent and Ritz-derived tests. §24.9 concedes the cells rest on three kinds of value and only two are independently checkable. The split is arithmetic on data already cited, and it would tell a referee how much of the verification is circular. E.4 The grid, read on the current set §2.16's grid asks, for each item, what its **first** unmet requirement is — everything below it is downstream. It was read once at eight items and the reading was never printed, only described. **It is printed here, read when the set stood at eleven.**

item	question	first unmet requirement
A	the Sc VI level, measured	does the datum exist – it does
not		
B	Paschen & Götze §III, read	is it reachable from here
C	the nine named literatures, entered	is the work bounded – it is not
D	the Kreuzer-Skarke Hodge pairs, read	is it reachable from here
F	the reorderability decision problem, searched	is it reachable from here
G	the complexity literature, entered	is it reachable from here
H	whether Sc VI has already been measured	is it reachable from here
I	the spectroscopic v-check per channel	is it reachable from here
M	the numbers index, nine stable cells	is the work bounded – it is not
N	whether the seniority scheme has been ordered	is it reachable from here
P	the 1,442 split by kind of value	is it reachable from here

Eight of eleven stop at the same requirement, and it is the third: *is it reachable from here*. One stops earlier — the datum does not exist. Two stop later — the work is unbounded.

That is the grid doing what §2.16.1 claims for it. Eight items do not present eight obstacles; they present one, and it is neither the hardest nor the most interesting. What blocks this book's open set is overwhelmingly **access** — a scan not exposed, a file not readable, a literature not entered — rather than difficulty or cost. **Two of the eleven are genuinely unbounded and one is genuinely absent; the other eight are fetching.** Registers 385, 386.

E.4.1 Q, enumerated and closed at eight

Historical. §32.2 asserted $|Q| = 7$. An assertion is not an enumeration, so the seven were indexed — blocks, obstacle, cost, depends, fibred by domain — and closed at eight. The same computation on the current thirteen is at E.1.2: **E(Q) = 0 in all four domain fibres and 1 unfibred.**

E.4.2 And the column that matters

Every item in E.1.2 has depends = no. Not one open question blocks a result this book relies on; Q is a list of what would be added, not of what is missing. That held when Q stood at eight and holds at thirteen — the only column whose value has never changed.

E.5 Precedence, pushed as far as it goes

Every component of §30.3 was found in print after being derived from scratch. The register carries the corrections; this is the ledger.

what was derived what already had it

the recovered bound φ_{ij} Rival 1973/74 — sublattices are the complement of a closed family of irreducible intervals, and Lemma 3.5(ii) of Siggers states φ_{ij} exactly

"subdirect" – every the book needed value used	Larson 1998; named, with the subdirect/tight distinction
tightness $\Leftrightarrow$ acyclic constraint digraph	Siggers 2016, Lemma 5.1(iii)
sublattices $\leftrightarrow$ preorder set by the empty extensions	Siggers 2016, Cor. 4.2 – verified here on four boxes, off- sublattice
the axis system as a chain decomposition of $J(\Lambda)$	Birkhoff 1937 · Dilworth 1950 · Larson 1998

the step law, $3 \cdot 2^{d-2}$ Rival 1973 — $K \leq L$ for a $(3 \text{ maximal } / 2 \text{ sublattice})$

the subcube-removal construction	Chen-Koh-Tan – maximal sublattices are $L - [a, b]$
-------------------------------------	---

the large sublattices of catalogued; the maximal ones in finite topology, Sharp, Stephen ²

the unbounded obstruction families	Tucker 1972
C1P and the linear algorithm	Booth & Lueker 1976 · Fulkerson & Gross 1965

the constraint-language Schaefer 1978 dichotomy

Nine components, nine precedents. The subcube-removal construction is Rival's interval removal; the "step law" is his $3/2$ bound sharpened to the Boolean case, which was already catalogued elsewhere.

E.6 And the complexity question itself may be settled

Ryter & Schmid, Deciding Frattini is NP-complete, Order 11 (1994). Adams, Dwinger & Schmid, Maximal sublattices of finite distributive lattices, Algebra Universalis 36 (1996) 488–504.

So decision problems about sublattices of finite distributive lattices have a complexity literature, and at least one of them is settled NP-complete.

This book has not read it. §30.3's residue may already be closed there, and until that is checked the claim "open" is a claim about this author's reading, not about the state of knowledge.

That is recorded as item G rather than left implicit.

E.7 What precedence costs the book

Nothing in §30.3 is withdrawn. The derivations agree with the literature wherever both exist, and the agreement is itself a check — Siggers' bijection was verified against an independent enumeration, and Rival's $3/2$ bound contains the measured $3/4$.

What is withdrawn is the implicit claim of novelty. The section states results; it does not claim to state them first, and §E.5 is where the difference is written down.

§2.10 says: enumerate targets before searching. It should have said, and now does: SEARCH BEFORE DERIVING. Nine components and a day, against nine citations.

E.8 What closed, and what it cost

Two items closed during this work. The target-spin axis was taken on the grid's advice — its first empty cell was reachable by labour alone — and Λ_9 passed six independent tests. §30.3's characterisation closed at $d = 2$ and its law at every d .

The cost is in Chapter 28. Sixty-odd corrections come from §30.3 alone.

An index that records its own rules must also record when it failed to consult them. This chapter is where that is done.

Appendix F — The numbers, indexed

This appendix indexes the **kinds of quantity** this work produces and the **ordered grounds** it produces them over. It does not index the numbers this book prints about itself. That was the domain the appendix first took, and it was the wrong one: an account of a volume's own printed figures belongs to the making of the volume and not to what the volume is about. The construction survived the change of domain; the readings did not.

What is indexed here is therefore a structure and not a table of results. Every figure this work states lives in its own chapter, and the Register carries the entry that established it. What this appendix supplies is the standard those statements are held to, the shape they take when a quantity is fibred over an order, and the one case where a quantity may not be stated as a figure at all.

F.1 The population

A quantity in this work is never loose. It arrives as a **kind** carried at a **point of a ground**, and both must be named before the number means anything. Six kinds recur across the volumes.

kind	what it counts	the ground it is carried over
extent	admitted cells of an index	the tower, as a chain
defect	the departure of a level from its hydrogenic position	the series ground, ordered by principal number
bound	the occupancy a shell admits	the periodic ground, ordered by proton number
share	a subset of a population against that population	whichever ground carries the population
depth	distance from a root in a dependence order	the dependence order of the compendia
grade	the warrant a statement carries	the audit hierarchy

The list is closed by construction rather than by survey: a kind enters it when some ground in this work carries that kind at every one of its points, and leaves it when no ground does. A seventh candidate is refused below at F.4.3.

F.2 The fibring of a quantity-kind over an ordered ground

Let a ground be a set P with a partial order $\leq$, and let K be a quantity-kind. The **fibring** of K over P assigns to each p in P the set $K(p)$ of values of that kind admitted at p . The assignment is not free. It is a fibring of K over P only when the order is carried:

1. **the fibre is total** — every point of the ground admits a value, so that a gap in the $>$ fibre is a fact about the ground and never about the reading;
2. **the order is respected** — where $p \leq q$, the admissions at p constrain those at q in a way stated with the fibring and not discovered afterwards;
3. **the fibre is a function of the ground alone** — two points that are indistinguishable in $>$ the ground admit the same values.

Clause (iii) is the load-bearing one, and it is a claim that can fail. It holds on the periodic ground for the multiplicities admitted at a species: the values admitted at a proton number and charge are a function of the electron count alone, on every pair the coordinate file carries and with no exception (register 1779). It fails wherever a quantity is carried at a point by a source rather than by the ground — which is why provenance is a column of the coordinate table and not a remark about it.

A fibring that satisfies (i)–(iii) is what this work means by an **index**. A fibring that fails (i) is a survey. One that fails (ii) is a list. One that fails (iii) is a table of readings, and it may be perfectly correct without being an index at all. The distinction is not pedantry: only an index composes with another index, and composition is what the tower is built out of.

F.3 The index, built

A quantity in this work is stated to one standard, and the standard has four columns. A statement that carries fewer than four is not a weaker version of the same claim; it is a different kind of object, and the chapters that cite this section cite it for that reason.

column	what it requires	what its absence costs
the method	the procedure named, not described	the reading cannot be repeated
the input set	the population named and bounded	the reading has no denominator
the inputs	every input printed, not summarised	the reading cannot be checked
the refutation	the test that would kill the claim	the reading cannot be wrong, and so says nothing

The fourth column is the one this work added and the one it defends. A claim that names no test that would refute it is not a claim about the ground; it is a description of the reading. Chapters 6, 12 and 24 state their central measurements to this standard explicitly and say so where they do.

A constraint refuted by its own extent is dead. This is F.3's rule and it is the fourth column turned on the constraint itself. If a constraint is asserted over a ground, and computing its extent on that ground returns the empty set or the whole of it, the constraint has been refuted by the only test that could reach it — not weakened, not pending, dead. The work applies this without exception, and several of the corrections the Register carries are exactly this rule firing.

F.3.1 The closure, represented as it must be

One class of quantity cannot be stated as a figure, and the reason is structural rather than practical.

Let a closure be computed over a domain, and let the statement of that count enter the domain. Then the count is a quantity whose extent includes its own counting. Writing it changes the thing counted: the figure is correct at the instant before it is written and false at the instant after, and no amount of care in the measuring repairs it, because the fault is in the domain and not in the reading. Such a count is **unstable by construction**.

The representation such a quantity must take instead is its **constraint** together with its **refutations**. The rule, stated as the chapters cite it:

Assert the law, compute the extent.

The law is stable because it does not live in the domain it governs. The extent is recomputed whenever it is wanted and is never printed as a property of the object. Where no law is available, a closure may still be carried **by the stable structure** — the shape that holds still under the motion of the count — and this work names the cells where that structure holds rather than the number they hold. Chapter 2's distinction between what is held by the law and what is held by the stable structure rests here, and the two routes are not interchangeable: the first is exact, the second is only as good as the stillness it claims.

The test of whether a quantity belongs to this class is short. **Would writing the number down change it?** If it would, the number is not the object; the constraint is.

F.4 The readings

F.4.1 Two ratios the book had never measured about itself

Retained as a reference, withdrawn as content. Two ratios of the book's own shape stood here. They were measurements of the volume rather than of its subject, and under the ruling that separates the record of the work from the work they do not return to this appendix. The entries that established them stand in the Register, which is where an account of the making belongs; the label is kept so that every reference written to it still resolves.

F.4.2 The ideal withdrawal ratio

Retained as a reference, withdrawn as content, on the same ground as F.4.1 and by the same ruling. Its census and its ruled row are in the Register.

F.4.3 Time, which this work has and does not read

A date is admitted nowhere in Λ , and the refusal is the seventh candidate kind of F.1 being turned away.

A date is a coordinate of the **recording**, not of the thing recorded. It orders the acts of measurement; it does not order the ground. Admitting it as a kind would satisfy F.2 (i) and (ii) trivially — every reading has a time, and times are totally ordered — and would fail (iii) absolutely, since two points indistinguishable in the ground routinely carry different dates for no reason in the ground at all. A date therefore indexes the observer, and an index of the observer composed with an index of the object yields neither.

The consequence is a prohibition and not a preference: **no cell of any index in this work is keyed by a date**. Where the work needs to read an order of events, it reads the Register, which is the record of the making and is ordered by exactly that clock. The two are kept apart on purpose. This is the one place where the work states what it will not index, and it states it here because F.2's third clause is the reason.

Appendix G — Transitions, indexed

Transitions v3.0 is the source of thirty-seven objects in this work, and until now it was an address no reader could follow. This appendix removes that. The thirty sections on which this book actually rests are set out below, each with the statement relied on and the objects that rest on it. The forty-seven sections of *Transitions* that this book does not use are not reproduced; the paper remains the origin and is named as such, and this table is the address.

The eight citations to *Transitions* in the Register are left exactly as they were written. Register entry bodies are never edited, and what a past entry recorded is part of the record.

§	what it states	what rests on it
1.2	$X \subseteq R(X)$, and therefore $E \geq 0$. Extensivity, proved.	A.ext
1.3	R is a closure operator — extensive, monotone, idempotent. The closed sets form a Moore family.	A.clos
1.4	R is the closure of a binary constraint network under monotone binary projections, and $E(X) = 0$ exactly when the network is globally consistent.	A.gc
1.6	The two failure modes — order and arity — and the third case: a vocabulary may not be indexable at all.	A.modeA, A.modeB
1.7	Six blindnesses of the closure. A derived coordinate cannot repair it: the box grows by that coordinate's value count while $ X $ stays fixed.	A.derived
1.8	The closure rule. A tightening preserves $E = 0$ if and only if it binds one coordinate by a monotone function of one other. One-sided as stated; the operator is two-sided (§2.15.2; register 402).	A.rule
1.9	E requires density, asymmetrically. $E = 0$ is informative at any density; $E > 0$ at low density measures sparsity and nothing else.	A.dens
1.10	The two named theorems covering Λ — Freuder 1982 and Montanari 1974. Neither reaches either violation index.	A.freuder, A.montanari
2.1	The tower closes at every level for every statistics order. Parastatistics of order m gives capacity $m(4\ell+2)$; $m = 1$ reproduces the canonical tower exactly.	T.para
2.1b	Closure is not scheme-contingent. Rebuilt in LS, LK, jK and jj: the coupling scheme is not a coordinate.	T.scheme
2.2	Λ_9 's constraint graph is a tree of nine nodes and eight edges, so Freuder applies. The admissible Pauli cut gains a cycle and loses the guarantee. Λ_9 is the last tree level.	T.a9p
2.4	Closure and exactness are incompatible. The exact coupling triangle is ternary, outside the closure-preserving class: Λ_{12} falls $70,905 \rightarrow 22,275$ and E goes $0 \rightarrow 35,570$.	T.trad
4.1	Λ_9 is the transit level. 1,169 of 1,654 cells are composable — the first level at which a transition has a defined endpoint, and the last that is a tree.	K.window
4.2	The composition graph is the line digraph of a quiver on 33 atomic states, with 27,027 edges. A loop sits at every vertex, so return is available in one step.	K.quiver
4.3	Girth exactly 4. Unit-step adjacency coincides with the covering relation at 6,658 edges either way, so the lattice is gap-free.	K.girth
4.4		K.helly

§	what it states	what rests on it
	Molecular transit. Feasibility decays geometrically at 2.51 per atom; the Helly number is at least 5 and at most 144.	
5.1	The coordinates of the violation index, and the position this work occupies in it — two components fixed by measurement rather than choice.	Mathematical Compendium, the coordinate set
5.3	2,370 cells, $E = 30$, all of it genuine global-consistency failure. The excess is exactly 1×30 and the core is a single cell.	W.core9
5.4	The failing triple is the unique minimal support. Arity exactly 3, one generator; no single literature pays for it.	W.sup
5.7	The defect does not scale. The core is one cell at every resolution, and the multiplicity moves non-arithmetically.	W.scale
6.5	The corrected alphabet — what becomes expressible once the conflated letters are split.	Mathematical Compendium, the split alphabet
7.1	Every structural result survives at fifteen letters: $E = 816$, one cell, arity 3.	W.core15
8.2	A jurisdicted forcing and an unjurisdicted disjunction are the same object. A theorem holding only under a scope condition is inherently ternary, and no further physics removes the scope.	W.jur ; Chapter 12; Register 543, 584, 585, 588, 592
8.4	The vocabularies. A locally covariant QFT is a functor with exactly four parts, which replaces the seven-vocabulary partition.	W.rel ; Chapter 12; Register 543, 544
10.1	Every vocabulary index closes; the charger index does not. The defect sits in V1, on the V1–V3 edge, and in the partition itself.	Mathematical Compendium, the index vocabularies
10.4	Six independent curved-space routes to the ANEC. Five require a Killing field or a horizon generated by one; the sixth yields a weighted bound instead.	M.sorce
10.4c	C1, computed. The presymplectic potential on a null surface contains no transverse derivative, so Ω is block diagonal and the algebra factorises over generators.	M.C1 ; Register 549, 550
10.4d	C2, the open condition — whether the state respects the decomposition. Independent of the Hadamard condition, which says nothing at finite transverse separation.	M.C2
10.4e	Half-sided modular inclusion is the converter from algebra to geometry: Borchers and Wiesbrock return the affine group acting on a line.	M.ledger ; Register 1035, 1483, 1484
10.5	Status is one bit per vocabulary and the partition forces one-hot — every term is physical in exactly one vocabulary, or in none.	W.onehot

Thirty sections, thirty resolved locators, thirty-four citations re-sourced inward.

END MATTER

Index

This index satisfies the law of this book.

Construction Terms are ordered by specificity: $s \sqsubseteq t$ when s is a specialisation of t . The index is closed iff

$$s \sqsubseteq t \Rightarrow \text{loc}(s) \subseteq \text{loc}(t)$$

every location of a specialisation being also a location of its generalisation. That is a down-set condition in §16.4 form, so **a closed index is a sublattice of (terms $\times$ locations)**.

Verified: 57 terms, 311 entries, 44 specialisation relations, **0 violations** — **E(index) = 0**. *Six terms and four relations entered with Chapters 35–36. Regenerated from the text on 2026-08-24: a term's locations are the sections whose heading names it or in which it occurs at density, and every generalisation carries the union of its specialisations' locations, which is what the down-set condition requires; before that union 90 locations of specialisations were absent from their generalisations. The previous index (57 terms, 139 entries, 44 relations) carried locators on the numbering the book had before Part I and Part IV were inserted — §32.1, §32.3, §24.2 throughout — and was regenerated rather than repaired.*

What closure buys. The term list is recoverable by projection (S1); the specificity order is recoverable by tree propagation, a term hierarchy being a tree (S2); $\text{loc}(\cdot)$ is recoverable as the union over specialisations (S3); and every (term, location) pair is present or absent — **no see also limbo, no partial entries** (D3).

An index with $E > 0$ requires the reader to already know where to look — which is the periodic table's failure at the scale of a back-matter page.

Entries **bracket** §12.11 · §22 · §22.1 · §22.2 · §22.3 · §22.4 · §22.6 · §23.1 · §23.2 · §23.6 · §23.9 · §23.10 · §24.9 · §25.1 · §25.3 · §25.6 · §26.7 · §28.7 · §29.5 · §32.5 · App A · App D bracket failure §25.1 · §25.3 · App A bracket width §22.1 · §23.1 · §23.2 · §23.6 · §23.10 · §28.7 · §29.5 · App D interiority §22.2 limit-free §22.3 · §22.4 · App A · App D the four rules §22.2 · §25.6 · §28.7 *interiority* §22.2

three bodies §12.11 · §18.4 · §21.5 · §25.6 · §34.2 · §34.10 · §36 · §36.3 · App D Λ_3 §25.6 · App D shape sphere §36.3 · App D

closure §1 · §2.23 · §6 · §6.2 · §12.10 · §12.11 · §14 · §14.2 · §14.5 · §14.6 · §15.2 · §15.5 · §16.8 · §18.4 · §27.4 · §29.2 · §29.8 · §29.11 · §29.12 · §32.7 · App A · App C · App D · App E closed index §2.23 · §6 · §12.11 · §14.6 · §15.2 · App D · App E closure defect §6.2 · §12.10 · §14.2 · §29.2 · §29.8 · §32.7 closure operator §14.2 · §14.5 · §16.8 · App D $\mathcal{R}(X) = X$ §12.11 · §15.5 · §27.4 · §29.11 · §29.12 · App A · App C · App D

cost of a guarantee §2.3 · §16.8 · §18.5 · §23.2 · §23.3 · §23.7 · §23.9 · §23.14 · §23.15 · §28.7 · §29.4 · §31.1 · §31.2 · §32.7 · App D $V = 4v/3$ §23.3 · §28.7 · §29.4 · §32.7 · App D the floor §16.8 · §23.2 the pole §18.5 · §23.7 · §23.9 · §31.1 · §31.2 v_V §2.3 · §23.14 · §23.15

extension §3.7 · §12.11 · §17 · §17.4 · §17.5 addable cells §17.5 adjoinable axes §17.5 imposable constraints §17.5

Löwdin §33.5 · §34 · §35 · App D challenge §33.5 · §34 · App D solution §35 · App D

prediction §2.20 · §11.1 · §12.11 · §16.7 · §18.6 · §22.1 · §25.2 · §25.6 · §31.3 · §32.1 · §32.5 · §36.3 · App C · App E · App F $E(X)$ as budget §2.20 · §12.11 · §18.6 · §22.1 · §32.1 Sc VI §25.2 · §25.6 · App C · App E second route §11.1 · §16.7 · §25.6 · §36.3 · App F

quantum defect §12.5 · §12.11 · §16.7 · §22.1 · §22.2 · §24.3 · §24.7 · §32 · App B isoelectronic law §22.2 · §32 penetration §24.3

retrieval §1 · §2.10 · §18.6 · §19 · §19.1 · §19.2 · §19.4 · §19.5 · §19.6 retrieval redundancy §1 · §18.6 · §19.1 · §19.2 · §19.5 route set §2.10 · §19.1 · §19.6

self-defence §1 · §2.8 · §2.13 · §12.11 · §16 · §16.2 · §16.4 · §16.5 · §18.7 · §19 · §25.6 · §28.2 · §28.7 · §28.8 · §32 · §32.3 · §32.4 · §32.5 · App D totality §16 · §16.5 · §28.2 process index §2.13 · §28.8 two routes §2.8 · §25.6 · App D $\mathcal{D}_{\text{def}}$ §16.4 · §32.3 $\mathcal{D}_{\text{phys}}$ §1 · §16.2 · §28.7 · §28.8 · §32.3 · §32.5

self-reference §1 · §15 · §15.1 · §15.2 · §15.3 · §15.5 · §16.8 · §19.4 · §30 · §30.1 · §32.2 · §32.4 · §32.7 · App C · App D *alphabet recovery* §15.1 · §15.5 *bound recovery* §15.2 · App D *order recovery* §15.3 · §30 · §30.1 · App C · App D

the collection §2.5 · §2.11 · §12.11 · §18.4 · §23.4 · §23.11 · §24 · §24.1 · §24.6 · §24.9 · §24.11 · §24.12 · §24.13 · §25.4 · §30.3 · §32 · §34.8 · §35.5 · App B *decline modes* §24.9 · §24.11 · §32 *isoelectronic pairs* §24.1 · §24.6 · §24.13 *provenance* §2.11 · §12.11 · §18.4 · §34.8 · §35.5 · App B *the census* §2.5 · §24.12 · §30.3

the index Index *index, self-referencing* Index

the record §28 · App C · App F *margins* App C *recomputed* App F *withdrawals* §28

index, self-referencing *this page*

The last entry is a cell of the index referring to the index. It is not a joke; it is the fixed point made visible, and its presence is what makes $\mathcal{R}(\text{index}) = \text{index}$ rather than merely $\text{index} \subseteq \mathcal{R}(\text{index})$.

References

Atomic coupling schemes and the provenance of level values. Entered at registers 433 and 436.

· Cowan, R. D. and Andrew, K. L. (1965). Coupling considerations in two-electron spectra. *JOSA* **55**(5), 502–516. Cited for the four types of pure coupling — LS, LK(Ls), jK(jl) and jj — as a single set, which is the set §12.11.4 rebuilds. A precedent for the grouping, not only for the members.

· Cowan, R. D. (1981). *The Theory of Atomic Structure and Spectra*. University of California Press. Cited for the physical conditions under which each coupling scheme applies.

· Kramida, A., Ralchenko, Yu., Reader, J. and NIST ASD Team (2024). *NIST Atomic Spectra Database* (ver. 5.12). <https://physics.nist.gov/asd> DOI 10.18434/T4W30F. Queried 3 August 2026 for H I and He I. Cited for the three-way provenance marking of level values — plain, [] and () — and for the per-species notes §24.9 now rests on. **[F] the query output was read in full for both species.**

· Drake, G. W. F. (2023), in *Springer Handbook of Atomic, Molecular and Optical Physics*; and Kandula, D. Z. et al. (2010), *Phys. Rev. Lett.* **105**, 063001. Cited at one remove, through the ASD’s He I provenance note, for the construction of He I’s level values: theoretical ionization energies anchored to a measured ground level below $n = 10$, and the extended Ritz quantum-defect expansion above it. **[S] not read directly; the note is the source used.**

Constraint satisfaction, consistency and closure. Entered at registers 400 and 401.

· Freuder, E. C. (1982). A sufficient condition for backtrack-free search. *JACM* **29**(1), 24–32. Cited for backtrack-free search at strong $(w+1)$ -consistency on a graph of width w . **Not** for global consistency, which this book previously attributed to it.

- Montanari, U. (1974). Networks of constraints. *Information Sciences* **7**, 95–132. Cited for path consistency implying global consistency under monotone constraints.
- Dechter, R. (1992). From local to global consistency. *Artificial Intelligence* **55**(1), 87–107. Cited for strong (w^*+1) -consistency on induced width w^* giving decomposability — the result §14.1 had been crediting to Freuder.
- Deville, Y., Barette, O. and Van Hentenryck, P. (1999). Constraint satisfaction over connected row-convex constraints. *Artificial Intelligence* **109**, 243–271. Cited for the *staircase* constraint, (α, β) -monotone with $\alpha, \beta \in \{\leq, \geq\}$, which is the class ϕ recovers.
- van Beek, P. and Dechter, R. (1995). On the minimality and global consistency of row-convex constraint networks. *JACM* **42**(3), 543–561; and (1997), Constraint tightness and looseness versus local and global consistency, *JACM* **44**(4), 549–566. Cited for row-convexity, and — since §14.5.9 — for **constraint tightness as a COUNT**: the number of (constraint, value) places where a binary constraint restricts rather than permits everything. **That count is exactly twice this book’s envelope-step number S**, and it is the tight-pair count rather than the tightness LEVEL that the seed cost tracks. *The tightness half of this entry went unused for four cycles.*
- Dilworth, R. P. (1950). A decomposition theorem for partially ordered sets. *Annals of Mathematics* **51**, 161–166. Cited for the order dimension of a finite distributive lattice as the width of its join-irreducible poset — §8.6, register 409. Every work cited in this book, with what it is cited for. Entries marked [unread] were named but not obtained; §32.2 lists them as open.

R.1 Spectroscopic data

Kaufman, V. & Martin, W. C. (1991). *J. Phys. Chem. Ref. Data* **20**,

1. — Al I, Al II levels. Kimura, K., Makino, K., Yamada, H. & Yoshizumi, Y. (2024). — characterising the integer points of 2-decomposable polyhedra by closedness under operations. **$E(\mathcal{R}) = 0$ IMPLIES 2-decomposability and is not implied by it** — §14.5.3 gives an 840-cell witness determined by its pairwise projections with $E(\mathcal{R}) = 750$. The class here is defined by closure under operations **including min and max**, which is a SUBLATTICE condition and is §14’s; determination by two-fold projections is Bergman’s CONCLUSION from that hypothesis, not an equivalent of it. *An earlier draft read $E(X) = 0$ is 2-decomposability.* §29.12 U1.

Kramida, A. E. & Martin, W. C. (1997). *J. Phys. Chem. Ref. Data* **26**,

1. — Be I levels. NIST Atomic Spectra Database. — Ar II, Be II, Bi I, C II, Ca II, Cd II, Ga I, He I, He II, Hg II, K II, Li I, Li II, Mg II, N II, Na II, Ne I ($\frac{3}{2}$), Ne I ($\frac{1}{2}$), Si I, Si II, Zn II.

Sansonetti, J. E. (2008). *J. Phys. Chem. Ref. Data* **37**, 1659. — Na I.

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Sugar, J. & Musgrove, A. (1990). *J. Phys. Chem. Ref. Data* **19**, 527; (1995) **24**, 1803. — further ionisation limits.

R.2 Exotic systems and collective quantities

Hori, M. et al. (2011). *Nature* **475**, 484; and (2006) *Phys. Rev. Lett.* **96**, 243401. — antiprotonic helium, both accessed via CODATA 2010, Table XII (arXiv:1203.5425) and arXiv:1304.4330.

Korobov, V. I. (2008). *Phys. Rev. A* **77**, 042506. — antiprotonic helium transition energies.

Singer, K., Stanojevic, J., Weidemüller, M. & Côté, R. (2005). *J. Phys. B* **38**, S295. — long-range Rydberg interaction coefficients.

R.3 Series structure and quantum defects

Lach, M. & Claude (Anthropic) (2026). *Indexing Electromagnetism: reconstruction and complete definition of Λ , electromagnetic mode indices, a lattice bracket for Rydberg spectra, and the sharpening of the law of realised closure*. — the companion paper that reconstructed Λ from §7.1 alone and located the two definitional defects of register 301; source of §18.4.1's two realisation modes and of the cap-arity criterion. Registers 301–303.

Lach, M. (2026). *Muon-Catalysed Fusion: definition, procedure, and two gaps*, draft v1.1. — the method applied to a nuclear-physics subject; source of §22.1.1's singleton bracket. Every computed claim in it was recomputed here and reproduces: the window [119, 918] m_e, the harmonic-sum cycle cap at $2.6 \times 10^8 \text{ s}^{-1}$, breakeven sticking at 0.203/0.262/0.292%, decay lengths 11.2 m and 623 m, flux at $1.2 \times 10^{12}/10^{15}/10^{18} \text{ s}^{-1}$, and the ash bracket of width 7.29. Register 319.

Lach, M. & Claude (Anthropic) (2026). *On the Matter of Time Travel: the clock in the constraint*. — the third companion paper; 25,000 words, 43 sections. Source of §12.11.0.1's occupancy clock and of the three defects repaired. Its computed claims were recomputed here and reproduce: 33 objects, 739 of 1,089 joined pairs, zero steps raising occupancy, three components of 8/15/10 pure in k, 389 reversible morphisms, 904 conservative cells at 54.7%, and 675,606 of 2,735,716 cell pairs joined by a composition path. Registers 344–346.

Racah, G. (1942). *Phys. Rev.* **62**, 438–462. — jK coupling; the scheme §12.11 uses for Rydberg channels where core splitting dominates exchange. §29.9.

Racah, G. (1943). *Phys. Rev.* **63**, 367–382. — seniority; the tenth axis of the tower, and the exact sets that fix its density. §§12.11.1, 22.9.

Rydberg, J. R. (1890). The term formula $T = Z^2 R/v^2$. — the object of Part V.

Ritz, W. (1903). The quantum-defect expansion $\delta = \delta_0 + \delta_2/n^2$. — used for every two-point solve.

Edlén, B. (1960). *Handbuch der Physik* **27**, 80. — δ falls with ℓ ; S and P penetrate; footnote 78 is the 1960 precedent for $\mathbb{D} = 0$ when parameters equal observations (§16.4.1).

Edlén, B. (1964). *Encyclopedia of Physics*. — examined at one remove (§29.5).

Paschen, F. & Götze, R. (1922). *Seriengesetze der Linienspektren*. Berlin: Springer. — **§III, Wie findet man eine Serie und ihre Grenze?**, [unread]: Springer Book Archives, scanned with OCR and not exposed.

Runge, C. (1926). *Die Seriengesetze in den Spektren der Elemente*, in Sommerfeld (ed.), *Encyklopädie der Mathematischen Wissenschaften*, 783–820. — [unread], found in the closing search; a second survey of the same subject.

R.4 Bounds, acceleration and the cost of rigour

Boole, G.; Nörlund, N. E.; Steffensen, J. F. — the ratio of first to second central differences, in the finite-difference literature (§23.1).

Aitken, A. C. (1926). *Proc. R. Soc. Edinburgh* **46**, 289. — Δ^2 acceleration; $\lambda^2 = y''/y''$ is its correction term (§23.8.1).

Seki Kōwa (c. 1680). — the same acceleration, three centuries earlier.

Richardson, L. F. (1911). — extrapolation with no $(p-1)$ denominator; the exception in §26.6.

- Shanks, D.** (1955). *J. Math. Phys.* **34**, 1. — iterated Δ^2 ; pays $(p-1)^{-m}$.
- Moore, R. E.** (1966). *Interval Analysis*. Prentice-Hall. — **the cost-of-rigour framing, stated plainly** (§23.8.4).
- IEEE Std 1788-2015.** — interval arithmetic; requires enclosures with bounded excess width.
- Manski, C. F.** (1989, 2003). *Partial Identification of Probability Distributions*. — **the law of decreasing credibility**, the same trade in econometrics.
- Shannon, C. E.** (1956). *IRE Trans. Inf. Theory* **2**, 8. — zero-error capacity: the same trade in bits.
- Körner, J. & Orlitsky, A. (1998). *IEEE Trans. Inf. Theory* **44**,
1. — zero-error information theory, surveyed. Nesterov, Y. & Nemirovskii, A. (1994). *Interior-Point Polynomial Algorithms in Convex Programming*. SIAM. — the Newton decrement and self-concordance, which λ^2 and §23.8.2 rediscover.

R.5 Order, lattices and closure

- Birkhoff, G. (1937). *Duke Math. J.* **3**, 443. — the representation theorem: a finite distributive lattice is the order ideals of its poset of join-irreducibles. §12.9’s maximal-chain count rests on it, and §E.3’s chain decomposition of $J(\Lambda)$.
- Dilworth, R. P. (1950). *Ann. Math.* **51**, 161. — chain decomposition, and the width bound §30.3 uses. (1960) *Combinatorial Analysis*, 85–90 — the lattice of maximum antichains.
- Rota, G.-C. (1964). *Z. Wahrscheinlichkeitstheorie* **2**, 340. — the Möbius function of a locally finite poset and the crosscut theorem, which §A.9 uses to show μ vanishes off Boolean intervals of $J(\Lambda)$.
- Rival, I. (1973). Maximal sublattices of finite distributive lattices. *Proc. Amer. Math. Soc.* **37**, 417–420. (1974) *ibid.* **II**, **44**, 263–268. — every sublattice of a finite distributive lattice is the complement of a closed family of IRREDUCIBLE INTERVALS. That family is the recovered bound ϕ_{ij} of Chapter 15, and the $3/2$ bound on maximal sublattices contains §30.3’s measured $3/4$.
- Chen, C. C., Koh, K. M. & Tan, S. K. (1975). *Algebra Universalis* **5**, 88–97. — the maximal proper sublattices are exactly $L - [a, b]$ with $[a, b] \cap J = \{a\}$ and $[a, b] \cap M = \{b\}$. §30.3’s subcube-removal construction is this.
- Larson, R. (1998). Embeddings of finite distributive lattices into products of chains. *Semigroup Forum* **56**, 70–77. — tight embeddings correspond to chain decompositions of $J(L)$; “subdirect” is the recovered-alphabet condition, named.
- Siggers, M. (2016). On the representations of finite distributive lattices (arXiv:1412.0011). — sublattices of a product of chains correspond one-to-one with spanned preorder extensions of C^∞ (verified here on four boxes); tight $\iff$ the digraph is acyclic; and the explicit warning that subdirect does not imply tight, which §30.3 walked into.
- Adams, M. E., Dwinger, P. & Schmid, J. (1996). Maximal sublattices of finite distributive lattices. *Algebra Universalis* **36**, 488–504. — [unread], §E.6.
- Ryter, Ch. & Schmid, J. (1994). Deciding Frattini is NP-complete. *Order* **11**, 257–279. — [unread]. §30.3’s residue may be settled here.
- Tucker, A. (1972). A structure theorem for the consecutive 1’s property. *J. Combin. Theory B* **12**, 153–162. — the unbounded forbidden submatrices; §30.3’s obstruction route closes on them.

- Brylawski, T. (1973). *The lattice of integer partitions*. Discrete Mathematics 6(3), 201–219. — the dominance order on integer partitions, isomorphic to an infimum subsemilattice of concave nondecreasing tuples; the object of §18.4.2, located after the derivation rather than before it. Register 355.
- Baker, K. A. & Pixley, A. F. (1975). Math. Z. 143, 165–174. — polynomial interpolation and the Chinese remainder theorem in universal algebra; an algebra with a majority term has every subalgebra of a product determined by its binary projections. **A.2 is this theorem specialised to products of chains**; see §29.11.
- Bergman, G. M. — the Double-projection Theorem: a sublattice of a finite product of lattices is determined by its two-fold projections. The lattice form of Baker–Pixley, the median $m(x,y,z) = (x \wedge y) \vee (x \wedge z) \vee (y \wedge z)$ being the majority term. §29.11.
- Booth, K. S. & Lueker, G. S. (1976). J. Comput. Syst. Sci. 13, 335–379. — PQ-trees; consecutive-ones in linear time, which is why unbounded obstructions do not preclude a polynomial procedure.
- Dunz, — *Seriengesetze der Linienspektren* (Leipzig, 1911). — earliest complete compilation of the series laws, chronologically closest to Ritz; §29.5’s route set and Q item B. Retrieved as a scan, unread: the volume is in Q, not in evidence.
- Fulkerson, D. R. & Gross, O. A. (1965). Pacific J. Math. 15, 835–855. — interval matrices, the foundation of the above.
- Schaefer, T. J. (1978). The complexity of satisfiability problems. STOC, 216–226. — the boolean CSP dichotomy. §30.3’s arity law is this theorem applied to a constraint language read off a cell set.
- Sperner, E.** (1928). Math. Z. 27, 544. — the antichain bound; §8.4’s largest level.
- Galois connections and formal concept analysis. — $E(X)$ as a closure defect; the general setting for $\mathcal{R}$. **This entry named a field and no work, alone among the thirteen in this section, until §2.23’s own requirement was turned on it.** · · **Anstee, R. P. & Farber, M.** (1984). Characterizations of totally balanced matrices. *J. Algorithms* 5, 215–230; **Hoffman, A. J., Kolen, A. & Sakarovitch, M.** (1985). Totally-balanced and greedy matrices. *SIAM J. Alg. Disc. Meth.* 6, 721–730; **Lubiw, A.** (1982). *Γ -free matrices*, Master’s thesis, University of Waterloo; and (1987) Doubly lexical orderings of matrices, *SIAM J. Computing* 16, 854–879. — **a matrix admits a Γ -free ordering iff it is totally balanced**, and recognition is polynomial. Cited for Q item F: §30.3 places reorderability strictly between chain and totally balanced, so these bound the question without settling it. **And they carry the distinction §30.3 may conflate** — *Γ -free* is a property of a given ordering, *admits a Γ -free ordering* one under permutation. [S] · · **Chandrasekaran, V. & Flanagan, É. É.** (2026). Subregion algebras in classical and quantum gravity. arXiv:2601.07915. — half-sided supertranslation generators from doubled corner edge modes; Type II ∞ horizon subalgebras; the GSL for **non-stationary linearised perturbations of Killing horizons**; quantum focusing in perturbative quantum gravity. **Cited for the verification at register 557**: their eq. (1.7b) shows the two-sided term $(Y_0^+ - Y_0^-)$ multiplied by $\mathcal{L}_\ell(\mu\Theta)$, so the generator’s two-sidedness enters through the expansion alone and vanishes on a non-expanding horizon. [S] — **abstract and body read; the body checked at their section 1.1 for eqs. (1.6) and (1.7b).** · · **Van Isacker, P.** (2014). Seniority in quantum many-body systems. arXiv: 1406.4094 and in2p3-00525728; **Racah, G.** (1954). A search for new quantum numbers in nuclear configurations, Glasgow conference on Nuclear and Mesonic Physics. — **cited for Q item N**. Seniority is classified by group-subgroup reduction chains, $U(4\ell+2) \supset U(2\ell+1) \otimes SU_S(2)$ and $U(2j+1) \supset Sp(2j+1)$, **not by a lattice**, and Racah recorded in 1954 that the classification still needed three running indices α, β, γ to separate states sharing the conventional quantum numbers. **Also cited for Λ ’s v-axis**: the allowed values $v = n, n-2, \dots, 1$ or 0 are Λ_{10} ’s parity congruence, and the type $[1^\wedge v]$ under $Sp(2j+1)$ underwrites $2S \leq v$. [S] · **Caspard, N. & Monjardet, B.** (2003). The lattices of closure systems, closure operators, and implicational systems on a finite set: a survey. *Discrete Applied Mathematics* 127(2), 241–269. — the standard survey of the setting; cited for the cryptomorphism

between closure operators, Moore families and closure systems, which §14.5 uses. **[S] located through the enumeration literature, not read in full.** · **Colomb, P., Irlande, A. & Raynaud, O.** (2010). Counting of Moore families for $n = 7$. *Formal Concept Analysis*, LNCS 5986, 72–87. — cited for the distinction §14.5 turns on: that literature counts **how many closure systems exist** on n elements, 1, 2, 7, 61, 2 480, 1 385 552, 75 973 751 474, where §14.5 counts **how many closed sets one operator has**. Different questions, and theirs is the harder. **[S] · Complexity results for implication bases of convex geometries** (arXiv:2211.08524). — **Lemma 2: the minimum cardinality generator problem is NP-HARD**, by reduction from Boolean CNF satisfiability. §14.5.7’s seed is that problem, so its exact computation is intractable and §2.24’s heuristic spread is the correct treatment rather than a concession. **[S] · Kuznetsov, S. O.** (2004). On the intractability of computing the Duquenne–Guigues base. *J. Universal Computer Science* **10**(8), 927–933; and **Kuznetsov & Obiedkov**, Counting pseudo-intents and #P-completeness (2006). — cited for Q item R: counting the closed sets of a closure system is #P-complete in general, which confirms that item’s *unbounded* grading by complexity rather than by estimate. **[S] · Carathéodory number / convexity spaces.** — the least c such that a point of a hull lies in the hull of c generators, equal to the largest irredundant set; for a semilattice with subsemilattices as convex sets it is the **breadth**, and the Helly number is the height. §14.5.1’s eleven-cell generating set is an irredundant set in this sense. **Located through the enumeration literature; no single work is cited because the result used is standard and appears in several.** **[S]**

Stahl, J. & Wille, R. (1984). — reorderability and scale-measurability of a context; §30.3’s original citation, withdrawn and reinstated.

Yannakakis, M. (1982). *SIAM J. Alg. Disc. Meth.* **3**, 351. — the complexity of order-dimension problems. Also withdrawn and reinstated.

Janet, C. (1928). — the left-step periodic table; $E(\text{Janet}) = 0$.

Sharp, H. and Stephen, C. — maximal sublattices of the Boolean lattice studied in the setting of finite topologies; cited via the catalogue of large sublattices of $2^{\mathbb{N}}$. [unread]

R.6 Cited in the transfer tests

Poincaré, H. (1890). — non-integrability of the three-body problem; §31.1 states that nothing here touches it.

Gödel, K. (1931). — incompleteness; P20’s bound on what a language can close.

Lagrange, J.-L. (1772) and Euler, L. (1767). — the five equilibrium points; §31.1.1’s bracket test.

Hill, G. W. (1878). — the Hill sphere; **the gravitational analogue of a capacity bound** (§23.14.1).

Mardling, R. A. & Aarseth, S. J. (2001). *MNRAS* **321**, 398. — the hierarchical-triple stability criterion; monotone and convex, $V \approx 100$.

Batyrev, V. V. (1994). *J. Alg. Geom.* **3**, 493 (arXiv:alg-geom/9310003). — dual polyhedra and mirror symmetry; the theorem behind §31.3.1.

Kreuzer, M. & Skarke, H. (2000). *Adv. Theor. Math. Phys.* **4**, 1209 (arXiv:hep-th/0002240). — all 473,800,776 reflexive 4-polytopes; 30,108 distinct Hodge pairs. The 30,108-pair file is published and was not read in this work (§32.2).

Huang, W. & Taylor, W. (2019). *JHEP* (arXiv:1809.05160). — the counts 248,305 / 248,305 / 495,515 used in §31.3.1.

Candelas, P., de la Ossa, X. & Rodriguez-Villegas, F. (2007). *Triadophilia* (arXiv:0706.3134). — the thinly populated tip of the Hodge plot.

R.7 What is not cited, and why

The Löwdin solution. Entered at registers 1701–1712. The companion paper carries the full list; these are the works entered with Chapter 35 (register 1810).

· Löwdin, P.-O. (1969). Some comments on the periodic system of the elements. *Int. J. Quantum Chem.* **3**(S3A), 331–334. — the challenge. · Löwdin, P.-O. (1950). On the non-orthogonality problem. *J. Chem. Phys.* **18**, 365. — the multiplier identity is an instance. · Koelling, D. D. & Harmon, B. N. (1977). *J. Phys. C* **10**, 3107. — the scalar-relativistic kernel; c the one entered number. · Griffin, D. C., Andrew, K. L. & Cowan, R. D. (1969). *Phys. Rev.* **177**, 62–71; Griffin, Cowan & Andrew (1971). — orbital collapse; the double well the collapse condition reads. · Pulay, P. (1969). *Mol. Phys.* **17**, 197. — the occupation-parameter term that is the walk’s one defect. · Gerratt, J. & Mills, I. M. (1968). *J. Chem. Phys.* **49**, 1719. — the orthogonal-complement response. · Madelung, E. (1936). *Die Mathematischen Hilfsmittel des Physikers*, 3rd ed. Springer. — the rule as a rule. · Lach, M. (2026). *The Löwdin Solution*. — the companion paper; the formal statement.

The three-body problem. Entered at registers 1713–1724. Hill 1878, Poincaré 1890, Freuder 1982 and Mardling & Aarseth 2001 are cited above and reused.

· Montgomery, R. (2014). The three-body problem and the shape sphere. *Amer. Math. Monthly* **122** (2015), 299–321. — shape space, the metric, the potential; **every structural object of Chapter 36 is his or older**. · Montgomery, R. (1998). *Nonlinearity* **11**, 363; (2002). *Arch. Ration. Mech. Anal.* **164**, 311. — braids and syzygies. · Chenciner, A. & Montgomery, R. (2000). *Ann. Math.* **152**, 881. — the figure-eight. · Hsiang, W.-Y. & Straume, E. (2006). arXiv:math-ph/0608060. — kinematic geometry of triangles. · Euler, L. (1767). *Novi Comm. Acad. Sci. Petrop.* **11**, 144; Lagrange, J.-L. (1772). *Prix Acad. Roy. Sci. Paris* **9**. — the five fixed points. · Lagrange, J.-L. (1770–71). *Mém. Acad. Berlin*. — the resolvent; N_8 is not new mathematics. · Maupertuis (1744); Jacobi, C. G. J. (1837). *C. R. Acad. Sci.* **5**, 61; (1842–43). *Vorlesungen über Dynamik*. — the metric whose geodesics are the trajectories. · Chazy, J. (1922). *Ann. Sci. École Norm. Sup.* **39**, 29. — the asymptotic classes; the five cells. · Saari, D. G. (1971). *Trans. AMS* **162**, 267; (1973) **181**, 351; (1984) *J. Diff. Eq.* **55**, 300. — collisions improbable; the strata disjoint up to measure zero. · Fleischer, S. & Knauf, A. (2019). *Arch. Ration. Mech. Anal.* **234**, 1007. — the same for n bodies. · Painlevé, P. (1897). *Leçons sur la théorie analytique des équations différentielles*. — the natural boundary; $n = 3$. · McGehee, R. (1974). *Invent. Math.* **27**, 191. — triple collision. · Kolmogorov (1954); Arnold (1963); Moser (1962, 1973). — the KAM stratum. · Alekseev, V. M. (1968–69). *Math. USSR Sbornik* 5–7. — quasirandom motion; the chaotic stratum. · Brudno, A. A. (1983). *Trans. Moscow Math. Soc.* **2**, 127. — **entropy equals complexity: why a complete index predicts nothing on the chaotic stratum**. · Marchal, C. & Saari, D. G. (1975). *Celest. Mech.* **12**, 115; Marchal & Bozis (1982). *Celest. Mech.* **26**, 311. — Hill regions and Hill stability. · Monaghan, J. J. (1976a, b). *MNRAS* **176**, 63; **177**, 583; Nash & Monaghan (1978). *MNRAS* **184**, 119. — the statistical disruption theory. · Stone, N. C. & Leigh, N. W. C. (2019). *Nature* **576**, 406; Kol, B. (2021). *CMDA* **133**, 17; (2023) **135**, 29. — the statistical solution and the flux; “microcanonical ergodic flux” is Kol’s term. · Moore, C. (1993). *Phys. Rev. Lett.* **70**, 3675. — braids in classical dynamics. · Montanari, U. (1974). *Inf. Sci.* **7**, 95; Baker & Pixley (1975). *Math. Z.* **143**, 165; Dechter, R. (1992). *Artif. Intell.* **55**, 87. — local-to-global consistency; the deficit’s owners. · Lach, M. (2026). *The Three-Body Problem for Unknown Masses*. — the companion paper.

No source is cited for the lattice itself. Λ is constructed in Chapter 7 from the four quantum numbers and the exclusion principle, both standard, and every structural result about it in this book was computed rather than looked up. §29 states what was searched for and not found.

Physical constants — $R = 109,737.31568 \text{ cm}^{-1}$ — are used without citation, as a metre is.

The companions' own bibliographies. The Mathematical Compendium names a work for every one of its objects — 156 works, 1669 to 2026, ordered by year with the objects each carries. Fifty-one of them are the works listed above (fifty-two of the 162 rows before five were removed and two merged at register 1891; fifty-nine of the 172 rows matched at register 1736, before ten rows were removed; register 1809); the remainder are the attributions of standard results — Newton 1669, Leibniz 1682, Euler 1748, Gauss 1809, Moore 1910, Birkhoff 1940 and their kind — that the objects rest on and this volume uses without citing, as it uses a metre. Those live in the compendium and are not repeated here (register 1736). The two companion papers carry their own reference lists; §R.7's two blocks are the works entered with the chapters at registers 1701–1712 and 1713–1724, not those lists (register 1810).

1. [r](#)

2. [r](#)

3. [r](#)